

Werk

Titel: Metrika

Verlag: Physica-Verlag

Jahr: 1997

Kollektion: Mathematica

Digitalisiert: Niedersächsische Staats- und Universitätsbibliothek Göttingen

Werk Id: PPN358794056_0045

PURL: http://resolver.sub.uni-goettingen.de/purl?PPN358794056_0045

Übergeordnetes Werk

Werk Id: PPN358794056

PURL: <http://resolver.sub.uni-goettingen.de/purl?PPN358794056>

Terms and Conditions

The Goettingen State and University Library provides access to digitized documents strictly for noncommercial educational, research and private purposes and makes no warranty with regard to their use for other purposes. Some of our collections are protected by copyright. Publication and/or broadcast in any form (including electronic) requires prior written permission from the Goettingen State- and University Library.

Each copy of any part of this document must contain there Terms and Conditions. With the usage of the library's online system to access or download a digitized document you accept the Terms and Conditions.

Reproductions of material on the web site may not be made for or donated to other repositories, nor may be further reproduced without written permission from the Goettingen State- and University Library.

For reproduction requests and permissions, please contact us. If citing materials, please give proper attribution of the source.

Contact

Niedersächsische Staats- und Universitätsbibliothek Göttingen
Georg-August-Universität Göttingen
Platz der Göttinger Sieben 1
37073 Göttingen
Germany
Email: gdz@sub.uni-goettingen.de

Metrika

Volume 45

1997

International Journal
of Theoretical and Applied Statistics

Editors

URSULA GATHER, Lehrstuhl Mathematische Statistik und industrielle Anwendungen, Universität Dortmund, 44221 Dortmund, Germany

E-mail: gather@omega.statistik.uni-dortmund.de

FRIEDRICH PUKELSHEIM, Institut für Mathematik, Universität Augsburg, 86135 Augsburg, Germany

E-mail: pukelsheim@uni-augsburg.de

Book Review Editor:

HERBERT HEYER, Mathematisches Institut, Universität Tübingen, Auf der Morgenstelle 10, 72076 Tübingen, Germany

Referees are at present:

J. Bartoszewicz, Wrocław; *D. A. Belsley*, Chestnut Hill; *W. Bischoff*, Karlsruhe; *H. Büning*, Berlin; *K. Chaloner*, St. Paul; *J. Chen*, Waterloo; *S.-C. Chow*, Princeton; *R. D. Cook*, St. Paul; *H. A. David*, Ames; *A. M. Dean*, Columbus; *K. Doksum*, Berkeley; *N. R. Draper*, Madison; *R. M. Dudley*, Cambridge; *D. Ferger*, Gießen; *W. Fieger*, Karlsruhe; *E. B. Fosam*, Sheffield; *N. Gaffke*, Magdeburg; *L. Gajek*, Łódź; *J. Galambos*, Philadelphia; *S. Ghosh*, Riverside; *A. Gnedin*, Göttingen; *J. Hartung*, Dortmund; *B. Heiligers*, Magdeburg; *N. Henze*, Karlsruhe; *F. Hering*, Dortmund; *C. Hesse*, Stuttgart; *G. Hübner*, Hamburg; *C. T. Isaki*, Washington; *A. Janssen*, Düsseldorf; *U. Jensen*, Ulm; *W. Kahle*, Magdeburg; *U. Kamps*, Dortmund; *A. C. Kimber*, Surrey; *C. Klüppelberg*, Mainz; *J.-P. Kreiß*, Braunschweig; *J. Kunert*, Dortmund; *L. R. LaMotte*, Baton Rouge; *F. Liese*, Rostock; *D. Majumdar*, Chicago; *L. Mattner*, Hamburg; *K.-J. Mieschke*, Chicago; *D. Morgenstern*, Hemmingen; *S. Morgenthaler*, Lausanne; *C. Müller*, Berlin; *U. Müller-Funck*, Münster; *R. Mukerjee*, Calcutta; *A. Owen*, Stanford; *J. Pawlitschko*, Dortmund; *J. Pfanzagl*, Köln; *I. Pigeot*, München; *D. Plachky*, Münster; *L. Pronzato*, Gif-Sur-Yvette Cedex; *H. Pruscha*, München; *D. Richards*, Charlottesville; *H. Rieder*, Bayreuth; *D. M. Rocke*, Davis; *L. Rüschendorf*, Freiburg i. B.; *S. Schach*, Dortmund; *W. Schmid*, Frankfurt/Oder; *N. Schmitz*, Münster; *R. Schwabe*, Berlin; *T. Sellke*, West Lafayette; *P. K. Sen*, Chapel Hill; *R. Shimizu*, Tokyo; *B. K. Sinha*, Baltimore; *U. Stadtmüller*, Ulm; *P. Stabilecker*, Hamburg; *W. Stute*, Gießen; *G. Székely*, Bowling Green; *G. Trenkler*, Dortmund; *A. Unwin*, Augsburg; *R. Valliant*, Cleveland; *K. Vännman*, Luleå; *E. von Collani*, Würzburg; *V. B. Waikar*, Oxford; *L. A. Wasserman*, Pittsburgh; *S. S. Yang*, Manhattan; *S. Zwanzig*, Hamburg



Physica-Verlag

A Springer-Verlag Company

All Rights Reserved. No part of this publication may be reproduced, stored in a retrieval system or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without the permission of the publisher.

© Physica-Verlag, Tiergartenstraße 17, 69121 Heidelberg, Germany, 1997

Printed in Germany

ISBN 0026-1335

Typesetting: Asco Trade Typesetting Limited, Hong Kong

Printing: Betz-Druck, Darmstadt

Binding: T. Gansert GmbH, Weinheim-Sulzbach



Contents

I. Publications

<i>Emad-Eldin A. A. Aly, Subhash C. Kochar: Change Point Tests Based on U-Statistics with Applications in Reliability</i>	259
<i>A. Arcos Cebrián, M. Rueda García: Variance Estimation Using Auxiliary Information: An Almost Unbiased Multivariate Ratio Estimator.....</i>	171
<i>K. Balasubramanian, Aloke Dey: Distributions Characterized Through Conditional Expectations.....</i>	189
<i>Isha Dewan, B. L. S. Prakasa Rao: Remarks on the Strong Law of Large Numbers for a Triangular Array of Associated Random Variables</i>	225
<i>Manuel Franco, Jose M. Ruiz: On Characterizations of Distributions by Expected Values of Order Statistics and Record Values with Gap</i>	107
<i>Rainer Göb: Tests of Significance for the Average Square Deviation from Target in a Finite Lot.....</i>	131
<i>Arjun K. Gupta, S. Kotz: A New Process Capability Index</i>	213
<i>Norbert Henze: Do Components of Smooth Tests of Fit have Diagnostic Properties?</i>	121
<i>Lynn Roy LaMotte: On Limits of Uniquely Best Linear Estimators.....</i>	197
<i>Shuangzhe Liu, Heinz Neudecker: Experiments with Mixtures: Optimal Allocations for Becker's Models</i>	53
<i>J. P. Morgan: Optimal Design for Interacting Blocks with OAVS Incidence.....</i>	67
<i>N. Mukhopadhyay, A. R. Padmanabhan, T. K. S. Solanky: On Estimating the Reliability After Sequentially Estimating the Mean: The Exponential Case...</i>	235
<i>N. Mukhopadhyay, T. K. S. Solanky: Estimating After Sequential Selection and Ranking.....</i>	95
<i>Daniel Rost: A Note on Compact Differentiability and the δ-Method</i>	39
<i>Rainer Schwabe, Weng Kee Wong: A Relationship between Efficiencies of Marginal Designs and the Product Design</i>	253
<i>M. Luisa Targhetta: A Two-Stage Sampling by Unequal Size Samples from Two Normal Populations with Unknown Variances</i>	31
<i>Axel Tenbusch: Nonparametric Curve Estimation with Bernstein Estimates.....</i>	1

II. Book Reviews

<i>S. Albeverio, W. Luxemburg, M. Wolff (Eds.): Advances in Analysis, Probability and Mathematical Physics (P. A. Loeb)</i>	88
<i>L. Alonso, R. Schott: Random Generation of Trees (H. Prodinger)</i>	272
<i>E. B. Andersen: The Statistical Analysis of Categorical Data (St. E. Fienberg) ...</i>	278
<i>V. S. Borkar: Probability Theory. An Advanced Course (H. Rost).....</i>	275
<i>A. Buja, P. A. Tukey (Eds.): Computing and Graphics in Statistics (P. Huber)....</i>	87
<i>R. J. Carroll, D. Ruppert, L. A. Stefanski: Measurement Error in Nonlinear Models (J. P. Buonaccorsi)</i>	182
<i>M. Grabisch, H. T. Nguyen, E. A. Walker: Fundamentals of Uncertainty Calculi with Applications to Fuzzy Inference (P. E. Kloeden)</i>	93
<i>P. Guttorp: Statistical Inference for Branching Processes (C. C. Heyde).....</i>	85
<i>N. Kazamaki: Continuous Exponential Martingales and BMO (D. L. Burkholder)</i>	89
<i>M. A. Lifshits: Gaussian Random Functions (N. Obata)</i>	179
<i>J. K. Lindsey: Modelling Frequency and Count Data (O. O. Aalen).....</i>	86
<i>J. S. Maritz: Distribution-free Statistical Methods (E. von Collani)</i>	188
<i>A. M. Mathai, S. B. Provost, T. Hayakawa: Bilinear Forms and Zonal Polynomials (Y. Chikuse)</i>	274
<i>N. Mukhopadhyay, T. K. S. Solanky: Multistage Selection and Ranking Procedures (Kh. Alam).....</i>	90
<i>D. Nualart: The Malliavin Calculus and Related Topics (L. Arnolds)</i>	181
<i>A. Pázman: Nonlinear Statistical Models (O. E. Barndorff-Nielsen).....</i>	91
<i>C. R. Rao: Was ist Zufall? Statistik und Wahrheit (J. Stark)</i>	187
<i>H. Rinne: Taschenbuch der Statistik für Wirtschafts- und Sozialwissenschaften (G. Ronning).....</i>	84
<i>M. J. Schervish: Theory of Statistics (H. Witting)</i>	184
<i>D. Stauffer, A. Aharony: Perkolationstheorie. Eine Einführung (H.-O. Georgii).....</i>	271
<i>H. Wackernagel: Multivariate Geostatistics (L. M. Bloom)</i>	185
<i>F. Weisz: Martingale Hardy Spaces and Their Application in Fourier Analysis (P. Imkeller).....</i>	279

Nonparametric Curve Estimation With Bernstein Estimates

AXEL TENBUSCH

Fachbereich Mathematik/Informatik der Universität, Albrechtstraße 28, 49069 Osnabrück, Germany

Abstract: In this paper we propose a Bernstein type estimate of the regression function $m(x) = E[Y|X = x]$. Various local and global asymptotic properties of this estimate are studied.

Key Words: Nonparametric regression estimation, smoothed regressograms, Bernstein polynomials.

1 Introduction to Nonparametric Regression Function Estimation

Within the past thirty years extended investigations have been devoted to the problem of regression function estimation. Assume that (X, Y) is a $\mathbb{R}^d \times \mathbb{R}$ -valued random vector. If Y has finite expectation $E(Y)$, the regression function $m(x) = E(Y|X = x)$, $x \in \mathbb{R}$, of Y on X exists and is (P_X almost surely in x) uniquely defined in view of the equation $m(X) = E(Y|X)$. In nonparametric regression function estimation one is provided with a sequence $(X_1, Y_1), (X_2, Y_2), \dots, (X_N, Y_N)$ of independent random observations with the same distribution as (X, Y) . It is required to construct an estimate $m_N(x)$ – a function of x and the data $(X_1, Y_1), \dots, (X_N, Y_N)$ – of $m(x)$, which behaves well even when only little information on the distribution of (X, Y) is available.

The most frequently used approach involves kernel functions. The kernel estimate

$$m_N^*(x) = \frac{\sum_{i=1}^N Y_i K((X_i - x)/h_N)}{\sum_{i=1}^N K((X_i - x)/h_N)} \quad (1)$$

was first proposed by Nadaraya [1964] and Watson [1964]. In (1) h_N is a sequence of positive numbers and $K \geq 0$ is an integrable function on $\mathbb{R}^d$. A variety of properties is known for m_N^* . The pointwise consistency of (1) was studied by Watson [1964], Nadaraya [1964, 1965], Rosenblatt [1969], Schuster [1972], Greblicki [1974], and Noda [1976]. The uniform consistency is treated in the papers by Nadaraya [1965, 1970], Greblicki [1974], and Devroye [1978]. Schuster [1972] showed under suitable assumptions that

$$(Nh_N)^{1/2} [m_N^*(x) - m(x)] \rightarrow \mathcal{N}(0, \sigma^2) \quad \text{in distribution ,}$$

where $\sigma^2 = \text{Var}(Y|X = x) \int K^2(u) du / g(x)$. Here g denotes the (marginal) density of X . See also Rosenblatt [1969]. In fact, even joint convergence in distribution of m_N^* at finitely many points $x_1, \dots, x_k$ holds, with $m_N^*(x_1), \dots, m_N^*(x_k)$ being asymptotically independent. The quantity

$$U_N = E \left(\int (m_N^*(x) - m(x))^2 g_N(x) h(x) dx \right)$$

was studied by Nadaraya [1974] and Konakov [1977]. Here $g_N(x)$ is the usual kernel density estimate of the density $g(x)$ of the random variable X and $h(x)$ is a weight function. The convergence in probability and the almost sure convergence of

$$\int |m_N^*(x) - m(x)| \mu(dx) ,$$

– here μ is the probability measure of X – to 0 was studied by Devroye and Wagner [1980a]. Stone [1977] has shown that a large class of regression function estimates including the nearest neighbor estimates satisfy

$$E \left(\int |m_N^*(x) - m(x)|^p \mu(dx) \right) \rightarrow 0$$

for all possible distributions of (X, Y) with $E(|Y|) < \infty$. For the kernel estimate, this was shown by Devroye and Wagner [1980b].

The kernel estimate is an intuitive estimate of a regression function. At each point x_0 the estimate of $m(x_0)$ is a weighted average of the sample Y_i with observations close to x_0 receiving the largest weights according to the kernel K . Unfortunately this simplicity has a drawback. At the boundary of the predictor space the kernel estimate may have substantial bias. A variety of kernel modifications have been proposed, see for example Gasser and Müller [1979], Rice [1984], and Müller [1991]. For an extensive discussion of various aspects of different kernel regression estimates see the articles of Chu and Marron [1991] and of Hastie and Loader [1993].

As an excellent reference for properties of kernel regression estimates we refer the reader to Nadaraya [1988]. Further see the monographs by Eubank [1988], Müller [1988], Härdle [1990], Wahba [1990] and Michels [1992] for a large variety of interesting details of nonparametric regression.

2 A Bernstein Nonparametric Regression Function Estimate

For $d = 1$, Vitale [1973] proposed an estimate of a univariate density function based on Bernstein polynomials, motivated by the classical theorem of Weierstraß on approximation of continuous functions by means of Bernstein polynomials. With a slight modification of the scaling parameter $\left(\frac{n}{N} \text{ instead of } \frac{n+1}{N}\right)$ this estimator is a special case of the smoothed histogram estimates developed by Gawronski and Stadtmüller [1981] for densities f concentrated on $[0, 1]$, $(-\infty, +\infty)$, and $[0, +\infty)$. For $d > 1$ Bernstein density estimates were introduced by Tenbusch [1994a]. For the Bernstein estimate a variety of properties are known. The pointwise convergence in the mean square is treated by Vitale [1973], Gawronski and Stadtmüller [1980, 1981], and by Tenbusch [1994a]. Uniform consistency properties and central limit theorems were investigated by Stadtmüller [1983] and by Tenbusch [1994a]. Consistency results in form of various strong laws were obtained by Gawronski [1985]. $L_1[0, 1]^d$ -properties of Bernstein estimates were studied by Tenbusch [1994b]. The object of this paper is to show that the same ideas can be used to obtain plausible estimates of regression function in the case of sampling from a multivariate distribution. For the nonparametric estimation of regression function with a one-dimensional design parameter this was done by Stadtmüller [1986]. He considered the following model:

$$Y_i = m(t_i) + \varepsilon_i \quad 1 \leq i \leq N \quad 0 \leq t_i \leq 1 ,$$

where $Y_1, \dots, Y_N$ are N observations at fixed but unordered design variables $t_1, \dots, t_n$, the ε_i are random variables which satisfy

$$E(\varepsilon_i) = 0 \quad E(\varepsilon_i \varepsilon_j) = \sigma_\varepsilon^2 \delta_{ij} \quad 1 \leq i, j \leq N$$

and $m(\cdot)$ is some smooth function defined on $[0, 1]$.

In this paper we present a regression function estimate based on Bernstein polynomials for the stochastic regression model. For the sake of exposition we shall consider sampling from a bivariate distribution. The assumptions, estimates, and conclusions are similar when the independent variable is d -dimensional ($d \geq 2$). Let (X, Y) be a two-dimensional random variable with density function $f(x, y)$ and let $g(x)$ be the marginal density function – concentrated on the interval $[0, 1]$ – of the random variable X . Let $E(|Y|) < \infty$ and denote the regression function of Y on X by $m(x)$, i.e. $m(x) = E(Y|X = x)$. Let $(X_1, Y_1), \dots, (X_N, Y_N)$ be a sample of size N (independent observations) from the population (X, Y) . In many statistical problems one does not know the functional form of $m(x)$. Our objective is to obtain an estimate of the regression function $m(x)$. As

an approximation to the regression curve we propose the following estimate:

$$m_{nN}(x) = \begin{cases} \frac{\sum_{i=1}^N \sum_{j=0}^n Y_i 1_{A_{n,j}}(X_i) p_{n,j}(x)}{\sum_{i=1}^N \sum_{j=0}^n 1_{A_{n,j}}(X_i) p_{n,j}(x)} & \text{if } \sum_{j=0}^n 1_{A_{n,j}}(X_i) p_{n,j}(x) \neq 0 , \\ 0 & \text{if } \sum_{j=0}^n 1_{A_{n,j}}(X_i) p_{n,j}(x) = 0 , \end{cases} \quad (2)$$

where as before $n = n(N)$ is a sequence of positive integers, $A_{n,j} = \{x \in [0, 1]: j/(n+1) \leq x < (j+1)/(n+1)\}$, $0 \leq j < n$, $A_{n,n} = \{x \in [0, 1]: j/(n+1) \leq x \leq (j+1)/(n+1)\}$, and $p_{n,j}(x) = \binom{n}{j} x^j (1-x)^{n-j}$.

Define

$$\varphi_{nN}(x) = \frac{n+1}{N} \sum_{i=1}^N \sum_{j=0}^n Y_i 1_{A_{n,j}}(X_i) p_{n,j}(x) ,$$

$$g_{nN}(x) = \frac{n+1}{N} \sum_{i=1}^N \sum_{j=0}^n 1_{A_{n,j}}(X_i) p_{n,j}(x) ,$$

$$\varphi(x) = \int y f(t, y) dy , \quad V(x) = \int y^2 f(t, y) dy ,$$

$$W(x) = \int |y|^{2+\delta} f(t, y) dy .$$

Note that $g_{nN}(x)$ can also be written as

$$\frac{n+1}{N} \sum_{j=0}^n m_{n,j}^N p_{n,j}(x) , \quad (3)$$

where $m_{n,j}^N$ is the number of X_i 's in $A_{n,j}$, i.e. the density estimate introduced by Vitale [1973]. We further note that

$$\text{Var}(Y|X=x) = V(x)/g(x) - \varphi^2(x)/g^2(x) .$$

Like the kernel smoother the Bernstein estimate is an intuitive estimate of a regression function or conditional expectation. At each point x_0 the estimate is a weighted mean of the sample Y_i . But contrary to the kernel estimate the Bernstein estimate uses not always the same weight function (kernel). The Bernstein estimate rather adjusts the weight function to the positions of the X_i 's in the predictor space. This adjustment of the weight function amounts in better

boundary behavior of the Bernstein estimate compared with the Nadaraya-Watson estimate. The objective of this paper is to show that for the Bernstein estimate we can achieve results comparable to the kernel estimate.

In Section 3 conditions are studied under which the estimate (2) is consistent (Theorem 3.1) and asymptotically normal (Theorem 3.3). Further the behavior of the bias is investigated (Theorem 3.2). Theorem 3.4 presents an asymptotic expression for $m_{nN}(x)$.

In Section 4 the quantity

$$U_{nN} = E(\int (m_{nN}(x) - m(x))^2 g_{nN}(x) h(x) dx)$$

is introduced. This quantity is used as a global measure of the quality of our estimate $m_{nN}(x)$. Here $g_{nN}(x)$ is an estimate of the type (3) of the density $g(x)$ of the random variable X and $h(x)$ is a weight function. The rate of convergence of U_{nN} towards 0 is also obtained (Theorem 4.1).

Further the distance in $L_1[0, 1]$

$$I_{nN} = \int_0^1 |m_{nN}(x) - m(x)| \mu(dx)$$

is introduced. Theorem 4.2 states the weak and strong consistency to 0 of I_{nN} for the estimate (2).

Finally, the deviation of estimate (2) from $m(x)$ in the uniform metric is studied. Sufficient conditions (Theorem 4.3) for a.s. convergence in $C[0, 1]$ of estimate (2) to $m(x)$ are presented.

3 Local Asymptotic Properties of Bernstein Regression Function Estimates

Our objective is to describe the asymptotic behavior of the Bernstein estimate (2). We will start with some local properties of our estimate.

Theorem 3.1: Let $E(Y^2) < \infty$, $n(N) \rightarrow \infty$, and $\frac{(n+1)^{1/2}}{N} \rightarrow 0$. Further let $g(x)$ and $V(x)$ be bounded on $[0, 1]$. Then for any $x \in (0, 1)$ for which $m(x)$, $g(x)$, and $V(x)$ are continuous at x and $g(x) > 0$, $m_{nN}(x)$ is a consistent estimate of $m(x)$. In order to obtain that $m_{nN}(x)$ is a consistent estimate at the endpoints of the interval $[0, 1]$ we have to require that $\frac{n+1}{N} \rightarrow 0$.

Proof: Since $g_{nN}(x)$ is a consistent estimate of $g(x)$ (see Vitale [1973]) it is sufficient to show that $\varphi_{nN}(x)$ is a consistent estimate of $\varphi(x)$. Let us verify that

$$E((\varphi_{nN}(x) - \varphi(x))^2) \rightarrow 0 \quad \text{as } N \rightarrow \infty .$$

We have

$$\begin{aligned} E(\varphi_{nN}(x)) &= (n+1) \sum_{j=0}^n E(Y 1_{A_{n,j}}(X)) p_{n,j}(x) \\ &= (n+1) \sum_{j=0}^n \int_0^1 \int_0^1 y 1_{A_{n,j}}(t) f(t, y) dt dy p_{n,j}(x) \\ &= (n+1) \sum_{j=0}^n \int_0^1 E(Y 1_{A_{n,j}}(X) | X = t) g(t) dt p_{n,j}(x) \\ &= (n+1) \sum_{j=0}^n \int_0^1 1_{A_{n,j}}(t) m(t) g(t) dt p_{n,j}(x) \\ &= (n+1) \sum_{j=0}^n \int_0^1 1_{A_{n,j}}(t) \varphi(t) dt p_{n,j}(x) \\ &= (n+1) \sum_{j=0}^n \int_{j/(n+1)}^{(j+1)/(n+1)} \varphi(t) dt p_{n,j}(x) . \end{aligned} \tag{4}$$

Since $\varphi(t)$ is integrable and continuous at the point x it follows from Theorem 2.1.1. of Lorentz [1986] that

$$E(\varphi_{nN}(x)) \rightarrow \varphi(x) \quad \text{as } N \rightarrow \infty . \tag{5}$$

Since

$$\begin{aligned} E(\varphi_{nN}^2(x)) &= \frac{(n+1)^2}{N^2} E \left(\sum_{i=1}^N \sum_{j=0}^n \sum_{k=0}^n Y_i^2 1_{A_{n,j}}(X_i) 1_{A_{n,k}}(X_i) p_{n,j}(x) p_{n,k}(x) \right) \\ &\quad + \frac{(n+1)^2}{N^2} E \left(\sum_{i=1}^N \sum_{l \neq i}^N \sum_{j=0}^n \sum_{k=0}^n Y_i Y_l 1_{A_{n,j}}(X_i) 1_{A_{n,k}}(X_l) p_{n,j}(x) p_{n,k}(x) \right) \\ &= \frac{(n+1)^2}{N} \sum_{j=0}^n E(Y^2 1_{A_{n,j}}(X)) p_{n,j}^2(x) \\ &\quad + \frac{(n+1)^2(N-1)}{N} \left(\sum_{j=0}^n E(Y 1_{A_{n,j}}(X)) p_{n,j}(x) \right)^2 , \end{aligned}$$

we have on the other hand

$$\begin{aligned}
 E((\varphi_{nN}(x) - E(\varphi_{nN}(x)))^2) &= \frac{(n+1)^2}{N} \sum_{j=0}^n E(Y^2 1_{A_{n,j}}(X)) p_{n,j}^2(x) \\
 &\quad - \frac{1}{N} E^2([\varphi_{nN}(x)]) \\
 &\sim \frac{(n+1)^2}{N} \sum_{j=0}^n \int_0^1 \int_0^1 y^2 1_{A_{n,j}}(t) f(t, y) dt dy p_{n,j}^2(x) \\
 &= \frac{(n+1)^2}{N} \sum_{j=0}^n \int_0^1 1_{A_{n,j}}(t) V(t) dt p_{n,j}^2(x) .
 \end{aligned}$$

Now, with the same arguments as used by Vitale [1973] to bound the variance term we get that the last term is of order

$$\sim \frac{(n+1)^{1/2}}{N} \frac{V(x)}{2(\pi x(1-x))^{1/2}} \quad (6)$$

for $x \in (0, 1)$, and if $x = 0$ we have

$$\begin{aligned}
 E((\varphi_{nN}(0) - E(\varphi_{nN}(0)))^2) &\sim \frac{(n+1)^2}{N} E(Y^2 1_{A_{n,0}}(X)) \\
 &= \frac{n+1}{N} \left((n+1)^{1/(n+1)} \int_0^1 V(t) dt \right) \\
 &\sim \frac{n+1}{N} V(0) . \quad (7)
 \end{aligned}$$

The case $x = 1$ is treated similarly:

$$E((\varphi_{nN}(1) - E(\varphi_{nN}(1)))^2) \sim \frac{n+1}{N} V(1) . \quad (8)$$

Relations (5), (6), (7), and (8) together with the assumptions of the theorem imply that

$$E((\varphi_{nN}(x) - \varphi(x))^2) \rightarrow 0 \quad \text{as } N \rightarrow \infty .$$

Thus Theorem 3.1 is proved. $\square$

For the proof of the next theorem we need the following lemma, which presents some asymptotic relations for the $p_{n,j}(x)$.

Part (i) was proved by Lorentz [1986] and part (ii) by Vitale [1973]. Part (iii) and (iv) follow along the same line.

Lemma 3.1: For fixed $0 < x < 1$ and $\alpha > 1/3$ we have for $\left| \frac{j}{n} - x \right| \leq n^{-\alpha}$:

(i)

$$p_{n,j}(x) \sim \left(\frac{n}{2\pi x(1-x)} \right)^{1/2} \int_{j/n}^{(j+1)/n} \exp \left[-\frac{n}{2x(1-x)}(u-x)^2 \right] du$$

(ii)

$$p_{n,j}^2(x) \sim \frac{1}{2\pi x(1-x)} \int_{j/n}^{(j+1)/n} \exp \left[-\frac{n}{x(1-x)}(u-x)^2 \right] du$$

(iii)

$$p_{n,j}^3(x) \sim \frac{1}{(2\pi x(1-x))^{3/2} n^{1/2}} \int_{j/n}^{(j+1)/n} \exp \left[-\frac{3n}{2x(1-x)}(u-x)^2 \right] du$$

(iv)

$$p_{n,j}^4(x) \sim \frac{1}{(2\pi x(1-x))^2 n} \int_{j/n}^{(j+1)/n} \exp \left[-\frac{2n}{x(1-x)}(u-x)^2 \right] du$$

Theorem 3.2: Suppose that $g(x)$ and $V(x)$ are continuous at x and bounded on $[0, 1]$.

(i) *If $|Y| \leq C_1 < \infty$ a.s., $n(N) \rightarrow \infty$, and $N^{-1}(n+1) \rightarrow 0$, and $g(x) > 0$ then for $x \in (0, 1)$*

$$E[m_{nN}(x)] = E[\varphi_{nN}(x)]/E[g_{nN}(x)] + O(N^{-1}(n+1)^{1/2}) ,$$

and for $x = 0$ or $x = 1$

$$E[m_{nN}(x)] = E[\varphi_{nN}(x)]/E[g_{nN}(x)] + O(N^{-1}(n+1)) ;$$

(ii) if $E(Y^2) < \infty$, $n(N) \rightarrow \infty$, and $N^{-1}(n+1) \rightarrow 0$, and $g(x) > 0$ then for $x \in (0, 1)$

$$E[m_{nN}(x)] = E[\varphi_{nN}(x)]/E[g_{nN}(x)] + O(N^{-1/2}(n+1)^{1/2}) ,$$

and for $x = 0$ or $x = 1$

$$E[m_{nN}(x)] = E[\varphi_{nN}(x)]/E[g_{nN}(x)] + O(N^{-1/2}(n+1)) .$$

Note that assertions (i) and (ii) imply that $m_{nN}(x)$ is an asymptotically unbiased estimate of $m(x)$ for any x at which $\varphi(x)$ is continuous.

Proof: Using the identity

$$1/z = 1/w - w^{-2}(z - w) + w^{-2}z^{-1}(z - w)^2 , \quad w \neq 0 , \quad z \neq 0 ,$$

with $z = g_{nN}(x)$ and $w = E[g_{nN}(x)]$ we obtain

$$\begin{aligned} E(m_{nN}(x)) &= \frac{E(\varphi_{nN}(x))}{E(g_{nN}(x))} - \frac{E([\varphi_{nN}(x) - E(\varphi_{nN}(x))][g_{nN}(x) - E(g_{nN}(x))])}{(E(g_{nN}(x)))^2} \\ &\quad + E\left(\frac{\varphi_{nN}(x)[g_{nN}(x) - E(g_{nN}(x))]^2}{g_{nN}(x)(E(g_{nN}(x)))^2}\right) \\ &= \frac{E(\varphi_{nN}(x))}{E(g_{nN}(x))} + (E(g_{nN}(x)))^{-2}[\alpha_{nN}^{(1)}(x) + \alpha_{nN}^{(2)}(x)] . \end{aligned} \quad (9)$$

We will first prove (i). Using the inequality of Schwarz we obtain

$$\alpha_{nN}^{(1)}(x) \leq (E([\varphi_{nN}(x) - E(\varphi_{nN}(x))]^2))^{1/2} (E([g_{nN}(x) - E(g_{nN}(x))]^2))^{1/2} .$$

Note that

$$Var(\varphi_{nN}(x)) \sim \begin{cases} \frac{(n+1)^{1/2}}{N} \frac{V(x)}{2(\pi x(1-x))^{1/2}} & x \in (0, 1) \\ \frac{n+1}{N} V(x) & x = 0 \text{ or } x = 1 \end{cases}$$

(see the proof of Theorem 3.1) and that

$$Var(g_{nN}(x)) \sim \begin{cases} \frac{(n+1)^{1/2}}{N} \frac{g(x)}{2(\pi x(1-x))^{1/2}} & x \in (0, 1) \\ \frac{n+1}{N} g(x) & x = 0 \text{ or } x = 1 \end{cases}$$

(see Vitale [1973]).

The above two relations imply that

$$\alpha_{nN}^{(1)}(x) \sim \begin{cases} O\left(\frac{(n+1)^{1/2}}{N}\right) & x \in (0, 1) \\ O\left(\frac{(n+1)}{N}\right) & x = 0 \text{ or } x = 1 \end{cases} . \quad (10)$$

Since the distribution of Y is bounded $\left(\left|\frac{\varphi_{nN}(x)}{g_{nN}(x)}\right|\right) \leq C_1$ we obtain

$$|\alpha_{nN}^{(2)}(x)| \leq C_1 E([g_{nN}(x) - E(g_{nN}(x))]^2) . \quad (11)$$

It follows that

$$\begin{aligned} & C_1 E([g_{nN}(x) - E(g_{nN}(x))]^2) \\ & \sim \begin{cases} C_1 \frac{(n+1)^{1/2}}{N} \frac{g(x)}{2(\pi x(1-x))^{1/2}} & x \in (0, 1) \\ C_1 \frac{n+1}{N} g(x) & x = 0 \text{ or } x = 1 \end{cases} . \end{aligned} \quad (12)$$

Relations (9), (10), (11), and (12) yield assertion (i).

To prove (ii) note that relation (10) remains valid also in case (ii), i.e. when Y is not bounded. Next observe that

$$\begin{aligned} |\alpha_{nN}^{(2)}(x)| &\leq E(\max\{|Y_1|, \dots, |Y_N|\} |g_{nN}(x) - E(g_{nN}(x))|^2) \\ &\leq \left(\sum_{i=1}^N E(Y_i^2) \right)^{1/2} (E[g_{nN}(x) - E(g_{nN}(x))]^4)^{1/2} \\ &= N^{1/2} (E(Y^2))^{1/2} (E[g_{nN}(x) - E(g_{nN}(x))]^4)^{1/2} . \end{aligned}$$

In order to prove (ii) we have to bound $(E([g_{nN}(x) - E(g_{nN}(x))]^4))^{1/2}$. First, note that

$$\begin{aligned} E([g_{nN}(x) - E(g_{nN}(x))]^4) &= E(g_{nN}^4(x)) - 4E(g_{nN}^3(x))E(g_{nN}(x)) \\ &\quad + 6E(g_{nN}^2(x))[E(g_{nN}(x))]^2 - 3[E(g_{nN}(x))]^4 . \end{aligned}$$

With some manipulations we get

$$\begin{aligned} &E([g_{nN}(x) - E(g_{nN}(x))]^4) \\ &= \frac{(n+1)^4}{N^3} \sum_{j=0}^n E(1_{A_{n,j}}(X)) p_{n,j}^4(x) \\ &\quad - 4 \frac{(n+1)^4}{N^3} \left(\sum_{j=0}^n E(1_{A_{n,j}}(X)) p_{n,j}^3(x) \right) \left(\sum_{j=0}^n E(1_{A_{n,j}}(X)) p_{n,j}(x) \right) \\ &\quad + 3 \frac{(n+1)^4}{N^3} (N-1) \left(\sum_{j=0}^n E(1_{A_{n,j}}(X)) p_{n,j}^2(x) \right) \left(\sum_{j=0}^n E(1_{A_{n,j}}(X)) p_{n,j}^2(x) \right) \\ &\quad + \frac{(n+1)^4}{N^3} (12-6N) \left(\sum_{j=0}^n E(1_{A_{n,j}}(X)) p_{n,j}^2(x) \right) \left(\sum_{j=0}^n E(1_{A_{n,j}}(X)) p_{n,j}(x) \right)^2 \\ &\quad + \frac{(n+1)^4}{N^3} (5N-6) \left(\sum_{j=0}^n E(1_{A_{n,j}}(X)) p_{n,j}(x) \right)^4 \\ &= I + II + III + IV + V . \end{aligned} \tag{13}$$

Our assumptions on g are sufficient to conclude that (see Vitale [1973] or Tenbusch [1994a])

$$\begin{aligned}
E([g_{nN}(x) - E(g_{nN}(x))]^4) &= O\left(\frac{1}{N^2}\right) + O\left(\frac{(n+1)^{3/2}}{N^3}\right) + O\left(\frac{(n+1)}{N^3}\right) \\
&\quad + O\left(\frac{(n+1)}{N^2}\right) + O\left(\frac{(n+1)^{1/2}}{N^2}\right) \\
&= O\left(\frac{(n+1)}{N^2}\right)
\end{aligned}$$

Thus,

$$\alpha_{nN}^{(2)}(x) = O\left(\frac{(n+1)^{1/2}}{N^{1/2}}\right) \quad (14)$$

So far we have assumed that $x \in (0, 1)$. Now we look at the endpoints of the interval $[0, 1]$ and start with the point 0 ($x = 1$ is treated similarly). Note that (13) reduces to

$$\begin{aligned}
E([g_{nN}(0) - E(g_{nN}(0))]^4) &= \frac{(n+1)^4}{N^3} E(1_{A_{n,0}}(X)) \\
&\quad - 4 \frac{(n+1)^4}{N^3} (E(1_{A_{n,j}}(X)))^2 \\
&\quad + 3 \frac{(n+1)^4}{N^3} (N-1) (E(1_{A_{n,j}}(X)))^2 \\
&\quad + \frac{(n+1)^4}{N^3} (12-6N) (E(1_{A_{n,j}}(X)))^3 \\
&\quad + \frac{(n+1)^4}{N^3} (5N-6) (E(1_{A_{n,j}}(X)))^4 \\
&= I(0) + II(0) + III(0) + IV(0) + V(0) . \quad (15)
\end{aligned}$$

Using now that $(n+1)E(1_{A_{n,0}}(X)) \sim g(0)$ we have

$$I(0) \sim \frac{(n+1)^3}{N^3} g(0) , \quad (16)$$

$$II(0) \sim (-4) \frac{(n+1)^2}{N^2} g^2(0) , \quad (17)$$

$$III(0) \sim 3 \frac{(n+1)^2}{N^2} g^2(0) , \quad (18)$$

$$IV(0) \sim \frac{(n+1)}{N^2} g^3(0) , \quad (19)$$

$$V(0) \sim \frac{1}{N^2} g^4(0) . \quad (20)$$

Combining relations (16), (17), (18), (19), and (20) we can conclude that

$$\alpha_{nN}^2(0) = O\left(\frac{n+1}{N^{1/2}}\right) . \quad (21)$$

Relations (10), (14), and (21) yield assertion (ii). This completes the proof of Theorem 3.2. $\square$

Theorem 3.3: Let $E(|Y|^{2+\delta}) < \infty$ for some $\delta > 0$. Set

$$W(x) = \int |y|^{2+\delta} f(x, y) dy .$$

Assume that $V(x)$ and $W(x)$ are continuous and let $\phi''(x)$ and $g''(x)$ exist. Then if

$$\frac{N}{(n+1)^{5/2}} \rightarrow 0, \frac{(n+1)^{3/2}}{N} \rightarrow 0, \text{ for example one may take } n(N) = N^\alpha, \alpha \in (2/5, 2/3),$$

$$\frac{N^{1/2}}{(n+1)^{1/4}} (m_{nN}(x) - m(x))$$

is asymptotically $N(0, \sigma^2(x))$ -distributed where

$$\sigma^2(x) = \frac{\text{Var}(Y|X=x)}{2g(x)\sqrt{\pi x(1-x)}} .$$

Proof: We have

$$\begin{aligned} \frac{N^{1/2}}{(n+1)^{1/4}}(m_{nN}(x) - m(x)) &= \frac{N^{1/2}}{(n+1)^{1/4}} \left(m_{nN}(x) - \frac{E(\varphi_{nN}(x))}{E(g_{nN}(x))} \right) \\ &\quad + \frac{N^{1/2}}{(n+1)^{1/4}} \left(\frac{E(\varphi_{nN}(x))}{E(g_{nN}(x))} - m(x) \right). \end{aligned} \quad (22)$$

Let us start with the second term on the r.h.s. of (22). Using (4) and our assumptions on φ and g we arrive at

$$E(\varphi_{nN}(x)) = \varphi(x) + \frac{1}{n+1} \left[\frac{\varphi'(x)(1-2x) + \varphi''(x)x(1-x)}{\alpha} \right] + o\left(\frac{1}{n+1}\right),$$

if $x \in (0, 1)$.

If $x = 0$ or $x = 1$ we get

$$E(\varphi_{nN}(x)) = \varphi(x) + \frac{\varphi'(x)}{2(n+1)} + o\left(\frac{1}{n+1}\right).$$

Thus

$$E(\varphi_{nN}(x)) = \varphi(x) + O\left(\frac{1}{n+1}\right). \quad (23)$$

Analogously,

$$E(g_{nN}(x)) = g(x) + O\left(\frac{1}{n+1}\right). \quad (24)$$

Relations (23) and (24) together with the assumption $\frac{N}{(n+1)^{5/2}} \rightarrow 0$ imply that

$$\frac{N^{1/2}}{(n+1)^{1/4}} \left(\frac{E(\varphi_{nN}(x))}{E(g_{nN}(x))} - m(x) \right) \rightarrow 0. \quad (25)$$

Hence, in order to prove Theorem 3.3 it is sufficient to show that

$$\frac{N^{1/2}}{(n+1)^{1/4}} \left(m_{nN}(x) - \frac{E(\varphi_{nN}(x))}{E(g_{nN}(x))} \right)$$

is asymptotically $N(0, \sigma^2(x))$ -distributed.

We have

$$m_{nN}(x) - E(\varphi_{nN}(x))/E(g_{nN}(x)) = Z_{nN}(x)g_{nN}(x)^{-1}[E(g_{nN}(x))]^{-1} , \quad (26)$$

where

$$Z_{nN}(x) = \frac{n+1}{N} \sum_{i=1}^N \sum_{j=0}^n (\beta_{nN}^{(1)}(x) Y_i - \beta_{nN}^{(2)}(x)) 1_{A_{n,j}}(X_i) p_{n,j}(x) ,$$

and moreover

$$E(Z_{nN}(x)) = 0 , \quad \beta_{nN}^{(1)}(x) = E(g_{nN}(x)) \rightarrow g(x) , \quad \text{and}$$

$$\beta_{nN}^{(2)}(x) = E(\varphi_{nN}(x)) \rightarrow \varphi(x) .$$

Next, we have

$$\begin{aligned} \text{Var}(Z_{nN}(x)) &= \left(\frac{n+1}{N} \right)^2 \sum_{i=1}^N \text{Var} \left(\sum_{j=0}^n (\beta_{nN}^{(1)}(x) Y_i - \beta_{nN}^{(2)}(x)) 1_{A_{n,j}}(X_i) p_{n,j}(x) \right) \\ &= \frac{(n+1)^2}{N} \left[E \left(\sum_{j=0}^n (\beta_{nN}^{(1)}(x) Y - \beta_{nN}^{(2)}(x))^2 1_{A_{n,j}}(X) p_{n,j}^2(x) \right) \right. \\ &\quad \left. - \left(E \left(\sum_{j=0}^n (\beta_{nN}^{(1)}(x) Y - \beta_{nN}^{(2)}(x)) 1_{A_{n,j}}(X) p_{n,j}(x) \right) \right)^2 \right] \\ &= \frac{(n+1)^2}{N} \sum_{j=0}^n \left[(\beta_{nN}^{(1)}(x))^2 \int_{j/(n+1)}^{(j+1)/(n+1)} V(t) dt p_{n,j}^2(x) \right. \\ &\quad \left. - 2\beta_{nN}^{(1)}(x)\beta_{nN}^{(2)}(x) \int_{j/(n+1)}^{(j+1)/(n+1)} \varphi(t) dt p_{n,j}^2(x) \right. \\ &\quad \left. + (\beta_{nN}^{(2)}(x))^2 \int_{j/(n+1)}^{(j+1)/(n+1)} g(t) dt p_{n,j}^2(x) \right] \end{aligned}$$

$$\begin{aligned}
& - \frac{(n+1)^2}{N} \left(\sum_{j=0}^n \left[\beta_{nN}^{(1)}(x) \int_{j/(n+1)}^{(j+1)/(n+1)} \varphi(t) dt p_{n,j}(x) \right. \right. \\
& \quad \left. \left. - \beta_{nN}^{(2)}(x) \int_{j/(n+1)}^{(j+1)/(n+1)} g(t) dt p_{n,j}(x) \right] \right)^2 \\
& \sim \frac{(n+1)^2}{N} \sum_{j=0}^n \left[(\beta_{nN}^{(1)}(x))^2 \int_{j/(n+1)}^{(j+1)/(n+1)} V(t) dt p_{n,j}^2(x) \right. \\
& \quad - 2\beta_{nN}^{(1)}\beta_{nN}^{(2)}(x) \int_{j/(n+1)}^{(j+1)/(n+1)} \varphi(t) dt p_{n,j}^2(x) \\
& \quad \left. + (\beta_{nN}^{(2)}(x))^2 \int_{j/(n+1)}^{(j+1)/(n+1)} g(t) dt p_{n,j}^2(x) \right].
\end{aligned}$$

Thus, in view of Lemma 3.1 and the proofs of Theorem 3.1 and Theorem 3.2 we can conclude that

$$\frac{N}{(n+1)^{1/2}} \text{Var}(Z_{nN}(x)) \rightarrow v(x) = \frac{g^3(x)}{2(\pi x(1-x))^{1/2}} \text{Var}(Y|X=x) . \quad (27)$$

We shall proceed by proving that $Z_{nN}(x)/\sqrt{\text{Var}(Z_{nN}(x))}$ is asymptotically $N(0, 1)$ -distributed. For this purpose we use the central limit theorem of Ljapunov. Therefore it is sufficient to show that

$$L_{nN}(x) = \frac{E(|z_{nN}(x)|^{2+\delta})}{N^{\delta/2}(\text{Var}[z_{nN}(x)])^{1+\delta/2}} \rightarrow 0 , \quad \text{for some } \delta > 0 , \quad (28)$$

where

$$z_{nN}(x) = (n+1) \sum_{j=0}^n (\beta_{nN}^{(1)}(x) Y - \beta_{nN}^{(2)}(x)) 1_{A_{n,j}}(X) p_{n,j}(x) .$$

Condition (28) is fulfilled. Indeed

$$\begin{aligned}
& E(|z_{nN}(x)|^{2+\delta}) \\
& \leq (n+1)^{2+\delta} E \left(\left| (\beta_{nN}^{(1)}(x) Y - \beta_{nN}^{(2)}(x)) \sum_{j=0}^n 1_{A_{n,j}}(X) p_{n,j}(x) \right|^{2+\delta} \right)
\end{aligned}$$

$$\begin{aligned}
&\leq 2^{1+\delta}(n+1)^{2+\delta} E \left(\left| \beta_{nN}^{(1)}(x) Y \sum_{j=0}^n 1_{A_{n,j}}(X) p_{n,j}(x) \right|^{2+\delta} \right) \\
&\quad + 2^{1+\delta}(n+1)^{2+\delta} E \left(\left| \beta_{nN}^{(2)}(x) \sum_{j=0}^n 1_{A_{n,j}}(X) p_{n,j}(x) \right|^{2+\delta} \right) \\
&\leq 2^{1+\delta}(n+1)^{2+\delta} E \left(|\beta_{nN}^{(1)}(x) Y|^{2+\delta} \left[\sum_{j=0}^n 1_{A_{n,j}}(X) p_{n,j}(x) \right]^2 \right) \\
&\quad + 2^{1+\delta}(n+1)^{2+\delta} E \left(|\beta_{nN}^{(2)}(x)|^{2+\delta} \left[\sum_{j=0}^n 1_{A_{n,j}}(X) p_{n,j}(x) \right]^2 \right) \\
&\leq 2^{1+\delta}(n+1)^{2+\delta} (\beta_{nN}^{(1)}(x))^{2+\delta} \sum_{j=0}^n \int_{j/(n+1)}^{(j+1)/(n+1)} |y|^{2+\delta} f(t, y) dt dy p_{n,j}^2(x) \\
&\quad + 2^{1+\delta}(n+1)^{2+\delta} (\beta_{nN}^{(2)}(x))^{2+\delta} \sum_{j=0}^n \int_{j/(n+1)}^{(j+1)/(n+1)} f(t, y) dt dy p_{n,j}^2(x) \\
&\leq 2^{1+\delta}(n+1)^{2+\delta} (\beta_{nN}^{(1)}(x))^{2+\delta} \sum_{j=0}^n \int_{j/(n+1)}^{(j+1)/(n+1)} W(t) dt p_{n,j}^2(x) \\
&\quad + 2^{1+\delta}(n+1)^{2+\delta} (\beta_{nN}^{(2)}(x))^{2+\delta} \sum_{j=0}^n \int_{j/(n+1)}^{(j+1)/(n+1)} g(t) dt p_{n,j}^2(x) \\
&\leq (n+1)^{1/2+\delta} C_2 ,
\end{aligned}$$

and

$$[(n+1)^{1/2} \text{Var}(z_{nN}(x))]^{1+\delta/2} \rightarrow v(x)^{1+\delta/2} .$$

Therefore

$$L_{nN}(x) = O \left(\frac{(n+1)^{3\delta/4}}{N^{\delta/2}} \right) .$$

Thus it follows from (27) and the preceding assertion that

$$\frac{N^{1/2}}{(n+1)^{1/4}} v^{-1/2}(x) Z_{nN}(x)$$

is asymptotically $N(0, 1)$ -distributed. Further, since $g_{nN}(x)E(g_{nN}(x))$ converges in probability to $g^2(x)$, we obtain that

$$\frac{N^{1/2}}{(n+1)^{1/4}} \sigma^{-1}(x) \left(m_{nN}(x) - \frac{E(\varphi_{nN}(x))}{E(g_{nN}(x))} \right) \quad (29)$$

is asymptotically $N(0, 1)$ -distributed. Relations (29) and (25) yield Theorem 3.3. $\square$

Remark: Under the following conditions we can also show a certain limit theorem: Let $|Y| \leq C_3$ a.s., and for a given x let V be continuous at x . Further let φ , g , and V be bounded on $[0, 1]$, and let $\varphi''(x)$ and $g''(x)$ exist. If $\frac{(n+1)^{5/2}}{N} \rightarrow \infty$ and $\frac{(n+1)^{1/2}}{N} \rightarrow 0$, then

$$\frac{N^{1/2}}{(n+1)^{1/4}} (m_{nN}(x) - m(x))$$

is asymptotically $N(0, \sigma^2(x))$ -distributed.

4 Global Asymptotic Properties of Bernstein Regression Function Estimates

So far we have studied local asymptotic properties of our estimate (2). Now we derive some global properties of (2). A simple way to characterize the error in estimating an unknown regression function is to use the mathematical expectation of the quadratic derivation

$$U_{nN} = E \left(\int_0^1 (m_{nN}(x) - m(x))^2 g_{nN}^2(x) h(x) dx \right), \quad (30)$$

where $h(x)$ is an integrable weight function. The quantity U_{nN} defines a ‘risk function’ for the estimate $m_{nN}(x)$.

Theorem 4.1: Let the second derivatives of the function $\varphi(x)$ and $g(x)$ be continuous, bounded, and square integrable with the weight $\frac{h(x)}{2\sqrt{\pi x(1-x)}}$ on $[0, 1]$. Fur-

ther let $\sup_{x \in [0, 1]} |V(x)| < \infty$, $\frac{m(x)h(x)}{2\sqrt{\pi x(1-x)}}$ and $\frac{m^2(x)h(x)}{2\sqrt{\pi x(1-x)}} \in L_1[0, 1]$. Then

$$U_{nN} \sim \frac{(n+1)^{1/2}}{N} A_1(f) + \frac{1}{(n+1)^2} A_2(f) , \quad (31)$$

where

$$A_1(f) = \int_0^1 \text{Var}(Y|X=x) \frac{g(x)h(x)}{2\sqrt{\pi x(1-x)}} dx ,$$

and

$$A_2(f) = \int_0^1 \left[\frac{1-2x}{2} (\varphi'(x) - m(x)g'(x)) + \frac{x(1-x)}{2} (\varphi''(x) - m(x)g''(x)) \right]^2 h(x) dx .$$

Proof: Fubini's theorem yields

$$\begin{aligned} U_{nN} &= \frac{(n+1)^2}{N} \int_0^1 E \left[\sum_{j=0}^n (Y - m(x))^2 1_{A_{n,j}}(X) p_{n,j}^2(x) \right] h(x) dx \\ &\quad + \left(1 - \frac{1}{N} \right) (n+1)^2 \int_0^1 \left(E \left[\sum_{j=0}^n (Y - m(x)) 1_{A_{n,j}}(X) p_{n,j}(x) \right] \right)^2 h(x) dx \\ &= D_{nN} + E_{nN} . \end{aligned}$$

Note that

$$\begin{aligned} D_{nN} &= \int_0^1 \sum_{j=0}^n \frac{(n+1)^2}{N} \int_{j/(n+1)}^{(j+1)/(n+1)} V(t) dt p_{n,j}^2(x) h(x) dx \\ &\quad - 2 \int_0^1 \sum_{j=0}^n \frac{(n+1)^2}{N} \int_{j/(n+1)}^{(j+1)/(n+1)} \varphi(t) dt p_{n,j}^2(x) m(x) h(x) dx \\ &\quad + \int_0^1 \sum_{j=0}^n \frac{(n+1)^2}{N} \int_{j/(n+1)}^{(j+1)/(n+1)} g(t) dt p_{n,j}^2(x) m^2(x) h(x) dx . \end{aligned}$$

Using Lebesgue's dominated convergence theorem we obtain

$$\begin{aligned}
 D_{nN} &\sim \frac{(n+1)^{1/2}}{N} \int_0^1 \frac{V(x)h(x)}{2\sqrt{\pi x(1-x)}} dx \\
 &\quad - 2 \frac{(n+1)^{1/2}}{N} \int_0^1 \frac{\varphi(x)m(x)h(x)}{2\sqrt{\pi x(1-x)}} dx \\
 &\quad + \frac{(n+1)^{1/2}}{N} \int_0^1 \frac{g(x)m^2(x)h(x)}{2\sqrt{\pi x(1-x)}} dx \\
 &= \frac{(n+1)^{1/2}}{N} \int_0^1 \text{Var}(Y|X=x) \frac{g(x)h(x)}{2\sqrt{\pi x(1-x)}} dx .
 \end{aligned}$$

We now consider E_{nN} :

$$\begin{aligned}
 E_{nN} &= \left(1 - \frac{1}{N}\right) \\
 &\quad \cdot \int_0^1 \left(\sum_{j=0}^n (n+1) \int_{j/(n+1)}^{(j+1)/(n+1)} (y - m(x))f(t, y) dt dy p_{n,j}(x) \right)^2 h(x) dx \\
 &= \left(1 - \frac{1}{N}\right) \\
 &\quad \cdot \int_0^1 \left(\sum_{j=0}^n (n+1) \int_{j/(n+1)}^{(j+1)/(n+1)} (\varphi(t) - m(x)g(t)) dt p_{n,j}(x) \right)^2 h(x) dx .
 \end{aligned}$$

Inserting Taylor expansions of φ and g we obtain

$$\begin{aligned}
 E_{nN} &= \left(1 - \frac{1}{N}\right) \int_0^1 \left[\left(\varphi(x) + \frac{\varphi'(x)(1-2x)}{2(n+1)} + \frac{\varphi''(x)x(1-x)}{2(n+1)} + o\left(\frac{1}{n+1}\right) \right) \right. \\
 &\quad \left. - m(x) \left(g(x) + \frac{g'(x)(1-2x)}{2(n+1)} + \frac{g''(x)x(1-x)}{2(n+1)} + o\left(\frac{1}{n+1}\right) \right) \right]^2 h(x) dx \\
 &\sim \int_0^1 \left[\frac{1-2x}{2(n+1)} (\varphi'(x) - m(x)g'(x)) \right. \\
 &\quad \left. + \frac{x(1-x)}{2(n+1)} (\varphi''(x) - m(x)g''(x)) \right]^2 h(x) dx ,
 \end{aligned}$$

where we used Lebesgue's theorem for the last relation.

These asymptotic relations for D_{nN} and E_{nN} prove our theorem. □

We shall now determine the optimal value $(n + 1)$, i.e. that value $n + 1 = n_0 + 1$ which minimizes the right hand side of (31). $n_0 + 1$ is given by

$$n_0 + 1 = (A_1(f)/A_2(f))^{2/5} N^{2/5} .$$

For this choice of $n + 1$ we have

$$U_{nN} \sim 2(A_1(f)[A_2(f)]^4)^{1/5} N^{-4/5} .$$

Another criterion measuring the closeness of m_{nN} to m is the distance in $L_1[0, 1]$,

$$I_{nN} = \int_0^1 |m_{nN}(x) - m(x)| \mu(dx) .$$

Here μ is the (unknown) probability measure of X . Theorem 4.2 states the weak and strong convergence to 0 of I_{nN} for our estimate (2) under the following conditions:

μ is an absolutely continuous (with respect to the Lebesgue measure) probability measure on $[0, 1]$, (32)

$|Y| \leq C_5 < \infty$ for some constant C_5 with probability one , (33)

$n = n(N) \rightarrow \infty$, (34)

$\frac{n + 1}{N} \rightarrow 0$. (35)

Theorem 4.2: Assume that (32)–(34) hold.

- (i) $I_{nN} \rightarrow 0$ in probability as $N \rightarrow \infty$, if (35) holds.
- (ii) $I_{nN} \rightarrow 0$ with probability one as $N \rightarrow \infty$, if

$$\sum_{N=1}^{\infty} \exp\left(-\alpha \frac{N}{n-1}\right) < \infty \quad \text{for all } \alpha > 0 . \quad (36)$$

Note that Theorem 4.2 is density-free. It is valid for all random variables X possessing a density on the interval $[0, 1]$. Moreover, no continuity condition is imposed on m , and the random variable Y need not have a density at all.

The proof of Theorem 4.2 is based on the following lemma due to Devroye and Wagner [1980a]. As in (2) we arbitrarily define $0/0$ to be 0 in all subsequent expressions.

Lemma 4.1 Devroye and Wagner [1980a]: Let m be a regression function and let m_N be a regression function estimate. Let further for some $C_6 < \infty$, $|m_N| < C_6$, $|m| < C_6$. Then $I_{nN} \rightarrow 0$ in probability (w.p.1) as $N \rightarrow \infty$ when $m_N(x) \rightarrow m(x)$ in probability (w.p.1) for almost all $x(\mu)$.

Proof of Theorem 4.2: If C_6 is the constant of (33) then $|m_{nN}| \leq C_6$, $|m| \leq C_6$. By Lemma 4.1 we only have to show that $m_{nN}(x) \rightarrow m(x)$ in probability (w.p.1) as $N \rightarrow \infty$ for almost all $x(\mu)$.

If we can show that under the conditions of Theorem 4.2 $\varphi_{nN}(x) \rightarrow m(x)g(x)$ as $N \rightarrow \infty$ in probability (w.p.1) for almost all $x(\mu)$, then we have also shown that $g_{nN}(x) \rightarrow g(x)$ in probability (w.p.1) for almost all $x(\mu)$, since $g_{nN}(x)$ is defined as $\varphi_{nN}(x)$ with $Y \equiv 1$ and thus $m \equiv 1$. Since $m_{nN}(x) = \varphi_{nN}(x)/g_{nN}(x)$ and since for almost all $x(\mu)g(x) > 0$ we can conclude that $m_{nN}(x) \rightarrow m(x)$ as $N \rightarrow \infty$ in probability (w.p.1) for almost all $x(\mu)$.

We split the proof into two parts according to the inequality

$$|\varphi_{nN}(x) - m(x)g(x)| \leq |\varphi_{nN}(x) - E[\varphi_{nN}(x)]| + |E[\varphi_{nN}(x)] - m(x)g(x)| .$$

First we consider the bias term. Since $|m| \leq C_5$ we have $mg \in L_1[0, 1]$. Next note that

$$\begin{aligned} E[\varphi_{nN}(x)] &= (n+1) \sum_{j=0}^n E[Y 1_{A_{n,j}}(X)] p_{n,j}(x) \\ &= (n+1) \sum_{j=0}^n E[E[Y 1_{A_{n,j}}(X)|X]] p_{n,j}(x) \\ &= (n+1) \sum_{j=0}^n \int_{j/(n+1)}^{(j+1)/(n+1)} m(t)g(t) dt p_{n,j}(x) \\ &=: P_n^{mg}(x) . \end{aligned}$$

Now, if $n \rightarrow \infty$ then $P_n^{mg}(x) \rightarrow m(x)g(x)$ for almost all (Lebesgue measure) x by Lemma 1.5 of Tenbusch [1994b]. Let us collect these x in a set A . Since g is a density, we obviously have $\mu(A) = 1$.

Theorem 4.2 is proved if we can show that for all $x \in A$ the variation term $|\varphi_{nN}(x) - E[\varphi_{nN}(x)]| \rightarrow 0$ as $N \rightarrow \infty$ in probability (w.p.1). Note that

$$\varphi_{nN}(x) - E[\varphi_{nN}(x)] = \frac{1}{N} \sum_{i=1}^N (T_i(n) - E[T_i(n)]) ,$$

where

$$T_i(n) = (n+1) \sum_{j=0}^n Y_i 1_{A_{n,j}}(X_i) p_{n,j}(x) .$$

Since

$$(n+1) \sum_{j=0}^n 1_{A_{n,j}}(X_i) p_{n,j}(x) \leq n+1$$

and

$$\sup_n P_n^{mg}(x) \leq C_7 \quad (C_7 \text{ is finite if } x \in A \text{ but depends on } x) ,$$

we have

$$|T_i(n)| \leq C_5(n+1) \quad \text{w.p.1} ,$$

and

$$E[T_i^2(n)] \leq C_5(n+1)E[|T_i(n)|] \leq C_5 C_7(n+1) .$$

Now we use an inequality due to Bennett [1962, pp. 39] (see also Hoeffding [1963, pp. 16]) which states:

If $U_1, \dots, U_N$ are independent random variables with $|U_i| \leq c$, $E[U_i] = 0$, and $E[U_i^2] \leq \sigma_i^2$, then

$$\begin{aligned} P\left(\left|\frac{1}{N} \sum_{i=1}^N U_i\right| \geq \varepsilon\right) &\leq 2 \exp\left\{-N \frac{\varepsilon}{2c} \left[\left(1 + \frac{\sigma^2}{2c\varepsilon}\right) \log\left(1 + \frac{2c\varepsilon}{\sigma^2}\right) - 1\right]\right\} \\ &\leq 2 \exp\left\{-N \frac{\varepsilon^2}{2(\sigma^2 + c\varepsilon)}\right\} , \end{aligned} \quad (37)$$

where $\sigma^2 = \frac{1}{N} \sum_{i=1}^N \sigma_i^2$ and the latter inequality follows from $\log(1+u) > 2u/(2+u)$ for all $u > 0$.

In our case we get

$$\begin{aligned} P(|\varphi_{nN}(x) - E[\varphi_{nN}(x)]| \geq \varepsilon) &\leq 2 \exp \left\{ -N \frac{\varepsilon^2}{2C_5 C_7 (n+1) + 2C_5 (n+1)\varepsilon} \right\} \\ &= 2 \exp \left\{ -\frac{\alpha N}{n+1} \right\}, \end{aligned} \quad (38)$$

where

$$\alpha = \frac{\varepsilon^2}{2C_5(C_7 + \varepsilon)}.$$

Condition (35) implies that (38) tends to 0 as $N \rightarrow \infty$ for all $\varepsilon > 0$. Utilizing (36) and the Borel-Cantelli lemma we can conclude that

$$|\varphi_{nN}(x) - E[\varphi_{nN}(x)]| \rightarrow 0 \quad \text{w.p.1} \quad \text{for all } x \in A.$$

Theorem 4.2 is proved. □

The next theorem gives sufficient conditions for the a.s. convergence in $C[0, 1]$ of our estimate (2) of the regression function to the theoretical $m(x)$.

Theorem 4.3: Assume that $g(x)$ and $m(x)$ are continuous on $[0, 1]$, $-\infty < A \leq Y \leq B < \infty$ a.s., $\min_{0 \leq x \leq 1} g(x) = g^ > 0$ and the series $\sum_{N=1}^{\infty} \exp(-\gamma N(n+1)^{-2})$ is convergent for any $\gamma > 0$. Then as $N \rightarrow \infty$,*

$$\sup_{0 \leq x \leq 1} |m_{nN}(x) - m(x)| \rightarrow 0 \quad \text{a.s.}$$

Without loss of generality we may assume $A = 0$ and $B = 1$. The theorem is proved with the aid of the following four lemmas.

Lemma 4.2: If $g(x)$ is continuous on $[0, 1]$, then

$$\sup_{0 \leq x \leq 1} |E(g_{nN}(x)) - g(x)| \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Proof: This lemma can be proven using the same arguments as Lorentz did proving the famous theorem on approximation of continuous functions by means of Bernstein polynomials. $\square$

Lemma 4.3:

$$P\left(\sup_{0 \leq x \leq 1} |g_{nN}(x) - E(g_{nN}(x))| > \varepsilon\right) \leq C_8 \exp\left(-\frac{\varepsilon}{2} \frac{N}{(n+1)^2}\right).$$

Proof: $g_{nN}(x)$ can be represented in the form

$$g_{nN}(x) = (n+1) \int_0^1 K_n(t, x) dG_N(t),$$

where $G_N(t)$ is the empirical distribution function of the sample $X_1, \dots, X_N$, and $K_n(t, x) = \sum_{j=0}^n 1_{A_{n,j}}(t) p_{n,j}(x)$.

Applying integration by parts we obtain

$$\sup_{0 \leq x \leq 1} |g_{nN}(x) - E(g_{nN}(x))| \leq 2(n+1) \sup_{0 \leq t \leq 1} |G_N(t) - G(t)|. \quad (39)$$

Using the Dvoretzky, Kiefer and Wolfowitz [1956] bound for one-dimensional empirical processes and (39) we obtain the assertion. $\square$

Remark: Note that, if the series $\sum_{N=1}^{\infty} \exp\left(-\gamma \frac{N}{(n+1)^2}\right)$ converges for any $\gamma > 0$, then Lemma 4.2 and Lemma 4.3 together with an application of the Borel-Cantelli lemma imply that

$$\sup_{0 \leq x \leq 1} |g_{nN}(x) - g(x)| \rightarrow \text{a.s.} \quad \text{as } N \rightarrow \infty.$$

Lemma 4.4: If $m(x)$ and $g(x)$ are continuous on $[0, 1]$, then

$$\sup_{0 \leq x \leq 1} |E(\varphi_{nN}(x)) - \varphi(x)| \rightarrow 0, \quad \text{as } N \rightarrow \infty.$$

Proof: Noting that

$$E(\varphi_{nN}(x)) = (n+1) \sum_{j=0}^n \int_{j/(n+1)}^{(j+1)/(n+1)} \varphi(t) dt p_{n,j}(x) ,$$

we can proceed as in the proof of Lemma 4.2. □

Lemma 4.5: Let $0 \leq Y \leq 1$ a.s. Then for every $\varepsilon > 0$ there is a constant C_{10} and an $a = a(\varepsilon)$ such that for all N large enough,

$$P\left(\sup_{0 \leq x \leq 1} |\varphi_{nN}(x) - E(\varphi_{nN}(x))| > 2\varepsilon\right) \leq C_{10} \exp\left(-a(\varepsilon) \frac{N}{(n+1)^2}\right) .$$

Proof: $\varphi_{nN}(x)$ can be represented in the form

$$\varphi_{nN}(x) = (n+1) \int_0^1 \int_0^1 y K_n(t, x) dF_N(t, y) ,$$

where $F_N(t, y)$ is the two-dimensional empirical distribution function constructed from the sample $(X_1, Y_1), \dots, (X_N, Y_N)$.

Two dimension partial integration leads to

$$\int_0^1 \int_0^1 F_N(u, v) dK_n(u, x) dv - \int_0^1 F_N^{(1)}(u) dK_n(u, x) + \int_0^1 v dF_N^{(2)}(v)$$

we obtain that

$$\begin{aligned} \sup_{0 \leq x \leq 1} |\varphi_{nN}(x) - E(\varphi_{nN}(x))| &= \sup_{0 \leq x \leq 1} \left| (n+1) \int_0^1 (F^{(1)}(t) - F_N^{(1)}(t)) dK_n(t, x) \right. \\ &\quad \left. + (n+1) \int_0^1 \int_0^1 (F_N(t, y) - F(t, y)) dK_n(t, x) dy \right| \\ &\quad + e(N) \\ &\leq 2(n+1) \sup_{0 \leq t \leq 1} |F_N^{(1)}(t) - F^{(1)}(t)| \\ &\quad + 2(n+1) \sup_{0 \leq t, y \leq 1} |F_N(t, y) - F(t, y)| + e(N) , \end{aligned} \tag{40}$$

where $F(t, y)$ is the distribution function of (X, Y) , $F^{(1)}(t)$ is the marginal distribution function of X and $F_N^{(1)}(t)$ is the empirical distribution function of the sample $X_1, \dots, X_N$, and $F^{(2)}(y)$ is the marginal distribution function of Y and $F_N^{(2)}(y)$ is the empirical distribution function of the sample $Y_1, \dots, Y_N$, and

$$e(N) = \left| \int_0^1 v dF_N^{(2)}(v) - \int_0^1 v dF^{(2)}(v) \right|.$$

Note that by Chernoff's inequality (Bauer [1991])

$$P(e(N) > \varepsilon) \leq 2 \exp(-C_{11}(\varepsilon)N)$$

Using now an analogue of the Dvoretzky, Kiefer and Wolfowitz bound for the multidimensional case (see Kiefer and Wolfowitz [1958]) completes the proof of Lemma 4.5. $\square$

We are now in a position to prove Theorem 4.3.

Proof Theorem 4.3: First note that

$$\begin{aligned} & P\left(\sup_{0 \leq x \leq 1} |m_{nN}(x) - m(x)| > \varepsilon\right) \\ & \leq P\left(\sup_{0 \leq x \leq 1} |\varphi_{nN}(x) - m(x)g_{nN}(x)| > \varepsilon(g^* - \varepsilon)\right) \\ & \quad + P\left(\sup_{0 \leq x \leq 1} |g_{nN}(x) - g(x)| > \varepsilon\right) \\ & \leq P\left(\sup_{0 \leq x \leq 1} |\varphi_{nN}(x) - m(x)g(x)| \right. \\ & \quad \left. + \sup_{0 \leq x \leq 1} |m(x)g(x) - m(x)g_{nN}(x)| > \varepsilon(g^* - \varepsilon)\right) \\ & \quad + P\left(\sup_{0 \leq x \leq 1} |g_{nN}(x) - g(x)| > \varepsilon\right) \\ & \leq P\left(\sup_{0 \leq x \leq 1} |\varphi_{nN}(x) - \varphi(x)| > \frac{\varepsilon(g^* - \varepsilon)}{2}\right) \end{aligned}$$

$$\begin{aligned}
& + P\left(\sup_{0 \leq x \leq 1} |g_{nN}(x) - g(x)| > \frac{\varepsilon(g^* - \varepsilon)}{2 \max_{0 \leq u \leq 1} |m(u)|}\right) \\
& + P\left(\sup_{0 \leq x \leq 1} |g_{nN}(x) - g(x)| > \varepsilon\right). \tag{41}
\end{aligned}$$

Applying Lemma 4.4 and Lemma 4.5 we obtain

$$\begin{aligned}
P\left(\sup_{0 \leq x \leq 1} |\varphi_{nN}(x) - \varphi(x)| > 2\varepsilon\right) & \leq P\left(\sup_{0 \leq x \leq 1} |\varphi_{nN}(x) - E(\varphi_{nN}(x))| > \varepsilon/2\right) \\
& \leq C_{10} \exp\left(-a(\varepsilon) \frac{N}{(n+1)^2}\right) \tag{42}
\end{aligned}$$

for $N > N_1(\varepsilon)$.

Further using Lemma 4.2 and Lemma 4.3 yields

$$P\left(\sup_{0 \leq x \leq 1} |g_{nN}(x) - g(x)| > \varepsilon\right) \leq C_9 \exp\left(-\beta_3 \frac{N}{(n+1)^2}\right), \tag{43}$$

for $N > N_2(\varepsilon)$.

Finally, in view of (42) and (43) we obtain from (41) that

$$P\left(\sup_{0 \leq x \leq 1} |m_{nN}(x) - m(x)| > \varepsilon\right) \leq C_{14} \exp\left(-C_{15} \frac{N}{(n+1)^2}\right) \tag{44}$$

for $N > N_3(\varepsilon) = \max\{N_1(\varepsilon), N_2(\varepsilon)\}$.

Inequality (44) together with an application of the Borel-Cantelli lemma completes the proof. $\square$

5 References

- Bauer H (1991) Wahrscheinlichkeitstheorie. Walter de Gruyter Berlin New York
 Bennett G (1962) Probability inequalities for the sum of independent random variables. J Amer Statist Assoc 57:33–45

- Chu CK, Marron (1991) Choosing a kernel regression estimator. *Statist Sciences* 6:404–436
- Davis PJ (1975) Interpolation and approximation. Dover Publications Inc New York
- Devroye LP (1978) The uniform convergence of the Nadaraya-Watson regression function estimate. *Canad J Statist* 6:179–191
- Devroye LP, Wagner TJ (1980a) On the L_1 convergence of kernel estimators of regression functions with applications in discrimination. *Z Wahrscheinlichkeitstheorie verw Gebiete* 25:15–25
- Devroye LP, Wagner TJ (1980b) Distribution-free consistency results in nonparametric discrimination and regression function estimation. *Annals of Statistics* 8:231–239
- Dvoretzky A, Kiefer J, Wolfowitz J (1956) Asymptotic minimax character of the sample distribution and of the classical multinomial estimator. *Ann Math Statist* 27:642–669
- Eubank RA (1988) Spline smoothing and nonparametric regression. Marcel Dekker New York
- Gasser T, Müller HG (1979) Kernel estimation of regression functions. In: *Smoothing techniques for curve estimation. Lecture Notes in Math* 757:23–68. Springer New York
- Gawronski W, Stadtmüller U (1980) On density estimation by means of Poisson's distribution. *Scand J Statist* 7:90–94
- Gawronski W, Stadtmüller U (1981) Smoothing histograms by lattice and continuous distributions. *Metrika* 28:155–164
- Gawronski W, Stadtmüller U (1984) Linear combinations of iterated generalized Bernstein functions with an application to density estimation. *Acta Sci Math* 47:205–221
- Gawronski W (1985) Strong laws for density estimators of Bernstein type. *Per Math Hungar* 16:23–43
- Greblicki WD (1974) Asymptotically optimal probabilistic algorithms for pattern recognition and identification. Monografie No. 3 Prace Naukowe Instytutu Cybernetyki Technicznej Politechniki Wrocławskiej No. 18, Poland
- Härdle W (1990) Applied nonparametric regression. Cambridge University Press New York
- Hastie T, Loader C (1993) Local regression: Automatic kernel carpentry. *Statist Sciences* 8:120–143
- Hoeffding W (1963) Probability inequalities for the sum of bounded random variables. *J Amer Statist Assoc* 58:13–30
- Kiefer J, Wolfowitz J (1958) On the deviations on the empiric distribution function of vector chance variables. *Trans Amer Math Soc* 87:173–186
- Konakov VD (1977) On a global measure of deviation for an estimate of the regression line. *Theory Prob Appl* 22:858–868
- Lorentz GG (1986) Bernstein polynomials. Chelsea Publishing Company New York NY
- Michels P (1992) Nichtparametrische Analyse und Prognose von Zeitreihen. *Arbeiten zur angewandten Statistik* 36. Physica-Verlag Heidelberg
- Müller HG (1988) Nonparametric regression analysis of longitudinal data. *Lecture Notes in Statistics*. Springer Verlag New York
- Müller HG (1991) Smooth optimum kernel estimators near endpoints. *Biometrika* 78:521–530
- Nadaraya EA (1964) On estimating regression. *Theory Prob Appl* 9:141–142
- Nadaraya EA (1965) On nonparametric estimates of density functions and regression curves. *Theory Prob Appl* 10:186–190
- Nadaraya EA (1970) Remarks on some nonparametric estimates of density functions and regression curves. *Theory Prob Appl* 15:134–137
- Nadaraya EA (1974) The limit distribution of the quadratic deviation of nonparametric estimates of the regression function. *Soobshch Akad Nauk Gruz SSR* 74,1:33–36
- Nadaraya EA (1988) Nonparametric estimation of probability densities and regression curves. *Mathematics and its applications (Soviet Series)*, Kluwer Academic Publishers Dordrecht/Boston/London
- Noda K (1976) Estimation of a regression function by the Parzen kernel-type density estimators. *Ann Inst Statist Math* 28:221–234
- Rice J (1984) Boundary modifications for kernel regression. *Comm Statist Theory Methods* 13:893–900
- Rosenblatt M (1969) Conditional probability density and regression estimators. In: Krishnaiah PR (ed) *Multivariate Analysis II* 25–31, Academic Press New York

- Schuster EF (1972) Joint distribution of the estimated regression function at a finite number of distinct points. *Ann Math Statist* 43:84–88
- Stadtmüller U (1983) Asymptotic distributions of smoothed histograms. *Metrika* 30:145–158
- Stadtmüller U (1986) Asymptotic properties of nonparametric curve estimates. *Per Math Hung* 17:83–108
- Stone CJ (1977) Consistent nonparametric regression. *Ann Statist* 5:595–645
- Tenbusch A (1994a) Two-dimensional Bernstein polynomial density estimators. *Metrika* 41:233–253
- Tenbusch A (1994b) The equivalence of weak, strong, and complete convergence in $L_1[0, 1]^d$ for Bernstein polynomial density estimates. Submitted to *Statistics*
- Watson GS (1964) Smooth regression analysis. *Sankhyā Series A* 26:359–372
- Vitale RA (1973) A Bernstein polynomial approach to density estimation. *Comm Stat* 2:493–506
- Wheeden L, Zygmund A (1977) *Measure and integral: An introduction to real analysis*. Marcel Dekker Inc New York and Basel
- Wahba G (1990) *Spline models for observational data*. SIAM Philadelphia

Received 15.12.1994

Revised version 09.05.1995

A Two-Stage Sampling by Unequal Size Samples from Two Normal Populations with Unknown Variances

M. LUISA TARGHETTA

Department of Mathematics, University of Cagliari, Via Ospedale 72, I-09124 Cagliari, Italy

Abstract: A two-stage procedure based on the statistic introduced by Scheffé (1943) to solve the Behrens-Fisher problem is considered in estimating the difference between the mean values of two normal distributions having unequal and unknown variances. Samples of unequal size are considered at each stage. A numerical example is given where the two-stage procedure is compared with the Satterthwaite (1946) approximate method.

Key Words: Behrens-Fisher problem; Scheffé test statistic; Stein's two-stage procedure.

1 Introduction

Let X_1 and X_2 be two independent normal random variables with unknown mean values μ_1 , μ_2 and unequal and unknown variances σ_1^2 , σ_2^2 respectively. Denote by X_{1i} , $i = 1, \dots, n_1$ and X_{2i} , $i = 1, \dots, n_2$ the random variables of two samples taken on X_1 and X_2 respectively and $\bar{X}_1$, $\bar{X}_2$ the corresponding sample means.

It is a problem of sample size methodology to choose n_1 and n_2 by controlling the absolute error of the usual estimator $\bar{X}_1 - \bar{X}_2$ with high probability, that is

$$P\{ |(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)| \leq d \} \geq 1 - \alpha. \quad (1)$$

where d and α ($0 < \alpha < 1$) are prechosen positive constants.

Stein (1945) considered a double sampling procedure to test hypotheses concerning the mean of normal variables with power independent of the unknown variances. Chapman (1950) adapted the Stein procedure to a two sample case by introducing a test statistic which is the difference of two independent Student variables. Koopmans and Qualls (1971) suggested a two-stage procedure to satisfy (1) by introducing a linear cost function to be minimized at the second stage of the sampling. Ghosh (1975) proposed a simple procedure which is based on a combination of Scheffé's (1943) and Stein's (1945) arguments in the case of equal sample sizes.

Here, we extend the Ghosh procedure to the case of unequal numbers of observations from each of the two populations at each stage to derive n_1 and n_2 in order to meet (1). More precisely, we consider a linear combination of X_{1i} and X_{2i} (see Scheffé, 1943) and make use of Stein's two-stage procedure (Stein, 1945; Desu and Raghavarao, 1990) for estimating the mean value of a single normal population with unknown variance.

Finally, we exemplify on a problem considered in Snedecor and Cochran (1989) and compare our numerical results with corresponding results derived by a sufficiently accurate yet approximate method due to Satterthwaite (1946).

2 The Main Result

a) First Stage

Take initial samples of size n_1° and n_2° ($n_1^\circ \leq n_2^\circ$, $n_1^\circ \geq 2$) from populations X_1 and X_2 respectively; let $\bar{X}_1^\circ$ and $\bar{X}_2^\circ$ be the corresponding sample means.

Regarding the choice of n_1° and n_2° , one can evaluate $E(n_1)$ and $E(n_2)$ for different values of n_1° and n_2° obtained in correspondence to different fixed values of σ_1^2 , σ_2^2 by the formula (Desu and Raghavarao, 1990)

$$n_1^\circ = [kn'] + 1, \quad n_2^\circ = [n'] + 1$$

where $[r] = \text{largest integer} \leq r$ and

$$n' = \frac{z_{\alpha/2}^2 \left(\frac{\sigma_1^2}{k} + \sigma_2^2 \right)}{d^2} \quad \text{for suitable } k \leq 1,$$

being $z_{\alpha/2}$ the upper $100(\alpha/2)$ percentile point of the standard normal distribution. For a detailed discussion of this problem see Moshman (1958) and Seelbinder (1953).

Now consider the random variables

$$U_i^\circ = X_{1i} - \sqrt{(n_1^\circ/n_2^\circ)} X_{2i}, \quad i = 1, \dots, n_1^\circ \quad (2)$$

and

$$\Sigma_0^2 = \sum_{i=1}^{n_1^\circ} \frac{(U_i^\circ - \bar{U}^\circ)^2}{n_1^\circ - 1} \equiv \frac{Q_0^2}{n_1^\circ - 1}, \quad (3)$$

where $\bar{U}^\circ$ is the sample mean of $U_1^\circ, \dots, U_{n_1^\circ}^\circ$.

As known (see Scheffé, 1943), the statistic

$$T_0 = \frac{\bar{X}_1^\circ - \bar{X}_2^\circ - (\mu_1 - \mu_2)}{\Sigma_0} \sqrt{n_1^\circ}$$

follows a Student distribution with $n_1^\circ - 1$ degrees of freedom.

If we find that

$$d \geq \frac{\Sigma_0 t_{n_1^\circ-1; \alpha/2}}{\sqrt{n_1^\circ}} , \quad (4)$$

where $t_{n_1^\circ-1; \alpha/2}$ is the upper $100(\alpha/2)$ percentile of the Student distribution, then (1) is satisfied. On the contrary, if

$$n_1^\circ < \frac{\Sigma_0^2 t_{n_1^\circ-1; \alpha/2}^2}{d^2} \equiv n^* , \quad (5)$$

we proceed to the second stage.

b) Second Stage

Take $n_1 - n_1^\circ$ additional observations from the population X_1 being $n_1 = [n^*] + 1$ with n^* as defined in (5). Suppose that $n_1 n_2^\circ / n_1^\circ$ is an integer number (note that this is always possible by choosing at the first stage $n_2^\circ = m n_1^\circ$, m positive integer number). Take $n_2 - n_2^\circ$ additional elements from the population X_2 being $n_2 = n_1 n_2^\circ / n_1^\circ$. Define the random variables

$$Z_i = X_{1i} - \sqrt{n_1/n_2} X_{2i} + \sum_{h=1}^{n_1} \frac{X_{2h}}{\sqrt{n_1 n_2}} - \frac{1}{n_2} \sum_{h=1}^{n_2} X_{2h} , \quad i = 1, \dots, n_1$$

with means

$$E(Z_i) = \mu_1 - \mu_2$$

and covariances

$$\text{cov}(Z_i, Z_j) = \begin{cases} 0 & i \neq j \\ \sigma_1^2 + (n_1/n_2)\sigma_2^2 & i = j \end{cases} .$$

Since $n_2 = n_1 n_2^\circ / n_1^\circ$ the variance of Z_i is also

$$\text{var } Z_i = \text{var } U_i^\circ = \sigma_1^2 + (n_1^\circ / n_2^\circ) \sigma_2^2, \quad i = 1, \dots, n_1. \quad (6)$$

Now, being Z_i a linear combination of normal random variables, the conditional distribution of the sample mean $\bar{Z}$, given Q_0^2 , is normal with mean value $\mu_1 - \mu_2$ and variance $(\sigma_1^2 + (n_1^\circ / n_2^\circ) \sigma_2^2) / n_1$, that is

$$f(\bar{z} | Q_0^2) = N(\mu_1 - \mu_2, (\sigma_1^2 + (n_1^\circ / n_2^\circ) \sigma_2^2) / n_1). \quad (7)$$

Let $\bar{X}_1, \bar{X}_2$ be the new sample means of X_{1i} ($i = 1, \dots, n_1$) and X_{2i} ($i = 1, \dots, n_2$) respectively and

$$T = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\Sigma_0} \sqrt{n_1} = \frac{\bar{Z} - (\mu_1 - \mu_2)}{\Sigma_0} \sqrt{n_1}.$$

It follows from (7) that the conditional distribution of T , given Q_0^2 , is

$$g(t | Q_0^2) = Q_0 \left\{ 2\pi(n_1^\circ - 1) \left(\sigma_1^2 + \frac{n_1^\circ}{n_2^\circ} \sigma_2^2 \right) \right\}^{-1/2} \exp \left\{ -\frac{t^2 Q_0^2}{2(n_1^\circ - 1)(\sigma_1^2 + (n_1^\circ / n_2^\circ) \sigma_2^2)} \right\}.$$

Since the distribution of the random variable $Q_0^2 / (\sigma_1^2 + (n_1^\circ / n_2^\circ) \sigma_2^2)$ is $\chi^2(n_1^\circ - 1)$, the unconditional distribution of T is a Student distribution with $n_1^\circ - 1$ degrees of freedom.

In conclusion, we have the following

Theorem: The sample sizes n_1 and n_2 required to meet the inequality (1) are

$$n_1 = \max \{ n_1^\circ, [n^*] + 1 \}, \quad n_2 = n_1 n_2^\circ / n_1^\circ \quad (8)$$

provided that $n_1 n_2^\circ / n_1^\circ$ is an integer positive number and n^* is defined as in (5).

Remark: If n_2 is not integer we can round it to the nearest integer number; in this case, the relation (6) holds only approximately but it should be adequate for practical purposes.

Example: A standard method which is precise but slow is compared to a quick but imprecise method by measuring the concentration of a chemical in a vat. The data by two methods are (Snedecor and Cochran, 1989)

Standard (X_1)	Quick (X_2)
25	23
24	18
25	22
26	28
	17
	25
	19
	16

Let us take these data as initial samples of size $n_1^o = 4$ and $n_2^o = 8$ respectively; we get as values of U_i^o

$$8.74, \quad 11.27, \quad 9.44, \quad 6.20$$

and consequently

$$\Sigma_0^2 = 4.41.$$

Assuming in (1) that $\alpha = 0.05$, so that $t_{3,0.025} = 3.182$, and substituting in the right hand of (4), we find that (1) is satisfied provided that

$$d \geq 3.34. \quad (9)$$

We report below the values of n_1 and n_2 as result from (8) for some values of d (< 3.34):

$$\begin{array}{llll}
 d = 1 & n^* = 44.66 & n_1 = 45 & n_2 = 90 \\
 d = 1.5 & n^* = 19.85 & n_1 = 20 & n_2 = 40 \\
 d = 2 & n^* = 11.16 & n_1 = 12 & n_2 = 24 \\
 d = 3 & n^* = 4.96 & n_1 = 5 & n_2 = 10.
 \end{array} \quad (10)$$

In Snedecor and Cochran (1989) approximate 95% confidence limits for the difference $\mu_1 - \mu_2$ of the mean values of X_1 and X_2 respectively, are calculated

by introducing the quantity

$$T' = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{(\Sigma_1^2/n_1) + (\Sigma_2^2/n_2)}} ,$$

where $\bar{X}_i$, Σ_i^2 are the sample mean and the sample variance of X_i , ($i = 1, 2$). An approximation, due to Satterthwaite (1946), assigns the following approximate number of degrees of freedom to T'

$$v' = \frac{(v_1 + v_2)^2}{(v_1^2/v_1) + (v_2^2/v_2)} , \quad \begin{cases} v_i = \Sigma_i^2/n_i \\ v_i = n_i - 1 \end{cases} \quad i = 1, 2 \quad (11)$$

so that the ordinary t tables may be used. For the data above we have

$$\Sigma_1^2 = 0.67 , \quad \Sigma_2^2 = 17.71 , \quad v' = 8 \quad (12)$$

hence the confidence limits $\bar{X}_1 - \bar{X}_2 \pm d'$ with $d' = 3.56$.

If we require $d = 3.56$ in (1) we find that the inequality (9) is satisfied, thus $n_1 \equiv n_1^\circ = 4$ and $n_2 \equiv n_2^\circ = 8$.

Now, suppose that four new samples of size n_1 and n_2 as given in (10) are taken on X_1 and X_2 respectively. Assume that the corresponding values of Σ_1^2 and Σ_2^2 remain approximately equal to those in (12). Thus, calculating the new values of v' (rounding (11) down to the nearest integer) and the corresponding values of d' we find that

$$v' = 101 , \quad d' = 0.91 \quad (n_1 = 45, n_2 = 90)$$

$$v' = 44 , \quad d' = 1.39 \quad (n_1 = 20, n_2 = 40)$$

$$v' = 26 , \quad d' = 1.83 \quad (n_1 = 12, n_2 = 24)$$

$$v' = 10 , \quad d' = 3.07 \quad (n_1 = 5, n_2 = 10) .$$

Comparison for each value of (n_1, n_2) of the values of d' as above with those of d as given in (10) shows that the agreement is better for “extreme” values of (n_1, n_2) .

References

- Chapman DG (1950) Some two-sample tests. *Ann Math Statist* 21:601–606
- Desu MM, Raghavarao D (1990) *Sample size methodology*. Academic Press
- Ghosh BK (1975) A two-stage procedure for the Behrens-Fisher problem. *J Amer Statist Assoc* 70:457–462
- Koopmans L, Qualls C (1971) Fixed length confidence intervals for parameters of the normal distribution based on two-stage sampling procedures. *Rocky Mountain J Math* 1:587–602
- Moshman J (1958) A method for selecting the size of the initial sample size in the Stein's two-sample procedure. *Ann Math Statist* 29:1271–1275
- Satterthwaite FE (1946) An approximate distribution of estimates of variance components. *Biom Bull* 2:110–114
- Scheffé H (1943) On solutions of the Behrens-Fisher problem based on the t -distribution. *Ann Math Statist* 14:35–44
- Seelbinder BM (1953) On Stein's two-stage procedure. *Ann Math Statist* 24:640–649
- Snedecor GW, Cochran WG (1989) *Statistical methods*. Iowa State University Press
- Stein C (1945) A two-sample test for a linear hypothesis whose power is independent of variance. *Ann Math Statist* 16:243–258

Received 05.05.95

Revised version 07.07.95

A Note on Compact Differentiability and the δ -Method

DANIEL ROST

Mathematisches Institut, LMU Muenchen, Theresienstrasse 39, 80333 Muenchen, Germany

Abstract: Let $\eta_n, n \in \mathbb{N}$, be arbitrary functions defined on a probability space $(\Omega, \mathcal{A}, P)$ with values in a normed vector space B_1 , $\mu \in B_1$ and ξ_0 a separable random element in B_1 such that $\xi_n := \sqrt{n}(\eta_n - \mu)$ converges weakly to ξ_0 in the sense of Hoffmann-Jørgensen. Then with $(B_2, \|\cdot\|_2)$ being another normed vector space and $\phi: B_1 \rightarrow B_2$ compactly differentiable at μ with derivate D_μ , the random variable $\|\sqrt{n}(\phi(\eta_n) - \phi(\mu)) - D_\mu(\sqrt{n}(\eta_n - \mu))\|_2^*$ converges to 0 P -stochastically where “ $*$ ” denotes the measurable cover. We show that the classical δ -method extends to the non-measurable case where in the proof we shall not make use of any representation theorems but only of a slight refinement of the usual characterisation of compact differentiability, due to the fact that we will not assume $\{\xi_n: n \in \mathbb{N}\}$ being tight.

Key Words: Weak convergence in the Hoffmann-Jørgensen sense, asymptotical tightness, compact differentiability tangentially to a subspace.

1 Introduction

Let $\eta_n, n \in \mathbb{N}$, be random elements defined on some probability space $(\Omega, \mathcal{A}, P)$ with values in a normed vector space $(B_1, \|\cdot\|_1)$. Further suppose that we have a limit theorem for the η_n 's properly normalised, i.e. there is a $\mu \in B_1$ and a random element ξ_0 such that

$$\sqrt{n}(\eta_n - \mu) \xrightarrow{\mathcal{L}} \xi_0 \quad \text{as } n \rightarrow \infty \quad (1)$$

where “ $\xrightarrow{\mathcal{L}}$ ” denotes weak convergence. Many statistics of interest can be written in the form $\phi(\eta_n)$, where $\phi: B_1 \rightarrow B_2$ is a measurable function from $(B_1, \|\cdot\|_1)$ into another normed vector space $(B_2, \|\cdot\|_2)$. We shall not prevail upon a discussion of examples (bootstrap, jackknife, quantil-function etc.) here but refer instead to the literature (see e.g. Bickel, Freedman (1981), Fernholz (1983), Gill (1989), Reeds (1978)). It suffices to say that one often encounters $B_1 = D[0, 1]$ or similar function spaces, where $D[0, 1]$ is the space of right continuous real functions on $[0, 1]$ having left hand limits, endowed with the supremum-norm $\|\cdot\|_1 := \|\cdot\|_\infty$. Then for η_n we may take (neglecting measurability questions)

$\eta_n := F_n$. Here F_n stands for the empirical distribution function pertaining to independent and identically distributed random variables $X_1, \dots, X_n$ with values in $[0, 1]$. For simplicity take $X_i \stackrel{\mathcal{L}}{\sim} F$ with F the uniform distribution on $[0, 1]$. Then (1) is given by Donsker's Theorem, saying

$$\sqrt{n}(F_n - F) \xrightarrow{\mathcal{L}} B_0 \quad (2)$$

where B_0 is the Brownian Bridge. A very simple choice for ϕ is

$$\phi(G) := \int_0^1 xG(dx)$$

for any distribution function $G \in D[0, 1]$, with ϕ properly extended to be defined on the whole $D[0, 1]$.

Returning to the general setting one is interested in a limit theorem for $\phi(\eta_n)$. Note that in our example above one has:

$$\sqrt{n}(\phi(\eta_n) - \phi(\mu)) = \sqrt{n}\left(\frac{1}{n} \sum_{i=1}^n X_i - E(X_1)\right) \xrightarrow{\mathcal{L}} \mathcal{N}\left(0, \frac{1}{12}\right).$$

In general, weak convergence of

$$\sqrt{n}(\phi(\eta_n) - \phi(\mu))$$

will follow from (1) if one imposes some smoothness conditions on ϕ : Under the assumption that ϕ is compactly differentiable at μ one can show

$$\sqrt{n}(\phi(\eta_n) - \phi(\mu)) \xrightarrow{\mathcal{L}} D_\mu(\xi_0) \quad (3)$$

where D_μ is the derivative of ϕ at μ . (3) follows easily from

$$\|\sqrt{n}(\phi(\eta_n) - \phi(\mu)) - D_\mu(\sqrt{n}(\eta_n - \mu))\|_2 \xrightarrow{P} 0 \quad (4)$$

by help of (1) and a Cramer-type result. (4) can be proved in a straightforward way only using the definition of compact differentiability if in addition one assumes the sequence $(\sqrt{n}(\eta_n - \mu); n \in \mathbb{N})$ being tight. Without tightness one can still obtain (4) via an application of the extended continuous mapping theorem and the Skorokhod-Dudley-Wichura almost-sure representation theorem (cf. e.g. Gill (1989)).

We shall show that (4) can still be obtained in the first straightforward way assuming only a weaker form of a tightness condition for the sequence $(\sqrt{n}(\eta_n - \mu): n \in \mathbb{N})$ which in view of (1) is fulfilled if ξ_0 is separable. For this we shall need some conditions equivalent to the definition of compact differentiability as given in the next section.

As to measurability questions we note that in (2) only the limit element B_0 but not the $\sqrt{n}(F_n - F)$ on the left hand side are measurable with respect to the Borel σ -algebra on $(D[0, 1], \|\cdot\|_\infty)$. So instead of introducing sub- σ -algebras to retain measurability of the $\sqrt{n}(\eta_n - \mu)$, $n \in \mathbb{N}$, we shall work with a concept of weak convergence in the Hoffmann-Jørgensen sense where measurability assumptions only come up in connection with the limit element and not with the members of the sequence.

2 Notations and Definitions

Let us first recall the definition and some basic properties of the concept of weak convergence in the Hoffmann-Jørgensen sense.

Let (S, d) be a metric space, $(\Omega, \mathcal{A}, P)$ a probability space, $\xi_n: \Omega \rightarrow S$, $n \in \mathbb{N}$, functions and $\xi_0: \Omega \rightarrow S$ a random element in S . The sequence $(\xi_n: n \in \mathbb{N})$ is said to *converge weakly* to ξ_0 , we write $\xi_n \xrightarrow{\mathcal{L}} \xi_0$, if for every bounded and continuous function $f: S \rightarrow \mathbb{R}$

$$E_P^*(f \circ \xi_n) \rightarrow E_P(f \circ \xi_0) \quad (5)$$

where E_P^* denotes the outer expectation, that is for $g: \Omega \rightarrow \mathbb{R}$

$$E_P^*(g) := \inf \left\{ \int_{\Omega} h \, dP: h \text{ is } P - \text{quasi-integrable and } h \geq g \right\}.$$

The main point of this concept is that only the limit element ξ_0 and not the ξ_n 's are assumed to be measurable.

From the Portmanteau Theorem (cf. v.d. Vaart (1991), Theorem 1.10) we pick up two conditions, each equivalent to (5):

$$\limsup_{n \rightarrow \infty} P^*(\xi_n \in F) \leq P(\xi_0 \in F) \quad \text{for all closed } F \subset S \quad (6)$$

$$\liminf_{n \rightarrow \infty} P_*(\xi_n \in G) \geq P(\xi_0 \in G) \quad \text{for all open } G \subset S \quad (7)$$

Here P^* , resp. P_* denotes the outer, resp. inner measure pertaining to P . Note that for an arbitrary $A \subset \Omega$ we have

$$P^*(A) + P_*(\complement A) = 1 \quad (8)$$

with $\complement A$ abbreviating $\Omega \setminus A$.

The sequence $\xi_1, \xi_2, \dots$ is said to be *tight*, if for every $\varepsilon > 0$ there is a compact $K \subset S$ with

$$\limsup_{n \rightarrow \infty} P^*(\xi_n \notin K) < \varepsilon, \quad (9)$$

which according to (8) is equivalent to

$$\liminf_{n \rightarrow \infty} P_*(\xi_n \in K) \geq 1 - \varepsilon. \quad (10)$$

A single function $\xi: \Omega \rightarrow S$ is said to be *tight*, if the sequence $\xi, \xi, \dots$ is tight.

Weak convergence does not in general imply tightness, but $\xi_n \xrightarrow{\mathcal{L}} \xi_0$ together with ξ_0 tight implies a slightly weaker form of tightness for the $\xi_1, \xi_2, \dots$, the so called *asymptotic tightness*. Here is the definition: The sequence $\xi_1, \xi_2, \dots$ is called *asymptotically tight*, if for every $\varepsilon > 0$ there is a compact $K \subset S$ such that for every $\delta > 0$ we have

$$\limsup_{n \rightarrow \infty} P^*(\xi_n \notin K^\delta) < \varepsilon \quad (11)$$

where $K^\delta := \{x \in S: d(x, K) < \delta\}$.

Under the condition $\xi_n \xrightarrow{\mathcal{L}} \xi_0$ we even have (cf. v.d. Vaart (1991), Lemma 1.14)

$$\xi_0 \text{ tight} \Leftrightarrow \xi_1, \xi_2, \dots \text{ asymptotically tight}. \quad (12)$$

Now we turn to the definition of compact differentiability: Let B_1, B_2 be normed vector spaces and $\phi: B_1 \rightarrow B_2$ a function. Fix $x \in B_1$. Then ϕ is said to be *compactly* or *Hadamard differentiable* at x if there is a continuous linear function $D_x: B_1 \rightarrow B_2$ such that

$$\limsup_{t \rightarrow 0 \ h \in K} \left\| \frac{Rem(x + th)}{t} \right\| = 0 \quad \text{for all compact } K \subset B_1 \quad (13)$$

where $Rem(x + th)$ stands for

$$Rem(x + th) := \phi(x + th) - \phi(x) - D_x(th) .$$

We just remark that if we have (13) with K ‘bounded’ instead of ‘compact’, ϕ is called *boundedly* or *Fréchet differentiable*, while in the case of K ‘finite’, ϕ is said to be *directionally* or *Gateaux differentiable*. In all three cases the linear function D_x is the same and uniquely determined. Clearly bounded differentiability implies compact differentiability and that implies directional differentiability. Now in view of (12) and the fact that the definitions of both (asymptotical) tightness and compact differentiability deal with compact sets one can get the idea that compact differentiability is just the right condition in order to get (4). Indeed, directional differentiability turns out to be too weak while on the other hand there are hardly any boundedly differentiable functionals ϕ , unless one switches over to other norms on B_1 .

3 Compact Differentiability and the δ -Method

We start by giving some equivalent statements for (13) some of them more resembling the concept of asymptotical tightness. Note that in the definition (11) we have K^δ and since we shall deal with asymptotically tight sequences later on we would also like to have K^δ in the definition (13) instead of only K . Now (15) in the next proposition shows how this can be done (cf. also Lohse (1987)). In the proof of the main theorem we shall work on the basis of (16) and (21) resp., while (14) is only a standard presentation of compact differentiability often also taking the role of the definition. We will give detailed proofs here.

Proposition: Let $(B_1, \|\cdot\|_1)$, $(B_2, \|\cdot\|_2)$ be normed vector spaces, $x \in B_1$ and $\phi: B_1 \rightarrow B_2$. Then each of the following statements is equivalent to the compact differentiability of ϕ at x :

$$\limsup_{n \rightarrow \infty} \left\| \frac{Rem(x + t_n h_n)}{t_n} \right\|_2 = 0$$

for all sequences $(h_n)_{n \in \mathbb{N}} \subset B_1$ and $(t_n)_{n \in \mathbb{N}} \subset \mathbb{R} \setminus \{0\}$ with $t_n \xrightarrow{n \rightarrow \infty} 0$ and $(h_n)_{n \in \mathbb{N}}$ converging to some $h \in B_1$,

(14)

$$\limsup_{n \rightarrow \infty} \limsup_{m \rightarrow \infty} \sup_{h \in K^{\delta_n}} \left\| \frac{\text{Rem}(x + t_m h)}{t_m} \right\|_2 = 0$$

for all sequences $(\delta_n)_{n \in \mathbb{N}} \subset \mathbb{R}_+$, $(t_m)_{m \in \mathbb{N}} \subset \mathbb{R} \setminus \{0\}$
 with $\delta_n \searrow 0$, $t_m \xrightarrow{m \rightarrow \infty} 0$ and all compact $K \subset B_1$,

(15)

$$\limsup_{n \rightarrow \infty} \sup_{h \in K^{\delta_n}} \left\| \frac{\text{Rem}(x + t_n h)}{t_n} \right\|_2 = 0$$

for all sequences $(\delta_n)_{n \in \mathbb{N}} \subset \mathbb{R}_+$ and $(t_n)_{n \in \mathbb{N}} \subset \mathbb{R} \setminus \{0\}$ with $\delta_n \searrow 0$,
 $t_n \xrightarrow{n \rightarrow \infty} 0$ and all compact $K \subset B_1$,

(16)

$$\limsup_{t \rightarrow 0} \sup_{n \in \mathbb{N}} \left\| \frac{\text{Rem}(x + t h_n)}{t} \right\|_2 = 0$$

for all sequences $(h_n)_{n \in \mathbb{N}} \subset B_1$ with $d_1(h_n, K) \xrightarrow{n \rightarrow \infty} 0$
 and all compact $K \subset B_1$.

(17)

Here d_1 denotes the metric induced by $\|\cdot\|_1$.

Proof: For simplicity we put

$$\psi_t(h) := \left\| \frac{\text{Rem}(x + t h)}{t} \right\|_2, \quad t \in \mathbb{R} \setminus \{0\}, \quad h \in B_1. \quad (18)$$

(13) $\Rightarrow$ (14):

Let $t_n \rightarrow 0$. For $h_n \rightarrow h$ let $K := \{h_n : n \in \mathbb{N}\} \cup \{h\}$. Then K is compact and we have for $t_n \rightarrow 0$

$$0 \leq \limsup_{n \rightarrow \infty} \psi_{t_n}(h_n) \leq \limsup_{n \rightarrow \infty} \sup_{h' \in K} \psi_{t_n}(h') = 0$$

according to (13).

(14) $\Rightarrow$ (15):

Suppose (15) does not hold. Then there exists $\varepsilon > 0$, a compact $K \subset B_1$ and $\delta_n \searrow 0$, $t_m \rightarrow 0$ with

$$\limsup_{n \rightarrow \infty} \limsup_{m \rightarrow \infty} \sup_{h \in K^{\delta_n}} \psi_{t_m}(h) > \varepsilon.$$

As $\left(\sup_{h \in K^{\delta_n}} \psi_{t_m}(h) \right)_{n \in \mathbb{N}}$ forms a nonincreasing sequence for fixed $m \in \mathbb{N}$, we have

$$\limsup_{m \rightarrow \infty} \sup_{h \in K^{\delta_n}} \psi_{t_m}(h) \geq \varepsilon \quad \text{for all } n \in \mathbb{N} ,$$

from which we can construct a subsequence $(s_n)_{n \in \mathbb{N}}$ of $(t_n)_{n \in \mathbb{N}}$ with

$$\sup_{h \in K^{\delta_n}} \psi_{s_n}(h) \geq \frac{\varepsilon}{2} \quad \text{for all } n \in \mathbb{N} .$$

So, for each $n \in \mathbb{N}$ there is a $h_n \in K^{\delta_n}$ with

$$\psi_{s_n}(h_n) \geq \frac{\varepsilon}{4}$$

and according to construction

$$d_1(h_n, K) \xrightarrow{n \rightarrow \infty} 0 .$$

K being compact there is a subsequence $(h_{n'})_{n'}$ of $(h_n)_n$ converging to some $h \in K$ and for this subsequence we have

$$\limsup_{n' \rightarrow \infty} \psi_{s_{n'}}(h_{n'}) > 0 ,$$

in contradiction to (14).

(15) $\Rightarrow$ (16):

Let $K \subset B_1$ be compact, $\delta_n \searrow 0$, $t_m \rightarrow 0$ and $\varepsilon > 0$.

Choose $n_0 \in \mathbb{N}$ such that

$$\limsup_{m \rightarrow \infty} \sup_{h \in K^{\delta_n}} \psi_{t_m}(h) \leq \varepsilon \quad \text{for all } n \geq n_0$$

and $m_0 \in \mathbb{N}$ such that

$$\sup_{h \in K^{\delta_{n_0}}} \psi_{t_m}(h) \leq 2\varepsilon \quad \text{for all } m \geq m_0 .$$

As

$$\sup_{h \in K^{\delta_n}} \psi_{t_m}(h) \leq \sup_{h \in K^{\delta_{n_0}}} \psi_{t_m}(h) \quad \text{for all } n \geq n_0, \quad m \in \mathbb{N}$$

we have

$$\sup_{h \in K^{\delta_n}} \psi_{t_n}(h) \leq 2\varepsilon \quad \text{for all } n \geq \max\{n_0, m_0\}.$$

$\varepsilon > 0$ being arbitrary, (16) follows.

(16) $\Rightarrow$ (17):

Let $K \subset B_1$ be compact and $d_1(h_n, K) \xrightarrow{n \rightarrow \infty} 0$. Then $K_1 := K \cup \{h_n; n \in \mathbb{N}\}$ is compact and we have for every $\delta > 0$ and $t \in \mathbb{R} \setminus \{0\}$

$$0 \leq \sup_{n \in \mathbb{N}} \psi_t(h_n) \leq \sup_{h \in K_1} \psi_t(h) \leq \sup_{h \in K_1^\delta} \psi_t(h).$$

Applying (16) with an arbitrary sequence $(\delta_n)_{n \in \mathbb{N}}$ with $\delta_n \searrow 0$ gives (17).

(17) $\Rightarrow$ (13):

Suppose there is a compact $K \subset B_1$ and $t_n \rightarrow 0$ with

$$\limsup_{n \rightarrow \infty} \sup_{h \in K} \psi_{t_n}(h) \geq \varepsilon > 0.$$

Then there exists an infinite subset $I \subset \mathbb{N}$ for which

$$\sup_{h \in K} \psi_{t_n}(h) \geq \frac{\varepsilon}{2} \quad \text{for all } n \in I.$$

For $n \in I$ choose $h_n \in K$ with

$$\psi_{t_n}(h_n) \geq \frac{\varepsilon}{4}$$

and put $h_n := h \in K$ for $n \notin I$ with an arbitrary $h \in K$. Then we have

$$d_1(h_n, K) = 0 \quad \text{for all } n \in \mathbb{N}$$

and

$$\limsup_{n \rightarrow \infty} \sup_{m \in \mathbb{N}} \psi_{t_n}(h_m) \geq \limsup_{n \rightarrow \infty} \psi_{t_n}(h_n) \geq \frac{\varepsilon}{4},$$

in contradiction to (17).

This completes the proof of the proposition.

Following Gill (1989), p. 102, we can use the proposition in order to introduce a weaker kind of compact differentiability. In view of (14) we say that $\phi: B_1 \rightarrow B_2$ is compactly differentiable at $x \in B_1$ *tangentially to some subspace* $H \subset B_1$ if there is a continuous linear function $D_x: B_1 \rightarrow B_2$ such that

$$\limsup_{n \rightarrow \infty} \left\| \frac{Rem(x + t_n h_n)}{t_n} \right\|_2 = 0$$

$$\begin{aligned} &\text{for all sequences } (h_n)_{n \in \mathbb{N}} \subset B_1 \text{ and } (t_n)_{n \in \mathbb{N}} \subset \mathbb{R} \setminus \{0\} \text{ with } t_n \xrightarrow{n \rightarrow \infty} 0 \\ &\text{and } (h_n)_{n \in \mathbb{N}} \text{ converging to some } h \in H. \end{aligned} \quad (19)$$

Note that the h_n in (19) still can be chosen to lie in B_1 while the limit h has to be an element of H only. Gill (1989), p. 109, gives examples of ϕ 's which fit in this new, slightly generalized concept of compact differentiability.

In order to adjust the conditions in (15), (16) and (17) one is tempted to replace the set K by the intersection $K \cap H$. Doing so, (15), (16) and (17) turn into

$$\limsup_{n \rightarrow \infty} \limsup_{m \rightarrow \infty} \sup_{h \in (K \cap H)^{\delta_n}} \left\| \frac{Rem(x + t_m h)}{t_m} \right\|_2 = 0$$

$$\begin{aligned} &\text{for all sequences } (\delta_n)_{n \in \mathbb{N}} \subset \mathbb{R}_+, (t_n)_{n \in \mathbb{N}} \subset \mathbb{R} \setminus \{0\} \\ &\text{with } \delta_n \searrow 0, t_m \xrightarrow{m \rightarrow \infty} 0 \text{ and all compact } K \subset B_1, \end{aligned} \quad (20)$$

$$\limsup_{n \rightarrow \infty} \sup_{h \in (K \cap H)^{\delta_n}} \left\| \frac{Rem(x + t_n h)}{t_n} \right\|_2 = 0$$

$$\begin{aligned} &\text{for all sequences } (\delta_n)_{n \in \mathbb{N}} \subset \mathbb{R}_+, (t_n)_{n \in \mathbb{N}} \subset \mathbb{R} \setminus \{0\} \text{ with } \delta_n \searrow 0, \\ &t_n \xrightarrow{n \rightarrow \infty} 0 \text{ and all compact } K \subset B_1, \end{aligned} \quad (21)$$

$$\lim_{t \rightarrow 0} \sup_{n \in \mathbb{N}} \left\| \frac{\text{Rem}(x + th_n)}{t} \right\|_2 = 0$$

$$\begin{aligned} & \text{for all sequences } (h_n)_{n \in \mathbb{N}} \subset B_1 \text{ with } d_1(h_n, K \cap H) \xrightarrow{n \rightarrow \infty} 0 \\ & \text{and all compact } K \subset B_1, \end{aligned} \quad (22)$$

while (13) now would read

$$\lim_{t \rightarrow 0} \sup_{h \in K \cap H} \frac{\text{Rem}(x + th)}{t} = 0 \quad \text{for all compact } K \subset B_1. \quad (23)$$

The next lemma shows the relationship between these statements:

Lemma: For $\phi: B_1 \rightarrow B_2$, $x \in B_1$ and H a subspace of B_1 , we have

$$(22) \Rightarrow (19) \Leftarrow (20) \Leftrightarrow (21) \Rightarrow (23)$$

and $(19) \Rightarrow (20)$ if H is closed.

Proof: For $(20) \Rightarrow (21)$ we refer to the proof of $(15) \Rightarrow (16)$.

$(21) \Rightarrow (20)$:

Suppose (20) does not hold. Then, with $\psi_t(h)$ defined as in (18) and following the lines of the proof of $(14) \Rightarrow (15)$ we find $\varepsilon > 0$, $s_n \rightarrow 0$ and $\delta_n \searrow 0$ such that

$$\sup_{h \in (K \cap H)^{t_n}} \psi_{s_n}(h) \geq \frac{\varepsilon}{2} \quad \text{for all } n \in \mathbb{N}.$$

contradicting (21).

$(21) \Rightarrow (19)$:

We have for $t_n \rightarrow 0$, $h_n \rightarrow h \in H$

$$\psi_{t_n}(h_n) \leq \sup_{h' \in (K \cap H)^{t_n}} \psi_{t_n}(h')$$

with $K := \{h\}$ and $\delta_n := \max \left\{ \sup_{m \geq n} d_1(h_m, h), \frac{1}{n} \right\}$. Now $\delta_n \searrow 0$ and (19) follows.

(21) $\Rightarrow$ (23) is obvious, also (22) $\Rightarrow$ (19), where we once again take $K := \{h\}$.
 (19) $\Rightarrow$ (20):

Following the proof of (14) $\Rightarrow$ (15) with $K \cap H$ taking the place of K , we note that for the limit element h of the sequence $(h_{n'})_{n' \in \mathbb{N}}$ we have $h \in K \cap H$ since H is closed.

Of course the conditions (19) to (23) are equivalent for $H = B_1$ what is just what the proposition states.

Now we turn to the main theorem:

Theorem (δ -Method): Let $(\Omega, \mathcal{A}, P)$ be a probability space, B_1, B_2 normed vector spaces, H a closed subspace of B_1 and $\phi: B_1 \rightarrow B_2$ compactly differentiable at $\mu \in B_1$ tangentially to H . Consider functions $\eta_n: \Omega \rightarrow B_1$, $n \in \mathbb{N}$, for which $\xi_n := \sqrt{n}(\eta_n - \mu) \xrightarrow{\mathcal{L}} \xi_0$ which ξ_0 is a separable random element, i.e. the induced probability measure $P \circ \xi_0^{-1}$ has a separable support. Suppose further that $P(\xi_0 \notin H) = 0$. Then we have

$$\|\sqrt{n}(\phi(\eta_n) - \phi(\mu)) - D_\mu(\sqrt{n}(\eta_n - \mu))\|_2^* \xrightarrow{P} 0, \quad (24)$$

and

$$\sqrt{n}(\phi(\eta_n) - \phi(\mu)) \xrightarrow{\mathcal{L}} D_\mu(\xi_0). \quad (25)$$

Proof: ξ_0 being separable, we can assume w.l.o.g. that ξ_0 is tight (cf. v.d. Vaart (1991), p. 7). Then for each $\varepsilon > 0$ there is a compact $K \subset B_1$ with

$$P(\xi_0 \in K) \geq 1 - \varepsilon.$$

Since $(K \cap H)^\delta$ is open for all $\delta > 0$, (7) gives

$$\liminf_{n \rightarrow \infty} P_*(\xi_n \in (K \cap H)^\delta) \geq P(\xi_0 \in (K \cap H)^\delta)$$

and using $P(\xi_0 \notin H) = 0$ we further have

$$\begin{aligned} P(\xi_0 \in (K \cap H)^\delta) &\geq P(\xi_0 \in K \cap H) = P(\xi_0 \in K) - P(\xi_0 \in K \cap \complement H) \\ &\geq P(\xi_0 \in K) - P(\xi_0 \notin H) \\ &\geq 1 - \varepsilon. \end{aligned} \quad (26)$$

So $\liminf_{n \rightarrow \infty} P_*(\xi_n \in (K \cap H)^\delta) \geq 1 - \varepsilon$ or equivalently

$$\limsup_{n \rightarrow \infty} P^*(\xi_n \notin (K \cap H)^\delta) \leq \varepsilon .$$

It follows that there is a sequence $(\delta_n)_{n \in \mathbb{N}}$ of positive real numbers with $\delta_n \searrow 0$ such that

$$\limsup_{n \rightarrow \infty} P^*(\xi_n \notin (K \cap H)^{\delta_n}) \leq 2\varepsilon .$$

Now, ϕ being compactly differentiable at μ tangentially to the closed subspace H of B_1 , (21), being equivalent to (19) via the Lemma, by substituting $\frac{1}{\sqrt{n}}$ for t_n and μ for x , yields

$$\sup_{h \in (K \cap H)^{\delta_n}} \left\| \sqrt{n} \left(\phi \left(\frac{h}{\sqrt{n}} + \mu \right) - \phi(\mu) \right) - D_\mu(h) \right\|_2 \xrightarrow{n \rightarrow \infty} 0$$

and with $L_n := \{\omega \in \Omega: \xi_n(\omega) \in (K \cap H)^{\delta_n}\}$ we also have

$$\sup_{\omega \in L_n} \left\| \sqrt{n} \left(\phi \left(\frac{\xi_n(\omega)}{\sqrt{n}} + \mu \right) - \phi(\mu) \right) - D_\mu(\xi_n(\omega)) \right\|_2 \xrightarrow{n \rightarrow \infty} 0 ,$$

resp.

$$\sup_{\omega \in L_n} \zeta_n(\omega) \xrightarrow{n \rightarrow \infty} 0 \tag{27}$$

with

$$\zeta_n(\omega) := \left\| \sqrt{n}(\phi(\eta_n(\omega)) - \phi(\mu)) - D_\mu(\sqrt{n}(\eta_n(\omega) - \mu)) \right\|_2 .$$

$(\zeta_n)_{n \in \mathbb{N}}$ is a sequence of real (not necessarily measurable) functions for which we want to show $\zeta_n^* \xrightarrow{P} 0$ with ζ_n^* denoting the measurable cover of ζ_n (cf. v.d. Vaart (1991), Lemma 1.1). Since for each $\delta > 0$ we have (cf. v.d. Vaart (1991), Lemma 1.2)

$$P(\zeta_n^* > \delta) = P^*(\zeta_n > \delta) .$$

We get, the outer measure P^* being subadditive, for each $\delta > 0$

$$\begin{aligned}
 \limsup_{n \rightarrow \infty} P(\zeta_n^* > \delta) &= \limsup_{n \rightarrow \infty} P^*(\zeta_n > \delta) \\
 &\leq \limsup_{n \rightarrow \infty} P^*(\{\zeta_n > \delta\} \cap L_n) + \limsup_{n \rightarrow \infty} P^*(\{\zeta_n > \delta\} \cap \complement L_n) \\
 &\stackrel{(27)}{=} \limsup_{n \rightarrow \infty} P^*(\{\zeta_n > \delta\} \cap \complement L_n) \\
 &\leq 2\varepsilon .
 \end{aligned}$$

$\varepsilon > 0$ being arbitrary we have shown $\zeta_n^* \xrightarrow{P} 0$, which is (24).

Now (25) follows as an easy application of the well known Cramer-type result (cf. v.d. Vaart (1991), Lemma 1.48, which is a direct consequence of (6)) and the Continuous mapping theorem 1.12 in v.d. Vaart (1991).

Note that in the proof compact differentiability tangentially to H only comes up through (21) via the Lemma. So if we worked with (21) instead of the slightly weaker condition (19) as the definition of compact differentiability tangentially to H , we would have the same proof while in the formulation of the Theorem we could drop the assumption that H is closed. Since then H need no longer be measurable either and we have not assumed the underlying probability space being complete, $P(\xi_0 \notin H) = 0$ has to be replaced by $P^*(\xi_0 \notin H) = 0$ and in the proof we also have to write P^* instead of P on the right hand side of (26).

Acknowledgement: The author would like to thank Prof. Dr. P. Gaenssler for helpful comments and suggestions.

References

- Bickel PJ, Freedman DA (1981) Some asymptotics results for the bootstrap. *Annals of Statistics* 9:1196–1217
- Fernholz LT (1983) Von Mises calculus for statistical functionals. *Lecture Notes in Statistics* 19. Springer Verlag New York
- Gill RD (1989) Non- and Semi-parametric maximum likelihood estimators and the von Mises method (Part 1). *Scandinavian Journal of Statistics* 16:97–128
- Lohse K (1987) The consistency of the bootstrap. *Statistics & Decisions* 5:353–366
- Reeds JA (1978) Jackknifing maximum likelihood estimates. *Annals of Statistics* 6:727–739
- v.d. Vaart A (1991) Stochastic convergence. Unpublished draft, Free University Amsterdam

Experiments with Mixtures: Optimal Allocations for Becker's Models

SHUANGZHE LIU AND HEINZ NEUDECKER

Institute of Actuarial Science and Econometrics, University of Amsterdam, Roetersstraat 11,
1018 WB Amsterdam, The Netherlands

Abstract: Extending Scheffé's simplex-centroid design for experiments with mixtures, we introduce a *weighted* simplex-centroid design for a class of mixture models. Becker's homogeneous functions of degree one belong to this class. By applying optimal design theory, we obtain *A*-, *D*- and *I*-optimal allocations of observations for Becker's models.

Key Words: Experiments with mixtures, Becker's models, Scheffé's canonical polynomials, weighted simplex-centroid design, *A*-, *D*- and *I*-optimal allocations.

1 Introduction

In experiments with mixtures, the response is usually assumed not to depend on the total amount in the mixture, but only on the proportions of the components. For q -component mixtures, the region of interest is restricted to the $(q - 1)$ -dimensional simplex

$$S_{q-1} = \left\{ x = (x_1, \dots, x_q)' \mid \sum_{i=1}^q x_i = 1, x_i \geq 0, i = 1, \dots, q \right\}$$

Suppose the general response model on S_{q-1} is

$$y = X\beta + \varepsilon, \quad (1.1)$$

where y is an $N \times 1$ vector of observables; X is an $N \times p$ extended design matrix (see e.g. Atkinson and Donev [1992] and Pukelsheim [1993] for this and other terms), whose elements are the mixture component proportions and functions thereof; β is a $p \times 1$ vector of (unknown) parameters; ε is an $N \times 1$ vector of random errors with $E(\varepsilon) = 0$ and $D(\varepsilon) = \sigma^2 I$.

There exist a number of specific model forms. For an extensive survey, see Cornell [1990]. Becker [1968] considered mixture experiments involving com-

ponents with additive effects. He presented the following three response models with homogeneous functions for q components

$$H_1: E(y) = \sum_{i=1}^q \beta_i x_i + \sum_{i < j} \beta_{ij} \min(x_i, x_j) + \cdots \\ + \sum_{i_1 < \cdots < i_n} \beta_{i_1 \dots i_n} \min(x_{i_1}, \dots, x_{i_n}) , \quad (1.2)$$

$$H_2: E(y) = \sum_{i=1}^q \beta_i x_i + \sum_{i < j} \beta_{ij} \frac{x_i x_j}{x_i + x_j} + \cdots \\ + \sum_{i_1 < \cdots < i_n} \beta_{i_1 \dots i_n} \frac{x_{i_1} \cdots x_{i_n}}{(x_{i_1} + \cdots + x_{i_n})^{n-1}} , \quad (1.3)$$

$$H_3: E(y) = \sum_{i=1}^q \beta_i x_i + \sum_{i < j} \beta_{ij} (x_i x_j)^{1/2} + \cdots \\ + \sum_{i_1 < \cdots < i_n} \beta_{i_1 \dots i_n} (x_{i_1} \cdots x_{i_n})^{1/n} , \quad (1.4)$$

where $2 \leq n \leq q$.

If in H_2 any denominator is zero, the value of the corresponding term is taken to be zero.

The three models are homogeneous of degree one. Snee [1973] discussed the types of curvature implied by the models and supplied plots for the three curvatures. Cornell [1990] drew contour plots of estimated surfaces for comparing H_1 , H_2 and H_3 .

Kiefer [1961], Farrell et al. [1967], Laake [1975], Draper and St. John [1977], Donev [1989], Lim [1990], Chan [1992], Czitrom [1992], Mikaeili [1993] and Vining et al. [1993] studied optimal designs for a variety of models in the context of experiments with mixtures. Becker [1978] specifically considered optimal allocations for models H_1 , H_2 and H_3 for $n = 2$.

In this paper, we concentrate our attention on determining optimal allocations for Becker's models. We apply A -, D -, and I -optimality criteria that are often used in experiments with mixtures (see e.g. Vining et al. [1993]). For these and other criteria, see e.g. Krafft [1978], Silvey [1980], Atkinson and Donev [1992] and Pukelsheim [1993]. Also we refer to Magnus and Neudecker [1991] for related matrix calculus. In Section 2, we introduce the weighted simplex-centroid design (WSCD). This is an extension of Scheffé's simplex-centroid design (see Scheffé [1958, 1963]). In Section 3, we present A -optimal allocations for models H_1 , H_2 and H_3 under WSCD. In Section 4, we examine D -optimal allocations and evaluate them for models H_1 , H_2 and H_3 for $n = 2$. In Section 5, we examine I -optimal allocations for models H_1 , H_2 and H_3 . Section 6 contains some concluding comments.

2 Weighted Simplex-Centroid Design and Matrix Forms

For studying the mixture response on S_{q-1} , the simplex-lattice and simplex-centroid designs are classical (see Scheffé [1958, 1963]). Cornell [1990] gave some alternatives. Discussing the modelling of the additivity property of the mixture components, Becker [1978] suggested the radial design for testing whether a component has an additive effect in a mixture.

It is well known that Scheffé's simplex-centroid design has optimal properties for Scheffé's linear and 2nd-order canonical polynomial models (see e.g. Atkinson and Donev [1992]). Also notice that the 3rd-order centroid points are a subset of support points of an optimal design for the full cubic model (see e.g. Mikaeili [1993]). It seems that simplex-centroid support points are suitable for "symmetric" model terms (such as $x_i x_j x_k$ and $\min(x_i, x_j)$, $1 \leq i < j < k \leq q$ for components $x_1, \dots, x_q$).

We give the following definition for WSCD on S_{q-1} (see also Liu and Neudecker [1995]): Let $x^{(k)}$ denote $(1/k, \dots, 1/k, 0, \dots, 0)' \in S_{q-1}$, $k = 1, \dots, n$. The component $1/k$ clearly has multiplicity k . In a WSCD, we assign a weight r_1 to each of the $\binom{q}{1}$ points formed by permutations of the q coordinates of $x^{(1)}$, a weight r_2 to each of the $\binom{q}{2}$ points corresponding to $x^{(2)}$, and so on, and finally, a weight r_n to each of the $\binom{q}{n}$ points corresponding to $x^{(n)}$. We then have

$$\binom{q}{1} r_1 + \dots + \binom{q}{n} r_n = 1. \quad (2.1)$$

As the simplex is symmetrically invariant with respect to the components $x_1, \dots, x_q$, it is legitimate to apply the same weight r_k to each of the points defined by the $\binom{q}{k}$ permutations of $x^{(k)}$, $k = 1, \dots, n$. (see also symmetric invariance discussed in Pukelsheim [1987, 1993] and Mikaeili [1988]). It should be noted that our WSCD can be used for a class of "symmetric" models. This class of models, which contain Becker's models H_1 , H_2 and H_3 , are symmetrically invariant with respect to the components $x_1, \dots, x_q$.

We shall now introduce the structures of the extended design matrices for model (1.1). The idea is to write the extended design matrix generically as $X = [f(x_{(1)}), \dots, f(x_{(N)})]'$, where $f(x_{(1)}), \dots, f(x_{(N)})$ are N distinct $p \times 1$ vectors, $x_{(m)}$ ($m = 1, \dots, N$) are N design points chosen from S_{q-1} . Note that $y = (y_{(1)}), \dots, y_{(N)}'$, where $y_{(m)}$ ($m = 1, \dots, N$) are N observables ($N \geq p \geq q \geq n \geq 2$).

For H_1 , the extended design matrix is $X_1 = [f_1(x_{(1)}), \dots, f_1(x_{(N)})]'$, (2.2)

where $f'_1 = (f'_{11}, f'_{12}, \dots, f'_{1n})$, (2.3)

$$f'_{11} = [x_1, \dots, x_q] , \quad f'_{12} = [\min(x_1, x_2), \dots, \min(x_{q-1}, x_q)], \dots ,$$

$$f'_{1n} = [\min(x_1, \dots, x_n), \dots, \min(x_{q-n+1}, \dots, x_q)] .$$

For H_2 , the extended design matrix is $X_2 = [f_2(x_{(1)}), \dots, f_2(x_{(N)})]'$, (2.4)

where $f'_2 = (f'_{21}, f'_{22}, \dots, f'_{2n})$, (2.5)

$$f'_{21} = [x_1, \dots, x_q] , \quad f'_{22} = \left[\frac{x_1 x_2}{x_1 + x_2}, \dots, \frac{x_{q-1} x_q}{x_{q-1} + x_q} \right], \dots ,$$

$$f'_{2n} = \left[\frac{x_1 \dots x_n}{(x_1 + \dots + x_n)^{n-1}}, \dots, \frac{x_{q-n+1} \dots x_q}{(x_{q-n+1} + \dots + x_q)^{n-1}} \right] .$$

For H_3 , the extended design matrix is $X_3 = [f_3(x_{(1)}), \dots, f_3(x_{(N)})]'$, (2.6)

where $f'_3 = (f'_{31}, f'_{32}, \dots, f'_{3n})$, (2.7)

$$f'_{31} = [x_1, \dots, x_q] , \quad f'_{32} = [(x_1 x_2)^{1/2}, \dots, (x_{q-1} x_q)^{1/2}], \dots ,$$

$$f'_{3n} = [(x_1 \dots x_n)^{1/n}, \dots, (x_{q-n+1} \dots x_q)^{1/n}] .$$

Observe that $f_{11} = f_{21} = f_{31}$. Under the WSCD, the q -dimensional design points $x_{(m)}$ ($m = 1, \dots, N$) are defined in the following (lexicographical) way: $x_{(1)} = (1, 0, \dots, 0)'$, ..., $x_{(q)} = (0, \dots, 0, 1)'$, ..., $x_{(N)} = (0, \dots, 0, 1/n, \dots, 1/n)'$.

Clearly $N = \binom{q}{1} + \dots + \binom{q}{n}$. It is easy to see that $p = N$. The extended design

matrices X_1, X_2 and X_3 can be expressed in a convenient way by using matrices M_{jk} ($j > k, j, k = 1, \dots, n$) and (the identity matrix) $I_{jj} = M_{jj}$ defined below.

The order of M_{jk} is $\binom{q}{j} \times \binom{q}{k}$. The matrix M_{j1} is based on $x^{(j)} = (1/j, \dots, 1/j, 0, \dots, 0)' = 1/j(1, \dots, 1, 0, \dots, 0)'$, where $1/j$ has multiplicity j . The $\binom{q}{j}$ rows of M_{j1} display the $\binom{q}{j}$ permutations of $x^{(j)}$. The matrices M_{jk} ($j > k > 1$) are then induced on M_{j1} according to the model specifications (see Appendix I(a) for an example). Finally $M_{jk} = 0$ for $k > j$.

This then leads to the (nonsingular) extended design matrices

$$X_1 = X_3 = \begin{bmatrix} I_{11} & 0 & 0 & \dots & 0 \\ 2^{-1}M_{21} & 2^{-1}I_{22} & 0 & \dots & 0 \\ 3^{-1}M_{31} & 3^{-1}M_{32} & 3^{-1}I_{33} & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ n^{-1}M_{n1} & n^{-1}M_{n2} & n^{-1}M_{n3} & \dots & n^{-1}I_{nn} \end{bmatrix}, \quad (2.8)$$

$$X_2 = \begin{bmatrix} I_{11} & 0 & 0 & \dots & 0 \\ 2^{-1}M_{21} & (2 \times 2)^{-1}I_{22} & 0 & \dots & 0 \\ 3^{-1}M_{31} & (2 \times 3)^{-1}M_{32} & (3^2 \times 3)^{-1}I_{33} & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ n^{-1}M_{n1} & (2n)^{-1}M_{n2} & (3^2n)^{-1}M_{n3} & \dots & (n^{n-1}n)^{-1}I_{nn} \end{bmatrix}. \quad (2.9)$$

For further details, see the Appendix of the paper and Liu and Neudecker [1995]. We then define the weight matrix

$$A = \begin{bmatrix} r_1 I_{11} & 0 & 0 & \dots & 0 \\ 0 & r_2 I_{22} & 0 & \dots & 0 \\ 0 & 0 & r_3 I_{33} & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & r_n I_{nn} \end{bmatrix}. \quad (2.10)$$

The models are fitted to the response by the method of least squares. The estimated response is $f'(x)b$, where $b = (X'AX)^{-1}X'A^{1/2}y$. Clearly $D(b) = \sigma^2(X'AX)^{-1}$. We shall now find optimal allocations for models H_1 , H_2 and H_3 . This means finding *optimal* weights r_k^* ($1 \leq k \leq n$), subject to (2.1), such that the WSCD is A -, D - or I -optimal, respectively.

3 A -Optimal Allocation

For A -optimality, the average variance of the parameter estimators $\text{tr}(X'AX)^{-1}$ is to be minimized. For our A -optimal allocation, $\text{tr}(X'AX)^{-1}$ is minimized subject to (2.1).

We have the following result:

Theorem 1: Under the WSCD, A -optimal allocations for models H_1 , H_2 and H_3 are

$$r_k^* = \frac{g(k)}{\sum_{k=1}^n \binom{q}{k} g(k)}, \quad (3.1)$$

where

$$\begin{aligned} \text{(a) for } H_1, g(k) &= \sqrt{k^2 \sum_{i=k}^n \binom{q-k}{i-k}}, k = 1, \dots, n. \\ \text{(b) for } H_2, g(k) &= \sqrt{k^2 \sum_{i=k}^n i^{2i-2} \binom{q-k}{i-k}}, k = 1, \dots, n. \\ \text{(c) for } H_3, g(k) &= \sqrt{k^2 \sum_{i=k}^n \binom{q-k}{i-k}}, k = 1, \dots, n. \end{aligned}$$

Proof: For model H_1 , under the WSCD the extended design matrix is X_1 , and the variance matrix is $(X_1'AX_1)^{-1}$. We then have from Appendix II(c)

$$\text{tr}(X_1'AX_1)^{-1} = \sum_{k=1}^n \left[\frac{k^2}{r_k} \sum_{i=k}^n \text{tr } M_{ik} M_{ik}' \right] = \sum_{k=1}^n \left[\frac{k^2}{r_k} \sum_{i=k}^n \binom{q-k}{i-k} \binom{q}{k} \right].$$

Formulate the optimization problem for optimal allocation as

$$\text{Min } \sum_{k=1}^n \frac{m_k}{r_k},$$

$$\text{subject to } \sum_{k=1}^n \binom{q}{k} r_k = 1,$$

$$\text{where } m_k = k^2 \sum_{i=k}^n \binom{q-k}{i-k} \binom{q}{k}.$$

Solving this problem by the Lagrangian function method, we get (3.1) and (a).

Note that $\text{tr}(X_1'AX_1)^{-1}$ is a convex function of A . As $(X_1'AX_1)^{-1} = X_1^{-1}A^{-1}X_1'^{-1}$, this matrix-valued function is convex (this is a special case of statement A.1 on page 69 in Silvey [1980]; for further extensions, see e.g. Krafitt [1978] and Pukelsheim [1993]), and hence $\text{tr}(X_1'AX_1)^{-1}$ is convex (for properties of matrix traces, see e.g. Magnus and Neudecker [1991]).

The proof for (c) is the same as above. In a similar way, we can verify (3.1) and (b) by using Appendix II(g). $\square$

4 *D*-Optimal Allocation

For *D*-optimality, the generalized variance of the parameter estimators is minimized, or equivalently $\ln[\det(X'AX)]$ is maximized. A version of the equivalence theorem for *D*-optimality (see e.g. Silvey [1980] and Atkinson and Donev [1992]) is that for any $x \in S_{q-1}$

$$f'(x)(X'AX)^{-1}f(x) \leq p, \quad (4.1)$$

where X and A are the provided extended design and weight matrices respectively, the number of parameters $p = \binom{q}{1} + \cdots + \binom{q}{n}$, for models H_1 , H_2 and H_3 . The *D*-optimality carries over to the entire simplex if and only if (4.1) holds.

We proceed as follows. We first obtain a *D*-optimal allocation by maximizing $\ln[\det(X'AX)]$ subject to (2.1). Then, we further consider the case when $n = 2$.

Theorem 2: The WSCD with weights $r_1 = \cdots = r_n = 1/p$ is a *D*-optimal allocation for models H_1 , H_2 and H_3 .

Proof: Model H_1 case: the *D*-optimal allocation problem is $\text{Max } \ln[\det(X'_1AX_1)]$ subject to (2.1). Because X_1 is fixed, we consider instead $\text{Max } \ln(\det A)$ subject to (2.1).

By using (2.10), this leads to the problem

$$\begin{aligned} \text{Max} \quad & \sum_{k=1}^n \binom{q}{k} \ln r_k \\ \text{subject to} \quad & \sum_{k=1}^n \binom{q}{k} r_k = 1. \end{aligned}$$

It is easy to establish $r_k^* = 1/p$, $k = 1, \dots, n$.

By the same method, we get the same results for the models H_2 and H_3 . $\square$

We now examine the case $n = 2$ to derive the following theorem.

Theorem 3: Under the WSCD, D -optimal allocations for models H_1 , H_2 and H_3 for $n = 2$ are not D -optimal designs over S_{q-1} when $q \geq 3$; but they are D -optimal designs over S_{q-1} when $q = 2$.

Proof: Model H_1 case. When $q \geq 3$, we show that (4.1) does *not* occur at the q th-centroid point $x^{(q)} = (1/q, \dots, 1/q)'$.

Let $\phi(x) = f'(x)(X_{(2)}A_{(2)}X_{(2)})^{-1}f(x)$, where $f'(x) = [x_1, \dots, x_q, \min(x_1, x_2), \dots, \min(x_{q-1}, x_q)]$, $p = q + \binom{q}{2}$, $X_{(2)}$ and $A_{(2)}$ are the corresponding $p \times p$ design extended and weight matrices respectively,

$$X_{(2)} = \begin{bmatrix} I_{11} & 0 \\ 2^{-1}M_{21} & 2^{-1}I_{22} \end{bmatrix}, \quad A_{(2)} = \begin{bmatrix} r_1^* I_{11} & 0 \\ 0 & r_2^* I_{22} \end{bmatrix},$$

blocks I_{11} , I_{22} and M_{21} are of order $\binom{q}{1} \times \binom{q}{1}$, $\binom{q}{2} \times \binom{q}{2}$ and $\binom{q}{2} \times \binom{q}{1}$ respectively, $r_1^* = r_2^* = 1/p$.

At the q th-centroid point $x^{(q)} = (1/q, \dots, 1/q)'$, we have $1 \times p$ vector $f'(x^{(q)}) = (1/q, \dots, 1/q)$ and easily find

$$\begin{aligned} \phi(x^{(q)}) &= f'(x^{(q)})X_{(2)}^{-1}A_{(2)}^{-1}X_{(2)}'^{-1}f(x^{(q)}) \\ &= \left[q + \binom{q}{2} \right] [q - 2 + 2/q] \not\leq p, \end{aligned}$$

where $q - 2 + 2/q > 1$, $q \geq 3$.

When $q = 2$, we show (4.1) *does* hold for any $x \in S_{q-1}$. Clearly, $f'(x) = [x_1, x_2, \min(x_1, x_2)]$, $p = 3$,

$$X_{(2)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1/2 & 1/2 & 1/2 \end{bmatrix}, \quad A_{(2)} = \begin{bmatrix} 1/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/3 \end{bmatrix}.$$

We then have for any $x = (x_1, x_2)' \in S_{q-1}$

$$\begin{aligned}
& [x_1, x_2, \min(x_1, x_2)] X_{(2)}^{-1} A_{(2)}^{-1} X_{(2)}'^{-1} [x_1, x_2, \min(x_1, x_2)]' \\
& = 3\{[x_1 - \min(x_1, x_2)]^2 + [x_2 - \min(x_1, x_2)]^2 + [2 \min(x_1, x_2)]^2\} \\
& \leq 3 = p ,
\end{aligned}$$

and hence (4.1) is established.

The proofs for H_2 and H_3 are similar and omitted here. $\square$

5 *I*-Optimal Allocation

Let

$$W = \int \cdots \int \text{var}(\hat{y}) \, dx_1 \dots dx_{q-1} \quad (5.1)$$

denote the integrated variance of the parameter estimators over S_{q-1} , where $\hat{y}$ is the estimated response $f'(x)b$, $b = (X'AX)^{-1}X'A^{1/2}y$. The *I*-optimality criterion requires that W be minimized (see Lambrakis [1968] and Laake [1975]). Note that *I*-optimality (also called *V*-optimality) is closely related to Linear optimality (see e.g. Atkinson and Donev [1992] and Pukelsheim [1993]).

In experiments with mixtures, Lambrakis [1968] and Laake [1975] discussed *I*-optimal allocations for several canonical polynomials. Mikaeili [1988] discussed the allocations for the full cubic canonical polynomial. We now study *I*-optimal allocations for Becker's models H_1 , H_2 and H_3 under the WSCD.

Replacing $\text{var}(\hat{y})$ by $(X'AX)^{-1}$, we can write

$$W = \int \cdots \int f'(x)(X'AX)^{-1}f(x) \, dx_1 \dots dx_{q-1} = \text{tr}[(X'AX)^{-1}F] , \quad (5.2)$$

where $F = \int \cdots \int f(x)f(x)' \, dx_1 \dots dx_{q-1}$, and $f(x)$ is the model term vector.

The problem to be solved is to minimize (5.2) subject to (2.1). The solution is an *I*-optimal allocation.

Theorem 4: Under the WSCD, the *I*-optimal allocations for the n th-order models H_1 , H_2 and H_3 are

$$r_k^* = \frac{g(k)}{\sum_{k=1}^n \binom{q}{k} g(k)} , \quad (5.3)$$

where

$$(a) \text{ for } H_1, g(k) = \sqrt{\frac{\binom{q}{n} a_k}{\binom{q}{k} a_n}},$$

$$a_k = \sum_{i,j=k}^n (-1)^{i+j} k^2 \operatorname{tr}(M_{ik} M'_{jk} A_{ji}), \quad k = 1, 2, \dots, n. \quad (5.4)$$

$A_{ji} = \int \cdots \int f_{1j} f'_{1i} dx_1 \dots dx_{q-1}$ is the (j, i) -th block of $F_1 = \int \cdots \int f_1 f'_1 dx_1 \dots dx_{q-1}$, f_1 and f_{1i} ($i = 1, \dots, n$) are the same as in (2.3).

$$(b) \text{ for } H_2, g(k) = \sqrt{\frac{\binom{q}{n} b_k}{\binom{q}{k} b_n}},$$

$$b_k = \sum_{i,j=k}^n (-1)^{i+j} k^2 i^{i-1} j^{j-1} \operatorname{tr}(M_{ik} M'_{jk} B_{ji}), \quad k = 1, 2, \dots, n.$$

$B_{ji} = \int \cdots \int f_{2j} f'_{2i} dx_1 \dots dx_{q-1}$ is the (j, i) -th block of $F_2 = \int \cdots \int f_2 f'_2 dx_1 \dots dx_{q-1}$, f_2 and f_{2i} ($i = 1, \dots, n$) are the same as in (2.5).

$$(c) \text{ for } H_3, g(k) = \sqrt{\frac{\binom{q}{n} c_k}{\binom{q}{k} c_n}},$$

$$c_k = \sum_{i,j=k}^n (-1)^{i+j} k^2 \operatorname{tr}(M_{ik} M'_{jk} C_{ji}), \quad k = 1, 2, \dots, n.$$

$C_{ji} = \int \cdots \int f_{3j} f'_{3i} dx_1 \dots dx_{q-1}$ is the (j, i) -th block of $F_3 = \int \cdots \int f_3 f'_3 dx_1 \dots dx_{q-1}$, f_3 and f_{3i} ($i = 1, \dots, n$) are the same as in (2.7).

Proof: (a) From Appendix II(d) we have

$$\operatorname{tr}[(X'_1 A X_1)^{-1} F_1] = \sum_{k=1}^n \frac{a_k}{r_k}.$$

Hence we consider the problem

$$\text{Min } \sum_{k=1}^n \frac{a_k}{r_k} \quad (5.5)$$

$$\text{subject to } \sum_{k=1}^n \binom{q}{k} r_k = 1 ,$$

where a_k is the same as in (5.4) .

It is easy to get the optimal allocation by using the Lagrange multiplier method.

Similarly we obtain (b) and (c) by using II(h) and II(i) of the Appendix respectively. $\square$

6 Comments

(I). Design points: For more discussions on choosing the design points for certain models in experiments with mixtures, see Cornell [1990], Fang and Wang [1994], Prescott et al. [1993] and Vining et al. [1993]. As we have seen, our approach was based on Scheffé's simplex-centroid design points.

(II). Optimal allocations: It is worth noting that the optimal allocations were obtained at the simplex-centroid points. We may examine whether they are indeed optimal designs over the whole simplex by using the equivalence theorem for A -, D - and I -optimalities (see also Theorem 3 in Section 4). For versions of the equivalence theorem for different optimalities and further studies, see e.g. Krafft [1978], Silvey [1980], Atkinson and Donev [1992] and Pukelsheim [1993]. For I -optimal allocations only, see Lambrakis [1968], Laake [1975], Becker [1978] and Mikaeili [1988].

(III). Matrix forms: In this paper, we express the variance functions in matrix form, then construct constrained optimization problems in accordance with the respective optimality criteria, and finally give optimal allocations. Using matrix calculus turns out to be very efficient to treat the problems in this paper. For other relevant considerations and approaches, see e.g. Becker [1978], Lim [1990] and Mikaeili [1993].

Appendix

We list some basic properties and results for M_{ik} ($k \leq i$) and X .

I. Examples and Properties.

(a) If $q = 4$, we have $\binom{q}{1} = \binom{4}{1} = 4$, $\binom{q}{2} = \binom{4}{2} = 6$, $\binom{q}{3} = \binom{4}{3} = 4$, and

$$M_{21} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \quad I_{22} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

$$M_{31} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}, \quad M_{32} = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}.$$

- (b) The number of ones in each row of M_{ik} is $\binom{i}{k}$, the number of ones in each column of M_{ik} is $\binom{q-k}{i-k}$;
- (c) $M_{ij}M_{jk} = \binom{i-k}{j-k}M_{ik}$;
- (d) $\text{tr } M_{ik}M'_{ik} = \binom{i}{k}\binom{q}{i} = \binom{q-k}{i-k}\binom{q}{k}$, where, in general, $\binom{i}{k} = \frac{i!}{k!(i-k)!}$, $k \leq j \leq i \leq q$, and $\binom{i}{0} = 1, i \geq 0$.

For related matters, see also Liu and Neudecker [1995].

II. Results for $X_1 = X_3$ and X_2 .

Let $\{\cdot\}_{ij}$ denote the (i, j) -th block of $\{\cdot\}$. We then have

- (a) $\{X_1^{-1}\}_{ij} = (-1)^{i+j}M_{ij}$;
- (b) $\{(X'_1AX_1)^{-1}\}_{ij} = \sum_{k=1}^{\min(i,j)} (-1)^{i+j} \frac{k^2}{r_k} M_{ik}M'_{jk}$;
- (c) $\text{tr}(X'_1AX_1)^{-1} = \sum_{k=1}^n \sum_{i=k}^n \frac{k^2}{r_k} \text{tr } M_{ik}M'_{ik} = \sum_{k=1}^n \sum_{i=k}^n \frac{k^2}{r_k} \binom{q-k}{i-k} \binom{q}{k}$;
- (d) $\text{tr}(X'_1AX_1)^{-1}F_1 = \sum_{k=1}^n \sum_{i,j=k}^n (-1)^{i+j} \frac{k^2}{r_k} \text{tr } M_{ik}M'_{jk}A_{ji}$.

- (e) $\{X_2^{-1}\}_{ij} = (-1)^{i+j} j i^{i-1} M_{ij}$;
 (f) $\{(X_2' A X_2)^{-1}\}_{ij} = \sum_{k=1}^{\min(i,j)} \frac{(-1)^{i+j} k^2 i^{i-1} j^{j-1}}{r_k} M_{ik} M'_{jk}$;
 (g) $\text{tr}(X_2' A X_2)^{-1} = \sum_{k=1}^n \sum_{i=k}^n \frac{k^2 i^{2i-2}}{r_k} \text{tr} M_{ik} M'_{ik} = \sum_{k=1}^n \sum_{i=k}^n \frac{k^2 i^{2i-2}}{r_k} \binom{q-k}{i-k} \binom{q}{k}$;
 (h) $\text{tr}(X_2' A X_2)^{-1} F_2 = \sum_{k=1}^n \sum_{i,j=k}^n \frac{(-1)^{i+j} k^2 i^{i-1} j^{j-1}}{r_k} \text{tr} M_{ik} M'_{jk} B_{ji}$;
 (i) $\text{tr}(X_3' A X_3)^{-1} F_3 = \sum_{k=1}^n \sum_{i,j=k}^n \frac{(-1)^{i+j} k^2}{r_k} \text{tr} M_{ik} M'_{jk} C_{ji}$.

We now prove (a). Let $Z_k = \begin{bmatrix} Z_{k-1} & 0 \\ k^{-1} G_k & k^{-1} I_{kk} \end{bmatrix}$, where $Z_1 = I_{11}$, $Z_n = X_1$ and $G_k = [M_{k1}, \dots, M_{k(k-1)}]$, ($k = 2, \dots, n$). We then have $Z_k^{-1} = \begin{bmatrix} Z_{k-1}^{-1} & 0 \\ -G_k Z_{k-1}^{-1} & k I_{kk} \end{bmatrix}$.

We apply the mathematical induction method with respect to k to establish (a) by using $\{X_1\}_{ik} = i^{-1} M_{ik}$ (see (2.8), note that $M_{ik} = 0$, if $k > i$) and I(c).

We obtain (b) from (a) and (2.10). Clearly (c) follows from (b) and I(d). It is easy to get (d) from (b).

Similarly to proving (a) we verify (e) by using (2.9) and I(c). From (e) and (2.10), (f) follows. From (f) and I(d), (g) follows. From (f), (h) follows. Noting $X_3 = X_1$, we obtain (i) from (b).

Acknowledgement: We would like to thank the referee for his critical and helpful remarks. A preliminary version of the paper was read at the 5th Tartu Conference on Multivariate Statistics, Otepää, Estonia (1994).

References

- Atkinson AC, Donev AN (1992) Optimum experimental designs. Clarendon Press, Oxford
 Becker NG (1968) Models for the response of a mixture. JR Stat Soc B 30:349–358
 Becker NG (1978) Models and designs for experiments with mixtures. Austral J Stat 20:195–208
 Chan LY (1992) D-optimal design for a quadratic log contrast model for experiments with mixtures. Commun Stat-Theor Meth 21:2909–2930
 Cornell JA (1990) Experiments with mixtures. 2nd edn. Wiley, New York
 Czitrom V (1992) Note on a mixture experiment with process variables. Commun Stat-Simul Comp 21:493–498
 Donev AN (1989) Designs of experiments with both mixture and qualitative factors. JR Stat Soc B 51:297–302
 Draper NR, St John RC (1977) A mixture model with inverse terms. Technometrics 19:37–46
 Fang KT, Wang Y (1994) Number-theoretic methods in statistics. Chapman & Hall, London
 Farrell RH, Kiefer J, Walbran A (1967) Optimum multivariate designs. In Proc 5th Berkeley Symp, Vol 1 LeCam LM, Neyman J (Eds) University of California Press, Berkeley 113–138

- Kiefer J (1961) Optimum designs in regression problems, II. *Ann Math Stat* 32:298–325
- Krafft O (1978) *Lineare statistische Modelle und optimale Versuchspläne*. Vandenhoeck und Ruprecht, Göttingen
- Laake P (1975) On the optimal allocation of observations in experiments with mixtures. *Scand J Stat* 2:153–157
- Lambrakis DP (1968) Experiments with p -component mixtures. *JR Stat Soc B* 30:137–144
- Lim YB (1990) D -optimal design for cubic polynomial regression on the q -simplex. *J Stat Plan Inf* 25:141–152
- Liu S, Neudecker H (1995) A V -optimal design for Scheffé's polynomial model. *Stat Prob Letters* 23:253–258
- Magnus JR, Neudecker H (1991) *Matrix differential calculus with applications in statistics and econometrics*. Wiley, Chichester
- Mikaeili F (1988) Allocations of measurements in experiments with mixtures. *Keio Sci Tech Reports* 41:25–37
- Mikaeili F (1993) D -optimum design for full cubic on q -simplex. *J Stat Plan Inf* 35:121–130
- Prescott P, Draper NR, Dean AM, Lewis SM (1993) Mixture designs for five components in orthogonal blocks. *J Appl Statist* 20:105–117
- Pukelsheim F (1987) Information increasing orderings in experimental design theory. *Int Stat Rev* 55:203–219
- Pukelsheim F (1993) *Optimal design of experiments*. Wiley, New York
- Scheffé H (1958) Experiments with mixtures. *JR Stat Soc B* 20:344–360
- Scheffé H (1963) The simplex-centroid design for experiments with mixtures. *JR Stat Soc B* 25:235–263
- Silvey SD (1980) *Optimal design*. Chapman & Hall, London
- Snee RD (1973) Techniques for the analysis of mixture data. *Technometrics* 15:517–528
- Vining GG, Cornell JA, Myers RH (1993) A graphical approach for evaluating mixture designs. *Appl Stat* 42:127–138

Received 02.11.1993

Revised version 05.07.1995

Optimal Design for Interacting Blocks with OAVS Incidence

J. P. MORGAN

Department of Mathematics and Statistics, Old Dominion University, Norfolk, Virginia 23529, USA

Abstract: Optimal design is studied for multiway cross classifications in which the blocking factors exhibit interaction and in which the incidence structure of the blocking factors describes an orthogonal array. Assuming the orthogonal array to be of sufficient strength (enough to allow orthogonal estimation of all of the interaction terms), easily used forms of the information matrix for treatment estimation are derived, and optimality conditions are stated. Some illustrative construction examples are given.

Some Key Words: balanced incomplete block design; nested row-column design; crossed blocking structure; optimal design; orthogonal array.

1 Introduction

The search for good experimental designs in settings with crossed nuisance factors dates back at least as far as the Fisher study of Latin squares, while the formal study of optimality questions in settings with two crossed nuisance factors has its roots in the papers of Kiefer (1958, 1975). For the general m -way heterogeneity setting the more recent starting point is Cheng (1978), who examined the situation of placing a fixed number of treatments at each combination of the m nuisance factors, assuming no interaction among those factors. Mukhopadhyay and Mukhopadhyay (1984) generalized Cheng's (1978) results by relaxing the complete crossing requirement on the nuisance factors; rather, they required only that the incidence structure of those factors describe an orthogonal array with variable number of symbols (OAVS) of strength two. In a further extension, the same authors (Bagchi and Mukhopadhyay, 1989) allowed two-way interaction among the nuisance factors, simultaneously increasing the strength requirement on the OAVS.

In this paper, the m -way heterogeneity setting with t -way interaction ($t < m$) is explored. Following Mukhopadhyay and Mukhopadhyay (1984) and Bagchi and Mukhopadhyay (1989), the strength of the OAVS described by the blocking structure will be allowed to depend on t , considerably simplifying the calculations. Using a notation due to Tjör (1984) and Bailey (1984, 1985), the

information matrix for estimation of treatment effects is derived (section 2) and manipulated to indicate optimal treatment assignments (section 3). The above-cited papers of Cheng (1978), Mukhopadhyay and Mukhopadhyay (1984), and Bagchi and Mukhopadhyay (1989) all include special cases of the more general formulation given here.

2 The Setting and the Information Matrix

As indicated in the introduction, there will be m crossed nuisance factors whose incidence structure can be written as an OAVS of strength h (say). Orthogonal arrays of variable strength, also called asymmetrical orthogonal arrays, were introduced by Rao (1947, 1973) and have been studied more recently by, among others, Mukhopadhyay (1981), Wang and Wu (1991), and Mukerjee and Wu (1996). Formally, an OAVS of strength h is an $n \times m$ matrix in which column j contains entries from a set of b_j elements, arranged so that selection of any h columns produces in rows every distinct h -tuple of symbols from those h sets with equal frequency. Here the m columns correspond to the m nuisance factors and the n rows to n experimental units, so that for an array $B = (B_{ij})$, the entry B_{ij} is the level of nuisance factor j on plot i . The model to be fitted will include additive effects for a set of v treatments, for the m nuisance factors, and for all interactions of the nuisance factors of order no greater than t , t being a fixed integer, $t < m$. The problem is to determine, in the face of this blocking structure, an optimal assignment of the v treatments to the n plots. A particular treatment assignment or design d will be stated via an $n \times v$ incidence matrix A_d . It is required that the strength h of B satisfy $h \geq \min(m, 2t)$. So for 2-way interactions, for instance, $h = 3$ for $m = 3$ and $h \geq 4$ for $m \geq 4$ (cf. Bagchi and Mukhopadhyay, 1989). While the bound placed on the strength somewhat limits this approach, it does not exclude the most important special case, that of the m factors being completely crossed.

To state the model and derive the relevant information matrix, the notations and concepts attendant to what Bailey (1984) has called *Tjur block structures* will be used. Readers to whom these are unfamiliar are referred to Tjur (1984) and Bailey (1984, 1985); the terminological and notational conventions adopted here are in accordance with the latter's recommendations. For $j = 1, \dots, m$ let F_j be the factor defined by the j^{th} column of B , that is, $F_j(i) = B_{ij}$. More generally, for each subset I of $\{1, 2, \dots, m\}$, $|I| \leq t$, define a factor G_I as the infimum (cf. Tjur, 1984, section 2, and Bailey, 1984, section D3) of the F_j 's identified by I . For $I = \{i_1, i_2, \dots, i_{t'}\}$ and $t' \leq t$,

$$G_I = F_{i_1} \wedge F_{i_2} \wedge \dots \wedge F_{i_{t'}}$$

and for $I = \emptyset$, $G_I = U$, the universal factor (for which $U(i) = 1$ for all units i). In terms of B , levels of the factor G_I correspond to distinct rows in the $n \times t'$ subarray defined by the columns with numbers in I .

Write J_m for the set of indices $\{1, 2, \dots, m\}$ identifying the m nuisance factors, and for any $q \leq m$, write J_q for the first q of these. Fitting a model with all interactions of order $\leq t$ for the blocking factors is exactly fitting a model for the set of factors

$$\mathcal{G} = \{G_I: I \subseteq J_m, |I| \leq t\} .$$

Now each G_I can be alternatively expressed in terms of its $n \times \prod_{j \in I} b_j$ incidence matrix X_I :

$$(X_I)_{il} = \begin{cases} 1 & \text{if } G_I(i) = l \\ 0 & \text{otherwise} \end{cases} ,$$

with l running over the levels of G_I . From these the model is written

$$Y = A_d \alpha + \sum_{|I| \leq t} X_I \beta_I + e ,$$

α , β_I being unknown parameter vectors and e a random vector satisfying the usual assumptions (uncorrelated, equivariable, mean 0 components).

Let $P_{\mathcal{G}}$ be the matrix which projects onto the space spanned by the columns of the X_I . Then the information, or C -, matrix for estimation of α is

$$C_d = A_d'(I - P_{\mathcal{G}})A_d \quad (1)$$

The first goal is to find an explicit expression for C_d in terms of the projection matrices $P_{G_I} = X_I(X_I'X_I)^{-}X_I'$ of the individual factors. To do so requires two properties relating the G_I , which are shown next.

Lemma 2.1: The factors G_I are pairwise geometrically orthogonal.

Proof: For the definition of geometric orthogonality, see Tjur (1984, section 3.2). Let $I_1 = \{i_1, i_2, \dots, i_{t_1}\}$ and $I_2 = \{j_1, j_2, \dots, j_{t_2}\}$ ($t_1, t_2 \leq t$). Also write $I_0 = I_1 \cap I_2$, $I_3 = I_1 \cup I_2$, $t_0 = |I_0|$, and $t_3 = |I_3|$. Note that $t_3 = t_1 + t_2 - t_0 \leq \min(m, 2t) \leq h$.

A sufficient condition for geometric orthogonality is that each cell of $G_{I_1} \wedge G_{I_2}$ has the same number of experimental units. Now

$$G_{I_1} \wedge G_{I_2} = F_{i_1} \wedge F_{i_2} \wedge \cdots \wedge F_{i_{t_1}} \wedge F_{j_1} \wedge \cdots \wedge F_{j_{t_2}} = G_{I_3}.$$

Levels of G_{I_3} are defined by distinct columns in the $t_3 \times n$ subarray of B identified by I_3 . Since $t_3 \leq h$, the OAVS property says that each of these occurs with the same frequency. $\square$

Lemma 2.2: For any two factors G_{I_1} and G_{I_2} with $I_1 \cap I_2 = I_0$, $P_{G_{I_1}} P_{G_{I_2}} = P_{G_{I_0}}$.

Proof: By lemma 2.1, $P_{G_{I_1}} P_{G_{I_2}} = P_{G_{I_1} \vee G_{I_2}}$, so it must be shown that $G_{I_1} \vee G_{I_2} = (F_{i_1} \wedge F_{i_2} \wedge \cdots \wedge F_{i_{t_1}}) \vee (F_{j_1} \wedge \cdots \wedge F_{j_{t_2}}) = G_{I_0}$.

Recall that $G_{I_1} \vee G_{I_2}$ is the factor which is constant on all units connected by a chain of steps for which each step changes at most one of G_{I_1} or G_{I_2} . Now if two units at different levels of $G_{I_1}(G_{I_2})$ have the same level of $G_{I_2}(G_{I_1})$, they must necessarily have the same level of G_{I_0} . Hence $G_{I_0} \geq G_{I_1} \vee G_{I_2}$. Also, $t_3 = t_1 + t_2 - t_0 \leq h$, so the rows I_3 of B form an OAVS of strength t_3 . This says that for fixed G_{I_0} , any level of $G_{I_1 \cap I_0}$ occurs with every level of $G_{I_2 \cap I_0}$ (on the same number of experimental units). It follows that $G_{I_1} \vee G_{I_2} \geq G_{I_0}$. $\square$

Proposition 2.3: $P_{\mathcal{G}} = \sum_{u=0}^t \sum_{|I|=u} (P_{G_I} + \sum_{l=0}^{u-1} \sum_{|H|=l, H \subset I} (-1)^{u-l} P_{G_H})$

Proof: Write E for the factor that identifies each and every unit; $E(i) = i$ for all units i . Lemmas 2.1 and 2.2 say that Tjur's (1984, pg. 42) assumptions 2 and 3 are met, while his assumption 1 requires that E be in the collection of factors. So letting $\tilde{\mathcal{G}} = \mathcal{G} \cup E$, Tjur's (1984) Theorem 1 says that

$$P_{\tilde{\mathcal{G}}} = \sum_{G_I \in \tilde{\mathcal{G}}} Q_{G_I} + Q_E = P_{\mathcal{G}} + Q_E$$

where $Q_{G_I} = P_{G_I} \prod_{H \subset I} (P_{G_I} - P_{G_H})$. The Q 's are orthogonal projectors, and the second equality in the expression for $P_{\tilde{\mathcal{G}}}$ is due to the fact that every G_I nests E . It remains to simplify the Q_{G_I} . Since $H_2 \subset H_1 \subset I \Rightarrow (P_{G_I} - P_{G_{H_1}})(P_{G_I} - P_{G_{H_2}}) = (P_{G_I} - P_{G_{H_1}})$,

$$P_{G_I} \prod_{H \subset I} (P_{G_I} - P_{G_H}) = P_{G_I} \prod_{\substack{H \subset I \\ |H|=|I|-1}} (P_{G_I} - P_{G_H}).$$

Then repeatedly using $P_{G_I}P_{G_H} = P_{G_H}$ for every $H \subset I$ gives

$$\begin{aligned}
 Q_{G_I} &= P_{G_I} \prod_{\substack{H \subset I \\ |H|=|I|-1}} (P_{G_I} - P_{G_H}) \\
 &= P_{G_I} - \sum_{\substack{H \subset I \\ |H|=|I|-1}} P_{G_H} + \sum_{\substack{H_1 \neq H_2 \subset I \\ |H_1|=|H_2|=|I|-1}} P_{G_{H_1}} P_{G_{H_2}} \\
 &\quad - \sum_{\substack{H_1 \neq H_2 \neq H_3 \subset I \\ |H_1|=|H_2|=|H_3|=|I|-1}} P_{G_{H_1}} P_{G_{H_2}} P_{G_{H_3}} + \cdots \\
 &= P_{G_I} - \sum_{\substack{H \subset I \\ |H|=|I|-1}} P_{G_H} + \sum_{\substack{H \subset I \\ |H|=|I|-2}} P_{G_H} - \sum_{\substack{H \subset I \\ |H|=|I|-3}} P_{G_H} + \cdots \\
 &= P_{G_I} + \sum_{l=0}^{|I|-1} \sum_{\substack{H \subset I \\ |H|=l}} (-1)^{|I|-l} P_{G_H}
 \end{aligned}$$

and the stated expression for $P_{\mathcal{G}}$ follows immediately. $\square$

It is now not hard to show that $P_{\mathcal{G}}$ may be compactly expressed as

$$P_{\mathcal{G}} = \sum_{u=0}^t \sum_{|I|=u} (-1)^{t-u} \binom{m-u-1}{t-u} P_{G_I},$$

which may prove useful in other investigations. For the current endeavor, the seemingly more complicated form of proposition 2.3 will be preferred.

3 Optimality Results

Because products of the P_{G_I} 's will not be needed, to begin this section the notation will be simplified: write P_I for P_{G_I} , it being henceforth understood that the subscripted set I implicitly refers to the factor G_I . Then $P_{\mathcal{G}}$ as given in Proposition 2.3 is

$$P_{\mathcal{G}} = \sum_{u=0}^t \sum_{|I|=u} \left(P_I + \sum_{l=0}^{u-1} \sum_{|H|=l, H \subset I} (-1)^{u-l} P_H \right) = \sum_{u=0}^t \sum_{|I|=u} P_I^* \quad (2)$$

The P_I^* are easily recognizable: absent treatments, $Y'P_I^*Y$ is the sum of squares for the u -factor interaction of the u factors involved in I . Specifically, the first few component sums of $P_{\mathcal{G}}$ as given in (2) are

$$P_U \quad \text{for } u = 0$$

$$\sum_{i=1}^m (P_i - P_U) \quad \text{for } u = 1$$

$$\sum_{1 \leq i < j \leq m} (P_{ij} - P_i - P_j + P_U) \quad \text{for } u = 2$$

$$\sum_{1 \leq i < j < k \leq m} (P_{ijk} - P_{ij} - P_{ik} - P_{jk} + P_i + P_j + P_k - P_U) \quad \text{for } u = 3$$

and so on.

To identify optimality conditions, Kiefer's (1975) result can be employed: any design d for which (1) is completely symmetric of maximum trace, is universally optimum.

Theorem 3.1: An equireplicate design d^ for which $A'_{d^*}P_I^*A_{d^*} = 0$ for every $I \subseteq J_m$, $1 \leq |I| \leq t$, is universally optimum.*

Proof: Since each P_I^* is nonnegative definite, $C_d = A'_d(I - P_{\mathcal{G}})A_d \leq A'_d(I - P_U)A_d$ ($\leq$ in the nonnegative definite sense), and d^* makes $A'_d(I - P_U)A_d$ completely symmetric of maximum trace. $\square$

The condition $A'_dP_I^*A_d = 0$ has an interesting interpretation. For $|I| = u$, and for each treatment l , form a u -way table with cells corresponding to levels of G_I and entries equal to the replication of l at those levels. Then $A'_dP_I^*A_d = 0$ iff for each l , the u -way table for l exhibits no u -way interaction; this will be termed 'no G_I -interaction among the treatment counts.' Evidently, in an equireplicate design, this holds for all I with $1 \leq |I| \leq t$ iff replications are constant in each of these tables. Hence a necessary condition for a design satisfying the Theorem 3.1 conditions is that the number of plots at each level of G_I , for every $1 \leq |I| \leq t$, is a multiple of v . Analogous to Cheng (1978), Mukhopadhyay and Mukhopadhyay (1984), and Bagchi and Mukhopadhyay (1989), this is called the *completely regular setting*.

To construct a design satisfying Theorem 3.1, start with an OAVS with $m + l$ columns and of strength $2t$, where $l \leq t$. The first m columns define the blocking structure as usual. Distinct l -tuples formed by the rows of the last l columns are the treatments.

More complicated, potentially more useful, and, at least to the author's sensibilities, more interesting, are settings where the divisibility conditions do not hold for all of the factors. The remainder of this section is devoted to identifying a series of settings where the Theorem 3.1 approach can still be applied despite a partial lack of this divisibility.

Recall that b_i is the number of levels of F_i . For any $I \subseteq J_m$ let $p_I = n / \prod_{i \in I} b_i$. The completely regular setting is identified by the condition $v|p_I$ for all I with $|I| \leq t$. A setting will be called *partly regular of order f* , defined for $f = 0, 1, \dots, t-1$, if over those I with $|I| \leq t$, $v|p_I \Leftrightarrow J_{t-f} \not\subseteq I$. In a partly regular setting of order f , divisibility fails for the number of cells at the levels of any G_I for which I includes $1, 2, \dots, t-f$ (up to the order f , this is an arbitrary but convenient choice of a particular $t-f$ factors), and as f increases so too does the extent of departure from complete regularity.

To find a Theorem 3.1-like result for the partly regular setting, the P_I^* in $P_{\mathcal{G}}$ for which $J_{t-f} \subseteq I$ need to be separated out and recombined in some fashion. This will result in some new terms $P_{I,s}^*$ defined by

$$P_{I,s}^* = \left(P_{I \cup J_s} + \sum_{l=0}^{|I|-1} \sum_{|H|=l, H \subset I} (-1)^{|I|-l} P_{H \cup J_s} \right)$$

(when $s = 0$, $P_{I,s}^* = P_I^*$). The manipulation is sketched here; proofs for the various steps, which mainly involve writing collections of subsets in different ways, may be found in Appendix A. Sums with lower limits exceeding upper limits are to be taken as 0. Starting with (2),

$$\begin{aligned} P_{\mathcal{G}} &= \sum_{u=0}^{t-1} \sum_{|I|=u} P_I^* + \sum_{|I|=t} P_I^* \\ &= \sum_{u=0}^{t-1} \sum_{|I|=u} P_I^* + \sum_{s=0}^t \sum_{\substack{|I|=t, J_s \subseteq I, \\ s+1 \notin I}} P_I^* \quad (\text{A.1 with } f = 0 \text{ and adding } J_t) \\ &= \sum_{u=0}^{t-1} \sum_{|I|=u} P_I^* + \sum_{s=0}^{t-f-1} \sum_{\substack{|I|=t, J_s \subseteq I, \\ s+1 \notin I}} P_I^* + \sum_{s=t-f}^t \sum_{\substack{|I|=t, J_s \subseteq I, \\ s+1 \notin I}} P_I^* \\ &= \sum_{u=0}^{t-1} \sum_{|I|=u} P_I^* + \sum_{\substack{|I|=t \\ |I \cap J_{t-f}| < t-f}} P_I^* \\ &\quad + \sum_{s=t-f}^t \left\{ \sum_{\substack{|K|=t-s \\ J_{s+1} \cap K = \emptyset}} \left[P_{K,s}^* - \sum_{q=0}^{s-1} \sum_{|M|=q, M \subset J_s} P_{M \cup K}^* \right] \right\} \quad (\text{A.1 and A.4}) \end{aligned}$$

$$\begin{aligned}
&= \sum_{u=0}^{t-1} \sum_{|I|=u} P_I^* + \sum_{\substack{|I|=t \\ |I \cap J_{t-f}| < t-f}} P_I^* \\
&\quad + \sum_{s=t-f}^t \sum_{\substack{|K|=t-s \\ J_{s+1} \cap K = \emptyset}} P_{K,s}^* - \sum_{s=t-f}^t \sum_{\substack{|K|=t-s \\ J_{s+1} \cap K = \emptyset}} \sum_{q=0}^{s-t+f} \sum_{|M|=q, M \subset J_s} P_{M \cup K}^* \\
&\quad - \sum_{s=t-f}^t \sum_{\substack{|K|=t-s \\ J_{s+1} \cap K = \emptyset}} \sum_{q=s-t+f+1}^{s-1} \sum_{|M|=q, M \subset J_s} P_{M \cup K}^* \\
&= \sum_{u=0}^{t-1} \sum_{|I|=u} P_I^* + \sum_{\substack{|I|=t \\ |I \cap J_{t-f}| < t-f}} P_I^* + \sum_{s=t-f}^t \sum_{\substack{|K|=t-s \\ J_{s+1} \cap K = \emptyset}} P_{K,s}^* \\
&\quad - \sum_{u=0}^f \sum_{|I|=u} P_I^* - \sum_{u=f+1}^{t-1} \sum_{\substack{|I|=u \\ |I \cap J_{t-f}| \geq u-f}} P_I^* \quad (\text{A.3 and A.2}) \\
&= \sum_{u=f+1}^t \sum_{\substack{|I|=u \\ |I \cap J_{t-f}| < u-f}} P_I^* + \sum_{s=t-f}^t \sum_{\substack{|K|=t-s \\ J_{s+1} \cap K = \emptyset}} P_{K,s}^* .
\end{aligned}$$

Hence

$$I - P_{\mathcal{G}} = (I - P_{J_t}) - \sum_{u=f+1}^t \sum_{\substack{|I|=u \\ |I \cap J_{t-f}| < u-f}} P_I^* + \sum_{s=t-f}^{t-1} \sum_{\substack{|K|=t-s \\ J_{s+1} \cap K = \emptyset}} P_{K,s}^* \quad (3)$$

Theorem 3.2: The design d^ is optimal if*

$$\text{the } G_{J_t} - \text{component block design is optimal} , \quad (4)$$

$$A'_{d^*} P_I^* A_{d^*} = 0 \quad \text{for all } I \text{ with } |I \cap J_{t-f}| < |I| - f \text{ and } f+1 \leq |I| \leq t , \quad (5)$$

and

$$A'_{d^*} P_{I,s}^* A_{d^*} = 0 \quad \text{for } s = t-f, \dots, t-1 \text{ and all } I \text{ with}$$

$$|I| = t-s \quad \text{and} \quad J_{s+1} \cap I = \emptyset . \quad (6)$$

By 'optimal' is meant with respect to any nonincreasing criterion $\Phi(C_d)$, in the sense of preserving the nonnegative definite ordering, that is, if $C_1 - C_2$ is

nonnegative definite then $\Phi(C_1) \leq \Phi(C_2)$. The extended universal optimality of Shah and Sinha (1989, pg. 7) falls within this framework, and the same bounding argument used in the proof of Theorem 3.1 goes through here.

In Theorem 3.2 the terms P_I^* are a subset of those in Theorem 3.1, there being fewer now due to the failure of 'full' divisibility. In their stead are found terms $P_{I,s}^*$, which are P_I^* -like terms with the addition of a fixed set of factors J_s . The condition $A_{d*}P_I^*A_{d*} = 0$ is that of no G_I -way interaction among the treatment counts, and analogous to this, the condition $A_{d*}P_{I,s}^*A_{d*} = 0$ is stated 'at each level of G_{J_s} , there is no G_I -way interaction among the treatment counts'. The accomplishment of the expression (3) for $I - P_{\mathcal{G}}$ is that all of the terms P_I^* are such that $v \nmid p_I$, so that it is at least feasible to achieve (5) by having each treatment occur equally often at each level of G_I (cf. Theorem 3.1), that is, by making the G_I -component block design *uniform* on the treatments. Factors G_I for which $v \nmid p_I$ are reflected in the $P_{I,s}^*$, and (6) says that the orthogonality of treatments with respect to those factors is achieved by an appropriate nesting of treatments within blocking levels. This can be seen more clearly with an examination of Theorem 3.2 for particular t , and as Bagchi and Mukhopadhyay (1989) have covered $t = 2$, attention is here focused on the case $t = 3$.

For $t = 3$ the OAVS must be of strength $h \geq \min(m, 6)$. The expressions for $P_{\mathcal{G}}$, depending on the order of the partial regularity, are in Table 1. It is easy to see that here, as for any t , the $P_{I,s}^*$ impose *nesting conditions* on the treatment assignments if (6) is to be satisfied. For example, $A_d'(P_{12i} - P_{12})A_d = 0$ requires that treatment replications at different levels of F_i are constant *within* each level of $G_{12} = F_1 \wedge F_2$. Likewise $A_d'(P_{1ij} - P_{1i} - P_{1j} + P_U)A_d = 0$ iff *within* each level of F_1 , the $F_i \wedge F_j$ interaction of treatment counts is zero. These do *not* imply that treatments be equally replicated within those levels; indeed, a particular treatment need not even appear at any particular level of one of those nestings.

The essential features of constructing designs for the $t = 3$ setting can be seen by specializing to $m = 4$, which is now done. This demands that the four blocking factors are completely crossed, that is, there are no empty cells, so assume one experimental unit at each combination of levels of those factors.

Table 1. Projection matrices for $t = 3$

f	divisibility	$P_{\mathcal{G}}$
0	$v \nmid p_{123}$	$P_{123} + \sum_{i=4}^m (P_i - P_U) + \sum_{\substack{i < j \\ (i,j) \neq (1,2), \\ (1,3), (2,3)}} (P_{ij} - P_i - P_j + P_U)$ $+ \sum_{\substack{i < j < k \\ (i,j,k) \neq (1,2,3)}} (P_{ijk} - P_{ij} - P_{ik} - P_{jk} + P_i + P_j + P_k - P_U)$
1	$v \nmid p_{12i}, i \geq 3$	$P_{123} + \sum_{i=4}^m (P_{12i} - P_{12}) + \sum_{3 \leq i < j \leq m} (P_{ij} - P_i - P_j + P_U)$ $+ \sum_{\substack{i < j < k \\ (i,j) \neq (1,2)}} (P_{ijk} - P_{ij} - P_{ik} - P_{jk} + P_i + P_j + P_k - P_U)$
2	$v \nmid p_{1ij}, 1 < i < j$	$P_{123} + \sum_{i=4}^m (P_{12i} - P_{12}) + \sum_{3 \leq i < j \leq m} (P_{1ij} - P_{1i} - P_{1j} + P_1)$ $+ \sum_{1 < i < j < k} (P_{ijk} - P_{ij} - P_{ik} - P_{jk} + P_i + P_j + P_k - P_U)$

Further demanding that the numbers of levels be as small as possible within the constraints of each partially regular setting, possible values of the b_i 's are

f	b_1	b_2	b_3	b_4
0	v	v	v	b_4
1	v	v	b_3	b_4
2	v	b_2	b_3	b_4

A construction will be given for each of these cases. Regardless of f , stratifying by the levels of F_1 gives v separate $b_2 \times b_3 \times b_4$ hyperrectangles each with the same combinations of levels of F_2 , F_3 , and F_4 . Denote the j^{th} of these by H_j .

3.1 Construction for $t = 3, f = 0$

Inspection of Table 1 shows this setting to be only marginally more complicated than that of complete regularity. If d^* is such that the $F_1 \wedge F_2 \wedge F_3$ -component block design is a BIBD, and the G_I -component blocks are multiples of complete blocks for every $I \not\subseteq J_3$, then d^* is optimal.

Begin with a $v^3 \times 4$ orthogonal array of strength 3 on v symbols (these exist for every v), written as a $v \times v \times v$ Latin cube L say, and a square BIBD D_0 for v treatments in blocks of size $k = b_4$. Denote by L_1, L_2, L_3 the three blocking factors forming the cube, each with v levels. Then $L_1 \wedge L_2, L_1 \wedge L_3$, and $L_2 \wedge L_3$ are uniform on the v symbols. Associate the levels of L_3 with the treatments of D_0 . H_j is obtained from L by retaining only the L_3 -blocks corresponding to treatments in the j^{th} block of D_0 .

Example 1: The Latin cube

L	1 2 3 4	2 3 4 1	3 4 1 2	4 1 2 3
	2 3 4 1	3 4 1 2	4 1 2 3	1 2 3 4
	3 4 1 2	4 1 2 3	1 2 3 4	2 3 4 1
	4 1 2 3	1 2 3 4	2 3 4 1	3 4 1 2

combined with the BIBD for 4 treatments in blocks of 3 yields the $4 \times 4 \times 4 \times 3$ design

H_1	1 2 3 4	2 3 4 1	3 4 1 2	4 1 2 3
	2 3 4 1	3 4 1 2	4 1 2 3	1 2 3 4
	3 4 1 2	4 1 2 3	1 2 3 4	2 3 4 1

H_2	1 2 3 4	2 3 4 1	3 4 1 2	4 1 2 3
	2 3 4 1	3 4 1 2	4 1 2 3	1 2 3 4
	4 1 2 3	1 2 3 4	2 3 4 1	3 4 1 2
H_3	1 2 3 4	2 3 4 1	3 4 1 2	4 1 2 3
	3 4 1 2	4 1 2 3	1 2 3 4	2 3 4 1
	4 1 2 3	1 2 3 4	2 3 4 1	3 4 1 2
H_4	2 3 4 1	3 4 1 2	4 1 2 3	1 2 3 4
	3 4 1 2	4 1 2 3	1 2 3 4	2 3 4 1
	4 1 2 3	1 2 3 4	2 3 4 1	3 4 1 2

In this display, H_j is an F_1 -block. The 3×4 subrectangles are $F_1 \wedge F_2$ -blocks, rows of 16 are $F_1 \wedge F_4$ -blocks, and columns of 3 are $F_1 \wedge F_2 \wedge F_3$ -blocks fulfilling (4).

3.2 Construction for $t = 3, g = 1$

Needed here will be a special type of nested row and column design. A $\text{BNRC}(v, b, p, q)$ is a design with b blocks of size $p \times q$ for which rows within blocks are the same multiset of treatments, and for which the bq columns are collectively a BIBD with blocksize p ; see Bagchi, Mukhopadhyay, and Sinha (1990), and Morgan and Uddin (1993), for details.

With $f = 1$, condition (6) is $A'_d(P_{124} - P_{12})A_d = 0$, i.e. F_4 -blocks are identical within levels of $F_1 \wedge F_2$. Also requiring the $F_1 \wedge F_2 \wedge F_3$ -component design to be a BIBD (it must be some optimal block design), then the blocks of the $F_1 \wedge F_2$ -component design, when written as rectangles with rows given by F_4 and columns by F_3 , are a $\text{BNRC}(v, b = v^2, p = b_4, q = b_3)$. The remaining conditions can be met by making the $F_3 \wedge F_4, F_1 \wedge F_3$, and $F_2 \wedge F_3 \wedge F_4$ component designs uniform on the treatments (either of the latter two implies the first). This suggests the following construction.

Start with a $\text{BNRC}(v, v, p, q)$ for which the blocks are generated from some initial block with elements from a group $\{g_1, \dots, g_v; +\}$ (e.g. Theorems 4 and 7 of Morgan and Uddin, 1993). Make these blocks the $p \times q$ layers of a $v \times p \times q$ hyperrectangle H . Then $H_j = H + g_j$ gives the design. Development over the group in building H has given uniformity of the $F_1 \wedge F_3 \wedge F_4$ component, and the development that constructs the H_j gives uniformity of the $F_2 \wedge F_3 \wedge F_4$ component.

Example 2: The initial block $\begin{bmatrix} 0 & 1 & 3 \\ 1 & 3 & 0 \\ 3 & 0 & 1 \end{bmatrix}$ over $\mathcal{L}_7$ generates a BNRC(7, 7, 3, 3), here written as the $7 \times 3 \times 3$ hyperrectangle H :

H	0 1 3	1 2 4	2 3 5	3 4 6	4 5 0	5 6 1	6 0 2
	1 3 0	2 4 1	3 5 2	4 6 3	5 0 4	6 1 5	0 2 6
	3 0 1	4 1 2	5 2 3	6 3 4	0 4 5	1 5 6	2 6 0

If one makes the identification $g_j = j$, then $H_1 = H + 1$ is

H_1	1 2 <u>4</u>	2 3 <u>5</u>	3 4 <u>6</u>	4 5 <u>0</u>	5 6 <u>1</u>	6 0 <u>2</u>	0 1 <u>3</u>
	2 4 1	3 5 2	4 6 3	5 0 4	6 1 5	0 2 6	1 3 0
	4 1 2	5 2 3	6 3 4	0 4 5	1 5 6	2 6 0	3 0 1

and so on for $H_2, \dots, H_7$ to get a $7 \times 7 \times 3 \times 3$ design for 7 treatments which is optimal in the face of 3-way interactions among the four blocking factors. The identification of the levels of $F_1, \dots, F_4$ in the display is completely analogous to that of example 1. One of the nine $F_1 \wedge F_3 \wedge F_4$ -blocks in H_1 is indicated by underscoring.

3.3 Construction for $t = 3, f = 2$

Now $b_2 \neq v$ and compared to $f = 1$ the key change is that the $F_3 \wedge F_4$ component design cannot be uniform on the treatments. Condition (6) thus says to make $A'_d(P_{134} - P_{13} - P_{14} + P_1)A_d = 0$ (see Table 1), which can be achieved by allowing F_1 blocks to contain only subsets of the v treatments, and arranging for $F_3 \wedge F_4$ uniformity within levels of F_1 for those subsets. Using subsets of the entire treatment set allows us to get around the divisibility problem.

This construction requires a k -resolvable BIBD(v, b, k). Such a BIBD can be written as $b^* = \frac{b}{v}$ rectangles of order $k \times v$ such that columns are together the BIBD and rows within each rectangle are uniform on the treatments. Join these $k \times v$ rectangles into a $b^* \times k \times v$ hyperrectangle D_1 . D_{1lji} will refer to the row l , column j symbol in rectangle i .

Also required is a BNRC($v_0 = k, b_0, p, q$) generated from $\frac{b_0}{v_0}$ initial blocks based on some group $\{g_1, \dots, g_{v_0}; +\}$ of order v_0 . Combine these b_0 blocks into a $b_0 \times p \times q$ hyperrectangle H (compare the $f = 1$ construction above). From H

create a hyperrectangle H_{ij} which is simply H with g_l replaced by D_{1lji} , $l = 1, 2, \dots, v_0 = k$. H_j is found by joining $H_{1j}, H_{2j}, \dots, H_{kj}$ at their $p \times q$ faces into a $b_0 b^* \times p \times q$ hyperrectangle. The parameters of the optimal 4-dimensional design are $b_1 = v$, $b_2 = b_0 b^*$, $b_3 = q$, and $b_4 = p$.

By construction, the $p \times q$ $F_1 \wedge F_2$ -component blocks are a BNRC, so the $F_1 \wedge F_2 \wedge F_3$ -component design is optimal, and $F_1 \wedge F_2 \wedge F_4$ blocks are permutations of one another within $F_1 \wedge F_2$. Since treatment assignment is uniform for a subset in the $F_3 \wedge F_4$ -component blocks of H , it is so for each H_{ij} and hence the permutation property holds for those blocks in each H_j , i.e. for $F_1 \wedge F_3 \wedge F_4$. Finally, uniformity of the treatment assignment in $F_2 \wedge F_3 \wedge F_4$ blocks follows from that property holding for rows within the $k \times v$ rectangles of D_1 .

Example 3: A simple example of the above construction is a $15 \times 7 \times 3 \times 3$ design for 15 treatments. The design D_1 , a hyperrectangle of order $1 \times 7 \times 15$, is the Youden square based on a square BIBD for 15 treatments in blocks of size 7. The BNRC is that of example 2, with $v_0 = b_0 = 7$ and $p = q = 3$, and H is the same as displayed there. The index i of the construction takes on only one value, so $H_j = H_{1j}$ is simply the $7 \times 3 \times 3$ hyperrectangle H constructed using the symbols of the j^{th} column of D_1 . The 15 F_1 -layers of the design are $H_1, H_2, \dots, H_{15}$.

4 Final Comments

The information matrix for the multiway crossed blocking structure, with orthogonal array incidence, has been derived and manipulated to obtain optimality conditions depending on the numbers of levels $b_1, b_2, \dots, b_m$ of the blocking factors. In the completely regular setting ($v|b_j$ for every j) optimal treatment assignment is simply achieved and gives no unexpected results. In some partly regular settings (some b_j 's and their products are not divisible by v), optimality has imposed nesting conditions on the treatment assignment, a perhaps unexpected result for a set of crossed nuisance factors. That bit of surprise is diminished, however, once one realizes that there are obvious nesting relationships among the G_l .

A salient lesson to be taken from this work is that optimality of the overall design may bear no relationship to, may even be contradictory to, optimality of the marginal designs for the individual blocking factors, an observation recently made also by Bagchi and Mukhopadhyay (1989), Bagchi, Mukhopadhyay, and Sinha (1990), and Morgan and Uddin (1993) in related multifactor settings. Equally important is that lower dimensional component designs nonetheless play a vital role in the optimality argument, as seen from Theorem 3.2 and

exhibited by the constructions for $t = 3$, the latter of which are special arrangements of known optimal designs with fewer blocking factors. Thus do advances in optimality, construction, and structural knowledge of designs in settings that have been until recently neglected, such as that of nested rows and columns, promise to be valuable far beyond themselves and their particular applications.

A Collections of Subsets

The set relations used in the manipulation of P_g in section 3 are derived below. In lemmas A.1, A.2, and A.3 it is left to the reader to convince him/herself that the two collections contain no sets more than once.

Lemma A.1: $\bigcup_{s=0}^{t-f-1} \{I: |I| = t, J_s \subseteq I, s+1 \notin I\} = \{I: |I| = t, |I \cap J_{t-f}| < t-f\}$

Proof: Let the left-hand collection be called S_1 , and the right-hand S_2 . If $A \in S_1$ then $|A| = t$ and $|A \cap J_{t-f}| < t-f$, the latter since $J_s \in A$, $s+1 \notin A$, and $s \in \{0, 1, \dots, t-f-1\}$. Hence $A \in S_2$.

If $A \in S_2$ then $|A \cap J_{t-f}| < t-f \Rightarrow s \notin A$ for some $s \in J_{t-f}$. Let $s_0 = \min\{s: s \in J_{t-f}, s \notin A\}$. Then $J_{s_0-1} \subseteq A$, $s_0 \notin A$, and $1 \leq s_0 \leq t-f \Rightarrow A \in S_1$. $\square$

Lemma A.2: For any $j = 1, 2, \dots, t-f-1$,

$$\begin{aligned} & \bigcup_{s=t-f}^t \{M \cup K: |M| = s-t+f+j, M \subseteq J_s, |K| = t-s, K \cap J_{s+1} = \emptyset\} \\ &= \{I: |I| = f+j, |I \cap J_{t-f}| \geq j\} \end{aligned}$$

Proof: Let the left-hand collection be called S_1 , and the right-hand S_2 . If $A \in S_1$ then $|A| = f+j$ and A can be written as $M \cup K$ with the stated conditions. Now $|M \cap J_s| = s-t+f+j \Rightarrow |M \cap J_{t-f}| = |M \cap J_s| - |M \cap (J_s - J_{t-f})| \geq (s-t+f+j) - (s-t+f) = j \Rightarrow A \in S_2$.

If $A \in S_2$ then $|A| = f+j$ and $|A \cap J_{t-f}| \geq j$. With $B_l = A \cap J_{t-f+l}$, $|B_0| \geq j$ and if $|B_l| > j+l$ then $|B_l| \geq j+l+1$. So letting $l_1 = \min\{l \geq 0: |B_l| = j+l\}$, it must be that $l_1 \leq f$, for otherwise $f+j = |A| \geq |A \cap J_t| = |B_f| \geq j+f+1$.

Next let $q_1 = \min\{q \geq 0: t-f+l_1+q+1 \notin A\}$. Then $q_1 \leq f-l_1$, for otherwise $f+j \geq |A \cap J_{t-f+l_1+q_1}| = j+l_1+q_1 > f+j$.

Finally write $M = A \cap J_{t-f+l_1+q_1}$, $K = A - M$, and $s = t - f + q_1 + l_1$. Using what has already been shown, it is easily checked that $t - f \leq s \leq t$, $|M| = s - t + f + j$, $M \subseteq J_s$, and $K \cap J_{s+1} = \emptyset$. $\square$

Lemma A.3:

$$\begin{aligned} & \bigcup_{s=t-f}^t \bigcup_{q=0}^{s-t+f} \{M \cup K: |M| = q, M \subseteq J_s, |K| = t - s, K \cap J_{s+1} = \emptyset\} \\ &= \bigcup_{u=0}^f \{I: |I| = u\} \end{aligned}$$

Proof: Let the left-hand collection be called S_1 , and the right-hand S_2 . Reversing the order of the $\bigcup$ in S_1 , it can be written

$$\begin{aligned} & \bigcup_{j=0}^f \bigcup_{s=t-f+j}^t \{M \cup K: |M| = s - t + f - j, M \subseteq J_s, |K| = t - s, K \cap J_{s+1} = \emptyset\} \\ &= \bigcup_{j=0}^f S_{1j} . \end{aligned}$$

And S_2 can be written as

$$\bigcup_{j=0}^f \{I: |I| = f - j\} = \bigcup_{j=0}^f S_{2j} .$$

One need only prove that $S_{1j} = S_{2j}$, which after observing that certainly $A \in S_{1j} \Rightarrow A \in S_{2j}$, is done similarly to the proof of lemma A.2.

Take any $A \in S_{2j}$ and let $A_l = A \cap J_{t-f+j+l}$. Then $|A_0| \geq 0$ and if $|A_l| > l$ then $|A_l| \geq l + 1$. Let $l_1 = \min\{l \geq 0: |A_l| = l\}$. Kf $l_1 > f - j$ then $f - j \geq |A \cap J_t| = |A_{f-j}| > f - j$, so $l_1 \leq f - j$.

Next let $q_1 = \min\{q \geq 0: t - f + j + l_1 + q + 1 \notin A\}$. Then $q_1 \leq f - j - l_1$, for otherwise $f - j \geq |A \cap J_{t-f+j+l_1+q_1}| \geq l_1 + q_1 > f - j$.

So $M = A \cap J_{t-f+j+l_1+q_1}$, $K = A - M$, and $s = t - f + j + q_1 + l_1$ give the required decomposition of A . $\square$

$$\text{Lemma A.4: } \sum_{\substack{|I|=t, J_s \subseteq I, \\ s+1 \notin I}} P_I^* = \sum_{\substack{|K|=t-s \\ J_{s+1} \cap K = \emptyset}} [P_{K,s}^* - \sum_{q=0}^{s-1} \sum_{|M|=q, M \subseteq J_s} P_{M \cup K}^*]$$

Proof: Consider any set A with $|A| = a$, $|A \cap J_s| = b$, $c = b - a$, and $s + 1 \notin A$. The matrix P_A occurs in the left hand side with coefficient

$(-1)^{t-a} \binom{m-s-1-c}{t-s-c}$, since it occurs once for every set I satisfying $|I| = t$, $J_s \subseteq I$, $s+1 \notin I$, $I \supseteq A$.

To derive the coefficient for P_A in the right hand side, first consider the case $b = s$, for which P_A appears only in terms written as $P_{K,s}^*$. Its coefficient is determined by the number of ways of selecting $J_s \cup K$ to contain A , and is thus $(-1)^{t-s-c} \binom{m-s-1-c}{t-s-c} = (-1)^{t-a} \binom{m-s-1-c}{t-s-c}$.

If $b < s$ then P_A appears only in terms $P_{M \cup K}^*$. Its coefficient other than the (-1) term is (number of ways to choose an M containing $A \cap J_s$) $\times$ (number of ways to choose a K containing $A \cap J_{s+1}^c$). It is thus

$$\begin{aligned} & \sum_{q=b}^{s-1} \binom{s-b}{q-b} \binom{m-s-1-c}{t-s-c} (-1)^{q+t-s-a+1} \\ &= \binom{m-s-1-c}{t-s-c} (-1)^{t-s-a+b+1} \sum_{q=b}^{s-1} \binom{s-b}{q-b} (-1)^{q-b} \\ &= \binom{m-s-1-c}{t-s-c} (-1)^{t-s-a+b+1} (-1)^{s-b-1} \\ &= \binom{m-s-1-c}{t-s-c} (-1)^{t-a}. \end{aligned}$$

□

Acknowledgements: The author was supported by National Science Foundation grant DMS92-03920, and further expresses his gratitude to Professor R. A. Bailey for helpful discussions on this topic. Thanks are also due a referee, whose suggestions led to an improved proof of proposition 2.3.

References

- Bagchi S, Mukhopadhyay AC (1989) Optimality in the presence of two factor interactions among the nuisance factors. *Commun Statist Theory Meth* 18:1139–1152
- Bagchi S, Mukhopadhyay AC, Sinha BK (1990) A search for optimal nested row-column designs. *Sankhya B* 52:93–104
- Bailey RA (1984) Discussion of paper by T. Tjur. *ISI Review* 52:65–77
- Bailey RA (1985) Factorial design and Abelian groups. *Linear Algebra Appl* 70:349–368
- Cheng CS (1978) Optimal designs for the elimination of multi-way heterogeneity. *Ann Statist* 6:1262–1272
- Kiefer J (1958) On the nonrandomized optimality and randomized nonoptimality of symmetrical designs. *Ann Math Statist* 3:675–699

- Kiefer J (1975) Construction and optimality of generalized Youden designs. In: Srivastava JN (Ed) *A Survey of Statistical Design and Linear Models* 333–353. North Holland Amsterdam
- Morgan JP, Uddin N (1993) Optimality and construction of nested row and column designs. *J Statist Plann Inf* 37: 81–93
- Mukerjee R, Wu CFJ (1996) On the existence of saturated and nearly saturated asymmetrical orthogonal arrays. *Ann Statist*, to appear
- Mukhopadhyay AC (1981) Orthogonal main effects plans with variable numbers of levels for factors. In: Rao SB (Ed) *Combinatorics and Graph Theory, Lecture Notes in Statistics* 885:347–357, Springer-Verlag New York
- Mukhopadhyay AC, Mukhopadhyay S (1984) Optimality in a balanced incomplete multiway heterogeneity set up. In: Ghosh JK, Roy J (Ed) *Statistics: Applications and New Directions* 466–477, Indian Statistical Institute, Calcutta
- Rao CR (1947) Factorial experiments derivable from combinatorial arrangements of arrays. *J Roy Statist Soc (supp)* 9: 128–139
- Rao CR (1973) Some combinatorial problems of arrays and applications to design of experiments. In: Srivastava JN (Ed) *A Survey of Combinatorial Theory* 349–359, North Holland Amsterdam
- Shah KR, Sinha BK (1989) *Theory of optimal designs*. Springer-Verlag, Berlin
- Tjur T (1984) Analysis of variance models in orthogonal designs. *ISI Review* 52:33–65
- Wang JC, Wu CFJ (1991) An approach to the construction of asymmetrical orthogonal arrays. *J Amer Statist Assoc* 86:450–456

Received 23.09.1994

Book Reviews

Contents

H. Rinne: Taschenbuch der Statistik für Wirtschafts- und Sozialwissenschaften (G. Ronning) 84

P. Guttorp: Statistical Inference for Branching Processes (C. C. Heyde) 85

J. K. Lindsey: Modelling Frequency and Count Data (O. O. Aalen) ... 86

A. Buja, P. A. Tukey (Eds.): Computing and Graphics in Statistics (P. Huber) 87

S. Albeverio, W. Luxemburg, M. Wolff (Eds.): Advances in Analysis, Probability and Mathematical Physics (P. A. Loeb) 88

N. Kazamaki: Continuous Exponential Martingales and *BMO* (D. L. Burkholder) 89

N. Mukhopadhyay, T. K. S. Solanky: Multistage Selection and Ranking Procedures (Kh. Alam) 90

A. Pázman: Nonlinear Statistical Models (O. E. Barndorff-Nielsen) 91

M. Grabisch, H. T. Nguyen, E. A. Walker: Fundamentals of Uncertainty Calculi with Applications to Fuzzy Inference (P. E. Kloeden) 93

H. Rinne: Taschenbuch der Statistik für Wirtschafts- und Sozialwissenschaften, Deutsch, Frankfurt am Main, 1995, XIII/476 pp., DM 29.80

Reviewed by G. Ronning, Tübingen

Rinne's "*Pocketbook of Statistics for Economics and Social Sciences*" (using the direct translation of the German title) is neither a handbook nor a textbook, and it is not a book for statisticians in all fields, as the title indicates. However, for those working with applied statistics in economics and social sciences, the book serves as a useful collection of basic results which one would like to have to hand on the desk. I have been using the book for some months and found an astonishingly high number of results which are not easily accessible elsewhere in the German statistical literature. For example, it lists a great variety of theoretical distributions including the multinomial distribution (the natural extension of the binomial distribution) and the beta distribution. On the other hand it excludes, for example, the Dirichlet distribution which is the natural extension of the beta distribution. An extensive subject index eases the search for certain results. The book contains a list of references with special emphasis on German statistical literature and in particular to German textbooks. This seems to be acceptable for the readership at which the book aims. The structure of the book follows closely that of the basic statistical courses in social sciences. It is divided into three parts: Part A considers descriptive statistics and exploratory data analysis; part B contains results from probability theory; and part C states main results from statistical inference.

In part A, the usual methods for describing and presenting data are discussed. Besides univariate data sets, bivariate and multivariate data sets are also considered. The inclusion of index numbers, concentration measures, life tables and descriptive models for time series show the special clientele that the author has in mind. Part B contains results from probability theory. Besides the basic notions, it contains an extensive list of distributions. For each distribution a detailed description of properties and the relation to other distributions is given. A short chapter on stochastic processes and asymptotic theory concludes the second part. Part C starts with basic concepts for estimation and testing. Both parametric and nonparametric test procedures are considered. The last chapter gives results for the regression model.

In some places I found the structuring of the material unsystematic. For example, the gamma distribution is not necessarily a model for waiting times, and stochastic processes should be presented separately from asymptotic theory. I also missed some topics such as analysis of variance or the generation of random numbers, both of which are standard tools in applied statistics. Finally I sometimes found the use of statistical terms a bit unusual. In particular the use of the terms 'autoregression' and 'autocorrelation' in the case of cross-section data is not standard.

Summarising, I found the pocketbook of almost 500 pages surprisingly complete and very useful, especially for students of economics and social sciences as well as for applied researchers working in these fields.

P. Guttorp: Statistical Inference for Branching Processes, Wiley, New York, ISBN 0 471 82291 4, 1991, XII/212 pp., £ 45.50

Reviewed by C. C. Heyde, Canberra

Studies of branching processes, especially the Bienayme–Galton–Watson (BGW) process, and its extensions, such as to Bellman–Harris, Crump–Mode–Jagers, and random and varying environment models, and processes with immigration have been fruitful. A considerable amount of literature has been spawned, including many books dating from the classic by T. E. Harris (*The Theory of Branching Processes*), Springer, Berlin, 1963). The models have been used both in the biological (e.g. populations, epidemics, genetics) and in the physical sciences (e.g. particle physics, transport phenomena, queues), and questions of statistical inference have naturally arisen. Most of the books have included results relevant to inference, but the first to give this subject some prominence was by G. Sankaranarayanan (*Branching Processes and its Estimation Theory*), Wiley Eastern, New Delhi, 1989); the book under review is the second.

Sankaranarayanan's book focuses on the mathematical theory with a theorem and proof format. Only one of the ten chapters is specifically concerned with inference. The remainder of the text provides a catalogue of theory, and a discussion, as processes, of many of the extensions of the BGW model.

Guttorp's book, on the other hand, focuses on the BGW process alone. After a brief discussion of basic properties of the process, it covers what can and cannot be estimated, and in some detail, estimation of the mean and variance of the offspring distribution, the extinction probability and the age of the process. An interesting set of illustrative examples, often including real data, accompany the theory. It is an excellent introduction to the subject of population process modelling and inference for the graduate student.

J. K. Lindsey: *Modelling Frequency and Count Data*, Clarendon Press Oxford, 1995, ix/291 pp., £30.00

Reviewed by O. O. Aalen, Oslo

This book represents an interesting attempt to give a unified treatment of material which is usually regarded as belonging to different fields, such as loglinear analysis of tables on the one hand and survival analysis on the other.

The book is organized in two parts entitled "Frequency data" and "Count data". According to the preface, the distinction corresponds to whether events happen to independent individuals, producing frequencies, or repeatedly to the same individual, producing counts. It is not obvious that this is a fruitful distinction since many of the examples in the second part of the book are not of a repeated measurement type. Rather, it seems as if "Count data" refers to data generated in some sense by a stochastic process. Also, a distinction between "frequencies" and "counts" does not seem very suggestive, and better names might have been chosen.

The subjects covered by the book include the analysis of tables in various dimensions, and with various special features such as structural zeros. Next, various regression models are studied such as logistic and polytomous regression. Furthermore, counting processes (including survival models), Markov chains and overdispersion models are included. Some GLIM programs for special models are included.

The author assumes, in his preface, that the reader has knowledge of the theory of generalized linear models. Indeed, most of the models in the book are phrased in the terminology of generalized linear models software, and the author uses GLIM4 in the analysis of all examples. One important point made is that this approach can also be extended to include proportional hazards models and other continuous time methods for counting processes. The idea is that most of these approaches give Poisson-type likelihoods. This realization seems to have some potential.

The author intentionally does not go into much mathematical detail. Sometimes this might leave the reader a bit confused. For instance, in the analysis of the Cox model on pages 163–4, most people would automatically assume that the results are from a partial likelihood, while indeed a complete likelihood appears to be used. Another question to the same analysis is on the interpreta-

tion of the deviance. Certainly this requires valid asymptotics, but that hardly holds for the basic hazard rate parameters of the Cox model. For that model one would expect asymptotic normality to hold only for the estimates of parameters assessing the effect of covariates.

Using the GLIM notation is well enough for GLIM users, but other people, like myself, would have liked to see a more standard mathematical formulation of models and concepts. For instance, what is the precise definition of Cook's distance? This is used a lot, but is it defined in the book?

I have a couple of minor objections: Firstly, exercise (13) on p. 186 contains incomplete data which are not sufficient to solve the problem. Essential information in the original source is not included. (In fact, the source is a paper by this reviewer, which is how I discovered the omission.)

Secondly, in the preface the author says: "In recent years, it has become increasingly obvious that the basis of applied statistics is not the normal distribution, but the multinomial and Poisson distributions. Let us leave the former to those statisticians working on the asymptote." Personally, I think this is a rather unfounded statement. Apart from the fact that the methods in the book are quite heavily based on normal asymptotic theory, the author's statement also seems to severely underestimate the importance of standard normality-based approaches such as regression and analysis of variance. Of course, these methods are not of relevance for the book, but when a general statement is made, it should have general validity.

Still, in spite of these objections, this is undoubtedly an interesting and useful book containing a wealth of information. The book is very clearly written. One good aspect is the large number of examples and exercises, since it is very much easier to understand ideas when they are applied to good examples. Particularly, the exercises make the book well-suited as a text book. Another fine aspect is the consistent use of Akaike's information criterion. In sum, the book is a nice addition to the applied statistical literature, being "applied" in the true sense of the word.

A. Buja, P. A. Tukey (Eds.): Computing and Graphics in Statistics, The IMA Volumes in Mathematics and its Applications, Vol. 36, Springer, Berlin Heidelberg New York, 1991, 279 pp., DM 78.00

Reviewed by P. Huber, Bayreuth

This volume contains the proceedings of the last third of a 6-week IMA 1989 summer program on "*Robustness, Diagnostics, Computing and Graphics in Statistics*". As conference proceedings go, this one is remarkable in that, after 7 years, it can still offer intriguing and inspirational thoughts on a variety of subjects, ranging from the interface between symbolic computation and statistics, to computing environments, numerical procedures and nonstandard graphical techniques.

S. Albeverio, W. Luxemburg, M. Wolff (Eds): Advances in Analysis, Probability and Mathematical Physics, Contributions of Nonstandard Analysis, Kluwer Academic Publishers, 1995, x/251 pp., Dfl. 180.00/US\$115.00/UK£76.00

Reviewed by P. A. Loeb, Urbana

For more than two thousand years, infinitesimals have been used as an aid for mathematical intuition and discovery. During the first half of this century, they were gradually banished from mathematics as imprecise intruders in a realm of clear thinking. The reviewer recalls his calculus teacher in the middle 1950s saying that he would give a PhD to any member of that freshman class who could explain the idea of an infinitesimal number.

Just when their banishment was completed, the first step in making infinitesimals rigorous was taken in the article "*Eine Erweiterung der Infinitesimalrechnung*" by D. Laugwitz and C. Schmieden. Independently, using mathematical logic, A. Robinson completed the foundation for using infinitesimals in 1960 with his development of nonstandard analysis. From that time on, infinitesimally large and small numbers were legitimate tools for mathematical research. Their widespread acceptance, however, is still waiting for the next, unbiased, generation of mathematicians. When that time comes, a large body of literature will be ready and waiting.

The volume under review is an important contribution to that literature. It contains papers presented at the July 1992 meeting dedicated to Professor Laugwitz held at the Tübingen University Institute in Blaubeuren, Germany. These papers show the remarkable diversity of the contributions nonstandard analysis has made to the fields of analysis, probability theory, and mathematical physics.

The first section of the book contains papers in analysis, topology, and topological groups. The following topics are discussed:

- 1) Singular traces on the space of compact operators in a separable Hilbert space.
- 2) Weak solutions for general Navier-Stokes equations.
- 3) Harmonic analysis on arbitrary locally compact abelian groups which possess a compact open subgroup.
- 4) The identity of the myope and Lawson topologies on the collection of compact subsets of a locally compact Hausdorff space.
- 5) An overview of the Rosinger-Colombeau nonlinear theory of generalized functions.
- 6) An approach to the Pettis integral using liftings to obtain nonstandard functions for stochastic processes with values in a nonseparable locally convex space.
- 7) The nonvalidity of the spectral mapping theorem for the exponential function.
- 8) Nonstandard polynomials in several variables and complex function theory.

- 9) The existence of a solution for a class of linear partial differential equations with C^∞ -coefficients in a space of generalized functions.
- 10) Generalizations of topology using external equivalence relations.
- 11) Extensions of the Bochner integral.

The second section of the book contains papers in probability theory. The following topics are discussed:

- 1) A nonstandard approach to diffusions on manifolds and nonstandard heat kernels.
- 2) An intuitive nonstandard approach to the Malliavin calculus for the classical Wiener space.
- 3) Ergodic transformations in the framework of “alternative set theory.”
- 4) General invariance principles of probability theory.
- 5) Brownian motion and the infinite dimensional Ornstein-Uhlenbeck process.
- 6) An infinite dimensional multiplicative ergodic theorem for generalized cocycles.
- 7) Results in the theory of statistical experiments.

The third, final, section of the book contains applications of nonstandard analysis to mathematical physics with a discussion of the following topics:

- 1) Strong L^1 -convergence in time to Maxwellian limits for standard solutions of the Boltzmann equation starting far from equilibrium.
- 2) A nonstandard approach to the theory of (quantum mechanical) mean field systems in which the relevant algebras of observables are quotients of subalgebras of a hyperfinite tensor product of the relevant one-particle algebra.

N. Kazamaki: Continuous Exponential Martingales and *BMO*, Lecture Notes in Mathematics 1579, Springer, Berlin Heidelberg New York, 1994, VII/91 pp., DM 34.00/öS 265.20/sFr 34.00

Reviewed by D. L. Burkholder, Urbana

The theory of exponential martingales is an important tool in probability theory and has been so for many decades. It plays a role in sequential statistical analysis, Cameron–Martin–Girsanov theory, stochastic differential equations, and a host of other contexts. This valuable monograph discusses some of the recent progress made.

Let M be a continuous local martingale and define the local martingale $\mathcal{E}(M)$ by

$$\mathcal{E}(M) = \exp\left(M - \frac{1}{2}\langle M \rangle\right).$$

Here are some of the many topics considered: (1) conditions on M assuring that $\mathcal{E}(M)$ is uniformly integrable or, even more, L^p -bounded for some $p > 1$; (2) Girsanov's formula; (3) BMO martingales and the martingale analogues of the John-Nirenberg inequality, Fefferman's inequality and duality theorem, and the Garnett-Jones theorem; and (4) the martingale analogues of the reverse Hölder inequality, Gehring's inequality, and some of the classical weighted-norm inequalities.

One of the fundamental results of the theory is that the following three statements are equivalent for a continuous martingale M : (i) $M \in BMO$; (ii) $\mathcal{E}(M)$ is a uniformly integrable martingale satisfying the reverse Hölder inequality for some $p > 1$; (iii) $\mathcal{E}(M)$ satisfies the martingale analogue of Muckenhoupt's A_p condition for some $p > 1$.

This short book surveys a beautiful area of modern probability theory to which a great many mathematicians, including Norihiko Kazamaki and his students, have contributed. One of the strengths of the book is that it contains numerous examples and corollaries that effectively supplement the main theory.

N. Mukhopadhyay, T. K. S. Solanky: Multistage Selection and Ranking Procedures, Marcel Dekker Inc. New York, Basel, Hong-Kong, 1994, XIV/405 pp., \$125.00

Reviewed by Kh. Alam, Clemson

The area of ranking and selection in the statistical framework of multiple decision theory has gained considerable recognition and impetus in the past four decades, beginning with the pioneering work of Bechhofer (1954) and Gupta (1956). During this period there has been a steady flow of publications in the statistical literature, dealing with interesting problems of ranking and selection. Multistage and sequential sampling techniques have been applied to these problems at various levels of sophistication. This book focuses on the applications of two-stage, three-stage, accelerated sequential and purely sequential sampling methods to certain problems in ranking and selections, largely under the indifference zone framework of Bechhofer (1954), with emphasis on the second-order asymptotics. The second-order theory is unified and exploited in a number of problems, including the estimation of ordered parameters. This book is written for graduate students, researchers, and practitioners. Each chapter in the book provides a section on Notes and Overviews and lists interesting topics for further study and research and ends with Complements and Problems.

The book consists of eight chapters. The problem of ranking and selection, is introduced in Chap. 1, with the formulation of the indifference zone point of view and the subset selection approach. Chapter 2 gives a general formulation of two-stage, modified two-stage, three-stage, purely sequential, and accelerated sequential sampling procedures. Here, the first-order asymptotic properties are

presented in detail for the two-stage and modified two-stage procedures, results on the associated second-order asymptotics due to Mukhopadhyay and Solanky (1991) are synthesized, and important results in the purely sequential set-up are summarized. Various selection and ranking as well as estimation problems in the contexts of normal and two-parameter negative exponential populations having unknown means and unknown and common variance are discussed in Chaps. 3–7. Here, the emphasis lies in the derivations of the second-order asymptotic properties. Only nonelimination type procedures have been considered. Chapter 8 includes additional topics, such as elimination-type selection procedures, likelihood-based selection procedures, selection procedures when variances are unknown and unequal, and some nonparametric selection procedures.

References

- Bechhofer RE (1954) A single-sample multiple decision procedure for ranking means of normal populations with known variances. *Ann Math Statistics* 25:16–39
- Gupta SS (1956) On a decision rule for a problem in ranking means. Ph.D. dissertation, University of North Carolina, Chapel Hill
- Mukhopadhyay N, Solanky TKS (1991) Second-order properties of accelerated stopping times with applications in sequential estimation. *Sequential Anal* 10:99–123

A. Pázman: Nonlinear Statistical Models, Kluwer Academic Publishers, Dordrecht, 1993, IX/259 pp., Dfl. 195.00/US\$ 119.00/UK£ 78.00

Reviewed by O. E. Barndorff-Nielsen, Aarhus

CONTENTS:

Introduction

- 1 Linear regression models
- 2 Linear methods in nonlinear regression models
- 3 Univariate regression models
- 4 The structure of a multivariate regression model and properties of L_2 estimators
- 5 Nonlinear regression models: computation of estimators and curvatures
- 6 Local approximations of probability densities and moments of estimators
- 7 Global approximations of densities of L_2 estimators
- 8 Statistical consequences of global approximations especially in flat models
- 9 Nonlinear exponential families

References

Basic symbols

Subject index

This book is mainly concerned with some mathematical aspects of the nonlinear normal regression model, particularly the distribution of the maximum likeli-

hood estimator. There is, however, also some discussion of the least squares estimator for the case of nonnormal errors and, in the final chapter (Chap. 9), of the maximum likelihood estimator in curved exponential models.

To a large extent the book is based on a series of research papers by the author that appeared in the period 1984–1992.

The mean vector and variance matrix of the entire observation vector y are supposed to be of the form $\eta(\theta)$ and $\sigma^2 W$, respectively, where σ^2 is scalar and the matrix W is assumed to be known. (When the author uses the terms ‘univariate’ and ‘multivariate’, this refers to the dimension of the unknown parameter θ that determines the mean vector.) The emphasis is on what the author calls ‘global’ approximations of the unconditional distribution $\hat{\theta}$, the maximum likelihood estimator of θ . The meaning to be attached to the term ‘global’ is only vaguely specified, but it is used in contrast to local approximations of the Edgeworth type, and such local approximations are reviewed in Chap. 6.

The most widely known and used approximations of the global type are the saddlepoint approximations, and such approximations are considered briefly in Sect. 7.7 for the normal regression model, and in Sect. 9.4 for curved exponential models. In Sect. 7.7 and only there does the author discuss conditioning on an approximately ancillary statistic, more specifically the score ancillary, and the saddlepoint approximation is then a special version of the p^* formula.

What makes this book special is its emphasis on what the author calls the ‘almost exact’ density of the unconditional distribution of $\hat{\theta}$, assuming that σ^2 is known. It is shown that the maximum likelihood estimator exists uniquely with probability 1, but to get a handle on the distribution it is assumed that, except for a region of negligible probability, $\hat{\theta}$ is determined as the unique solution to the likelihood equation. While there is some discussion in geometric terms of when this happens, the criteria given are too general to be of much use in practice.

On the assumption that the region in question does in fact have small probability, what remains to be done is the calculation of the density of $\hat{\theta}$. Not surprisingly, the resulting expression given is essentially the (unconditional) saddlepoint approximation [a special case of a formula due to Fields (1982)] times a norming constant and a multivariate integral, both of which will have to be evaluated numerically in general.

Chapter 8 discusses the simplifications that occur when the model is flat, in the sense of differential geometry. What this means in statistical terms is that there exists a reparametrization under which the expected Fisher information is a constant matrix. This rarely happens, however.

As appears from the discussion above, the book has rather little new to offer from the practical point of view. Further, its scope is fairly restricted theoretically. It is pleasant to see the geometric viewpoint emphasized, but its role lies, in effect, mostly in providing compact and geometrically interpretable formulae for some of the terms occurring in the calculations; that is to say that geometry does not enter at a deeper statistical level (and perhaps never will).

The exposition is well-organized and lucid throughout. All in all, however, a book of quite specialized interest.

M. Grabisch, H. T. Nguyen, E. A. Walker: Fundamentals of Uncertainty Calculi with Applications to Fuzzy Inference, Kluwer Academic Press, Dordrecht, 1995, VIII/346 pp., Dfl185.00/US\$120.00/UK£78.00

Reviewed by P. E. Kloeden, Geelong

Decision-making under uncertainty has attracted considerable mathematical attention for a long time, with the probabilistic approaches of traditional inference, Bayesian inference, and the Schafer-Dempster decision functions being now well known. A more recent, approach though not without controversy, is that of fuzzy inference, which is based on Zadeh's theory of fuzzy sets. This book focuses on fuzzy inference, but attempts to place it in context with previous and more established methodologies. The authors emphasize the role of fuzzy inference as being complementary rather than competitive, in particular being appropriate and useful in situations where the mathematical assumptions of the other approaches are inapplicable or untestable. Nevertheless there are close mathematical structural relationships between the different approaches to inference, the main one being the lack of the subadditivity property of probabilities. In particular, Choquet capacity has an important role to play, both for comparison and unification.

The book consists of ten chapters. The first two outline the book and review the modelling of uncertainty. Chapter 3, the most mathematical deals with capacities and Choquet functionals, relating them to belief functions and Bayesian statistics, while Chap. 4 considers information measures. The next two chapters, Chaps. 5 and 6, introduce and expound fuzzy inference and the necessary background including t -norms and t -conorms, which have been borrowed from the theory of probabilistic metric spaces. Particularly useful here, given the confusion in the literature due to many competing definitions, is the unified treatment of fuzzy integrals due to Murofushi and Sugeno which includes the Choquet integral amongst its options. Applications to decision-making are then considered in Chaps. 7 and 8, the latter on subjective multicriteria evaluation. Interesting case studies comparing fuzzy, Bayesian, and other approaches are included. Chapter 9 introduces the fuzzy approach to pattern recognition and computer vision, concentrating on the how fuzzy integrals are used for the information fusion stage. The final chapter, Chap. 10, is devoted to the practical issues of identification, as well as the interpretation of fuzzy measures. The book provides a systematic overview of the subject, relates it to existing methodologies and illustrates its potential power in applications. Moreover, it highlights the level of mathematical sophistication that underlies fuzzy inference. A more thorough proofreading would have eliminated the mainly minor, but irritatingly numerous typographical errors.

Newly Accepted Papers

The following articles, in addition to those listed in previous issues, have been accepted for publication and will appear in future issues of *Metrika*.

Joachim Engel

The Multiresolution Histogram

Erhard Cramer and Udo Kamps

The UMVUE of $P(X < Y)$ Based on Type-II Censored Samples from Weinman Multivariate Exponential Distributions

M. Cramer

Convergence of a Branching Type Recursion with Non-Stationary Immigration

Naveen K. Bansal and Sudhir Gupta

On the Natural Selection Rule in General Linear Models

Holger Dette

A Note on the Uniform Distribution on the Arcsin Points

Estimation After Sequential Selection and Ranking

N. MUKHOPADHYAY

Department of Statistics, Univ. of Connecticut, 196 Auditorium Road, U-120, Storrs, CT 06269-3120, USA

T. K. S. SOLANKY

Department of Mathematics, University of New Orleans, New Orleans, LA 70148, USA

Abstract: Problems of estimating parameters from normal populations after the completion of the purely sequential selection and ranking methodology are addressed. Various second-order asymptotic characteristics associated with natural estimators are presented.

Key Words and Phrases: Sequential selection procedures; normal populations; common unknown scale; bias; mean square error; confidence regions; asymptotics.

1 Introduction

Consider independent populations $\pi_1, \dots, \pi_k$ with $k \geq 2$ and suppose that $X_{i1}, \dots, X_{in}, \dots$ are independent and identically distributed (i.i.d.) random variables from π_i having the probability density function (p.d.f.) $f(x; \mu_i, \sigma) = (2\pi\sigma^2)^{-1/2} \exp\{-\frac{1}{2}(x - \mu_i)^2/\sigma^2\}$ for $i = 1, \dots, k$. Throughout, $I(\cdot)$ will stand for the indicator function of $(\cdot)$ and let $\underline{\mu} = (\mu_1, \dots, \mu_k)$ and σ be all unknown parameters. Suppose that $\mu_{[1]} \leq \dots \leq \mu_{[k-1]} \leq \mu_{[k]}$ be the ordered location parameters and one implements the classical *indifference zone* approach of Bechhofer (1954) in order to select the *best* population, that is the population associated with $\mu_{[k]}$. In other words, given two preassigned numbers $\delta^*(> 0)$ and $P^* \in (k^{-1}, 1)$, one devises selection methodologies in such a way that

$$P(CS) \geq P^* \quad \text{whenever } \underline{\mu} \in \Omega(\delta^*)$$

where $\Omega(\delta^*) = \{\underline{\mu}: \mu_{[k]} - \mu_{[k-1]} \geq \delta^*\}$, the *preference zone*, and *CS* stands for *correct selection*.

In the absence of any knowledge about the unknown nuisance parameter σ , various types of multi-stage selection rules have been proposed in the literature. In the normal case, two-stage, modified two-stage, purely sequential,

accelerated sequential and three-stage selection methodologies have been put forth in Bechhofer et al. (1954), Mukhopadhyay (1983), Robbins et al. (1968), Mukhopadhyay and Solanky (1992), and Mukhopadhyay and Judge (1991) respectively. In each case, one obtains random samples $X_{i1}, \dots, X_{iN}$ from π_i with appropriately designed stopping variable N and the associated selection rule.

But, at this stage, suppose that the experimenter turns around and wishes to estimate some of the μ_i 's or the scale parameter σ . That is, having observed $X_{i1}, \dots, X_{iN}$ from π_i , one may like to estimate such parameters or parametric functions by natural estimators, to be defined in the sequel, where it is re-emphasized that " N " had been originally designed for the selection problem, and not the estimation problem. This has been the subject matter introduced in Chapter 7 of Mukhopadhyay and Solanky (1994).

We should point out that there is considerable amount of literature on estimating certain parameter of the "selected" population. One will find several other related references from Vellaisamy et al. (1988), Venter (1988), and Hsieh (1981). There are available recent contributions for handling the selection and estimation problems simultaneously. One should refer to Gupta and Miescke (1990, 1993) for such problems and other references. From the preceding paragraph, it is quite clear that there is no overlap between our present approaches and the existing literature of certain aspects of essentially estimating parameters of the "selected" population. Clearly, we are discussing different types of problems.

Our objective is to provide asymptotic (as $\delta^* \rightarrow 0$) second-order properties of the proposed estimators which take us way beyond what is now available in Chapter 7 of Mukhopadhyay and Solanky (1994). Sections 3, 4 and 5 discuss such extensions in the contexts of estimating the means, variance and standard errors of the means, after the termination of the full sequential selection process. Section 2 gives some preliminaries. We, however, note that analogous developments in the case of three-stage or accelerated sequential methodologies remain unresolved at this point.

2 Some Preliminaries

The i th population π_i is $N(\mu_i, \sigma^2)$ and having recorded $X_{i1}, \dots, X_{in}$ from π_i , one writes

$$\bar{X}_{in} = n^{-1} \sum_{j=1}^n X_{ij}, \quad S_{in}^2 = (n-1)^{-1} \sum_{j=1}^n (X_{ij} - \bar{X}_{in})^2$$

for $i = 1, \dots, k$ with $n \geq 2$. Let us write $U(n) = k^{-1} \sum_{i=1}^k S_{in}^2$. Let $C = h^2 \sigma^2 / \delta^{*2}$ where h satisfies the equation $\int_{-\infty}^{\infty} \Phi^{k-1}(y+h) \phi(y) dy = P^*$, ϕ being the p.d.f.

of $N(0, 1)$ and $\Phi(y) = \int_{-\infty}^y \phi(x) dx$, $y \in R$. If σ^2 were known, then C could be thought as the optimal fixed sample size required from each π in conjunction with the selection of the population associated with the corresponding largest sample mean.

Since C is unknown, various multi-stage selection methodologies have been proposed in the literature. For a review, one is referred to Chapter 3 of Mukhopadhyay and Solanky (1994). It may be remarked that such selection techniques have the additional property that $P\{\mu_{[k]} \in (\mu_s, \mu_s + \delta^*)\} \geq P^*$ for all $\mu \in \Omega(\delta^*)$ where μ_s stands for the mean of the selected population. In the present report, we focus only on the purely sequential selection procedure of Robbins et al. (1968).

One starts with $m(\geq 2)$ samples from each π and proceed by taking one observation at a time from each π according to the stopping rule

$$N = \inf\{n \geq m: n \geq h^2 U(n)/\delta^{*2}\} . \quad (2.1)$$

It is easy to see that $P(N < \infty) = 1$ for all fixed μ , σ^2 and δ^* . Finally one obtains $X_{i1}, \dots, X_{iN}$ from π_i , $i = 1, \dots, k$ and one implements the following selection rule.

$$\mathcal{P}_N: \text{Select } \pi_i \text{ as the best population if } \bar{X}_{iN} = \underset{1 \leq j \leq k}{\text{Max}} \bar{X}_{jN} . \quad (2.2)$$

We, however, focus on the estimation methodologies after the implementation of $\mathcal{P}_N$. At this step, it is quite natural to estimate σ^2 by means of $U(N)$. In Section 3, we provide second-order expansions for the bias and mean squared error of $U(N)$. In Section 4, we address the mean estimation problems and the associated second-order expansions of the estimated variance and the standard error of such estimators are briefly mentioned. Section 5 introduces the simultaneous fixed-width confidence interval approach for estimating μ as well as the associated second-order asymptotics. Before we proceed for details, we summarize some results in the form of a Lemma that is used repeatedly in the sequel.

Observe that $N = Q + 1$ where

$$Q = \inf\left\{n \geq m - 1: kn^2(1 + n^{-1})C^{-1} \geq \sum_{i=1}^n W_i\right\} , \quad (2.3)$$

W 's being i.i.d. χ_k^2 . Let $\eta = (v - 2)k^{-1} - 1$ where

$$v = \frac{1}{2}(k + 2) - \sum_{n=1}^{\infty} n^{-1} E[\max\{0, \chi_{nk}^2 - 2nk\}] . \quad (2.4)$$

Define $\mathcal{R} = kQ(Q + 1)C^{-1} - S_Q$, the overshoot for the stopping variable in (2.3), where one writes $S_Q = \sum_{i=1}^Q W_i$. Lemma 2.1 of Woodroffe (1977) shows that $\mathcal{R}^p$ is uniformly integrable for $p > 2$ and $E(\mathcal{R}) = v + o(1)$.

Lemma 1: Let $0 < \varepsilon < 1$, $\varpi(> 0)$ and $b(> 0)$ be arbitrary, but fixed numbers. Then, for the stopping rule (2.3), we have:

- (a) $P\{Q \leq \varepsilon C\} = O(C^{-k(m-1)/2})$;
- (b) $E(Q) = C + \eta + o(1)$ if $m > 1 + 2k^{-1}$;
- (c) $(C/Q)^\varpi$ is uniformly integrable if $m > 2\varpi k^{-1} + 1$;
- (d) $(Q/C)^\varpi$ and $|Q^{-1/2}(S_Q - kQ)|^\varpi$ are both uniformly integrable;
- (e) $E\{C^\varpi(Q + b)^{-\varpi}\} = 1 - \varpi C^{-1}\{b + \eta - (\varpi + 1)k^{-1}\} + o(C^{-1})$ if $m > 1 + 2(\varpi + 3)k^{-1}$.

Parts (a)–(c) and the first part of (d) follow from Woodroffe (1977). The second part of (d) follows from Chow et al. (1979). Part (e) can be proved along the lines of Lemma 1 in Ghosh and Mukhopadhyay (1990).

3 Estimation of Variance

We added earlier that at the stopped stage, we are interested in estimating σ^2 by $U(N)$. In Mukhopadhyay and Solanky (1994, pp. 291–292), it was shown that

$$E\{U(N)\} = \sigma^2 - 2\sigma^2(kC)^{-1} + o(C^{-1}) \quad (3.1)$$

if $m \geq 3$ when $k = 2$, and $m \geq 2$ when $k \geq 3$ as well as

$$E\{(U(N) - \sigma^2)^2\} = 2\sigma^2\delta^{*2}(kh^2)^{-1} + o(\delta^{*2}) \quad (3.2)$$

if $m \geq 2$. We are now in a position to strengthen (3.2) considerably.

Theorem 2: For the purely sequential selection rule (2.1), we have as $\delta^ \rightarrow 0$:*

$$E\{(U(N) - \sigma^2)^2\} = 2\sigma^2\delta^{*2}(kh^2)^{-1} + (kh^2)^{-2}\delta^{*4}[16 + 4k - 2v] + o(\delta^{*4})$$

if (i) $m \geq 3$ when $k = 2$, (ii) $m \geq 2$ when $k \geq 3$, where v comes from (2.4).

Proof: Observe that $U(N)$ and $\sigma^2 S_Q(kQ)^{-1}$ have identical distributions and hence

$$E[(U(N) - \sigma^2)^2] = \sigma^4 k^{-2} E\left[\frac{(S_Q - kQ)^2}{Q^2}\right]. \quad (3.3)$$

Now,

$$\begin{aligned} E[(S_Q - kQ)^2/Q^2] &= E[(S_Q - kQ)^2/C^2] + C^{-2} E\left[\left(\frac{C^2}{Q^2} - 1\right)(S_Q - kQ)^2\right] \\ &= 2kE(Q)C^{-2} + C^{-2}E(J), \quad \text{say}, \end{aligned} \quad (3.4)$$

by Wald's Lemma. See Chow et al. (1965). Next, we write

$$\begin{aligned} E(J) &= E\left[\left(1 - \frac{Q^2}{C^2}\right)(S_Q - kQ)^2\right] + E\left[\left(1 - \frac{Q^2}{C^2}\right)^2 \frac{C^2}{Q^2}(S_Q - kQ)^2\right] \\ &= E(J_0) + E(J_0^*), \quad \text{say}. \end{aligned} \quad (3.5)$$

Note that $Q^{-1/2}(S_Q - kQ) \xrightarrow{\mathcal{L}} N(0, 2k)$ and by Lemma 1, part (e) one has $E[(S_Q - kQ)^4/Q^2] = 12k^2 + o(1)$ if $m \geq 2$. Now, $1 - Q^2C^{-2} = (1 + QC^{-1})[1 - (\mathcal{R} + S_Q)(kQ)^{-1} + C^{-1}] = (1 + QC^{-1})[-(S_Q - kQ)(kQ)^{-1} - \mathcal{R}(kQ)^{-1} + C^{-1}]$, and hence

$$\begin{aligned} E(J_0^*) &= 2E[(S_Q - kQ)^4(kQ)^{-2}\{1 + CQ^{-1}\}] + o(1) \\ &= 2(12 + 12) + o(1) = 48 + o(1), \end{aligned} \quad (3.6)$$

if (i) $m \geq 3$ when $k = 2$, (ii) $m \geq 2$ when $k \geq 3$. This is so because the second term within the parenthesis in (3.6) can be evaluated as follows. First, $(S_Q - kQ)^4 Q^{-2}(C/Q)I(Q > \frac{1}{2}C) \leq 2(S_Q - kQ)^4/Q^2$ which is uniformly integrable when $m \geq 2$, by Lemma 1, part (d). Hence, $E[(S_Q - kQ)^4 Q^{-2}(C/Q)I(Q > \frac{1}{2}C)] = 12k^2 + o(1)$. On the other hand, use Holder's inequality to write (with $\delta > 0$)

$$\begin{aligned} &E[(S_Q - kQ)^4 Q^{-2}(C/Q)I(Q \leq \tfrac{1}{2}C)] \\ &\leq E^{\delta/(1+\delta)}[\{(S_Q - kQ)^4/Q^2\}^{(1+\delta)/\delta}] E^{1/(1+\delta)}[(C/Q)^{1+\delta}I(Q \leq \tfrac{1}{2}C)] \\ &= O(1)\{O(C^{1+\delta})O(C^{-k(m-1)/2})\} = o(1) \end{aligned}$$

if $m > 1 + 2(1 + \delta)k^{-1}$, in view of Lemma 1, part (a). Next, we write

$$\begin{aligned}
 J_0 &= (1 - Q^2 C^{-2})(S_Q - kQ)^2 \\
 &= (1 + QC^{-1})[1 - (\mathcal{R} + S_Q)(kQ)^{-1} + C^{-1}](S_Q - kQ)^2 \\
 &= (1 + QC^{-1}) \left[-\frac{(S_Q - kQ)^3}{kQ} - \frac{\mathcal{R}(S_Q - kQ)^2}{kQ} + \frac{(S_Q - kQ)^2}{C} \right] \\
 &= -J_1 - J_2 + J_3, \quad \text{say.} \tag{3.7}
 \end{aligned}$$

As before, one can verify that $E(J_3) = 4k + o(1)$ if $m \geq 2$, by combining with the indicators of events $\{Q \leq 2C\}$. In a similar fashion, we can show that $E(J_2) = 4v + o(1)$ if $m \geq 2$. Now, using Chow et al. (1965) we get $E[(S_Q - kQ)^3] = 8kE(Q) + 6kE[Q(S_Q - kQ)]$ and thus

$$\begin{aligned}
 E[kJ_1] &= 2E[(S_Q - kQ)^3/C] + E \left[\frac{C - Q}{CQ} (S_Q - kQ)^3 \right] \\
 &= 2E(J_{11}) + E(J_{12}), \quad \text{say.} \tag{3.8}
 \end{aligned}$$

But,

$$E[J_{11}] = E[8kQ/C] + 6kE[(Q + 1)(S_Q - kQ)/C]$$

while $E[(Q + 1)(S_Q - kQ)/C] = E \left[\frac{\mathcal{R}(S_Q - kQ)}{kQ} + \frac{(S_Q - kQ)^2}{kQ} \right] = 2 + o(1)$ if (i) $m \geq 3$ when $k = 2$, (ii) $m \geq 2$ when $k \geq 3$. We have repeatedly used Holder's inequality and tools from Chow et al. (1979). Also, $E(J_{12}) = E[(1 - QC^{-1})(S_Q - kQ)^3/Q] = E[\{1 - (\mathcal{R} + S_Q)(kQ)^{-1} + C^{-1}\}(S_Q - kQ)^3/Q] = E[-(S_Q - kQ)^4(kQ^2)^{-1} - \mathcal{R}(S_Q - kQ)^3(kQ^2)^{-1} + (S_Q - kQ)^3(CQ)^{-1}] = -12k + o(1)$ if (i) $m \geq 3$ when $k = 2$, (ii) $m \geq 2$ when $k \geq 3$, again by repeated use of Holder's inequality and part (b) of Lemma 1. Thus, we obtain from (3.8) and Lemma 1, part (b),

$$\begin{aligned}
 E(J_1) &= k^{-1}[2\{8kE(Q/C) + 6k(2 + o(1))\} + (-12k + o(1))] \\
 &= 16(1 + o(1)) + 12 + o(1) \\
 &= 28 + o(1), \tag{3.9}
 \end{aligned}$$

if (i) $m \geq 3$ when $k = 2$, (ii) $m \geq 2$ when $k \geq 3$. So, (3.7) and (3.9) imply

$$\begin{aligned} E(J_0) &= -28 + o(1) - 4v + o(1) + 4k + o(1) \\ &= -28 - 4v + 4k + o(1) , \end{aligned} \quad (3.10)$$

if (i) $m \geq 3$ when $k = 2$, (ii) $m \geq 2$ when $k \geq 3$. Now, combining (3.5), (3.6) and (3.10) we obtain $E(J) = 20 - 4v + 4k + o(1)$ under the same conditions on m . Hence, the result follows from (3.4). $\square$

4 Estimation of Means and Their Standard Errors

For a fixed i , one wishes to estimate μ_i by means of the unbiased estimator $\bar{X}_{iN}$, $i = 1, \dots, k$. Now, the variance of $\bar{X}_{iN}$ is given by $\sigma^2 E(1/N)$, which would then be estimated by means of $U(N)/N$. The following result provides the second-order expansion of the bias of $U(N)/N$.

Theorem 3: For the purely sequential selection rule (2.1), we have as $\delta^ \rightarrow 0$:*

$$E[U(N)/N] = \sigma^2 C^{-1} - \sigma^2 v (kC^2)^{-1} + o(\delta^{*4})$$

if (i) $m \geq 4$ when $k = 2$, (ii) $m \geq 3$ when $k = 3, 4$ and (iii) $m \geq 2$ when $k > 4$, where v comes from (2.4).

Proof: With our earlier notations, one can write

$$\begin{aligned} E[U(N)/N] &= \sigma^2 k^{-1} E \left[\frac{S_Q}{Q(Q+1)} \right] \\ &= \sigma^2 k^{-1} E \left[\left\{ \frac{kQ(Q+1)}{C} - \mathcal{R} \right\} / \{Q(Q+1)\} \right] \\ &= \sigma^2 C^{-1} - \sigma^2 (kC^2)^{-1} E(J) , \quad \text{say} , \end{aligned} \quad (4.1)$$

where $J = \mathcal{R}C^2 / \{Q(Q+1)\}$. Now, $J I(Q \geq \frac{1}{2}C) \leq 4\mathcal{R}$ and hence the l.h.s. is uniformly integrable if $m \geq 2$. Thus, $E[J I(Q \geq \frac{1}{2}C)] = v + o(1)$ if $m \geq 2$.

However, $E[JI(Q < \frac{1}{2}C)] \leq C^2 E[\mathcal{H}I(Q < \frac{1}{2}C)] \leq C^2 E^{1/r}[\mathcal{H}^r] P^{1/s}[Q < \frac{1}{2}C] = O(C^2)O(C^{-1/2k(m-1)/s}) = o(1)$ if $m > 1 + 4sk^{-1}$. Here, we used Holder's inequality with $r > 1, s > 1, r^{-1} + s^{-1} = 1$. Hence, $E[JI(Q < \frac{1}{2}C)] = o(1)$ if (a) $m \geq 4$ when $k = 2$, (b) $m \geq 3$ when $k = 3, 4$, and (c) $m \geq 2$ when $k > 4$. Thus, $E(J) = v + o(1)$ under similar conditions on m . Now, combining this with (4.1) leads to the result. $\square$

Also the estimated standard error of $\bar{X}_{iN}$ is given by $\{U(N)/N\}^{1/2}$. The second-order expansion of the bias of this estimated standard error is provided in the following.

Theorem 4: For the purely sequential selection rule (2.1), we have as $\delta^ \rightarrow 0$:*

$$E[\{U(N)/N\}^{1/2}] = \sigma C^{-1/2} - \frac{1}{4}\sigma k^{-1}(2v - 2)C^{-3/2} + o(C^{-3/2})$$

if (i) $m \geq 5$ when $k = 2$, (ii) $m \geq 4$ when $k = 3$, (iii) $m \geq 3$ when $k = 4, 5, 6, 7$, and (iv) $m \geq 2$ when $k \geq 8$, where v comes from (2.4).

Proof: First observe that $E[\{U(N)/N\}^{1/2}] = \sigma E\left\{(Q + 1)^{-1/2} \left(\frac{S_Q}{kQ}\right)^{1/2}\right\}$, and then use the Taylor expansion to write

$$\begin{aligned} E[\{U(N)/N\}^{1/2}] &= \sigma E\{(Q + 1)^{-1/2}\} + \frac{1}{2}\sigma k^{-1} E\left[(Q + 1)^{-1/2} \frac{(S_Q - kQ)}{Q}\right] \\ &\quad - \frac{1}{8}\sigma k^{-2} E\left[(Q + 1)^{-1/2} \frac{(S_Q - kQ)^2}{Q^2}\right] \\ &\quad + \frac{\sigma}{16} k^{-3} E\left[(Q + 1)^{-1/2} \frac{(S_Q - kQ)^3}{Q^3 \xi^{5/2}}\right] \end{aligned} \quad (4.2)$$

where ξ is some appropriate random variable between 1 and $(kQ)^{-1}S_Q$. Now, from part (e) of Lemma 1, we immediately obtain

$$E[C^{1/2}(Q + 1)^{-1/2}] = 1 - \frac{1}{2}C^{-1}(1 + \eta - 2k^{-1}) + o(C^{-1}) \quad (4.3)$$

if $m > 1 + 7k^{-1}$. Recall that $\eta = k^{-1}(v - 2) - 1$. Next, one shows that

$$E \left[(Q+1)^{-1/2} \frac{(S_Q - kQ)^2}{Q^2} \right] = 2kC^{-3/2} + o(C^{-3/2}) \quad (4.4)$$

if (i) $m \geq 3$ when $k = 2, 3$, (ii) $m \geq 2$ when $k \geq 4$. Now, we write

$$\begin{aligned} E \left[\frac{(S_Q - kQ)}{Q(Q+1)^{1/2}} \right] &= E \left[\frac{(S_Q - kQ)}{Q(Q+1)^{1/2}} \left\{ 1 - \frac{Q(Q+1)}{C(C+1)^{1/2}(Q+1)^{1/2}} \right\} \right] \\ &= E \left[\frac{(S_Q - kQ)}{Q(Q+1)^{1/2}} \left\{ 1 - \frac{(\mathcal{R} + S_Q)}{k(C+1)^{1/2}(Q+1)^{1/2}} \right\} \right] \\ &= E \left[\frac{(S_Q - kQ)}{Q(Q+1)^{1/2}} \left\{ 1 - \frac{(\mathcal{R} + S_Q)}{kQ} \right\} \right] + o(C^{-3/2}) \\ &= -E \left[\frac{(S_Q - kQ)^2}{kQ^2(Q+1)^{1/2}} \right] - E \left[\frac{(S_Q - kQ)}{Q^{1/2}} \frac{\mathcal{R}}{kQ^{3/2}(Q+1)^{1/2}} \right] \\ &\quad + o(C^{-3/2}) \\ &= -2C^{-3/2} - E(I) + o(C^{-3/2}), \quad \text{say,} \end{aligned} \quad (4.5)$$

in view of (4.4). At this stage, use Holder's inequality and part (c) from Lemma 1 to obtain

$$E[C^2 I] = O(1),$$

if (i) $m \geq 4$ when $k = 2$, (ii) $m \geq 3$ when $k = 3, 4$, and (iii) $m \geq 2$ when $k \geq 5$. Thus, (4.5) leads to

$$E \left[\frac{(S_Q - kQ)}{Q(Q+1)^{1/2}} \right] = -2C^{-3/2} + o(C^{-3/2}), \quad (4.6)$$

if (i) $m \geq 4$ when $k = 2$, (ii) $m \geq 3$ when $k = 3, 4$ and (iii) $m \geq 2$ when $k \geq 5$.

Now, let us evaluate the last term in (4.2). Observe that for some $\phi^* \in (0, 1)$, $\xi = \phi^* S_Q(kQ)^{-1} + (1 - \phi^*) > (1 - \phi^*)$. Hence, $\left| E \left[(Q+1)^{-1/2} \frac{(S_Q - Q)^3}{Q^3 \xi^{5/2}} \right] \right| \leq O(C^{-2}) E \left[\frac{C^2}{Q^2} \left| \frac{(S_Q - kQ)}{Q^{1/2}} \right|^3 \right] = O(C^{-2}) = o(C^{-3/2})$, as before, provided that (i) $m \geq 4$ when $k = 2$, (ii) $m \geq 3$ when $k = 3, 4$, and (iii) $m \geq 2$ when $k \geq 5$. Combine (4.2)–(4.6) and the last observation to write

$$\begin{aligned}
E[\{U(N)/N\}^{1/2}] &= \sigma\{C^{-1/2} - \frac{1}{2}C^{-3/2}(1 + \eta - \frac{3}{2}k^{-1}) + o(C^{-3/2})\} \\
&\quad + \frac{1}{2}\sigma k^{-1}\{-2C^{-3/2} + o(C^{-3/2})\} \\
&\quad - \frac{1}{8}\sigma k^{-2}\{2kC^{-3/2} + o(C^{-3/2})\} + o(C^{-3/2}) \\
&= \sigma C^{-1/2} - \frac{1}{4}\sigma C^{-3/2}\{2k^{-1} + 2(1 + \eta)\} + o(C^{-3/2}) \\
&= \sigma C^{-1/2} - \frac{1}{4}\sigma k^{-1}(2v - 2)C^{-3/2} + o(C^{-3/2}),
\end{aligned}$$

since $\eta + 1 = k^{-1}(v - 2)$, provided that (i) $m \geq 5$ when $k = 2$, (ii) $m \geq 4$ when $k = 3$, (iii) $m \geq 3$ when $k = 4, 5, 6, 7$, and (iv) $m \geq 2$ when $k \geq 8$. $\square$

5 Simultaneous Confidence Intervals for the Means

After implementing the purely sequential selection procedure (2.1), one may like to construct simultaneous fixed-length confidence intervals for $\mu_1, \dots, \mu_k$. In case one wishes to obtain fixed-size ellipsoidal confidence regions for $\mu_1, \dots, \mu_k$, one should refer to Chapter 7 in Mukhopadhyay and Solanky (1994) for the second-order expansion of the associated confidence coefficient. In this section, however, we address the simultaneous confidence intervals approach.

Let us assume that the experimenter has started with some preassigned positive numbers $a_1, \dots, a_k$. Having obtained $\bar{X}_{1N}, \dots, \bar{X}_{kN}$, we propose the simultaneous fixed-width confidence region

$$\mathcal{R}_N^* = \{\mu: \bar{X}_{iN} - a_i\delta^* \leq \mu_i \leq \bar{X}_{iN} + a_i\delta^*, i = 1, \dots, k\} \quad (5.1)$$

for the unknown means $\mu_1, \dots, \mu_k$. Note that $\delta^*(> 0)$ is also preassigned and hence the lengths of these simultaneous intervals are indeed fixed in advance. The associated confidence coefficient is now given by

$$\begin{aligned}
P\{\mu \in \mathcal{R}_N^*\} &= P\{|\bar{X}_{iN} - \mu_i| \leq a_i\delta^*, i = 1, \dots, k\} \\
&= \sum_{n=m}^{\infty} P\{|\bar{X}_{in} - \mu_i| \leq a_i\delta^*, i = 1, \dots, k\} P(\dot{N} = n) \\
&= E\left[\prod_{i=1}^k \{2\Phi(N^{1/2}a_i\delta^*/\sigma) - 1\}\right].
\end{aligned} \quad (5.2)$$

The following theorem provides the second-order expansion of the confidence coefficient given in (5.2). Observe that we can write

$$P\{\mu \in \mathcal{R}_N^*\} = E[g(N/C)] \quad (5.3)$$

where $g(x) = \prod_{i=1}^k \{2\Phi(a_i h x^{1/2}) - 1\}$, $x > 0$.

Theorem 5: For the purely sequential selection rule (2.1), we have as $\delta^* \rightarrow 0$:

$$P\{\mu \in \mathcal{R}_N^*\} = \prod_{i=1}^k \{2\Phi(a_i h) - 1\} + (kC)^{-1} \{(v-2)g'(1) + g''(1)\} + o(\delta^{*2})$$

if (i) $m \geq 4$ when $k = 2$, (ii) $m \geq 3$ when $k = 3, 4, 5$, and (iii) $m \geq 2$ when $k \geq 6$, where v comes from (2.4).

Proof: We shall derive the expansion from Theorem 1 of Mukhopadhyay and Datta (1994). Observe that conditions (2.1)–(2.6) listed there actually hold with $\eta^* = \eta$, $n_0^* = C$, $a = 2/k$, $s = 1$, $r = \frac{1}{2}k$, $A_1 = 1 + 2k^{-1}$ and $A_2 = 1 + 2k^{-1}$. By direct differentiation, one can verify equation (2.7) of Mukhopadhyay and Datta (1994) with $p = 3$ and $b_1 = \frac{1}{2}$, $b_2 = 1$, $b_3 = \frac{3}{2}$, that is $A_3 = 1 + 5k^{-1}$ in their Theorem 1. Hence, Theorem 1 of Mukhopadhyay and Datta (1994) implies that

$$P\{\mu \in \mathcal{R}_N^*\} = \prod_{i=1}^k \{2\Phi(a_i h) - 1\} + (kC)^{-1} \{(v-2)g'(1) + g''(1)\} + o(\delta^{*2}) ,$$

if $m > 1 + 5k^{-1}$. $\square$

Acknowledgement: We thank Professor S. Panchapakesan for providing some of the references to related papers. We also thank the referee for pointing out a few algebraic mishaps.

References

- [1] Bechhofer RE (1954) A single-sample multiple decision procedure for ranking means of normal populations with known variances. *Ann Math Statist* 25:16–39

- [2] Bechhofer RE, Dunnett CW, Sobel M (1954) A two-sample multiple decision procedure for ranking means of normal populations with a common unknown variance. *Biometrika* 41: 170–176
- [3] Chow YS, Hsiung C, Lai T (1979) Extended renewal theory and moment convergence in Anscombe's theorem. *Ann Prob* 7:304–318
- [4] Chow YS, Robbins H, Teicher H (1965) Moments of randomly stopped sums. *Ann Math Statist* 36:789–799
- [5] Ghosh M, Mukhopadhyay N (1990) Sequential estimation of the location parameter of an exponential distribution. *Sankhya* A52:303–313
- [6] Gupta SS, Miescke KJ (1990) On finding the largest normal mean and estimating the selected mean. *Sankhya* B52:144–157
- [7] Gupta SS, Miescke KJ (1993) Single-sample Bayes and empirical Bayes rules for ranking and estimating multinomial probabilities. *J Statist Plan Inf* 33:367–382
- [8] Hsieh HK (1981) On estimating the mean of the selected population with unknown variance. *Commun Statist Theor Meth* 10:1869–1878
- [9] Mukhopadhyay N (1983) Theoretical investigations of some sequential and two-stage procedures to select the larger mean. *Sankhya* A45:346–356
- [10] Mukhopadhyay N, Datta S (1994) Replicated piecewise multistage sampling with applications. *Sequential Analysis* 13:253–276
- [11] Mukhopadhyay N, Judge J (1991) Three-stage procedures for selecting the largest normal mean. *Modern Sequential Statistical Analysis*, H. Robbins Vol., Govindarajulu Z (ed) American Sciences Press Inc, Columbus, 241–262
- [12] Mukhopadhyay N, Solanky TKS (1992) Accelerated sequential procedure for selecting the largest mean. *Sequential Analysis* 11:137–148
- [13] Mukhopadhyay N, Solanky TKS (1994) Multistage selection and ranking procedures. Marcel Dekker Inc, New York
- [14] Robbins H, Sobel M, Starr N (1968) A sequential procedure for selecting the best of k populations. *Ann Math Statist* 39:88–92
- [15] Vellaisamy P, Kumar S, Sharma D (1988) Estimating the mean of the selected uniform population. *Commun Statist Theor Meth* 17:3447–3476
- [16] Venter JH (1988) Estimation of the mean of the selected population. *Commun Statist Theor Meth* 17:791–805
- [17] Woodroffe M (1977) Second-order approximations for sequential point and interval estimation. *Ann Statist* 5:984–995

Received 03.03.1995

Revised version 13.06.1995

On Characterizations of Distributions by Expected Values of Order Statistics and Record Values with Gap

MANUEL FRANCO AND JOSE M. RUIZ

Departamento de Matemática Aplicada y Estadística, Universidad de Murcia, Apdo. 4021,
30071 Murcia, Spain

Abstract: In this paper, we obtain some recurrence relationships for conditional expectations of nonadjacent order statistics and record values when the distribution function is absolutely continuous, and we prove that the distribution function is uniquely determined by the distribution of conditioned record values and by the expected values of these records. Further, different distributions are characterized by these relationships.

Key Words: Order statistics, order mean function, record values, record mean function, cumulative failure rate function.

1 Introduction

Let $X_{1,n} \leq X_{2,n} \leq \dots \leq X_{n,n}$ be the order statistics and let $R_0, R_1, \dots, R_n, \dots$ be the sequence of the upper record values from an absolutely continuous random variable (r.v.) X with distribution function (d.f.) F . Let h be a real, continuous and strictly monotone function. We define the *order mean function with gap of r over s* , by

$$\xi_{r/s;n}(x) = E(h(X_{r,n}) | X_{s,n} = x) \quad \text{where} \quad 1 \leq r \leq n, \quad 1 \leq s \leq n, \quad r \neq s;$$

and the *record mean function with gap i* , by

$$\varphi_{r/r+i}(x) = E(h(R_r) | R_{r+i} = x) \tag{1.1}$$

for $0 \leq r, 1 \leq i$, where both functions have as domain the set $D = (\alpha, \beta) = \{x \in \mathbb{R}: 0 < F(x) < 1\}$, where α and β may be infinite.

When r and s are adjacent values, the order mean function have been extensively studied by some authors. If $h(x) = x$ and the order mean function is linear, see Ferguson (1967) and Nagaraja (1988b). Furthermore, Khan and Abu-Salih (1989) characterize some specific distributions from some particular expressions

of the order mean function, and Franco and Ruiz (1995a) obtain necessary and sufficient conditions to be order mean function.

So, for the order mean functions with gap i , $\xi_{r+i/r;n}(x)$ and $\xi_{r/r+i;n}(x)$ where $1 \leq r < n$ and $1 \leq i \leq n - r$, it is simple to obtain the following identities

$$\xi_{r+i/r;n}(x) = E(h(X_{r+i,n})|X_{r,n} = x) = E(h(Y_{i,n-r})) \quad (1.2)$$

and

$$\xi_{r/r+i;n}(x) = E(h(X_{r,n})|X_{r+i,n} = x) = E(h(Z_{r,r+i-1})) \quad (1.3)$$

where $Y_{i,n-r}$ denotes the i -th order statistic from a sample size $n - r$ of a r.v. Y with d.f. $F_Y(y) = \frac{F(y) - F(x)}{1 - F(x)} \mathbf{1}_{(x,\beta)}(y)$, and $Z_{r,r+i-1}$ represents the r -th order statistic from a sample size $r + i - 1$ of a r.v. Z with d.f. $F_Z(z) = \frac{F(z)}{F(x)} \mathbf{1}_{(\alpha,x)}(z)$. See

Arnold et al. (1992) and Balakrishnan and Cohen (1991).

From (1.2) and (1.3) when $h(x) = x^k$ and for some particular distributions several recurrence relationships have been obtained for the order mean function based on relationship of moments of order statistics from Y and Z . See Khan et al. (1983), Khan and Masoom Ali (1987), Balakrishnan et al. (1988), Gajek and Gather (1991), Kamps (1991a) and Mohie et al. (1991).

In a different way, some distribution functions have been characterized using expected values of functions of order statistics, see Arnold (1977), Balasubramanian and Beg (1990) and (1992), Beg and Balasubramanian (1990), Gupta et al. (1993) and Kamps and Mattner (1993) among others.

On the other hand, the record values have had an extensive study in the literature. For an excellent review we refer to Galambos (1987), Nevzorov (1987), Ahsanullah (1988), Nagaraja (1988a), Arnold et al. (1992) and Franco and Ruiz (1995b).

Using relationships of expected values of record values, some specific distributions have also been characterized, see Nagaraja (1977) and (1988b), Ahsanullah (1982), (1990) and (1991), Gupta (1984), Lin (1988), Roy (1990), Ahsanullah and Kirmani (1991), Gajek and Gather (1991), Kamps (1991b), Balakrishnan et al. (1992) and Balakrishnan and Ahsanullah (1994).

We consider the families of distribution functions given by:

$$\mathcal{F}_- = \left\{ F \text{ d.f. } \left| \int_x^\beta h(y)(F(y) - F(x))^{i-1}(1 - F(y))^{n-r-i} dF(y) < \infty \quad \forall x \leq \beta \right. \right\}$$

$$\mathcal{F}_+ = \left\{ F \text{ d.f. } \left| \int_\alpha^x h(y)F(y)(F(y) - F(x))^{i-1} dF(y) < \infty \quad \forall x \geq \alpha \right. \right\}$$

and

$$\mathcal{F}_R = \left\{ F \text{ d.f.} \left| \int_{\alpha}^x h(y)H(y)^r(H(x) - H(y))^{i-1} dH(y) < \infty \quad \forall x \geq \alpha \right. \right\}$$

where $H(x) = -\log(1 - F(x))$ is the cumulative failure rate function. The study of these families of distributions $\mathcal{F}_-$, $\mathcal{F}_+$ and $\mathcal{F}_R$, enables us to ensure the existence of the mean functions with gap i , defined in (1.2), (1.3) and (1.1), respectively. Here it is easy to see that the families of distributions $\mathcal{F}_-$ and $\mathcal{F}_+$ are uniquely determined by their respective order mean functions, see for example Chan (1967), Huang (1975), Patil et al. (1975), Galambos and Kotz (1978) and Lin (1990) and (1991).

The aim of this paper is to prove that the family of distributions $\mathcal{F}_R$ is uniquely determined by the record mean functions with gap and to obtain recurrence relationships for the order mean functions with gap i , $\xi_{r+i/r;n}$ and $\xi_{r/r+i;n}$, as well as for the record mean function with gap i , $\varphi_{r/r+i}$, based on the mean functions between adjacent order statistics and between adjacent record values. As an application we characterize three subsets of the families $\mathcal{F}_-$, $\mathcal{F}_+$ and $\mathcal{F}_R$, from the order mean functions and the record mean function. Consequently, for different expressions of h we characterize some specific distributions.

2 Distributions and Mean Functions

Let us look at the characterization of F by the order mean functions with gap as given in the following proposition.

Proposition 2.1: If $F \in \mathcal{F}_-$ (and, likewise, $F \in \mathcal{F}_+$), then F is uniquely determined by the order mean function $\xi_{s/r;n}$, for some $r, s \in \{1, \dots, n\}$ with $r < s$ ($r > s$) and all $n \geq r$.

Proof: It is clear from (1.2) and Corollary 3 of Huang (1975) when $m = n - r$ and for $k = s - r$ with $s \leq n$. To prove the result for the case $F \in \mathcal{F}_+$, from (1.3) we use arguments following the lines for $F \in \mathcal{F}_-$. $\square$

We first prove a lemma, which will be useful for the characterization of the distribution by the record mean function.

Lemma 2.2: For all $r \geq 0$ and $i \geq 1$, the polynomial of degree $r + i$,

$$p(z) = \sum_{s=0}^{i-1} (-1)^s \binom{r+s}{s} \binom{r+i}{i-s-1} z^{r+s+1}$$

is strictly increasing on the open interval $(0, 1)$.

Proof: Taking the first derivative of $p(z)$ we obtain

$$p'(z) = \sum_{s=0}^{i-1} (-1)^s \binom{r+s}{s} \binom{r+i}{i-s-1} (r+s+1) z^{r+s} = \frac{(r+i)!}{r!(i-1)!} z^r (1-z)^{i-1}$$

and therefore, $p'(z) > 0$ for all $z \in (0, 1)$, i.e., $p(z)$ is strictly increasing. $\square$

Proposition 2.3: Let X and Y be two absolutely continuous r.v.'s such that $F(q) = G(q)$ for at least one $q \in D$. Let $\{R_n\}$ and $\{S_n\}$ for $n = 0, 1, \dots$ be the corresponding sequences of record values. If for any $r \geq 0$ and $i \geq 1$, the conditional r.v.'s $(R_r | R_{r+i})$ and $(S_r | S_{r+i})$ are identically distributed, then X and Y have the same d.f.

Proof: In fact, if we denote by $F_{r/r+i}$ the d.f. of the conditional r.v. $(R_r | R_{r+i})$, and let $f_{r/r+i}$ be its density function, from the joint density function of record values, see Arnold et al. (1992), we get

$$f_{r/r+i}(y|x) = \begin{cases} \frac{(r+i)!}{r!(i-1)!} \left(\frac{H(y)}{H(x)} \right)^r \left(1 - \frac{H(y)}{H(x)} \right)^{i-1} \frac{H'(y)}{H(x)} & \text{if } y \leq x \\ 0 & \text{otherwise} \end{cases} \quad (2.1)$$

where $H(x) = -\log(1 - F(x))$ for all $x \in D$, is the cumulative failure rate function of the d.f. F . Now (2.1) implies

$$F_{r/r+i}(y|x) = \sum_{s=0}^{i-1} (-1)^s \binom{r+s}{s} \binom{r+i}{i-s-1} \left(\frac{H(y)}{H(x)} \right)^{r+s+1}$$

for all $y < x$ and $y, x \in D$. Analogously, one can obtain the d.f. $G_{r/r+i}$ of the conditional r.v. $(S_r | S_{r+i})$, and taking into account the assumption of this proposi-

tion, that the r.v.'s $(R_r|R_{r+i})$ and $(S_r|S_{r+i})$, are identically distributed, it is equivalently to say that

$$p\left(\frac{H(y)}{\bar{H}(x)}\right) = p\left(\frac{K(y)}{\bar{K}(x)}\right)$$

where $p(z)$ is the polynomial of Lemma 2.2, and $K(x) = -\log(1 - G(x))$ is the cumulative failure rate function of d.f. G .

Consequently, from Lemma 2.2 and taking into account that

$$0 < \frac{H(y)}{H(x)} < 1 \quad \text{and} \quad 0 < \frac{K(y)}{K(x)} < 1$$

for all $y, x \in D$ with $y < x$, we get that

$$\frac{H(y)}{H(x)} = \frac{K(y)}{K(x)} \Leftrightarrow \frac{H(y)}{K(y)} = \frac{H(x)}{K(x)} = C$$

i.e., the cumulative failure rate functions are proportional. Hence, we have $C = 1$ when $x = q$, and therefore $H(x) = K(x)$, i.e., $F(x) = G(x)$ for all $x \in D$. $\square$

Proposition 2.4: Let $F \in \mathcal{F}_R$ and $q \in D$ a point such that $F(q)$ is known. Then F is uniquely determined by the record mean function $\varphi_{r/r+i}(x)$, for some $r \in \mathbb{N}$ and all $i \geq 1$.

Proof: It is simple to obtain

$$\varphi_{r/r+i}(x) = E(h(R_r)|R_{r+i} = x) = E(h(U_{r+1, r+i}))$$

where $U_{r+1, r+i}$ represents the $r + 1$ -th order statistics from a sample size $r + i$ of a r.v. U with d.f. $F_U(u) = \frac{H(u)}{H(x)} \mathbf{1}_{(a, x)}(u)$. Hence, using Corollary 3 of Huang (1975) when $m = r + i$, for $k = r + 1$ and $i \geq 1$, we have that F_U is uniquely determined by the record mean function, and taking into account that $F(q)$ is known, we finish the proof in the same way as Proposition 2.3. $\square$

Remark 2.5: Propositions 2.1 and 2.4 show the bijection between the families $\mathcal{F}_-$, $\mathcal{F}_+$ and $\mathcal{F}_R$, and their respective mean functions with gap.

3 Recurrence Relationships of Expected Values

In this section we obtain some relationships to determine the mean functions with gap i , which characterize the families $\mathcal{F}_-$, $\mathcal{F}_+$ and $\mathcal{F}_R$ by Section 2.

Firstly, we see the recurrence relationships based on the order mean functions, wherein the sample size stays fixed, which are trivial from (3.4.3) of David (1981) and expressions (1.2) and (1.3) respectively.

Proposition 3.1: If $1 \leq r < n$ and $1 \leq i \leq n - r$, then

$$\xi_{r+i/r;n}(x) = \sum_{s=0}^{i-1} (-1)^s \binom{n-r}{i-s-1} \binom{n-r-i+s}{s} \xi_{r+i-s/r+i-s-1;n}(x) .$$

Proposition 3.2: If $1 \leq r < n$ and $1 \leq i \leq n - r$, then

$$\xi_{r/r+i;n}(x) = \sum_{s=0}^{i-1} (-1)^s \binom{r+i-1}{i-s-1} \binom{r+s-1}{s} \xi_{r+s/r+s+1;n}(x) .$$

Let us see now, as we previously indicated, the recurrence relationship for the record mean function with gap i .

Proposition 3.3: If $0 \leq r$ and $1 \leq i$, then

$$\varphi_{r/r+i}(x) = \sum_{s=0}^{i-1} (-1)^s \binom{r+s}{s} \binom{r+i}{i-s-1} \varphi_{r+s/r+s+1}(x) .$$

Proof: It is immediate from (1.1) and (2.1). $\square$

4 Some Characterizations of Distributions by Mean Functions

Firstly, we give some technical lemmas, which will be useful to obtain the d.f. F from the recurrence relationships among conditional expectations of order statistics and record values with gap.

Lemma 4.1: If $1 \leq i$, then for all $j \geq 1$

$$\begin{aligned} \text{(i)} \quad & \sum_{s=0}^{i-1} (-1)^s \binom{j+i-1}{i-s-1} \binom{j+s-1}{s} = 1. \\ \text{(ii)} \quad & \sum_{s=0}^{i-1} \frac{(-1)^s}{j+s} \binom{j+i-1}{i-s-1} \binom{j+s-1}{s} = \sum_{s=0}^{i-1} \frac{1}{j+s}. \\ \text{(iii)} \quad & \sum_{s=0}^{i-1} \frac{(-1)^s}{j+s+1} \binom{j+i-1}{i-s-1} \binom{j+s-1}{s} = \frac{i}{j+i}. \end{aligned}$$

Lemma 4.2: If $1 \leq i$, then for all $j > 1$

$$\sum_{s=0}^{i-1} \frac{(-1)^s}{j+s-1} \binom{j+i-1}{i-s-1} \binom{j+s-1}{s} = \frac{i}{j-1}.$$

Theorem 4.3: Let X be a r.v. with $F \in \mathcal{F}_-$. Then $\xi_{r+i/r;n}(x) = a_n h(x) + b_n$ with

$$\begin{aligned} \text{(i)} \quad & a_n = \frac{n-r+1-i}{n-r+1} \quad \text{and} \quad b_n = \frac{ih(\beta-)}{n-r+1} \quad \text{if and only if} \quad F(x) = 1 - \frac{h(\beta-) - h(x)}{h(\beta-) - h(\alpha+)}. \\ \text{(ii)} \quad & a_n = 1 \quad \text{and} \quad b_n = b \quad \text{if and only if} \quad F(x) = 1 - e^{-\theta(h(x)-h(\alpha+))}, \quad \text{where} \quad \theta = \frac{1}{b} \sum_{s=0}^{i-1} \frac{1}{n-r-s} \quad \text{and} \quad h(\beta-) = \infty. \\ \text{(iii)} \quad & i < n-r, \quad a_n = \frac{n-r}{n-r-i} \quad \text{and} \quad b_n = \frac{i\delta}{n-r-i} \quad \text{if and only if} \quad F(x) = 1 - \frac{h(\alpha+) + \delta}{h(x) + \delta}, \quad \text{where} \quad h(\beta-) = \infty. \end{aligned}$$

Proof: The proof of the necessary part is obtained from Proposition 2.1. To prove the sufficient part, we consider the case (i). In this case, from the expression of F we obtain the order mean function of $r+i-s$ over $r+i-s-1$ with sample size n given by

$$\begin{aligned} \xi_{r+i-s/r+i-s-1;n}(x) &= \frac{n-r-i+s+1}{n-r-i+s+2} h(x) + \frac{h(\beta-)}{n-r-i+s+2} \\ &= h(x) + \frac{h(\beta-) - h(x)}{n-r-i+s+2}. \end{aligned}$$

Substituting this in the relationship of Proposition 3.1 and applying Lemma 4.1 for $j = n - r - i + 1$, we get

$$\begin{aligned}\xi_{r+i/r;n}(x) &= h(x) + \frac{i}{n-r+1}(h(\beta-) - h(x)) \\ &= \frac{n-r+1-i}{n-r+1}h(x) + \frac{ih(\beta-)}{n-r+1}.\end{aligned}$$

Analogously, (ii) and (iii) are proved applying Lemma 4.1 and 4.2. $\square$

Theorem 4.4: Let X be a r.v. with $F \in \mathcal{F}_+$. Then $\xi_{r/r+i;n}(x) = a_n h(x) + b_n$ with

- (i) $a_n = \frac{r}{r+i}$ and $b_n = -\frac{ih(\alpha+)}{r+i}$ if and only if $F(x) = \frac{h(x) - h(\alpha+)}{h(\beta-) - h(\alpha+)}$.
- (ii) $a_n = 1$ and $b_n = b$ if and only if $F(x) = e^{-\theta(h(x) - h(\beta-))}$, where $\theta = \frac{1}{b} \sum_{s=0}^{i-1} \frac{1}{r+s}$ and $h(\alpha+) = \infty$.
- (iii) $r > 1$, $a_n = \frac{r+i-1}{r-1}$ and $b_n = \frac{i\delta}{r-1}$ if and only if $F(x) = \frac{h(\beta-) + \delta}{h(x) + \delta}$, where $h(\alpha+) = \infty$.

Theorem 4.5: Let X be a r.v. with $F \in \mathcal{F}_R$ and $q = \inf\{x \in D: F(x) \geq (e-1)e^{-1}\}$. Then $\varphi_{r/r+i}(x) = ah(x) + b$ with

- (i) $a = \frac{r+1}{r+i+1}$ and $b = \frac{ih(\alpha+)}{r+i+1}$ if and only if $F(x) = 1 - \exp\left(-\frac{h(x) - h(\alpha+)}{h(q) - h(\alpha+)}\right)$, where $h(\beta-) = \text{sign}(h(q) - h(\alpha+))\infty$.
- (ii) $a = 1$ and $b = \delta \sum_{s=0}^{i-1} \frac{1}{r+s+1}$ if and only if $F(x) = 1 - \exp\left(-\exp\left(\frac{h(q) - h(x)}{\delta}\right)\right)$, where $h(\alpha+) = \text{sign}(\delta)\infty = -h(\beta-)$.
- (iii) $r > 0$, $a = \frac{r+i}{r}$ and $b = -\frac{ih(\beta-)}{r}$ if and only if $F(x) = 1 - \exp\left(-\frac{h(\beta-) - h(q)}{h(\beta-) - h(x)}\right)$, where $h(\alpha+) = -\text{sign}(h(\beta-) - h(q))\infty$.

Theorems 4.4 and 4.5 are proved in a similar way to Theorem 4.3 from Propositions 2.1 and 2.4 and Lemmas 4.1 and 4.2.

Remark 4.6: Note that the above theorems make it possible to characterize subfamilies of $\mathcal{F}_-$, $\mathcal{F}_+$ and $\mathcal{F}_R$ respectively, which contain many frequently used distributions.

5 Examples

Let us see now the characterization of some d.f.'s from former results.

Example 5.1: In the case $h(x) = x$, Theorems 4.3 and 4.4 are reduced to the characterizations of Ferguson (1967) when $i = 1$. Furthermore, if both order mean function, $\xi_{r/r+i;n}$ and $\xi_{r+i/r;n}$, are linear we characterize the uniform distribution, see Nagaraja (1988b) when $i = 1$.

Table I.

Distribution	$F(x)$	$h(x)$	Parameters
Exponential	$1 - e^{-\theta(x-\alpha)}$	x	$\theta > 0$
Rayleigh	$1 - \exp(-2^{-1}\theta^{-2}x^2)$	x^2	$\theta \neq 0$
Weibull	$1 - \exp(-\theta x^p)$	x^p	$\theta > 0, p > 0$
Burr type VII	$(1 + \text{Tanh}(x))/2$	$\text{Tanh}(x)$	
Burr type VIII	$2\pi^{-1} \arctan(e^x)$	$\arctan(e^x)$	
Burr type XII	$1 - (1 + x^p)^{-\theta}$	$(1 + x^p)^\theta$	$\theta > 0, p > 0$
Singh-Maddala	$1 - (1 + \lambda x^p)^{-\theta}$	$\log(1 + \lambda x^p)^\theta$	$\theta > 0, \lambda > 0, p > 0$
Beta 1st kind	$1 - (1 - x)^\theta$	$-\log(1 - x)$	$\theta > 0$
Beta 2nd kind	$1 - (1 + x)^{-1}$	x	
Extreme value I	$1 - \exp(-e^x)$	e^x	
Extreme value II	$1 - \exp(\theta/x)$	x^{-1}	$\theta > 0$
Logistic	$(1 + e^{-x})^{-1}$	e^x	
Log-logistic	$x^p/(\theta^p + x^p)$	x^p	$\theta > 0, p > 0$
Half-logistic	$(1 - e^{-x})/(1 + e^{-x})$	e^x	
Uniform	$(x - \alpha)/(\beta - \alpha)$	x	
Pareto	$1 - (\alpha/x)^\theta$	$\log(x)$	$\theta > 0$
Cauchy	$2^{-1} + \pi^{-1} \arctan(x)$	$\arctan(x)$	
Kappa	$x^p/(\theta + x^p)$	x^p	$\theta > 0, p > 0$

Example 5.2: If $h(x) = x$, from Theorem 4.3 we have that the d.f. is exponential if and only if $E(X_{r+i,n} - X_{r,n} | X_{r,n} = x) = \xi_{r+i/r,n} - x = c$. See Ouyang (1982) for $n = 3$ and $i = 2$.

Example 5.3: If we consider the order mean functions between adjacent order statistics, i.e., $i = 1$, then the parts (i) of Theorems 4.3 and 4.4 are the same as Theorem 2.1 and Lemma 2.1, respectively, of Khan and Abu-Salih (1989) with the relation $b = -\frac{b_n}{a_n(n-r+1)}$, $a = \frac{n-r}{n-r+1}$ and $c = 1$; and likewise, the parts (ii) of Theorems 4.3 and 4.4 are the same as Theorem 2.2 and Lemma 2.2, respectively, of Khan and Abu-Salih (1989).

Table II.

Distribution	$F(x)$	$h(x)$	Parameters
Exponential	$1 - e^{-\theta(x-\alpha)}$	$e^{-\theta x}$	$\theta > 0$
Rayleigh	$1 - \exp(-2^{-1}\theta^{-2}x^2)$	$\exp(-\theta^{-2}x^2/2)$	$\theta \neq 0$
Weibull	$1 - \exp(-\theta x^p)$	$\exp(-\theta x^p)$	$\theta > 0, p > 0$
Burr type II	$(1 + e^{-x})^{-\theta}$	$\log(1 + e^{-x})$	$\theta > 0$
Burr type III	$(x^p/(1 + x^p))^\theta$	$\log(1 + x^{-p})$	$\theta > 0, p > 0$
Burr type IV	$(1 + (\theta/x - 1)^{1/\theta})^{-1}$	$(\theta/x - 1)^{1/\theta}$	$\theta > 0$
Burr type V	$(1 + \theta e^{-\tan(x)})^{-1}$	$e^{-\tan(x)}$	$\theta > 0$
Burr type VII	$(1 + \tanh(x))/2$	$\tanh(x)$	
Burr type VIII	$2\pi^{-1} \arctan(e^x)$	$\arctan(e^x)$	
Logistic	$(1 + e^{-x})^{-1}$	e^{-x}	
Log-logistic	$x/(\theta + x)$	x^{-1}	$\theta > 0$
Uniform	$(x - \alpha)/(\beta - \alpha)$	x	
Cauchy	$2^{-1} + \pi^{-1} \arctan(x)$	$\arctan(x)$	
Pareto	$1 - (\alpha/x)^\theta$	$x^{-\theta}$	$\theta > 0$
Power function	$((x - \alpha)/(\beta - \alpha))^\theta$	$-\log(x - \alpha)$	$\theta > 0$
Gumbel	$\exp(-e^{-x})$	e^{-x}	
Extreme value II'	$\exp(-(\theta/x)^p)$	x^{-p}	$\theta > 0, p > 0$
Inverse Weibull	$\exp(-\theta x^p)$	x^p	$\theta > 0, p > 0$

Table III.

Distribution	$F(x)$	$h(x)$	Parameters
Exponential	$1 - e^{-\theta(x-\alpha)}$	x	$\theta = (q - \alpha)^{-1} > 0$
Rayleigh	$1 - \exp(-\theta^{-2}x^2/2)$	x^2	$\theta = q/2^{1/2} > 0$
Weibull	$1 - \exp(-\theta x^p)$	$-\log(x)$	$\theta = q^{-p} > 0, p > 0$
Singh-Maddala	$1 - (1 + \lambda x^p)^{-\theta}$	$\log(1 + \lambda x^p)^\theta$	$\lambda > 0, p > 0, \theta = (\log(1 + \lambda q^p))^{-1} > 0$
Beta 1st kind	$1 - (1 - x)^\theta$	$-\log(1 - x)$	$\theta = -\log(1 - q) > 0$
Beta 2nd kind	$1 - (1 + x)^{-\theta}$	$(\log(1 + x))^{-1}$	$\theta = \log(1 + q) > 0$
Lomax	$1 - \theta^p(\theta + x)^{-p}$	$(\log(1 + x/\theta))^{-1}$	$\theta > 0, p = \log(1 + x/\theta) > 0$
Extreme value I	$1 - \exp(-e^{-(x-q)/\theta})$	x	$\theta > 0$
Extreme value II	$1 - \exp(-\theta/(\beta - x))$	x	$\theta = \beta - q > 0$
Uniform	$(x - \alpha)/(\beta - \alpha)$	$-\log(\beta - x)$	
Pareto	$1 - (\alpha/x)^\theta$	$\log(x)$	$\theta = (\log(q/\alpha))^{-1} > 0$
Cauchy	$2^{-1} + \pi^{-1} \arctan(x)$	$-\log(2^{-1} - \pi^{-1} \arctan(x))$	$q = \pi/2 - \pi/e$

Example 5.4: Tables I, II and III show the corresponding d.f.'s and $h(x)$ as resulting from Theorem 4.3, 4.4 and 4.5 respectively.

Acknowledgement: The authors are grateful to the referee for the valuable comments, suggestions and careful reading of the manuscript, which have improved and clarified this paper.

References

- Ahsanullah M (1982) Characterizations of the exponential distribution by some properties of the record values. *Statist Hefte* 23:326–332
- Ahsanullah M (1988) Introduction to record statistics. Ginn Press, Needham Heights, MA
- Ahsanullah M (1990) Some characterizations of the exponential distribution by first moment of record values. *Pakistan J Statist* 6:183–188
- Ahsanullah M (1991) Some characteristic properties of the record values from the exponential distributions. *Sankhyā Ser B* 53:403–408
- Ahsanullah M, Kirmani SNUA (1991) Characterizations of the exponential distribution through a lower record. *Comm Statist Theory Meth* 20:1293–1299
- Arnold BC (1977) Recurrence relations between expectations of functions of order statistics. *Scand Actuarial J* 169–174
- Arnold BC, Balakrishnan N, Nagaraja HN (1992) A first course in order statistics. Wiley, New York
- Balakrishnan N, Malik HJ, Ahmed SE (1988) Recurrence relations and identities for moments of order statistics, II: Specific continuous distributions. *Commun Statist Theory Meth* 17:2657–2694
- Balakrishnan N, Cohen AC (1991) Order statistics and inference. Academic Press, Boston
- Balakrishnan N, Ahsanullah M, Chan PS (1992) Relations for single and product moments of record values from Gumbel distribution. *Statist & Prob Letters* 15:223–227
- Balakrishnan K, Ahsanullah M (1994) Recurrence relations for a single and product moments of record values from generalized Pareto distribution. *Commun Statist Theory Meth* 23:2841–2852
- Balasubramanian K, Beg MI (1990) On expectations of functions of order statistics. *Sankhyā Ser A* 52:103–114
- Balasubramanian K, Beg MI (1992) Distributions determined by conditioning on a pair of order statistics. *Metrika* 39:107–112
- Beg MI, Balasubramanian K (1990) Distributions determined by conditioning on a single order statistic. *Metrika* 37:37–43
- Chan LK (1967) On a characterization of distributions by expected values of extreme order statistics. *Amer Math Monthly* 74:950–951
- David HA (1981) Order statistics, 2nd ed. Wiley, New York
- Ferguson TS (1967) On characterizing distributions by properties of order statistics. *Sankhyā Ser A* 29:265–278
- Franco M, Ruiz JM (1995a) On characterization of continuous distributions with adjacent order statistics. *Statistics* 26:375–385
- Franco M, Ruiz JM (1995b) On characterization of continuous distributions by conditional expectation of record values. To appear in *Sankhyā Ser A*
- Gajek L, Gather U (1991) Moment inequalities for order statistics with applications to characterizations of distributions. *Metrika* 38:357–367

- Galambos J, Kotz S (1978) Characterizations of probability distributions. Lecture Notes in Mathematics 675. Springer, Berlin
- Galambos J (1987) The asymptotic theory of extreme order statistics, 2nd ed. Krieger, Melbourne, FL
- Gupta RC (1984) Relationships between order statistics and record values and some characterization results. *J Appl Prob* 24:425–430
- Gupta SD, Goswami A, Rao BU (1993) On a characterization of uniform distributions. *J Multivariate Analysis* 44:102–114
- Huang JS (1975) Characterization of distributions by expected values of order statistics. *Ann Inst Statist Math* 27:87–93
- Kamps U (1991a) A general recurrence relation for moments of order statistics in a class of probability distributions and characterizations. *Metrika* 38:215–225
- Kamps U (1991b) Recurrence relations for moments of order statistics and record values. In: Gritzmman P et al (eds) *Operations Research '91*:338–340. Physica, Heidelberg
- Kamps U, Mattner L (1993) An identity for expectations of functions of order statistics. *Metrika* 40:361–365
- Khan AH, Abu-Salih MS (1989) Characterizations of probability distributions by conditional expectation of order statistics. *Metron* 47:171–181
- Khan AH, Masoom Ali M (1987) Characterization of probability distributions through higher order gap. *Commun Statist Theory Meth* 16:1281–1287
- Khan AH, Yaqub M, Parvez S (1983) Recurrence relations between moments of order statistics. *Naval Res Logist Quart* 30:419–461
- Lin GD (1988) Characterizations of distributions via relationships between two moments of order statistics. *J Statist Plann and Inf* 19:73–80
- Lin GD (1990) Characterizations of continuous distributions via expected values of two functions of order statistics. *Sankhyā Ser A* 52:84–90
- Lin GD (1991) Characterizations of distributions via moments of order statistics when sample size is random. *SEA Bull Math* 15:139–144
- Mohie El-Din MM, Mahmoud MAW, Abo Youssef SE (1991) Moments of order statistics from parabolic and skewed distributions and a characterization of Weibull distribution. *Commun Statist Simula* 20:639–645
- Nagaraja HN (1977) On a characterization based on record values. *Austral J Statist* 19:70–73
- Nagaraja HN (1988a) Record values and related statistics. A review. *Comm Statist Theory Meth* 17:2223–2238
- Nagaraja HN (1988b) Some characterizations of continuous distributions based on regressions of adjacent order statistics and record values. *Sankhyā Ser A* 50:70–73
- Nevzorov VB (1987) Records. *Theory Probab Appl* 32:201–228
- Ouyang LY (1982) On characterizations of the exponential and its related distributions by order statistics. *Tamkang J Management Sci* 3:17–42
- Patil G, Kotz S, Ord JK (1975) Statistical distributions in scientific work, Vol. 3, Characterizations and applications. Reidel, Dordrecht, Holland
- Roy D (1990) Characterization through record values. *J Indian Statist Assoc* 28:99–103

Received 24.06.1994

Revised version 31.07.1995

Do Components of Smooth Tests of Fit have Diagnostic Properties?

NORBERT HENZE

Institut für Mathematische Stochastik, Universität Karlsruhe, Englerstr. 2, 76128 Karlsruhe, Germany

Abstract: Smooth goodness of fit tests were introduced by Neyman (1937). They can be regarded as a compromise between globally consistent (“omnibus”) tests of fit and procedures having high power in the direction of a specific alternative. It is commonly believed that components of smooth tests like, e.g., skewness and kurtosis measures in the context of testing for normality, have special diagnostic properties in case of rejection of a hypothesis H_0 in the sense that they constitute direct measures of the kind of departure from H_0 . Recent years, however, have witnessed a complete change of attitude towards the diagnostic capabilities of skewness and kurtosis measures in connection with normality testing. In this paper, we argue that any component of any smooth test of fit is strictly non-diagnostic when used conventionally. However, a proper rescaling of components does indeed achieve the desired “directed diagnosis”.

Key Words and Phrases: Goodness of fit; smooth tests; components; directed diagnosis.

1 Introduction

Since Karl Pearson’s pioneering paper on the celebrated χ^2 test (Pearson, 1900), the problem of assessing the goodness of fit of given observations with a hypothesized probabilistic model has always been a crucial aspect of data analysis (see, e.g., D’Agostino and Stephens, 1986). Another cornerstone in the development of goodness of fit tests is the work of Neyman (1937) who introduced the class of *smooth tests of fit*. An account of smooth tests of fit is given in Rayner and Best (1989).

Smooth goodness of fit tests are commonly viewed as a compromise between globally consistent (“omnibus”) tests and procedures having high power in the direction of a specific alternative, and it is believed that

the use of components (to be defined later) of the smooth tests overcomes one of the major criticism of Pearson’s χ^2 test in particular, and goodness of fit tests in general—that when the null hypothesis is rejected there is no alternative model apparent (Rayner and Best (1989), p. 139).

A prominent example of this conventional view of the presumed diagnostic capabilities of smooth goodness of fit tests is the first component of the smooth test for (univariate) normality. This component is a standardized version of the skewness coefficient, typically denoted by $\sqrt{b_1}$. Now, a test for normality based on $\sqrt{b_1}$ has been valued because of its presumed diagnostic properties, for it is usually argued that

Departure of $\sqrt{b_1}$ from the 'normal' value of zero is an indication of skewness in the frequency function of the sampled population ...
(Pearson and Hartley, 1970, p. 67).

In the same way, Mardia's affine invariant measure of multivariate skewness (Mardia 1970) which is also connected with a smooth goodness of fit test (see Koziol, 1987) was commonly regarded as a *direct measure of the kind of departure from normality*.

However, in recent years the attitude towards the diagnostic properties of tests for normality based on skewness (and kurtosis) measures has changed completely (see, e.g., Baringhaus and Henze (1992); Henze (1994); Horswell and Looney (1992, 1993); Rayner, Best and Mathews (1995)). For example, Rayner, Best and Mathews (1995, p. 594) write:

We conclude that the tests based on both the skewness and kurtosis coefficients are strictly non-diagnostic. Similarly, examination of the moment ratios (derived from smooth tests) as suggested in Best and Rayner (1986) or, equivalently, in Rayner and Best (1983), is suspect.

In this paper we will show that any component of any smooth goodness of fit test is strictly non-diagnostic when used conventionally. The reason for this (already commented on by Bera and John (1983, p. 104) in the special context of skewness and kurtosis tests for multivariate normality) is very simple.

Any smooth goodness of fit test statistic (and any component thereof), after rescaling, can (and must) be regarded as an estimator of a certain parameter $\mathfrak{J}$ (say) of the underlying unknown distribution P so that $\mathfrak{J} = \mathfrak{J}(P)$ and the estimator is $\hat{\mathfrak{J}}_n = \mathfrak{J}(\hat{P}_n)$, where $\hat{P}_n$ denotes the empirical measure of a random sample of size n from P . Then, typically, $\mathfrak{J}(P) = 0$ for all distributions P belonging to some hypothetical class $\mathcal{P}_0$, and often the distribution of $\mathfrak{J}(\hat{P}_n)$ does not depend on any specific P in $\mathcal{P}_0$. The crucial point, however, is that the equation $\mathfrak{J}(P) = 0$ does not characterize the class $\mathcal{P}_0$. This means that there are distributions P not belonging to $\mathcal{P}_0$ and satisfying $\mathfrak{J}(P) = 0$ (skewness and the class $\mathcal{P}_0$ of all normal distributions provide a simple example).

Now, rejection of $H_0: P \in \mathcal{P}_0$ for a two-sided test is for large values of $|\mathfrak{J}(\hat{P}_n)|$, and, in case of rejection, a "directed diagnosis" would be "rejection due to $\mathfrak{J}(P) \neq 0$ ". However, this diagnosis is not possible for the following reason. Although, typically, consistency is achieved against any fixed alternative distribution having $\mathfrak{J}(P) \neq 0$, the power behavior of the test based on $\mathfrak{J}(\hat{P}_n)$ for a

distribution P satisfying $\mathcal{J}(P) = 0$ depends entirely on the distribution of $\mathcal{J}(\hat{P}_n)$, and this distribution, although (in many cases) being constant over the subset of all P in $\mathcal{P}_0$ (the critical value is derived from this distribution!), may vary extremely over the set $\{P: P \notin \mathcal{P}_0, \mathcal{J}(P) = 0\}$. For example, the power of skewness tests for normality is inflated against non-skewed leptokurtic distributions and deflated against non-skewed platykurtic distributions (see, e.g., Horswell and Looney (1993)).

To put it shortly: Based on $\mathcal{J}(\hat{P}_n)$, we cannot expect to give an answer to the question “Do we have $\mathcal{J}(P) = 0$ or $\mathcal{J}(P) \neq 0$?”, and at the same time letting the answer depend solely on the distributional behavior of $\mathcal{J}(\hat{P}_n)$ for a (in general: “small”) subset $\mathcal{P}_0$ of all P satisfying $\mathcal{J}(P) = 0$.

As a way out, we propose a suitable rescaling of $\mathcal{J}(\hat{P}_n)$ in order to obtain a statistic which is asymptotically distribution free over a “large class” of distributions P satisfying $\mathcal{J}(P) = 0$.

The paper is organized as follows. In Section 2, Neyman’s smooth test for uniformity is briefly reviewed and motivated in the spirit of projection pursuit. Section 3 shows that any component of any smooth test of fit is strictly non-diagnostic when used conventionally. The remedy is given in Section 4.

It should be stressed that the restriction to tests for uniformity is only for the sake of lucidity. Smooth tests of fit for parametric families of distributions involve the additional technical problem that parameters have to be estimated. The message of this paper, however, remains completely unchanged.

2 Neyman’s Smooth Test for Uniformity

Let $X_1, X_2, \dots, X_n, \dots$ be a sequence of independent and identically distributed (i.i.d.) copies of a random variable X with unknown distribution P over (the Borel sets of) the unit interval $[0, 1]$, and let P_0 denote the uniform distribution on $[0, 1]$. In what follows, L^2 is the space of measurable functions defined on the unit interval $[0, 1]$ that are square integrable with respect to Lebesgue measure. For $f, g \in L^2$, let

$$\langle f, g \rangle := \int_0^1 f(t)g(t) dt, \quad |[f]|^2 := \langle f, f \rangle.$$

Furthermore, let

$$\mathcal{H} := \{h_1, h_2, \dots, h_k\} \tag{2.1}$$

be any subset of L^2 satisfying

$$\langle h_j, 1 \rangle = 0 \quad (j = 1, \dots, k) \quad (2.2)$$

and

$$\langle h_i, h_j \rangle = \delta_{i,j} \quad (i, j = 1, \dots, k) , \quad (2.3)$$

where $\delta_{i,j} = 1$ if $i = j$ and $\delta_{i,j} = 0$, otherwise. A restatement of (2.2) and (2.3) is that $\mathcal{H}$ constitutes an orthonormal subset of the so called *full tangent space* $H(P_0) := \{h \in L^2: \langle h, 1 \rangle = 0\}$ of P_0 . For testing the hypothesis

$$H_0: P = P_0$$

of uniformity, *Neyman's smooth test of order k* (with respect to $\mathcal{H}$) rejects H_0 for large values of

$$\Psi_{n,k}^2 := \sum_{i=1}^k U_{n,i}^2 , \quad (2.4)$$

where

$$U_{n,i} := \frac{1}{\sqrt{n}} \sum_{j=1}^n h_i(X_j) \quad (i = 1, \dots, k) . \quad (2.5)$$

$U_{n,1}, U_{n,2}, \dots, U_{n,k}$ are called *components* of $\Psi_{n,k}^2$. A common choice for the orthonormal set $\mathcal{H}$ figuring in (2.1) is to use the first k (nonconstant) normalized Legendre polynomials. The first four such polynomials are

$$\begin{aligned} h_1(t) &= \sqrt{3}(2t - 1) , & h_2(t) &= \sqrt{5}(6t^2 - 6t + 1) , \\ h_3(t) &= \sqrt{7}(20t^3 - 30t^2 + 12t - 1) , \\ h_4(t) &= 3(70t^4 - 140t^3 + 90t^2 - 20t + 1) . \end{aligned} \quad (2.6)$$

The test based on $\Psi_{n,k}^2$ may be regarded as a projection pursuit procedure in a local asymptotic setting. To this end, consider an i.i.d. triangular array $X_{n,1}, X_{n,2}, \dots, X_{n,n}$ and the testing problem

$$H_{0,n}: X_{n,1} \text{ has the uniform density } f_0 \text{ on } [0, 1]$$

versus

$$H_{1,n}: X_{n,1} \text{ has the density } f_0 + n^{-1/2}h, \quad (2.7)$$

where $h \in H(P_0)$. Expanding the loglikelihood ratio, it is seen that the (optimal) Neyman-Pearson test of $H_{0,n}$ vs. $H_{1,n}$ is asymptotically equivalent to the test which rejects $H_{0,n}$ for large values of the *central sequence*

$$L_n(h) := \frac{1}{\sqrt{n}} \sum_{j=1}^n h(X_{n,j}).$$

Now, suppose that $h \in \mathcal{H}$, i.e., we have $h = \sum_{i=1}^k c_i h_i$ for some vector $c = (c_1, \dots, c_k) \in \mathbb{R}^k$. As seen above, the asymptotically optimal test rejects $H_{0,n}$ for large values of

$$\begin{aligned} L_n\left(\sum_{i=1}^k c_i h_i\right) &= \sum_{i=1}^k c_i \cdot \frac{1}{\sqrt{n}} \sum_{j=1}^n h_i(X_{n,j}) \\ &= \sum_{i=1}^k c_i \cdot U_{n,i}, \end{aligned}$$

where $U_{n,i}$ (with X_j replaced by $X_{n,j}$) is given in (2.5). Let $||[h]||^2 = \|c\|^2 = 1$ (a normalizing condition on the “size” of h), where $\|\cdot\|$ denotes the Euclidean norm. In order to construct a “reasonable” test for $H_{0,n}$ against $H_{1,n}$ with **unknown h subject to $h \in \mathcal{H}$ and $||[h]||^2 = 1$** , it suggests itself to consider the “worst case for $L_n(\sum_{i=1}^k c_i h_i)$ ” and reject $H_{0,n}$ for large values of

$$\max_{c: \|c\|^2=1} \left(\sum_{i=1}^k c_i U_{n,i} \right)^2 = \Psi_{n,k}^2$$

which is nothing but Neyman’s smooth test, motivated in the spirit of projection pursuit.

An appealing property of the test based on $\Psi_{n,k}^2$ is the simplicity of its asymptotic power function, as $n \rightarrow \infty$, under the local asymptotic setting (2.7). Since, under (2.7) with $h \in H(P_0)$, the vector $(U_{n,1}, \dots, U_{n,k})$ converges in distribution to a k -variate normal distribution with expectation vector $(\langle h, h_1 \rangle, \dots, \langle h, h_k \rangle)$ and unit covariance matrix (this follows from the multivariate Lindeberg-Feller Central Limit Theorem), the asymptotic distribution of $\Psi_{n,k}^2$ is that of a noncentral χ^2 -variate $\chi^2(k, \delta^2(h))$ with k degrees of freedom and noncentrality

parameter

$$\delta^2(h) = \sum_{j=1}^k \langle h, h_j \rangle^2 .$$

Consequently, the asymptotic power function $g(h, \alpha)$ of an asymptotically level α test based on $\Psi_{n,k}^2$ is given by

$$g(h, \alpha) = P(\chi^2(k, \delta^2(h)) > \chi_{k; 1-\alpha}^2) ,$$

where $\chi_{k; 1-\alpha}^2$ is the $(1 - \alpha)$ -quantile of the χ_k^2 distribution. Since $\delta^2(h) = |[h]|^2$, if $h \in \mathcal{H}$, Neyman's smooth test has constant asymptotic power "over all directions" in $\mathcal{H}$. On the other hand, $g(h, \alpha) = \alpha$ if h belongs to the orthogonal complement of $\mathcal{H}$.

3 Diagnostic Limitations

To see the typical pitfall associated with an "interpretation" of a component $U_{n,i}$, we reprint below the values of $U_{n,k}^2$ and $\Psi_{n,k}^2$ for a data set ($n = 100$) of Mahalanobis (1934, p. 303), tested for standard normality (i.e., for uniformity after applying the probability integral transform) by Neyman (1937).

k	1	2	3	4
$U_{n,k}^2$	0.047	0.391	3.624	0.129
$\Psi_{n,k}^2$	0.047	0.438	4.062	4.191

Interpreting these results, Rayner and Best (1989, p. 46) write:

*In assessing these values, we claim that since U_r^2 is based on a polynomial of degree r , it reflects consistency of the data with the r^{th} moment of the null distribution. Although none of the $\Psi_{n,k}^2$ are significant, $U_{n,3}^2$ is sufficiently large that **a test to assess if the data were positively skewed is significant at the 5% level.***

There are two basic problems associated with the "interpretation of components". Firstly, components are examined after having observed the data. This

means that there would have been interest also in the second component (say) if this would have been “too large”. Since, under uniformity, $U_{n,1}, \dots, U_{n,4}$ are approximately independent and standard normal random variables, the p -value, under uniformity, associated with the event that “at least one of $U_{n,1}^2, \dots, U_{n,4}^2$ is at least 3.624”, is approximately

$$1 - (2 \cdot \Phi(\sqrt{3.624}) - 1)^4 = 0.209 \dots,$$

where Φ denotes the standard normal distribution function. Thus, the real significance, **under uniformity**, is much less than 5%.

The second basic problem is the starting point of this paper. Testing “whether the data are positively skewed” can only mean testing whether the underlying unknown distribution P has zero skewness or better, since interpretation is done by means of the “too large” observed value of $U_{n,3}^2$, whether the distributional parameter corresponding to $U_{n,3}^2$ is equal to zero.

To this end, let $\mathcal{Q}_1$ denote the set of all probability measures P on $[0, 1]$ such that $\int_{[0,1]} t^l P(dt) < \infty$. For $P \in \mathcal{Q}_3$, define

$$\begin{aligned} \mathfrak{J}(P) &:= \left(\int h_3 dP \right)^2 \\ &= 7 \cdot (20EX^3 - 30EX^2 + 12EX - 1)^2, \end{aligned}$$

where X has distribution P (recall h_3 from (2.6)). Note that

$$\mathfrak{J}(P_0) = 0$$

for the uniform distribution P_0 . Furthermore, we have

$$\mathfrak{J}(\hat{P}_n) = \left(\frac{1}{\sqrt{n}} U_{n,3} \right)^2,$$

where $\hat{P}_n$ denotes the empirical distribution of random variables $X_1, \dots, X_n$ which are independent and identically distributed according to P . Since, by the strong law of large numbers,

$$\frac{1}{n} U_{n,3}^2 \xrightarrow{n \rightarrow \infty} \mathfrak{J}(P) \quad \text{almost surely,}$$

we see that $n^{-1} U_{n,3}^2$ is a strongly consistent estimator of $\mathfrak{J}(P)$.

Let us now take a look at the real meaning of “ $U_{n,3}^2$ is sufficiently large ... at the 5% level”. Note that, **for the uniform distribution P_0** , we have $\int h_3 dP_0 = 0$ and $\int h_3^2 dP_0 = 1$ and thus

$$U_{n,3} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1) , \quad \text{as } n \rightarrow \infty ,$$

where “ $\xrightarrow{\mathcal{D}}$ ” denotes convergence in distribution and, generally, $\mathcal{N}(\mu, \sigma^2)$ is the normal distribution with expectation μ and variance σ^2 . The fact that the observed value of $U_{n,3}^2$ is 3.624 does only mean that, **under the assumption of a uniform distribution**, this value is near the 95% quantile (= 3.84) of the χ_1^2 distribution so that this value of $U_{n,3}^2$ **casts doubt on the assumption of uniformity** (and nothing more!).

Going a step further and claiming that the observed “large” value of $U_{n,3}^2$ gives the additional diagnosis (at the 5% level) that the unknown underlying distribution P has the property $\mathfrak{J}(P) \neq 0$ is not possible for the following reason.

Let $P \in \mathcal{Q}_6$ so that $EX^6 < \infty$. From the Central Limit Theorem, we have

$$\sqrt{n} \left(\int h_3 d\hat{P}_n - \int h_3 dP \right) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2(P)) ,$$

where

$$\sigma^2(P) = \int h_3^2 dP - \left(\int h_3 dP \right)^2 .$$

In particular, it follows that

$$U_{n,3} \xrightarrow{\mathcal{D}} \mathcal{N} \left(0, \int h_3^2 dP \right) \quad (3.1)$$

if $\int h_3 dP = 0$ or, equivalently, if $\mathfrak{J}(P) = 0$. Thus, $P \in \mathcal{Q}_6$ and $\mathfrak{J}(P) = 0$ imply

$$U_{n,3}^2 \xrightarrow{\mathcal{D}} \int h_3^2 dP \cdot \chi^2(1) , \quad (3.2)$$

where $\chi^2(1)$ is a random variable having the χ_1^2 distribution.

As a consequence, a “large value” of $U_{n,3}^2$ may be due to a “large value” of $\int h_3^2 dP = \text{Var}(h_3^2(X))$, although $\mathfrak{J}(P) = 0$. Since this reasoning, mutatis mutandis, is completely analogous for any component and any orthonormal system $\{h_1, \dots, h_k\}$, we conclude that any component of any smooth test of fit, at least the way it is commonly used, is strictly non-diagnostic.

4 A Remedy

There is a simple way out of the dilemma described above. Let

$$U_{n,i} = \frac{1}{\sqrt{n}} \sum_{j=1}^n h_i(X_j)$$

be a component of a smooth test for uniformity (recall that $\int h_i dP_0 = 0$ and $\int h_i^2 dP_0 = 1$). If the underlying unknown distribution P satisfies

$$\int h_i dP = 0 \tag{4.1}$$

and

$$0 < \int h_i^2 dP < \infty, \tag{4.2}$$

we have (cf. (3.2))

$$U_{n,i}^2 \xrightarrow[n \rightarrow \infty]{\mathcal{D}} \int h_i^2 dP \cdot \chi^2(1)$$

and thus

$$\frac{U_{n,i}}{\frac{1}{n} \sum_{j=1}^n h_i^2(X_j)} \xrightarrow[n \rightarrow \infty]{\mathcal{D}} \chi^2(1). \tag{4.3}$$

Consequently, rejecting the hypothesis (4.1) if the left hand side of (4.3) exceeds $\chi_{1; 1-\alpha}^2$ yields an asymptotically level α test of (4.1) within the class of distributions P satisfying (4.2).

Thus, $U_{n,i}^2$ is able to give a “directed diagnosis” concerning $\int h_i dP$ when properly adjusted by the factor $(n^{-1} \sum_{j=1}^n h_i^2(X_j))^{-1}$.

References

- Baringhaus L, Henze N (1992) Limit distributions for Mardia’s measure of multivariate skewness. *Ann Statist* 20: 1889–1902
- Bera A, John S (1983) Tests for multivariate normality with Pearson alternatives. *Commun Statist A – Theor Meth* 12: 103–117

- D'Agostino RB, Stephens MA (eds) (1986) Goodness-of fit-techniques. Marcel Dekker, New York
- Henze N (1994) Tests auf Normalverteilung (in German, English summary). *Allg Statist Archiv* 78:293–317
- Horswell RL, Looney SW (1992) A comparison of tests for multivariate normality that are based on measures of multivariate skewness and kurtosis. *Journ Statist Comp Simul* 42:21–38
- Horswell RL, Looney SW (1993) Diagnostic limitations of skewness coefficients in assessing departures from univariate and multivariate normality. *Commun Statist B – Simul Comp* 22:437–459
- Kozioł JA (1987) An alternative formulation of Neyman's smooth goodness of fit tests under composite alternatives. *Metrika* 34:17–24
- Mahalanobis PC (1934) Tables of random samples from a normal population. *Sankhya* 1:289–328
- Mardia KV (1970) Measures of multivariate skewness and kurtosis with applications. *Biometrika* 57:519–530
- Neyman J (1937) Smooth test for goodness of fit. *Skandinavisk Aktuarietidskrift* 20:150–199
- Pearson K (1900) On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can reasonably be supposed to have arisen from random sampling. *Philos Mag* 5th ser. 50:157–175
- Pearson ES, Hartley HO (1970) *Biometrika tables for statisticians*, Vol. 1 (3rd ed.). New York, Cambridge University Press
- Rayner JCW, Best DJ (1986) Neyman-type smooth tests for location-scale families. *Biometrika* 73:437–466
- Rayner JCW, Best DJ (1989) Smooth tests of goodness of fit. New York, Oxford University Press
- Rayner JCW, Best DJ, Mathews KL (1995) Interpreting the skewness coefficient. *Commun Statist A – Theor Meth* 24:593–600

Received: 13.07.1995

Tests of Significance for the Average Square Deviation from Target in a Finite Lot

RAINER GÖB

Lehrstuhl für Statistik, Sanderring 2, 97070 Würzburg, Germany

Abstract: Consider lots of discrete items $1, 2, \dots, N$ with quality characteristics $x_1, x_2, \dots, x_N$. Let a be a target value for item quality. Lot quality is identified with the average square deviation $z = \frac{1}{N} \sum_{i=1}^N (x_i - a)^2$ from target per item in the lot (lot average square deviation from target). Under economic considerations this is an appropriate lot quality indicator if the loss respectively the profit incurred from an item is a quadratic function of $x_i - a$. The present paper investigates tests of significance on the lot average square deviation z under the following assumptions: The lot is a subsequence of a process of production, storage, transport; the random quality characteristics of items resulting from this process are i.i.d. with normal distribution $N(\mu, \sigma^2)$; the target value a coincides with the process mean μ .

1 A Model of Statistical Lot Inspection

Consider discrete items $1, 2, 3, \dots$ of a certain type of product. The quality of each item i is supposed to depend on a continuous scale measure value x_i , the *quality indicator* of item i . In course of manufacture, storage, shipping the items are exposed to various random influences. Thus we consider the quality indicators $x_1, x_2, x_3, \dots$ as realizations of independent identically distributed random variables $\xi_1, \xi_2, \xi_3, \dots$. For a model of broad application we assume that ξ_i has normal distribution $N(\mu; \sigma^2)$, i.e. that the distribution function of ξ_i is a member of the family $(F_{\mu, \sigma})_{\substack{\mu \in \mathbb{R} \\ \sigma \in (0; +\infty)}}$ defined by the densities

$$f_{\mu, \sigma}(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right),$$

$$F_{\mu, \sigma}(x) = \int_{-\infty}^x f_{\mu, \sigma}(s) ds.$$

Consider the above assumptions in the framework of *process quality control*. An important object of interest of this discipline is a process (mostly a manufacturing process) of items $1, 2, 3, \dots$ with quality indicators $\xi_1, \xi_2, \xi_3, \dots$. In the

above model, process quality is identified with the *process mean* $\mu = E[\xi_i]$ and/or the *process variance* $\sigma^2 = V[\xi_i]$, or a suitable function thereof. The task of statistical process inspection is to evaluate random samples from the process for estimations or tests of significance on the process quality parameters μ and σ^2 .

In the sequel we shall consider a problem of *product quality control*. Product quality control is concerned with the control of finished product, mostly in the form of discrete items assembled in finite lots. In this situation product quality control coincides with the control of lot quality. From the point of view of the above model we can consider a finite lot $1, \dots, N$ of items with quality indicators $\xi_1, \dots, \xi_N$ as a subsequence of an underlying process of manufacture, storage, shipping. Lot quality is identified with a function $T(\xi_1, \dots, \xi_N)$ of the item quality indicators in the lot. A list of important *lot quality indicator functions* $T(\xi_1, \dots, \xi_N)$ is given in Göb (1996). The following two examples of broad application are considered in detail.

(LQ1): *Lot Proportion Nonconforming*. Let $C \subset \mathbb{R}$ be a specification range and consider an item i to be

$$\begin{cases} \text{conforming,} & \text{if } \xi_i \in C, \\ \text{nonconforming,} & \text{if } \xi_i \in \mathbb{R} \setminus C. \end{cases}$$

Then lot proportion nonconforming is given by

$$T_L(\xi_1, \dots, \xi_N) = \frac{1}{N} |\{i | 1 \leq i \leq N, \xi_i \in \mathbb{R} \setminus C\}|. \quad (1)$$

(LQ2): *Lot Mean*. The lot mean (average measure value per item in the lot) is

$$T_M(\xi_1, \dots, \xi_N) = \frac{1}{N} \sum_{i=1}^N \xi_i. \quad (2)$$

The task of statistical lot inspection is to evaluate random samples from the lot for estimations or tests of significance on the value

$$z = T(\xi_1, \dots, \xi_N)$$

of the lot quality indicator. Like the process quality parameters μ and σ^2 the lot quality indicator z can be considered as a distribution parameter by considering

the joint conditional distribution function

$$F_{\mu, \sigma|z}^{(N)}(x_1, \dots, x_N) = W_{\mu, \sigma|z}^{(N)}(\xi_1 \leq x_1, \dots, \xi_N \leq x_N | T(\xi_1, \dots, \xi_N) = z) \quad (3)$$

of the lot variables $\xi_1, \dots, \xi_N$ under the condition $z = T(\xi_1, \dots, \xi_N)$. In this sense a hypothesis on z is a hypothesis on the parameter z of the family $(F_{\mu, \sigma|z}^{(N)})_{\substack{\mu \in \mathbb{R} \\ \sigma \in (0; +\infty) \\ z \in I}}$ of distribution functions where I is a suitably chosen subset of $\mathbb{R}$,

for instance the minimum set such that $W_{\mu, \sigma|z}^{(N)}(T(\xi_1, \dots, \xi_N) \in I) = 1$.

From a methodological point of view the status of the process parameters μ and σ^2 and the lot parameter z is essentially different. Direct observation of the process mean μ or the process variance σ^2 is impossible. Hypotheses on μ and σ^2 can only be examined by statistical methods. The lot quality indicator, however, satisfies the functional relation $T(\xi_1, \dots, \xi_N) = z$. Thus, complete knowledge of the realizations $x_1, \dots, x_N$ of the lot variables $\xi_1, \dots, \xi_N$ entails complete knowledge of the lot parameter z . If the observations $\xi_1, \dots, \xi_N$ are reliable any hypothesis on z can be verified in a strict sense. On shopfloor, however, 100 percent inspection of the lot mostly is impossible because of economic, organizational and technical reasons. In these cases information on the lot is restricted to a random sample $(\xi_{i_1}, \dots, \xi_{i_n})$ of size $n < N$ and we have to use statistical methods (estimation or testing) for investigations on the lot quality indicator z . For process inspection, application of statistics is a methodological necessity, whereas in lot inspection application of statistics results from practical restrictions.

2 A New Lot Quality Indicator

Lot proportion nonconforming (1) and lot mean (2) are two standard lot quality indicators which cover a wide range of applications. Lot proportion nonconforming is the lot quality indicator of customary acceptance sampling by attributes as described in standards and textbooks of statistical quality control. As shown in Göb (1996) the lot mean is the appropriate lot quality indicator for a reasonable interpretation of sampling by variables (sampling by measurement).

Lot proportion nonconforming and lot mean stand for two competitive and mutually exclusive lot quality models. In model (LQ1) each item is rated separately as conforming or nonconforming. The original measure values ξ_i are absorbed in a discrete binary scheme. Large and small deviations from specifications cannot be distinguished. In model (LQ2) the lot quality indicator (2) is a continuous strictly increasing function of each ξ_i . The original continuous scale information does not get lost. (2) does not treat the items as completely isolated units, small values ξ_i can be compensated by large values ξ_j .

Some applications, however, require a concept of lot quality settled between the extremes (LQ1) and (LQ2). Consider the following example.

2.1 Example: Packages of a composite plant-protective agent are shipped to a wine-growers' cooperative. For protection of one acre of cultivated area one package of agent is dissolved in 20 liters of water, the resulting fluid is spread out over the area. For proper application a package is supposed to contain 100 cm^3 of agent. Deviations from the target value 100 cm^3 in both directions continuously diminish the grape output of the wine yard. If the package contains less than 100 cm^3 the plants are attacked by noxious parasites and pests. If the package contains more than 100 cm^3 the vines and the ecological balance of the wine yard are damaged by an excess of chemical substances. ■

Neither of the lot quality indicators (1) and (2) fits to the requirements of Example (2.1). As in (1) the deviations of the item quality indicators from target should have cumulative effect on lot quality, but the deviation measure should be a continuous and not a step function as (1). On the other hand the continuous lot quality indicator (2) obviously is inadequate since mutual compensation of values above and below target should not be admitted.

Presumably, examples like (2.1) form the intuitive background of *variables sampling for lot proportion nonconforming*. This approach reflects the need for a concept of lot quality which combines the aspect of specifications with the aspect of continuous scale measuring. The solution suggested by variables sampling for lot proportion nonconforming, however, is a strange hodgepodge of models (LQ1) and (LQ2) in an approach without a sound model, under vague and often unexpressed assumptions. This approach leads to absurdities, see *von Collani* (1990), *von Collani* (1991), *Hryniewicz* (1991), and it contradicts essential principles of statistical methodology, see *Göb* (1996).

Let us now try to establish a sound model to cope with the problems provoked by Example (2.1). An appropriate lot quality indicator should be a continuous strictly increasing function of each's package deviation from the target value $a = 100$. This requirement is met by the *average square deviation from target a per item in the lot (lot average square deviation)*

$$T_a(\xi_1, \dots, \xi_N) = \frac{1}{N} \sum_{i=1}^N (\xi_i - a)^2. \quad (4)$$

Clearly the interest is to have lots with $z = T_a(\xi_1, \dots, \xi_N)$ as small as possible. From an economic point of view (4) makes sense in the following situation. For each item i the cost incurred from the deviation of ξ_i from target a is a quadratic function $L \cdot (\xi_i - a)^2$ of the deviation $\xi_i - a$, where $L > 0$. The income from a lot

of size N is I_N . Then the profit incurred from a lot of size N is

$$I_N - \sum_{i=1}^N L \cdot (\xi_i - a)^2 = I_N - N \cdot L \cdot T_a(\xi_1, \dots, \xi_N) .$$

The study of statistical methods under this type of economic functions has been propagated by Taguchi and his scholars, see *Taguchi/Elsayed/Hsiang* (1989), for instance.

In the early days of modern mass production quality control often was reduced to quality control of finished product and statistical quality control was identified with statistical lot inspection. Nowadays main emphasis is on the control of processes which lead to finished product in the customer's incoming material department: above all the control of manufacturing processes but also the control of storage and shipping. In a modern quality control system satisfactory average product quality is achieved by process control under suitable application of statistics. However, even if stable expected quality is guaranteed item quality still varies at random around the mean. Thus lot inspection in product control is still advisable so as to prevent unwelcome large quality variation of items in a lot.

In our model we can reflect this relation among process control and product control in the following way. Expected item quality is determined by the process mean $\mu = E[\xi_i]$ of the underlying normal distribution. Process control and its statistical tools keep the process mean on target. Thus for lot inspection in product control the process mean μ can be assumed to be known and fixed. Under reasonable circumstances we then have to assume that this known and fixed item quality mean coincides with the intended target value a . Inserting $a = \mu$ in (4) we finally obtain the lot quality indicator

$$T_0(\xi_1, \dots, \xi_N) = T_\mu(\xi_1, \dots, \xi_N) = \frac{1}{N} \sum_{i=1}^N (\xi_i - \mu)^2 . \quad (5)$$

As well-known, (5) is sufficient for the process variance σ^2 . Hence the conditional distribution function (3) depends on the value of the lot quality indicator z and the process mean (target value) μ and not on σ . Thus we can rewrite (3) as

$$F_{\mu|z}^{(N)}(x_1, \dots, x_N) = W_{\mu|z}^{(N)}(\xi_1 \leq x_1, \dots, \xi_N \leq x_N | T_0(\xi_1, \dots, \xi_N) = z) . \quad (6)$$

Considering z as a parameter of the family $(F_{\mu|z}^{(N)})_{\substack{\mu \in \mathbb{R} \\ z \in (0; +\infty)}}$ of distribution functions we shall investigate tests of significance on z based on random samples $(\xi_1, \dots, \xi_n)$ from the lot. In detail we shall consider the problems of testing the

following one-sided and two-sided hypotheses H_i against alternatives K_i :

$$H_1: z \leq z_0, \quad K_1: z > z_0, \quad (7)$$

$$H_2: z \geq z_0, \quad K_2: z < z_0, \quad (8)$$

$$H_3: z \leq z_1 \text{ or } z \geq z_2, \quad K_3: z_1 < z < z_2, \quad (9)$$

$$H_4: z_1 \leq z \leq z_2, \quad K_4: z < z_1 \text{ or } z > z_2, \quad (10)$$

$$H_5: z = z_0, \quad K_5: z \neq z_0. \quad (11)$$

Clearly industrial quality control is mainly interested in the schemes (7) and (8) where z_0 is interpreted as a break even value for lot quality. In special cases, however, schemes (9), (10) and (11) are also interesting. For instance, assume that in ordinary course of business a consumer reckons upon lots of normal satisfactory quality with $z_1 \leq T_\mu(\xi_1, \dots, \xi_N) \leq z_2$. In such a case she will use scheme (10) to sort out lots of intolerable quality level with $T_\mu(\xi_1, \dots, \xi_N) > z_2$ as well as lots of extremely good quality level $T_\mu(\xi_1, \dots, \xi_N) < z_1$ which may be used for special purposes.

3 The Lot Sampling Distribution

As shown in Göb (1996), we may always consider the first n lot variables $\xi_1, \dots, \xi_n$ as a random sample without replacement from $\xi_1, \dots, \xi_N$. Hence the *lot sampling distribution* is the joint distribution of $\xi_1, \dots, \xi_n$ under the

condition $z = \frac{1}{N} \sum_{i=1}^N (\xi_i - \mu)^2$.

For calculating the joint conditional density $f_{\mu|z}^{(N,n)}$ of the sample variables $\xi_1, \dots, \xi_n$ we shall need χ^2 -distribution. The density of χ^2 -distribution with k degrees of freedom is

$$\chi_k(x) = \begin{cases} 0 & \text{for } x \in (-\infty; 0] , \\ \frac{1}{2^{k/2} \Gamma(k/2)} x^{(k/2)-1} \exp(-x/2) & \text{for } x \in (0; +\infty) . \end{cases} \quad (12)$$

Let $h: \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ be the joint unconditional density of $\xi_1, \dots, \xi_n$ and $T_0(\xi_1, \dots, \xi_N)$, and let $h_0: \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ be the joint unconditional density of

$\xi_1, \dots, \xi_n$ and $\sum_{i=n+1}^N (\xi_i - \mu)^2$. Evidently

$$h(x_1, \dots, x_n, z) = N \cdot h_0 \left(x_1, \dots, x_n, Nz - \sum_{i=1}^n (x_i - \mu)^2 \right). \quad (13)$$

$\xi_1, \dots, \xi_n$ and $\sum_{i=n+1}^N (\xi_i - \mu)^2$ are independent and, as well-known, $\frac{1}{\sigma^2} \sum_{i=n+1}^N (\xi_i - \mu)^2$ has χ^2 -distribution with $N - n$ degrees of freedom. Hence

$$h_0(x_1, \dots, x_n, u) = \prod_{i=1}^n f_{\mu, \sigma}(x_i) \cdot \frac{1}{\sigma^2} \chi_{N-n} \left(\frac{u}{\sigma^2} \right). \quad (14)$$

Combining (13) and (14) we obtain

$$h(x_1, \dots, x_n, z) = \frac{N}{\sigma^2} \cdot \prod_{i=1}^n f_{\mu, \sigma}(x_i) \cdot \chi_{N-n} \left[\frac{Nz - \sum_{i=1}^n (x_i - \mu)^2}{\sigma^2} \right]. \quad (15)$$

Since $\frac{1}{\sigma^2} \sum_{i=1}^N (\xi_i - \mu)^2$ has χ^2 -distribution with N degrees of freedom the unconditional density $h_{T_0}: \mathbb{R} \rightarrow \mathbb{R}$ of $T_0(\xi_1, \dots, \xi_N)$ satisfies

$$h_{T_0}(z) = \frac{N}{\sigma^2} \cdot \chi_N \left(\frac{Nz}{\sigma^2} \right). \quad (16)$$

Hence the conditional density $f_{\mu|z}^{(N,n)}$ of the sample vector $\xi = (\xi_1, \dots, \xi_n)$ under

$$T_0(\xi_1, \dots, \xi_N) = \frac{1}{N} \sum_{i=1}^N (\xi_i - \mu)^2 = z \text{ is}$$

$$f_{\mu|z}^{(N,n)}(\mathbf{x}) = f_{\mu|z}^{(N,n)}(x_1, \dots, x_n) = \frac{h(x_1, \dots, x_n, z)}{h_{T_0}(z)} =_{(15,16)} \begin{cases} \frac{\Gamma\left(\frac{N}{2}\right)}{\Gamma\left(\frac{N-n}{2}\right)} \left[1 - \frac{\sum_{i=1}^n (x_i - \mu)^2}{Nz} \right]^{(N-n)/2-1} \frac{1}{(Nz\pi)^{n/2}}, & \text{if } 0 \leq \sum_{i=1}^n (x_i - \mu)^2 < Nz, \\ 0, & \text{otherwise.} \end{cases} \quad (17)$$

Evidently

$$\tau = t(\xi) = \sum_{i=1}^n (\xi_i - \mu)^2 \quad (18)$$

is sufficient for z in the family $(f_{\mu|z}^{(N,n)})_{z>0}$. The conditional density $g_z^{(N,n)}$ of τ is determined through considerations analogous to the ones which lead to (17).

Observe that $\frac{1}{\sigma^2} \tau = \frac{1}{\sigma^2} \sum_{i=1}^n (\xi_i - \mu)^2$ has χ^2 -distribution with n degrees of freedom. We obtain

$$g_z^{(N,n)}(s) = \frac{\frac{1}{\sigma^2} \cdot \chi_n \left(\frac{s}{\sigma^2} \right) \cdot \frac{N}{\sigma^2} \cdot \chi_{N-n} \left(\frac{Nz - s}{\sigma^2} \right)}{h_{T_0}(z)} =_{(12), (16)} \left\{ \begin{array}{ll} \frac{\Gamma\left(\frac{N}{2}\right)}{\Gamma\left(\frac{n}{2}\right)\Gamma\left(\frac{N-n}{2}\right)} \left(\frac{s}{Nz}\right)^{n/2-1} \left(1 - \frac{s}{Nz}\right)^{(N-n)/2-1} \frac{1}{Nz}, & \text{if } 0 \leq s < Nz, \\ 0 & \text{otherwise.} \end{array} \right. \quad (19)$$

The distribution of τ can be expressed by beta-distribution. For $a, b > 0$ the density of beta-distribution of order (a, b) is

$$b_{a,b}(x) = \left\{ \begin{array}{ll} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}, & \text{if } 0 < x < 1, \\ 0 & \text{otherwise,} \end{array} \right. \quad (20)$$

with corresponding distribution function

$$B_{a,b}(y) = \int_0^y b_{a,b}(x) dx, \quad (21)$$

see Rényi (1970), page 205. Comparing (19) and (20) we see that

$$g_z^{(N,n)}(s) = \frac{1}{Nz} \cdot b_{n/2, (N-n)/2} \left(\frac{s}{Nz} \right). \quad (22)$$

Thus the conditional distribution of $\frac{1}{Nz}\tau$ is a beta-distribution of order $\left(\frac{n}{2}, \frac{N-n}{2}\right)$ and the conditional distribution function $G_z^{(N,n)}$ of τ satisfies

$$G_z^{(N,n)}(s) = B_{n/2, (N-n)/2} \left(\frac{s}{Nz} \right). \quad (23)$$

Using the moments of beta-distribution – see *Patel/Kapadia/Owen* (1976) – we can calculate expectation and variance of τ :

$$E[\tau] = nz, \quad V[\tau] = 2z^2 \frac{n(N-n)}{N+2}.$$

Hence the average square deviation

$$\frac{1}{n}\tau = \frac{1}{n}t(\xi) = \frac{1}{n} \sum_{i=1}^n (\xi_i - \mu)^2$$

from target μ per item in the sample is an unbiased sufficient estimator of z .

For investigations on tests for parameter z we can clearly restrict attention to tests φ which depend on the sample vector ξ only through the sufficient statistic $\tau = t(\xi)$, i.e. $\varphi(\xi) = \psi(t(\xi))$. Thus we can simplify notation in the proofs of the subsequent paragraphs by considering tests $\psi: \mathbb{R} \rightarrow [0; 1]$ for H_i against K_i under the family of distributions $(g_z^{(N,n)})_{z>0}$.

4 One-Sided Hypotheses

To establish uniformly most powerful (UMP) tests for the one-sided hypotheses H_1 and H_2 we shall consider the likelihood ratio for densities $g_{z_1}^{(N,n)}, g_{z_2}^{(N,n)}$.

To determine the UMP test for the scheme (7)

$$H_1: z \leq z_0 \quad \text{against} \quad K_1: z > z_0$$

let us first consider parameter values $0 < z_1 < z_2$.

For $s \leq 0$ and for $s \geq Nz_2$ we obtain from (19) $g_{z_1}^{(N,n)}(s) = 0 = g_{z_2}^{(N,n)}(s)$. Hence the likelihood ratio can be chosen arbitrarily on $(-\infty; 0]$ and on $[Nz_2; +\infty)$.

For $Nz_1 \leq s < Nz_2$ we obtain from (19) $g_{z_1}^{(N,n)}(s) = 0 < g_{z_2}^{(N,n)}(s)$. Hence the likelihood ratio is

$$L_{z_1, z_2}(s) = +\infty \quad \text{for } Nz_1 \leq s < Nz_2 .$$

For $0 < s < Nz_1$ we have $g_{z_1}^{(N,n)}(s) > 0, g_{z_2}^{(N,n)}(s) > 0$, hence

$$L_{z_1, z_2}(s) = \frac{g_{z_2}^{(N,n)}(s)}{g_{z_1}^{(N,n)}(s)} =_{(19)} \left(\frac{z_1}{z_2} \right)^{N/2-1} \left(\frac{Nz_2 - s}{Nz_1 - s} \right)^{(N-n)/2-1} .$$

Thus we can use the following version of the likelihood ratio:

$$L_{z_1, z_2}(s) = \begin{cases} 0 , & \text{if } s \leq 0 , \\ \left(\frac{z_1}{z_2} \right)^{N/2-1} \left(\frac{Nz_2 - s}{Nz_1 - s} \right)^{(N-n)/2-1} , & \text{if } 0 < s < Nz_1 , \\ +\infty , & \text{if } s \geq Nz_1 . \end{cases} \quad (24)$$

Evidently the likelihood ratio is increasing in s . Hence the UMP test for the scheme (7) can be determined by applying the well-known theorem on densities with monotone likelihood ratio, see *Lehmann* (1959), pp. 68–69:

The test

$$\text{Reject iff } \tau \geq c \quad (25)$$

with critical value c determined by

$$\alpha \stackrel{!}{=} 1 - G_{z_0}^{(N,n)}(c) \quad (26)$$

is UMP at level α for $H_1: z \leq z_0$ against $K_1: z > z_0$.

Obviously the solution for the dual scheme (8)

$$H_2: z \geq z_0 \quad \text{against} \quad K_2: z < z_0$$

is obtained in an analogous manner. The intuitively reasonable result is:

The test

$$\text{Reject iff } \tau \leq c \quad (27)$$

with critical value c determined by

$$\alpha \stackrel{!}{=} G_{z_0}^{(N,n)}(c) \quad (28)$$

is UMP at level α for $H_2: z \geq z_0$ against $K_2: z < z_0$.

5 The Two-Sided Hypothesis $z \leq z_1$ or $z \geq z_2$

In this paragraph we shall establish a test for the two-sided scheme (9)

$$H_3: z \leq z_1 \text{ or } z \geq z_2 \quad \text{against} \quad K_3: z_1 < z < z_2 .$$

From an intuitive point of view the solution is clear: An appropriate test for H_3 against K_3 should reject if the total square deviation $\tau = t(\xi)$ from target μ in the sample is neither small nor large enough. This intuitive conjecture is confirmed by the following theorem.

5.1 Theorem: Let $0 < \alpha < 1, 0 < z_1 < z_2, n \leq N - 5$.

(a) There is a unique pair $0 < c_1 < c_2$ satisfying

$$G_{z_1}^{(N,n)}(c_2) - G_{z_1}^{(N,n)}(c_1) = \alpha = G_{z_2}^{(N,n)}(c_2) - G_{z_2}^{(N,n)}(c_1) . \quad (29)$$

(b) Let the test $\psi^*: \mathbb{R} \rightarrow [0; 1]$ be defined by

$$\psi^*(s) = \begin{cases} 1 , & \text{if } c_1 \leq s \leq c_2 , \\ 0 , & \text{if } s < c_1 \text{ or } s > c_2 , \end{cases} \quad (30)$$

i.e. ψ^* represents the decision rule

$$\text{Reject iff } c_1 \leq \tau \leq c_2 .$$

The test $\psi^\star$ is UMP at level α for testing $H_3: z \leq z_1$ or $z \geq z_2$ against $K_3: z_1 < z < z_2$.

(c) For any test $\psi: \mathbb{R} \rightarrow [0; 1]$ with $E_{z_1}[\psi(\tau)] = \alpha = E_{z_2}[\psi(\tau)]$ we have

$$E_z[\psi^\star(\tau)] \leq E_z[\psi(\tau)] \quad \text{for } z \in (0; z_1) \cup (z_2; +\infty) .$$

(d) Let

$$\beta^\star(z) = E_z[\psi^\star(\tau)] = G_z^{(N,n)}(c_2) - G_z^{(N,n)}(c_1) \quad \text{for } z > 0$$

be the power function of $\psi^\star$ and let

$$z^\star = \frac{1}{N} \frac{c_2 - c_1 \left(\frac{c_1}{c_2} \right)^{n/(N-n-2)}}{1 - \left(\frac{c_1}{c_2} \right)^{n/(N-n-2)}} . \quad (31)$$

Then $\max\{z_1, c_2/N\} < z^\star < z_2$ and

$$\beta^\star(z) = 0 \quad \text{for } 0 < z \leq \frac{c_1}{N} ,$$

$$\beta^\star \text{ is strictly } \begin{cases} \text{increasing on} & \left(\frac{c_1}{N}; z^\star \right) , \\ \text{decreasing on} & (z^\star; +\infty) , \end{cases}$$

$$\lim_{z \rightarrow +\infty} \beta^\star(z) = 0 . \quad \blacksquare$$

Although $(g_z^{(N,n)})_{z>0}$ is not an exponential family the proof of Theorem 5.1 can use some elements of the corresponding theorem on tests of two-sided hypotheses under exponential families, see *Lehmann*, pp. 88–90.

As a first step we investigate the uniqueness of values satisfying (29). A first proposition follows from the investigations on the likelihood ratio in the preceding Paragraph 4, see (24).

5.2 Proposition: Let $0 < z_1 < z_2$, $n \leq N - 3$. The densities $g_{z_1}^{(N,n)}$ and $g_{z_2}^{(N,n)}$ have a unique intersection point s_{z_1, z_2} with

$$g_{z_2}^{(N,n)}(s) \begin{cases} < g_{z_1}^{(N,n)}(s) & \text{for } s \in (0; s_{z_1, z_2}) , \\ > g_{z_1}^{(N,n)}(s) & \text{for } s \in (s_{z_1, z_2}; +\infty) . \end{cases} \quad \blacksquare$$

Using (5.2) we can establish an inequality among the distribution functions $G_{z_1}^{(N,n)}, G_{z_2}^{(N,n)}$.

5.3 Proposition: Let $0 < z_1 < z_2, n \leq N - 3$. We have

$$G_{z_1}^{(N,n)}(c) > G_{z_2}^{(N,n)}(c) \quad \text{for } 0 < c < Nz_2 .$$

Proof: Let s_{z_1, z_2} be the intersection point of $g_{z_1}^{(N,n)}$ and $g_{z_2}^{(N,n)}$ defined in Proposition (5.2). For $c \in (0; s_{z_1, z_2}]$ we have

$$G_{z_1}^{(N,n)}(c) - G_{z_2}^{(N,n)}(c) = \int_{0 \leq s \leq c} (g_{z_1}^{(N,n)}(s) - g_{z_2}^{(N,n)}(s)) ds > 0 .$$

For $c \in (s_{z_1, z_2}; Nz_2)$ we have

$$\begin{aligned} G_{z_1}^{(N,n)}(c) - G_{z_2}^{(N,n)}(c) &= \int_{0 \leq s \leq c} (g_{z_1}^{(N,n)}(s) - g_{z_2}^{(N,n)}(s)) ds - \underbrace{\int_{\mathbb{R}} (g_{z_1}^{(N,n)}(s) - g_{z_2}^{(N,n)}(s)) ds}_{=0} \\ &= - \int_{s > c} (g_{z_1}^{(N,n)}(s) - g_{z_2}^{(N,n)}(s)) ds > 0 . \quad \blacksquare \end{aligned}$$

The next proposition also follows from the investigations on the likelihood ratio in the preceding Paragraph 4.

5.4 Proposition: Let $0 < z_1 < z_2, n \leq N - 3, 0 < c_1 < d_1 < c_2 < d_2, c_2 < Nz_1$, and let $\psi: \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$\psi(s) = \begin{cases} 0 , & \text{for } s \in (-\infty; c_1) \cup (d_1; c_2) \cup (d_2; +\infty) , \\ -1 , & \text{for } s \in (c_1; d_1) , \\ 1 , & \text{for } s \in (c_2; d_2) . \end{cases} \quad (32)$$

Then we have for the conditional expectations: $E_{z_1}[\psi(\tau)] \neq 0$ or $E_{z_2}[\psi(\tau)] \neq 0$.

Proof: For $z \in (z_1; z_2)$ let the likelihood ratios L_{z, z_2}, L_{z, z_1} be defined as in (24), Paragraph 4.

For $z \in (z_1; z_2)$ we have

$$\begin{aligned}
 E_{z_2}[\psi(\tau)] &= \int_{s < d_2} \psi(s) L_{z, z_2}(s) g_z^{(N, n)}(s) ds \\
 &> L_{z, z_2}(c_2) \int_{s < c_2} \psi(s) g_z^{(N, n)}(s) ds + L_{z, z_2}(c) \int_{c_2 < s < d_2} \psi(s) g_z^{(N, n)}(s) ds \\
 &= L_{z, z_2}(c_2) E_z[\psi(\tau),]
 \end{aligned} \tag{33}$$

and analogously

$$E_{z_1}[\psi(\tau)] < L_{z, z_1}(c_2) E_z[\psi(\tau)] . \tag{34}$$

For $z \in (z_1; z_2)$ we have $L_{z, z_2}(c_2) > 0$, $L_{z, z_1}(c_2) > 0$, and thus the assumption

$$E_{z_1}[\psi(\tau)] = 0 = E_{z_2}[\psi(\tau)]$$

is obviously incompatible with (33) and (34). ■

Applying Proposition (5.4) we can now prove the uniqueness of values satisfying (29).

5.5 Proposition: Let $0 < \alpha < 1$, $0 < z_1 < z_2$, $n \leq N - 3$. Then there exists at most one pair $0 < c_1 < c_2$ with

$$G_{z_1}^{(N, n)}(c_2) - G_{z_1}^{(N, n)}(c_1) = \alpha = G_{z_2}^{(N, n)}(c_2) - G_{z_2}^{(N, n)}(c_1) .$$

Proof: Let $0 < c_1 < c_2$, $0 < d_1 < d_2$ satisfy

$$\int_{c_1 < s < c_2} g_{z_i}^{(N, n)}(s) ds = \alpha = \int_{d_1 < s < d_2} g_{z_i}^{(N, n)}(s) ds \quad \text{for } i = 1, 2 , \tag{35}$$

and let s_{z_1, z_2} be the intersection point of $g_{z_1}^{(N, n)}$ and $g_{z_2}^{(N, n)}$ defined in Proposition (5.2). Clearly

$$c_1, d_1 < s_{z_1, z_2} < c_2, d_2 .$$

Obviously $c_i = d_i$ implies $c_j = d_j$.

Thus we can restrict attention to the case $0 < c_1 < d_1 < c_2 < d_2$, $c_2 < Nz_1$ considered in Proposition (5.4). Let $\psi: \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying (32) as in Proposition (5.4). Then

$$0 =_{(35)} \int_{c_1 < s < c_2} g_{z_i}^{(N,n)}(s) ds - \int_{d_1 < s < d_2} g_{z_i}^{(N,n)}(s) ds = -E_{z_i}[\psi(\tau)] \quad \text{for } i = 1, 2$$

in contradiction to Proposition (5.4). Hence $c_i = d_i$. ■

The existence of values satisfying (29) will result as a by-product of the proof of assertion (c) of Theorem (5.1). To prepare the proof of the theorem we shall establish the following technical proposition.

5.6 Proposition: Let $0 < z_1 < z_2$, $n \leq N - 5$, $0 < c_1 < c_2$ with $c_1 < Nz_1$, $c_2 < Nz_2$, and let for $u_1, u_2 \in \mathbb{R}$, $z > 0$, the function $e_{u_1, u_2, z}: [0; Nz] \rightarrow \mathbb{R}$ be defined by

$$\begin{aligned} e_{u_1, u_2, z}(s) = & u_1 \cdot \left(\frac{\max\{Nz_1 - s, 0\}}{Nz - s} \right)^{(N-n)/2-1} \\ & + u_2 \cdot \left(\frac{\max\{Nz_2 - s, 0\}}{Nz - s} \right)^{(N-n)/2-1}. \end{aligned} \quad (36)$$

(a) Let $z \in (0; z_1)$, $c_1 < Nz$. Then there is $u_1 > 0$ such that

$$e_{u_1, 0, z}(s) \begin{cases} < 1, & \text{for } 0 < s < c_1, \\ > 1, & \text{for } c_1 < s < Nz. \end{cases}$$

(b) Let $z \in (z_1; z_2)$. Then there is $u_1 > 0$ such that

$$e_{u_1, 0, z}(s) \begin{cases} > 1, & \text{for } 0 < s < c_1, \\ < 1, & \text{for } c_1 < s < Nz. \end{cases}$$

(c) Let $z \in (z_1; z_2)$, $c_2 < Nz$. Then there are $u_1 > 0, u_2 > 0$ such that

$$e_{u_1, u_2, z}(s) \begin{cases} > 1, & \text{for } s \in (0; c_1) \cup (c_2; Nz), \\ < 1, & \text{for } s \in (c_1; c_2). \end{cases}$$

- (d) Let $z \in (z_2; +\infty)$. The function $e_{u_1, u_2, z} - 1$ has at most two zeros on $(0; Nz)$.
If two zeros $0 < s_1 < s_2 < Nz$ exist, then $s_2 < Nz_2$ and

$$e_{u_1, u_2, z}(s) \begin{cases} < 1, & \text{for } s \in (0; s_1) \cup (s_2; Nz) , \\ > 1, & \text{for } s \in (s_1; s_2) . \end{cases}$$

- (e) For $k_1, k_2 \in \mathbb{R}$, $z > 0$, let the function $E_{k_1, k_2, z}: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$E_{k_1, k_2, z}(s) = k_1 \cdot g_{z_1}^{(N, n)}(s) + k_2 \cdot g_{z_2}^{(N, n)}(s) - g_z^{(N, n)}(s) . \quad (37)$$

Then with $u_i = k_i \left(\frac{z}{z_i} \right)^{N/2-1}$ we have

$$E_{k_1, k_2, z}(s) = g_z^{(N, n)}(s) \cdot (e_{u_1, u_2, z}(s) - 1) \quad \text{for } s \in (0; Nz) .$$

Proof of (a): We have

$$e_{u_1, 0, z}(s) = u_1 \cdot \left(\frac{Nz_1 - s}{Nz - s} \right)^{(N-n)/2-1} \quad \text{for } 0 < s < Nz .$$

In case of $u_1 > 0$ $e_{u_1, 0, z}$ is strictly increasing on $(0; Nz)$. Hence the assertion of (a) is satisfied with

$$u_1 = \left(\frac{Nz - c_1}{Nz_1 - c_1} \right)^{(N-n)/2-1} > 0$$

since then $e_{u_1, 0, z}(c_1) = 1$.

Proof of (b): Analogously to the proof of (a).

Proof of (c): Consider $e_{u_1, u_2, z}$ for arbitrary $u_1 > 0, u_2 > 0$.

From definition (36) it is obvious that $e_{u_1, u_2, z}$ is strictly increasing on $(Nz_1; Nz)$.

For $s \in (0; Nz_1)$ elementary calculation yields

$$e'_{u_1, u_2, z}(s) = N \cdot \left(\frac{N-n}{2} - 1 \right) \cdot (z_2 - z) \cdot \frac{(Nz_1 - s)^{(N-n)/2-2}}{(Nz - s)^{(N-n)/2}} \cdot u_2 \cdot \tilde{e}_{u_1, u_2, z}(s) \quad (38)$$

where $\tilde{e}_{u_2, u_2, z}: [0; Nz_1) \rightarrow \mathbb{R}$ is defined by

$$\tilde{e}_{u_1, u_2, z}(s) = \left(\frac{Nz_2 - s}{Nz_1 - s} \right)^{(N-n)/2-2} - \frac{u_1 \cdot z - z_1}{u_2 \cdot z - z} . \quad (39)$$

$\tilde{e}_{u_1, u_2, z}$ is strictly increasing on $(0; Nz_1)$ with $\lim_{s \uparrow Nz_1} \tilde{e}_{u_1, u_2, z}(s) = +\infty$. Hence $e_{u_1, u_2, z}$ is either strictly increasing on $(0; Nz_1)$ or $e'_{u_1, u_2, z}$ has exactly one change of sign from < 0 to > 0 on $(0; Nz_1)$.

Thus the maximum number of zeros of $e_{u_1, u_2, z} - 1$ on $(0; Nz)$ is 2. If two zeros $0 < s_1 < s_2$ exist, s_1 is a change of sign from > 0 to < 0 and s_2 is a change of sign from < 0 to > 0 .

Let

$$u_1^* := \frac{((Nz_2 - c_2)(Nz - c_1))^{(N-n)/2-1} - ((Nz_2 - c_1)(Nz - c_2))^{(N-n)/2-1}}{((Nz_1 - c_1)(Nz_2 - c_2))^{(N-n)/2-1} - ((Nz_2 - c_1) \max\{Nz_1 - c_0, 0\})^{(N-n)/2-1}} . \quad (40)$$

$$u_2^* := \frac{((Nz_1 - c_1)(Nz - c_2))^{(N-n)/2-1} - ((\max\{Nz_1 - c_2, 0\})(Nz - c_1))^{(N-n)/2-1}}{((Nz_1 - c_1)(Nz_2 - c_2))^{(N-n)/2-1} - ((Nz_2 - c_1) \max\{Nz_1 - c_2, 0\})^{(N-n)/2-1}} . \quad (41)$$

To prove the positivity of u_1^* and u_2^* we use

$$(a_1 - b_1) \cdot (a_2 - b_2) - (a_1 - b_2) \cdot (a_2 - b_1) = (a_2 - a_1) \cdot (b_2 - b_1) . \quad (42)$$

Applying (42) with

$$a_1 = Nz , \quad b_1 = c_1 , \quad a_2 = Nz_2 , \quad b_2 = c_2$$

we see that the numerator in (40) is positive.

Consider the case $c_2 \geq Nz_1$. In this case the denominators in (40) and (41) and the numerator in (41) are obviously positive. Thus $u_2^* > 0$ and because of the above proved positivity of the numerator in (40) also $u_1^* > 0$.

Now consider the case $c_2 < Nz_1$. Applying (42) with

$$a_1 = Nz_1 , \quad b_1 = c_1 , \quad a_2 = Nz_2 , \quad b_2 = c_2$$

we see that the denominators in (40) and (41) are positive. Applying (42) with

$$a_1 = Nz_1 , \quad b_1 = c_1 , \quad a_2 = Nz , \quad b_2 = c_2$$

we see that the numerator in (41) is positive. With the above proved positivity of the numerator in (40) both $u_1^* > 0$ and $u_2^* > 0$.

Elementary calculation shows

$$e_{u_1^*, u_2^*, z}(c_1) = 1 = e_{u_1^*, u_2^*, z}(c_2) .$$

Since, as shown above, the maximum number of zeroes of $e_{u_1, u_2, z} - 1$ is 2, the assertion of (c) is proved.

Proof of (d): In case of $u_1 \geq 0, u_2 > 0$ $e_{u_1, u_2, z}$ is obviously strictly decreasing on $(0; Nz_2)$ and $e_{u_1, u_2, z} = 0$ for $s \in (Nz_2; Nz)$. Thus $e_{u_1, u_2, z} - 1$ has at most one zero on $(0; Nz)$.

In case of $u_1 > 0 = u_2$ $e_{u_1, u_2, z}$ is obviously strictly decreasing on $(0; Nz_1)$ and $e_{u_1, u_2, z} = 0$ for $s \in (Nz_1; Nz)$. Thus $e_{u_1, u_2, z} - 1$ has at most one zero on $(0; Nz)$.

Consider $u_1 \geq 0 > u_2$. From (38) and (39) we see that either $e_{u_1, u_2, z}$ is strictly increasing on $(0; Nz_1)$ or $e'_{u_1, u_2, z}$ has a change of sign from < 0 to > 0 on $(0; Nz_1)$. On $(Nz_1; Nz)$ $e_{u_1, u_2, z}$ is strictly increasing with $e_{u_1, u_2, z}(s) < 0$. Thus $e_{u_1, u_2, z} - 1$ has at most one zero on $(0; Nz)$.

Now consider $u_1 < 0 < u_2$.

From definition (36) it is obvious that $e_{u_1, u_2, z}$ is strictly decreasing on $(Nz_1; Nz_2)$ and that $e_{u_1, u_2, z}(s) = 0$ for $s \in (Nz_2; Nz)$.

According to (39) the sign of $e'_{u_1, u_2, z}$ on $(0; Nz_1)$ is the sign of $-\tilde{e}_{u_1, u_2, z}$ which by (39) is strictly decreasing on $(0; Nz_1)$ with $\lim_{s \uparrow Nz_1} -\tilde{e}_{u_1, u_2, z}(s) = -\infty$.

Hence $e_{u_1, u_2, z}$ is either strictly decreasing on $(0; Nz_2)$ or $e'_{u_1, u_2, z}$ has exactly one change of sign from > 0 to < 0 on $(0; Nz_1)$.

Thus the maximum number of zeroes of $e_{u_1, u_2, z} - 1$ on $(0; Nz_2)$ is 2. If two zeroes $0 < s_1 < s_2$ exist, s_1 is a change of sign from < 0 to > 0 and s_2 is a change of sign from > 0 to < 0 .

Proof of (e): From definitions (19), (36), (37) by elementary calculation. ■

Finally we shall use two elements of the proof for Theorem 6, pp. 88–90, in *Lehmann* (1959), which for the case of exponential families corresponds to the present Theorem (5.1). Actually both of these elements are valid also without the assumption of exponential families. This is obvious for the first element which is Theorem 5 in *Lehmann* (1959), pp. 83–84. This generalization of the fundamental lemma of Neyman and Pearson does not require exponential families. The second element is established in *Lehmann* (1959) as a step in the proof of Theorem 6. To show that it is in fact independent of the assumption of a special family of distributions we shall give it as a separate proposition.

5.7 Proposition: Let $0 < z_1 < z_2$, $0 < \beta < 1$. Then (β, β) is an inner point of the set

$$M = \{ (E_{z_1}[\psi(\tau)], E_{z_2}[\psi(\tau)]) | \psi: \mathbb{R} \rightarrow [0; 1] \text{ test} \} .$$

Proof: By the argument of page 67 in *Lehmann* (1959) the set M is always convex. Obviously $(u, u) \in M$ for each $u \in (0; 1)$ since $(u, u) = (E_{z_1}[\psi_u(\tau)], E_{z_2}[\psi_u(\tau)])$ with $\psi_u(s) = u$.

Let $0 < u_1 < \beta < u_2 < 1$. By the generally valid Corollary 1 in *Lehmann* (1959), p. 67, there are values $\beta < v_2 < 1$, $u_1 < v_1 < 1$, with $(\beta, v_2) \in M$ and $(v_1, u_1) \in M$. By the convexity of M the convex quadrangle defined by the points (u_1, u_1) , (v_1, u_1) , (u_2, u_2) , (β, v_2) is a subset of M and (β, β) is an inner point of this quadrangle. Hence (β, β) is an inner point of M . ■

Let us now turn to the proof of the elements of Theorem (5.1). It is favourable to combine the proof of part (a) and part (c).

Proof of Theorem (5.1), Parts (a) and (c): Let $z \in (z_2; +\infty)$.

By Proposition (5.7) the point $(1 - \alpha, 1 - \alpha)$ is an inner point of

$$\{ (E_{z_1}[\psi(\tau)], E_{z_2}[\psi(\tau)]) | \psi: \mathbb{R} \rightarrow [0; 1] \text{ test} \} .$$

Hence by assertion (i) of Theorem 5 in *Lehmann* (1959), pp. 83–84, there exists a test $\psi_0: \mathbb{R} \rightarrow [0; 1]$ with

$$E_{z_i}[\psi_0(\tau)] = 1 - \alpha \quad \text{for } i = 1, 2 , \quad (43)$$

$$E_z[\psi_0(\tau)] \geq E_z[\psi(\tau)] \quad \text{for tests } \psi \text{ with } E_{z_i}[\psi(\tau)] = 1 - \alpha \quad \text{for } i = 1, 2 . \quad (44)$$

By assertion (iv) of Theorem 5 in *Lehmann* (1959), pp. 83–84, there are constants $k_1, k_2 \in \mathbb{R}$ such that

$$E_{k_1, k_2, z}(s) < 0 \Rightarrow 1 - \psi_0(s) = 0 ,$$

$$E_{k_1, k_2, z}(s) > 0 \Rightarrow 1 - \psi_0(s) = 1 .$$

By part (e) of Proposition (5.6) we can infer the existence of constants $u_1, u_2 \in \mathbb{R}$

such that for $s \in (0; Nz)$

$$\begin{aligned} e_{u_1, u_2, z}(s) < 1 &\Rightarrow 1 - \psi_0(s) = 0, \\ e_{u_1, u_2, z}(s) > 1 &\Rightarrow 1 - \psi_0(s) = 1. \end{aligned} \quad (45)$$

Because of (43) there is at least one $c \in (0; Nz)$ with $e_{u_1, u_2, z}(c) = 1$.

Consider the assumption that there is one and only one $c \in (0; Nz)$ with $e_{u_1, u_2, z}(c) = 1$. Because of (43) c is a change of sign of $e_{u_1, u_2, z} - 1$ either from < 0 to > 0 or from > 0 to < 0 . Thus ψ_0 is of one-sided type, either

$$\psi_0(s) = \begin{cases} 1 & \text{for } s \in (0; c) \\ 0 & \text{for } s \in (c, +\infty) \end{cases}, \quad \text{or} \quad \psi_0(s) = \begin{cases} 0 & \text{for } s \in (0; c) \\ 1 & \text{for } s \in (c, +\infty) \end{cases}.$$

Hence either

$$\begin{aligned} E_z[\psi_0(\tau)] &= G_z^{(N, n)}(c) \quad \text{for } i = 1, 2 \quad \text{or} \\ E_z[\psi_0(\tau)] &= 1 - G_z^{(N, n)}(c) \quad \text{for } i = 1, 2. \end{aligned}$$

With Proposition (5.3) this result implies a contradiction to (43).

Hence there are at least two values $0 < c_1 < c_2 < Nz$ with $e_{u_1, u_2, z}(c_i) = 1$. Applying part (d) of Proposition (5.6) we see that then $0 < c_1 < c_2 < Nz_2$ and

$$e_{u_1, u_2, z}(s) \begin{cases} < 1 & \text{for } s \in (0; c_1) \cup (c_2; Nz) \\ > 1 & \text{for } s \in (c_1; c_2) \end{cases}.$$

Hence by (45)

$$\psi_0(s) = \begin{cases} 1 & \text{for } s \in (0; c_1) \cup (c_2; Nz) \\ 0 & \text{for } s \in [c_1; c_2] \end{cases}. \quad (46)$$

Because of (43) we have

$$\alpha = 1 - E_{z_i}[\psi_0(\tau)] = G_{z_i}^{(N, n)}(c_2) - G_{z_i}^{(N, n)}(c_1).$$

Hence we have proved the existence of values $0 < c_1 < c_2 < Nz_2$ with (29). The uniqueness is shown by Proposition (5.5). Thus we have proved (a).

Let us continue with the proof of part (c). By (46) and (30) we have for $z \in (z_2; +\infty)$

$$\psi^\star(s) = 1 - \psi_0(s) \quad \text{for } s \in (0; Nz) ,$$

hence by (44)

$$E_z[\psi^\star(\tau)] \leq E_z[\psi(\tau)] \quad \text{for tests } \psi \text{ with } E_{z_i}[\psi(\tau)] = \alpha \quad \text{for } i = 1, 2 .$$

Now consider $z \in (0; z_1)$

Evidently from (19) and (37):

$$E_{k_1, 0, z}(s) = \begin{cases} k_1 g_{z_1}^{(N, n)}(s) - g_z^{(N, n)}(s) & \text{for } s \in (0; Nz) , \\ k_1 g_{z_1}^{(N, n)}(s) & \text{for } s \in [Nz; Nz_1) , \\ 0 & \text{for } s \in [Nz_1; +\infty) . \end{cases} \quad (47)$$

Case 1: $c_1 < Nz$.

By part (a) and (c) of Proposition (5.6) and by (47) there is $k_1 = u_1 \cdot \left(\frac{z_1}{z}\right)^{N/2-1}$ with

$$E_{k_1, 0, z}(s) \begin{cases} < 0 & \text{for } s \in (0; c_1) , \\ > 0 & \text{for } s \in (c_1; Nz_1) . \end{cases}$$

Case 2: $c_1 \geq Nz$.

From (47) we have

$$E_{0, 0, z}(s) = \begin{cases} -g_z^{(N, n)}(s) < 0 & \text{for } s \in (0; Nz) , \\ 0 & \text{for } s \in (Nz; +\infty) . \end{cases}$$

In both cases we can choose $k_1 \geq 0, k_2 \geq 0$ as above such that

$$E_{k_1, k_2, z}(s) < 0 \Rightarrow \psi^\star(s) = 0 ,$$

$$E_{k_1, k_2, z}(s) > 0 \Rightarrow \psi^\star(s) = 1 .$$

From assertion (iii) of Theorem 5 in *Lehmann*(1959), pp. 83–84, we see that

$$E_z[\psi^\star(\tau)] \geq E_z[\psi(\tau)] \quad \text{for tests } \psi \text{ with } E_{z_i}[\psi(\tau)] = 1 - \alpha \quad \text{for } i = 1, 2 .$$

Hence

$$E_z[\psi^\star(\tau)] \leq E_z[\psi(\tau)] \quad \text{for tests } \psi \text{ with } E_{z_i}[\psi(\tau)] = \alpha \quad \text{for } i = 1, 2 .$$

This completes the proof of part (c) of Theorem (5.1). ■

Proof of Theorem (5.1), Part (b): Let $z \in (z_1; z_2)$.

Evidently

$$E_{k_1, k_2, z}(s) = \begin{cases} k_2 g_{z_2}^{(N, n)}(s) & \text{for } s \in (Nz; Nz_2) , \\ 0 & \text{for } s \in [Nz_2; +\infty) . \end{cases} \quad (48)$$

Case 1: $c_2 \geq Nz$.

By part (b) and (e) of Proposition (5.6) and by (48) we can choose $k_1 > 0$ such that

$$E_{k_1, 0, z}(s) \begin{cases} > 0 & \text{for } s \in (0; c_1) , \\ < 0 & \text{for } s \in (c_1; Nz) , \\ = 0 & \text{for } s \in [Nz; +\infty) , \end{cases}$$

hence with

$$E_{k_1, 0, z}(s) > 0 \Rightarrow \psi^\star(s) = 0 ,$$

$$E_{k_1, 0, z}(s) < 0 \Rightarrow \psi^\star(s) = 1 .$$

Case 2: $c_2 < Nz$.

By part (c) and (e) of Proposition (5.6) and by (48) we can choose $k_1 > 0$, $k_2 > 0$ such that

$$E_{k_1, k_2, z}(s) \begin{cases} > 0 & \text{for } s \in (0; c_1) \cup (c_2; Nz_2) , \\ < 0 & \text{for } s \in (c_1; c_2) , \\ = 0 & \text{for } s \in [Nz_2; +\infty) , \end{cases}$$

hence with

$$E_{k_1, k_2, z}(s) > 0 \Rightarrow \psi^*(s) = 0 ,$$

$$E_{k_1, k_2, z}(s) < 0 \Rightarrow \psi^*(s) = 1 .$$

From part (a) we have

$$E_{z_1}[\psi^*(\tau)] = \alpha .$$

Hence from assertion (iii) of Theorem 5 in *Lehmann* (1959), pp. 83–84

$$E_z[\psi^*(\tau)] \geq E_z[\psi(\tau)] \quad \text{for tests } \psi \text{ with } E_{z_1}[\psi(\tau)] \leq \alpha .$$

Since $z \in (z_1; z_2)$ we chosen arbitrarily we have shown that ψ^* is UMP at level α for H_3 against K_3 . ■

Part (d) of Theorem (5.1) follows from the following Proposition (5.8) which we shall also make use of in Paragraph 7, below.

5.8 Proposition: Let $n \leq N - 3$, $0 < c_1 < c_2$ and let $\beta: (0; +\infty) \rightarrow \mathbb{R}$ be defined by

$$\beta(z) = G_z^{(N, n)}(c_2) - G_z^{(N, n)}(c_1)$$

and let

$$z^* = \frac{1}{N} \frac{c_2 - c_1 \left(\frac{c_1}{c_2} \right)^{n/(N-n-2)}}{1 - \left(\frac{c_1}{c_2} \right)^{n/(N-n-2)}} .$$

Then z^* satisfies

$$z^* > \frac{c_2}{N} , \quad \left(\frac{c_1}{c_2} \right)^{n/2} = \left(\frac{Nz^* - c_2}{Nz^* - c_1} \right)^{(N-n)/2-1}$$

and we have

$$\begin{aligned} \beta(z) &= 0 \quad \text{for } 0 < z \leq \frac{c_1}{N} , \\ \beta \text{ is strictly } &\begin{cases} \text{increasing on} & \left(\frac{c_1}{N}, z^\star \right) , \\ \text{decreasing on} & (z^\star; +\infty) , \end{cases} \\ \lim_{z \rightarrow +\infty} \beta(z) &= 0 . \end{aligned}$$

Proof: From (23) we have

$$\beta(z) = B_{n/2, (N-n)/2} \left(\frac{c_2}{Nz} \right) - B_{n/2, (N-n)/2} \left(\frac{c_1}{Nz} \right) .$$

Thus from (20) and (21)

$$\beta(z) = 0 \quad \text{for } 0 < z \leq \frac{c_1}{N} , \quad \lim_{z \rightarrow +\infty} \beta(z) = 0 ,$$

$$\beta \text{ strictly increasing on } \left(\frac{c_1}{N}, \frac{c_2}{N} \right) ,$$

and for $z > \frac{c_2}{N}$

$$\begin{aligned} \beta'(z) &= -\frac{1}{Nz^2} \cdot \left\{ c_2 b_{n/2, (N-n)/2} \left(\frac{c_2}{Nz} \right) - c_1 b_{n/2, (N-n)/2} \left(\frac{c_1}{Nz} \right) \right\} \\ &= \frac{\Gamma\left(\frac{N}{2}\right)}{\Gamma\left(\frac{n}{2}\right)\Gamma\left(\frac{N-n}{2}\right)} \frac{c_2^{n/2} (Nz - c_1)^{(N-n)/2-1}}{N^{N/2-1} z^{N/2}} \\ &\quad \cdot \left\{ \left(\frac{c_1}{c_2} \right)^{n/2} - \left(\frac{Nz - c_2}{Nz - c_1} \right)^{(N-n)/2-1} \right\} . \end{aligned}$$

Elementary calculation shows $\beta(z^*) = 0$, consequently

$$\beta'(z) \begin{cases} > 0 & \text{for } z \in \left(\frac{c_1}{N}; z^*\right) , \\ < 0 & \text{for } z \in (z^*; +\infty) . \end{cases} \quad \blacksquare$$

Proof of Theorem (5.1), Part (d): From (30) we have

$$\beta^*(z) = G_z^{(N,n)}(c_2) - G_z^{(N,n)}(c_1) \quad \text{for } z > 0 .$$

Thus (d) follows from application of Proposition (5.8) with $\beta = \beta^*$. Because of (29) we have $z_1 < z^* < z_2$. ■

6 The Two-Sided Hypothesis $z_1 \leq z \leq z_2$

We shall now consider the two-sided scheme (10)

$$H_4: z_1 \leq z \leq z_2 \quad \text{against} \quad K_4: z < z_1 \text{ or } z > z_2 .$$

For this scheme the following Theorem (6.1) gives an UMPU test which is analogous to the test (30) for the dual problem (9) H_3 against K_3 .

6.1 Theorem: Let $0 < \alpha < 1$, $0 < z_1 < z_2$, $n \leq N - 5$.

(a) There is a unique pair $0 < c_1 < c_2$ satisfying

$$G_{z_1}^{(N,n)}(c_1) + 1 - G_{z_1}^{(N,n)}(c_2) = \alpha = G_{z_2}^{(N,n)}(c_1) + 1 - G_{z_2}^{(N,n)}(c_2) . \quad (49)$$

We have $c_2 < Nz_2$.

(b) Let the test $\psi^*: \mathbb{R} \rightarrow [0; 1]$ be defined by

$$\psi^*(s) = \begin{cases} 1 , & \text{if } s \leq c_1 \text{ or } s \geq c_2 , \\ 0 , & \text{if } c_1 < s < c_2 , \end{cases} \quad (50)$$

i.e. $\psi^\star$ represents the decision rule

$$\text{Reject iff } \tau \leq c_1 \text{ or } \tau \geq c_2 .$$

Then for all tests $\psi: \mathbb{R} \rightarrow [0; 1]$ with $E_{z_1}[\psi(\tau)] = \alpha = E_{z_2}[\psi(\tau)]$ we have

$$E_z[\psi^\star(\tau)] \geq E_z[\psi(\tau)] \quad \text{for } z \in (0; z_1) \cup (z_2; +\infty) .$$

In particular, the test $\psi^\star$ is UMPU at level α for testing $H_4: z_1 \leq z \leq z_2$ against $K_4: z < z_1$ or $z > z_2$.

(c) Let

$$\beta^\star(z) = E_z[\psi^\star(\tau)] = G_z^{(N,n)}(c_1) + 1 - G_z^{(N,n)}(c_1) \quad \text{for } z > 0$$

be the power function of $\psi^\star$ and let $z^\star$ be defined by (31).

Then $\max\{z_1, c_2/N\} < z^\star < z_2$ and

$$\beta^\star(z) = 1 \quad \text{for } 0 < z \leq \frac{c_1}{N} ,$$

$$\beta^\star \text{ is strictly } \begin{cases} \text{decreasing on} & \left(\frac{c_1}{N}; z^\star\right) , \\ \text{increasing on} & (z^\star; +\infty) , \end{cases}$$

$$\lim_{z \rightarrow +\infty} \beta^\star(z) = 1 .$$

■

Theorem (6.1) is established as a consequence of (5.1). As an additional element we only need the fact that power functions are continuous:

6.2 Proposition: Let $0 < z_1 < z_2$, $n \leq N - 3$, let $\psi: \mathbb{R} \rightarrow [0; 1]$ be a test and let

$$\beta_\psi(z) = E_z[\psi(\tau)] = \int_{\mathbb{R}} \psi(s) g_z^{(N,n)}(s) ds \quad \text{for } z > 0 \tag{51}$$

be the power function of ψ .

Then β_ψ is continuous on $(0; +\infty)$.

Proof: Let $0 < z_1 < z_2$. From (19) we immediately infer

$$g_z^{(N,n)}(s) < \left(\frac{z_2}{z_1}\right)^{n/2} g_{z_2}^{(N,n)}(s) \quad \text{for } z \in [z_1; z_2] .$$

On $[z_1; z_2]$ the integrand of (51) is majorized by the integrable function $\psi(s) \left(\frac{z_2}{z_1}\right) g_{z_2}^{(N,n)}(s)$. Hence by the well-known theorem on parameter integrals β_ψ is continuous on $(z_1; z_2)$.

Since $0 < z_1 < z_2$ was chosen arbitrarily β_ψ is continuous on $(0; +\infty)$. ■

With Proposition (6.2) the proof of Theorem (6.1) can be reduced to an application of Theorem (5.1) in exactly the same manner as it is done in *Lehmann* (1959), p. 126, for the corresponding problem under exponential families.

7 The Two-Sided Hypothesis $z = z_0$

The last two-sided scheme to be considered is (11):

$$H_5: z = z_0 \quad \text{against} \quad K_5: z \neq z_0 .$$

From an intuitive point of view the rejection region of the UMPU test for this problem should have the same structure as the one of the UMPU test for the similar problem (10) H_4 against K_4 given in Theorem (6.1). The following Theorem (7.1) corresponds to this intuitive conjecture.

7.1 Theorem: Let $0 < \alpha < 1$, $z_0 > 0$, $n \leq N - 5$.

(a) There is a unique pair $0 < c_1 < c_2$ satisfying

$$G_{z_0}^{(N,n)}(c_1) + 1 - G_{z_0}^{(N,n)}(c_2) = \alpha \tag{52}$$

and

$$\left(\frac{c_1}{c_2}\right)^{n/2} = \left(\frac{Nz_0 - c_2}{Nz_0 - c_1}\right)^{(N-n)/2-1} . \tag{53}$$

(b) Let the test $\psi^*: \mathbb{R} \rightarrow [0; 1]$ be defined by

$$\psi^*(s) = \begin{cases} 1, & \text{if } s \leq c_1 \text{ or } s \geq c_2, \\ 0, & \text{if } c_1 < s < c_2, \end{cases} \quad (54)$$

i.e. ψ^* represents the decision rule

$$\text{Reject iff } \tau \leq c_1 \text{ or } \tau \geq c_2.$$

Then ψ^* is UMPU at level α for testing $H_5: z = z_0$ against $K_5: z \neq z_0$.

(c) Let

$$\beta^*(z) = E_z[\psi^*(\tau)] = G_z^{(N,n)}(c_1) + 1 - G_z^{(N,n)}(c_1) \quad \text{for } z > 0$$

be the power function of ψ^* .

Then

$$\beta^*(z) = 1 \quad \text{for } 0 < z \leq \frac{c_1}{N},$$

$$\beta^* \text{ is strictly } \begin{cases} \text{decreasing on} & \left(\frac{c_1}{N}, z_0 \right), \\ \text{increasing on} & (z_0; +\infty), \end{cases}$$

$$\lim_{z \rightarrow +\infty} \beta^*(z) = 1. \quad \blacksquare$$

As in the corresponding theorem for exponential families, see *Lehmann* (1959), pp. 126–127, the second requirement (53) of part (a) expresses the fact that, as asserted by part (c), the minimum of the power function of ψ^* coincides with z_0 .

The proof of Theorem (7.1) is based on applying Theorem (6.1) in a limiting process. The essential step is contained in the following Proposition (7.2).

7.2 Proposition: Let $0 < \alpha < 1$, $z_0 > 0$, $n \leq N - 5$, let $(z_{1,m})_{m \in \mathbb{N}}$, $(z_{2,m})_{m \in \mathbb{N}}$, $(\alpha_m)_{m \in \mathbb{N}}$ be sequences with

$$0 < z_{1,m} < z_{2,m}, \quad z_{1,m} \leq z_0 \leq z_{2,m}, \quad \lim_{m \rightarrow \infty} z_{1,m} = \lim_{m \rightarrow \infty} z_{2,m},$$

$$0 < \alpha_m < 1, \quad \lim_{m \rightarrow \infty} \alpha_m = \alpha,$$

and let for $m \in \mathbb{N}$ $0 < c_{1,m} < c_{2,m}$ be the unique values with

$$G_{z_{1,m}}^{(N,n)}(c_{1,m}) + 1 - G_{z_{1,m}}^{(N,n)}(c_{2,m}) = \alpha = G_{z_{2,m}}^{(N,n)}(c_{1,m}) + 1 - G_{z_{2,m}}^{(N,n)}(c_{2,m}) . \quad (55)$$

Then there exist converging subsequences $(c_{1,m_k})_{k \in \mathbb{N}}$, $(c_{2,m_k})_{k \in \mathbb{N}}$ such that the limits $c_i = \lim_{k \rightarrow \infty} c_{i,m_k}$ satisfy (52) and (53).

Proof: Because of part (a) of Theorem (6.1) we have $0 < c_{1,m} < c_{2,m} < Nz_{2,m}$ for $m \in \mathbb{N}$. Hence there are converging subsequences $(c_{1,m_k})_{k \in \mathbb{N}}$, $(c_{2,m_k})_{k \in \mathbb{N}}$.

Let $c_i = \lim_{k \rightarrow \infty} c_{i,m_k}$ and let for $k \in \mathbb{N}$

$$z_k^* = \frac{1}{N} \frac{c_{2,m_k} - c_{1,m_k} \left(\frac{c_{1,m_k}}{c_{2,m_k}} \right)^{n/(N-n-2)}}{1 - \left(\frac{c_{1,m_k}}{c_{2,m_k}} \right)^{n/(N-n-2)}}$$

be the minimizing argument of the power function

$$\beta_k^*(z) = G_z^{(N,n)}(c_{1,m_k}) + 1 - G_z^{(N,n)}(c_{2,m_k}) \quad \text{for } z > 0$$

introduced in Theorem (6.1), part (c).

By Proposition (5.8) we have

$$\left(\frac{c_{1,m_k}}{c_{2,m_k}} \right)^{n/2} = \left(\frac{Nz_k^* - c_{2,m_k}}{Nz_k^* - c_{1,m_k}} \right)^{(N-n)/2-1} . \quad (56)$$

By Theorem (6.1), part (c) we have

$$z_{1,m_k} < z_k^* < z_{2,m_k} ,$$

hence

$$\lim_{k \rightarrow \infty} z_k^* = z_0 ,$$

and thus from (56) by taking limits on both sides

$$\left(\frac{c_1}{c_2}\right)^{n/2} = \left(\frac{Nz_0 - c_2}{Nz_0 - c_1}\right)^{(N-n)/2-1}$$

which is (53).

From (55) we obtain by taking limits on both sides

$$\alpha = G_{z_0}^{(N,n)}(c_1) + 1 - G_{z_0}^{(N,n)}(c_2) ,$$

which is (52). ■

Proof of Theorem (7.1), Parts (a) and (c): The existence of values $0 < c_1 < c_2$ with (52), (53) obviously follows from Proposition (7.2). With these values we can define the test ψ^* as done in (54).

Obviously the power function β^* of ψ^* satisfies

$$\beta^*(z) = E_z[\psi^*(\tau)] = G_z^{(N,n)}(c_1) + 1 - G_z^{(N,n)}(c_2) \quad \text{for } z > 0 .$$

(53) is equivalent to

$$z_0 = \frac{1}{N} \frac{c_2 - c_1 \left(\frac{c_1}{c_2}\right)^{n/(N-n-2)}}{1 - \left(\frac{c_1}{c_2}\right)^{n/(N-n-2)}} .$$

Thus assertion (c) on the power function β^* follows from Proposition (5.8).

We have to prove that $0 < c_1 < c_2$ are unique. Let $\tilde{\alpha} \in (\alpha; 1)$. By the already proved assertion (c) and by Proposition (6.2) there exist $0 < z_1 < z_0 < z_2$ with

$$\tilde{\alpha} = \beta^*(z_i) = G_{z_i}^{(N,n)}(c_1) + 1 - G_{z_i}^{(N,n)}(c_2) .$$

Hence c_1, c_2 coincide with the unique values determined by (49) for significance level $\tilde{\alpha}$ and for $0 < z_1 < z_2$ and are therefore unique. ■

Proof of Theorem (7.1), Part (b): Let $\psi: \mathbb{R} \rightarrow [0; 1]$ be an arbitrary unbiased level α test for H_5 against K_5 , i.e.

$$E_{z_0}[\psi(\tau)] = \alpha , \quad E_z[\psi(\tau)] \geq \alpha \quad \text{for } z \neq z_0 . \quad (57)$$

Case 1: There is $\varepsilon \in (0; z_0)$ with $E_z[\psi(\tau)] = \alpha$ for $z \in (z_0 - \varepsilon; z_0)$.

Let

$$z_{2,m} = z_0, \quad \alpha_m = \alpha \quad \text{for } m \in \mathbb{N},$$

let $(z_{1,m})_{m \in \mathbb{N}}$ be a sequence with

$$z_0 - \varepsilon < z_{1,m} < z_0 = z_{2,m},$$

and let for $m \in \mathbb{N}$ $0 < c_{1,m} < c_{2,m}$ be the unique values with

$$G_{z_{1,m}}^{(N,n)}(c_{1,m}) + 1 - G_{z_{1,m}}^{(N,n)}(c_{2,m}) = \alpha = G_{z_{2,m}}^{(N,n)}(c_{1,m}) + 1 - G_{z_{2,m}}^{(N,n)}(c_{2,m}).$$

By Proposition (7.2) we can assume

$$\lim_{m \rightarrow \infty} c_{i,m} = c_i \quad (58)$$

where c_1, c_2 are uniquely determined by (52), (53).

For $m \in \mathbb{N}$ let $\psi_m^\star: \mathbb{R} \rightarrow [0; 1]$ be the level α test determined by the critical values $c_{1,m}, c_{2,m}$ as in Theorem (6.1), i.e.

$$\psi_m^\star(s) = \begin{cases} 1, & \text{if } s \leq c_{1,m} \text{ or } s \geq c_{2,m}, \\ 0, & \text{if } c_{1,m} < s < c_{2,m}. \end{cases} \quad (59)$$

By the assumption of the present Case 1 ψ is a test with

$$E_{z_{1,m}}[\psi(\tau)] = \alpha = E_{z_{2,m}}[\psi(\tau)]. \quad (60)$$

Hence by part (b) of Theorem (6.1)

$$E_z[\psi_m^\star(\tau)] \geq E_z[\psi(\tau)] \quad \text{for } z \in (0; z_{1,m}) \cup (z_{2,m}; +\infty). \quad (61)$$

Obviously from (58) and (59)

$$\lim_{m \rightarrow \infty} \psi_m^\star(s) = \psi^\star(s) \quad \text{for } s \in \mathbb{R},$$

hence by Lebesgue's dominated convergence theorem

$$\lim_{m \rightarrow \infty} E_z[\psi_m^\star(\tau)] = E_z[\psi^\star(\tau)] \quad \text{for } z > 0 . \quad (62)$$

From (61) and (62) finally

$$E_z[\psi^\star(\tau)] \geq E_z[\psi(\tau)] \quad \text{for } z \neq z_0, z > 0 .$$

Case 2: There is $\varepsilon \in (0; z_0)$ with $E_z[\psi(\tau)] = \alpha$ for $z \in (z_0; z_0 + \varepsilon)$.

The proof for this case is completely analogous to the proof in Case 1.

Case 3: For each $\varepsilon \in (0; z_0)$

$$\sup\{E_z[\psi(\tau)] \mid z_0 - \varepsilon < z < z_0\} > \alpha , \quad \sup\{E_z[\psi(\tau)] \mid z_0 < z < z_0 + \varepsilon\} > \alpha .$$

By Proposition (6.2) the power function $\beta_\psi(z) = E_z[\psi(\tau)]$ is continuous. Hence there are sequences $(z_{1,m})_{m \in \mathbb{N}}, (z_{2,m})_{m \in \mathbb{N}}$ with

$$0 < z_{1,m} < z_0 < z_{2,m} ,$$

$$\alpha < \beta_\psi(z_{1,m}) = E_{z_{1,m}}[\psi(\tau)] = E_{z_{2,m}}[\psi(\tau)] = \beta_\psi(z_{2,m}) ,$$

$$\lim_{m \rightarrow \infty} z_{1,m} = z_0 = \lim_{m \rightarrow \infty} z_{2,m} , \quad \lim_{m \rightarrow \infty} \beta_\psi(z_{1,m}) = \alpha = \lim_{m \rightarrow \infty} \beta_\psi(z_{2,m}) .$$

For $m \in \mathbb{N}$ let

$$\alpha_m = \beta_\psi(z_{1,m}) = \beta_\psi(z_{2,m})$$

and let $0 < c_{1,m} < c_{2,m}$ be the unique values with

$$G_{z_{1,m}}^{(N,n)}(c_{1,m}) + 1 - G_{z_{1,m}}^{(N,n)}(c_{2,m}) = \alpha_m = G_{z_{2,m}}^{(N,n)}(c_{1,m}) + 1 - G_{z_{2,m}}^{(N,n)}(c_{2,m}) .$$

Then the assumptions of Proposition (7.2) are fulfilled.

The further argument is now completely analogous to the argument of Case 1 if “ α ” in (60) is substituted by “ α_m ”. ■

8 Asymptotic Distribution of the Test Statistic

As mentioned in Paragraph 3 the unconditional distribution (process distribution) of $\frac{1}{\sigma^2} \tau$ is χ^2 -distribution with n degrees of freedom. From an intuitive point of view the lot distribution of τ for large lot size N should be of the same type as the process distribution of τ with the process variance σ^2 replaced by the lot variance z , i.e. we expect $\frac{1}{z} \tau$ to have χ^2 -distribution with n degrees of freedom.

Let us consider this in detail. Stirling's formula provides

$$\Gamma(x) = \sqrt{2\pi} x^{x-1/2} \exp(-x) \exp(l(x)) \quad \text{for } x > 0 \quad (63)$$

where

$$0 < l(x) < \frac{1}{12x}.$$

Applying this formula we obtain for $N > n$, $z < 0 < \frac{1}{N}$, $0 < s < Nz$

$$\begin{aligned} g_z^{(N,n)}(s) &=_{(19)} \frac{s^{n/2-1}}{\Gamma\left(\frac{n}{2}\right) z^{n/2}} \frac{\Gamma\left(\frac{N}{2}\right)}{\Gamma\left(\frac{N-n}{2}\right) N^{n/2}} \left(1 - \frac{s}{Nz}\right)^{(N-n)/2-1} \\ &=_{(63)} \frac{s^{n/2-1}}{2^{n/2} \Gamma\left(\frac{n}{2}\right) z^{n/2}} \exp\left(-\frac{n}{2}\right) \left(1 + \frac{n}{N-n}\right)^{(N-n) \cdot 1/2} \left(1 + \frac{n}{N-n}\right)^{-1/2} \\ &\quad \cdot \left(1 - \frac{(s/z)}{N}\right)^{N \cdot 1/2} \left(1 - \frac{(s/z)}{N}\right)^{-(n/2)-1} \frac{\exp\left(l\left(\frac{N}{2}\right)\right)}{\exp\left(l\left(\frac{N-n}{2}\right)\right)} \end{aligned}$$

and hence for $z > 0$, $s > 0$,

$$\lim_{N \rightarrow \infty} g_z^{(N,n)}(s) = \frac{1}{2^{n/2} \Gamma\left(\frac{n}{2}\right)} \left(\frac{s}{z}\right)^{n/2-1} \exp\left(-\frac{s}{2z}\right) \frac{1}{z}.$$

Since clearly

$$\lim_{N \rightarrow \infty} g_z^{(N,n)}(s) = 0 \quad \text{for } s \leq 0$$

we have by comparison with definition (12)

$$\lim_{N \rightarrow \infty} g_z^{(N,n)}(s) = \frac{1}{z} \chi_n \left(\frac{s}{z} \right) \quad \text{for } s \in \mathbb{R} .$$

As expected, for large lot size we can approximate the distribution of $\frac{1}{z} \tau$ by χ^2 -distribution with n degrees of freedom.

9 Computation of Critical Values

By (23) the computation of critical values for the tests (25), (27), (30), (50), (54), can be reduced to the computation of analogously defined values of the beta-distribution of order $\left(\frac{n}{2}, \frac{N-n}{2} \right)$. In detail:

The critical value c_α of the UMP level α test (25) for $H_1: z \leq z_0$ against $K_1: z > z_0$ satisfies $c_\alpha = Nz_0 x_\alpha$ where $0 < x_\alpha < 1$ is defined by

$$\alpha \stackrel{!}{=} 1 - B_{n/2, (N-n)/2}(x_\alpha) . \quad (64)$$

The critical value c_α of the UMP level α test (27) for $H_2: z \geq z_0$ against $K_2: z < z_0$ satisfies $c_\alpha = Nz_0 x_\alpha$ where $0 < x_\alpha < 1$ is defined by

$$\alpha \stackrel{!}{=} B_{n/2, (N-n)/2}(x_\alpha) . \quad (65)$$

The critical values $0 < c_{1,\alpha} < c_{2,\alpha}$ of the UMP level α test (30) for $H_3: z \leq z_1$ or $z \geq z_2$ against $K_3: z_1 < z < z_2$ satisfy $c_{1,\alpha} = Nz_1 x_{1,\alpha}$, $c_{2,\alpha} = Nz_1 x_{2,\alpha}$, where $0 < x_{1,\alpha} < x_{2,\alpha} < 1$ are defined by

$$\begin{aligned} & B_{n/2, (N-n)/2}(x_{2,\alpha}) - B_{n/2, (N-n)/2}(x_{1,\alpha}) \stackrel{!}{=} \alpha \\ & \stackrel{!}{=} B_{n/2, (N-n)/2} \left(x_{2,\alpha} \frac{z_1}{z_2} \right) - B_{n/2, (N-n)/2} \left(x_{1,\alpha} \frac{z_1}{z_2} \right) . \end{aligned} \quad (66)$$

The critical values $0 < c_{1,\alpha} < c_{2,\alpha}$ of the UMPU level α test (50) for $H_4: z_1 \leq z \leq z_2$ against $K_4: z < z_1$ or $z > z_2$ satisfy $c_{1,\alpha} = Nz_1x_{1,\alpha}$, $c_{2,\alpha} = Nz_1x_{2,\alpha}$, where $0 < x_{1,\alpha} < x_{2,\alpha} < 1$ are defined by

$$\begin{aligned} B_{n/2, (N-n)/2}(x_{1,\alpha}) + 1 - B_{n/2, (N-n)/2}(x_{2,\alpha}) &\stackrel{!}{=} \alpha \\ &\stackrel{!}{=} B_{n/2, (N-n)/2}\left(x_{1,\alpha} \frac{z_1}{z_2}\right) + 1 - B_{n/2, (N-n)/2}\left(x_{2,\alpha} \frac{z_1}{z_2}\right). \end{aligned} \quad (67)$$

The critical values $0 < c_{1,\alpha} < c_{2,\alpha}$ of the UMPU level α test (54) for $H_5: z = z_0$ against $K_5: z \neq z_0$ satisfy $c_{1,\alpha} = Nz_0x_{1,\alpha}$, $c_{2,\alpha} = Nz_0x_{2,\alpha}$, where $0 < x_{1,\alpha} < x_{2,\alpha} < 1$ are defined by

$$B_{n/2, (N-n)/2}(x_{1,\alpha}) + 1 - B_{n/2, (N-n)/2}(x_{2,\alpha}) \stackrel{!}{=} \alpha, \quad (68)$$

$$\left(\frac{x_{1,\alpha}}{x_{2,\alpha}}\right)^{n/2} = \left(\frac{1 - x_{2,\alpha}}{1 - x_{1,\alpha}}\right)^{(N-n)/2 - 1}. \quad (69)$$

Percentage points of the beta-distribution are tabulated in *Thompson* (1941), *Clark* (1953), and in *Harter* (1964). *Pearson* (1968) gives tables of the beta-distribution. Percentage points may also be calculated by applying a relationship among beta-distribution and F-distribution. The density of F-distribution with degrees of freedom $m_1, m_2 \in \mathbb{N}$ is

$$\begin{aligned} &f_{m_1, m_2}(x) \\ &= \begin{cases} 0 & \text{for } x \in (-\infty; 0] , \\ \frac{\Gamma\left(\frac{m_1 + m_2}{2}\right)}{\Gamma\left(\frac{m_1}{2}\right)\Gamma\left(\frac{m_2}{2}\right)} m_1^{m_1/2} m_2^{m_2/2} x^{m_1/2 - 1} (m_2 + m_1 x)^{-(m_1 + m_2)/2} & \text{for } x \in (0; +\infty) , \end{cases} \end{aligned}$$

with corresponding distribution function

$$F_{m_1, m_2}(y) = \int_0^y f_{m_1, m_2}(x) dx .$$

According to *Abramowitz/Stegun* (1970), p. 945, we have for $a, b > 0$, $2a, 2b \in \mathbb{N}$,

$$0 < x < 1$$

$$B_{a,b}(x) = 1 - F_{2b,2a}\left(\frac{a}{b} \frac{1-x}{x}\right).$$

Several approximate formulae for percentage points of the beta-distribution are collected by *Thomson* (1947).

A rough approximation of critical values is provided by the asymptotics of Paragraph 8. For large N the approximate distribution of $\frac{1}{z}\tau$ is χ^2 -distribution with n degrees of freedom. Under this approximation the parameter z can be considered as the variance of an underlying normal distribution $N(\mu, z)$ with known mean μ . Thus the tests for z can be treated as tests for the variance of this normal distribution with critical values calculated in well-known manner, see *Lehmann* (1959), p. 129, for instance.

10 Numerical Examples

Consider the following situations.

- (i) Let $z_0 = 1$ be the consumer's break even value for the average square deviation from target per item in a lot. In ordinary course of business the consumer can count upon lots of satisfactory quality. Hence she tests $H_1: z \leq z_0$ against $K_1: z > z_0$.
- (ii) Let $z_1 = 0.8, z_2 = 1.2$. The consumer considers lots with $z > z_2$ as intolerable. Lots with $z \leq z_2$ are distinguished in two classes: Lots with $z < z_1$ which can be used for especially refined purposes, and normal tolerable lots with $z_1 \leq z < z_2$ for ordinary purposes. In ordinary course of business the consumer expects tolerable but not extraordinarily good lots. Thus she tests $H_4: z_1 \leq z \leq z_2$ against $K_4: z < z_1$ or $z > z_2$. If the test rejects further inspection is performed to decide whether the lot is of high or of intolerable quality level.

Let the significance level be $\alpha = 0.01$ and let the sample size be $n = 20$. To study the effect of lot size let $N = 50, 100, 500, 1000$.

Consider situation (i), i.e. the test of $H_1: z \leq 1$ against $K_1: z > 1$. By (64) the critical value satisfies $c_\alpha = Nx_\alpha$ where $0 < x_\alpha < 1$ is determined from

$$0.99 \stackrel{!}{=} B_{10,(N-20)/2}(x_\alpha). \quad (70)$$

The approximate critical value c'_α calculated from χ^2 -distribution is determined by

$$0.99 \stackrel{!}{=} \int_0^{c'_\alpha} \chi_n(s) ds . \quad (71)$$

We shall also try an approximation of the percentage points of beta-distribution developed by *Cochran* (1940), reported about by *Thompson* (1941) and *Thomson* (1947). *Thompson* (1941) describes the approximation for solving

$$\beta \stackrel{!}{=} B_{a,b}(x') \quad \text{for values } 0 < \beta \leq 0.5 .$$

To solve (70) we solve

$$0.01 \stackrel{!}{=} B_{(N-20)/2, 10}(x')$$

and obtain the approximation $\tilde{x}_\alpha = 1 - x'$ by applying the symmetry relation

$$B_{a,b}(x) = 1 - B_{b,a}(1 - x) .$$

In the terminology of *Thompson* (1941) we identify

$$p = \frac{N - n}{2} = \frac{N - 20}{2} , \quad q = \frac{n}{2} = 10 ,$$

and we let

$$y = 2.3263 , \quad \lambda = \frac{y^2 + 3}{6} , \quad A = \frac{4pq}{p + q} ,$$

$$z = \frac{y}{\sqrt{A - \lambda}} + \frac{\left(\lambda - \frac{1}{6}\right)(A - 2p)}{pA} ,$$

$$x' = \frac{p}{p + q \exp(2z)} .$$

The following table gives the exact value x_α and $c_\alpha = Nz_0 x_\alpha = Nx_\alpha$ calculated by a computer program, the χ^2 -approximation c'_α , see (71), taken from the table

Table 10.1

N	50	100	500	1000
x_α	0.629507	0.345900	0.073959	0.037273
c_α	31.475349	34.589977	36.979571	37.273383
c'_α	37.566	37.566	37.566	37.566
$\tilde{x}_\alpha$	0.629969	0.345733	0.073677	0.037110
$\tilde{c}_\alpha$	31.498438	34.573543	36.838515	37.109658
c_α/n	1.573767	1.729499	1.848979	1.863669
c'_α/n	1.878	1.878	1.878	1.878
$\tilde{c}_\alpha/n$	1.574922	1.728677	1.841926	1.855483

Table 10.2

N	50	100	500	1000
$x_{1,\alpha}$	0.193509	0.088879	0.016748	0.008315
$x_{2,\alpha}$	0.946683	0.520961	0.111533	0.056219
$c_{1,\alpha}$	7.740346	7.110292	6.699249	6.652177
$c_{2,\alpha}$	37.867301	41.676850	44.613327	44.975158
$c_{1,\alpha}/n$	0.387017	0.355515	0.334962	0.332609
$c_{2,\alpha}/n$	1.893365	2.083843	2.230666	2.248758

of percentage points of χ^2 -distribution in *Harter* (1964), the above described approximation $\tilde{x}_\alpha = 1 - x'$ with the corresponding value $\tilde{c}_\alpha = Nz_0\tilde{x}_\alpha = N\tilde{x}_\alpha$, and finally the critical values c_α/n , c'_α/n , $\tilde{c}_\alpha/n$ for the average square deviation $\frac{1}{n}\tau$ from target per item in the sample.

Concluding from these examples, the approximation $\tilde{c}_\alpha$ seems to be sufficiently exact. For large N however, $\tilde{c}_\alpha$ is a bit smaller than c_α , and the actual size of the test based on $\tilde{c}_\alpha$ is slightly larger than the significance level α . The rough χ^2 -approximation should only be used if lot size N is large with respect to sample size n ; because of $c'_\alpha > c_\alpha$ the approximation at least does not lead to a violation of significance level α . Concluding from the values c_α/n , c'_α/n , $\tilde{c}_\alpha/n$ the test seems intuitively reasonable: The average square deviation $\frac{1}{n}\tau$ from target per item in the sample is an unbiased estimator for z , see Paragraph 3; in the above example the test rejects if this estimator is considerably above the break even value $z_0 = 1$.

Determination of critical values for the two-sided test of situation (ii) is less convenient. There are no tables for the values defined by (67) and even for χ^2 -distribution two-sided limits are not tabulated. In applications throughout, as in our example, n is considerably smaller than $N - n$ so that we cannot use considerations based on symmetry of the distribution function $B_{n/2, (N-n)/2}$. Thus the critical values have to be calculated by a computer program. For an ordinary modern computer the amount of time required to solve this problem is negligible. Under the parameters of situation (ii) the results are as follows.

As in Table (10.1) the critical values considerably depend on lot size N as long as N is not very large with respect to sample size n . Again the test seems intuitively reasonable inasmuch as it rejects if the average square deviation $\frac{1}{n} \tau$ from target per item in the sample is considerably outside the interval $[z_1; z_2] = [0.8; 1.2]$.

References

- Abramowitz M, Stegun IA (1970) Handbook of mathematical functions. National Bureau of Standards, Washington DC
- Clark RE (1953) Percentage points of the incomplete beta-function. *Journal of the American Statistical Association* 48:831–843
- Cochran WG (1940) Note on an approximate formula for significance levels of z . *The Annals of Mathematical Statistics* 11:93–95
- von Collani E (1990) ANSI/ASQC Z1.4 versus ANSI/ASQC Z1.9. *Economic Quality Control* 5:60–64
- von Collani E (1991) A note on acceptance sampling for variables. *Metrika* 38:19–36
- Göb R (1996) An elementary model for statistical lot inspection and its application to sampling by variables. *Metrika* 43:223–238
- Harter HL (1964) New tables of the incomplete gamma-function ratio and of percentage points of the chi-square and beta-distribution. Aerospace Research Laboratories, Wright-Patterson Air Force Base, Ohio
- Hryniewicz O (1991) A note on E. von Collanis Paper “ANSI/ASQC Z1.4 versus ANSI/ASQC Z1.9”. *Economic Quality Control* 6:16–18
- Lehmann EL (1959) Testing statistical hypotheses. John Wiley, New York
- Patel J, Kapadia CH, Owen DB (1976) Handbook of statistical distributions. Marcel Dekker, New York, Basel
- Pearson K (1968) Tables of the incomplete beta-functions, second edition. Cambridge University Press, London
- Rényi A (1970) Probability theory. North-Holland Publishing Company, Amsterdam, London
- Taguchi T, Elsayed EA, Hsiang TC (1989) Quality engineering in production systems. McGraw-Hill, New York
- Thompson CM (1941) Tables of percentage points of the incomplete beta-function. *Biometrika* 32:151–181
- Thomson DH (1947) Approximate formulae for the percentage points of the incomplete beta-function and of the χ^2 -distribution. *Biometrika* 34:368–372

Variance Estimation Using Auxiliary Information: An Almost Unbiased Multivariate Ratio Estimator

A. ARCOS CEBRIÁN AND M. RUEDA GARCÍA

Departamento de Estadística e Investigación Operativa, Facultad de Ciencias,
Campus de Fuentenueva, 18071 Granada, España

Abstract: The goal of this paper is to investigate the repeated substitution method (see *Srivastava*, 1967) estimating population variance in finite population sample surveys. We propose an almost unbiased multivariate ratio estimator that has a smaller mean squared error than the conventional biased multivariate ratio estimator (established by *Isaki* (1983)) and with the same precision as the multivariate regression estimator. Furthermore, it is a computationally much more interesting estimator since to compute it we only need to have knowledge of correlation among available variables, which it is common to have in several practical situations. A comparison of the multivariate ratio estimator proposed and the multivariate regression estimator is given.

1 Introduction

In finite populations, the classical methods of estimation are based on direct estimators, i.e., those which only use the response variable, y . Information provided by an auxiliary variable, x , highly correlated with the main variable and with simpler values for reporting all population units, is sometimes used as well. It is known that indirect estimators usually improve on precision with respect to direct ones. Among these indirect methods, two are especially important: the ratio methods of estimation and the regression method.

Examples abound in the literature in which these methods are used for estimating population means or totals. *Cochran* (1977) places as the year 1802 in which *Laplace's* procedure (with a ratio method) was used to estimate the population in France. Important references on indirect methods to estimate population means and totals are found in *Krisnaiah* and *Rao* (1988) and *Chaudhuri* and *Vos* (1988). The assumption that indirect methods can improve (under certain conditions) on direct procedures for estimating population variance is a reasonable one.

Estimation of population variance was considered in *Isaki* (1983), where ratio and regression estimators were proposed. Ratio estimators for population variance were also considered by *Prasad* and *Singh* (1990, 1992), improving on *Isaki's* estimator with respect to bias and precision. In *Rueda* and *Arcos* (1995), another ratio estimator was introduced, which also improved on *Isaki's* estimator in that it is almost unbiased and more precise.

Assuming simple random sampling, the usual ratio estimator of variance of the main variable y can be generalized to a multivariate ratio estimator as per *Olkin* (1958) when multiple auxiliary information of variables $(x_1, \dots, x_k)$ is available. Along this line, *Isaki* (1983) proposed a natural extension of the univariate ratio estimator in the multivariate case assuming that the population variance of each component of the vector $(x_1, \dots, x_k)$ is known. *Isaki* also proposed a multivariate regression estimator of variance and showed its properties under a multivariate normal model in which the correlations among available variables are equal ($\rho_{x_i x_j} = \rho_{y x_i} = \rho$ $i, j = 1, \dots, k$).

In the minimization process of variance of this multivariate regression estimator, a system of equations is present. To solve it, since it depends on unknown parameters, *Isaki* defined a new estimator whose variance is the same as that of the optimum multivariate regression estimator (omitting certain terms).

In this paper we present a multivariate ratio estimator of population variance in simple random sampling with replacement, which has the same precision as the multivariate regression estimator. Thus, under the specific multivariate normal model proposed by *Isaki*, we need not know the population variance of the main variable to compute it because the estimator we propose only requires a knowledge of correlation among available variables and it is common to have information about these correlations in several practical situations (for example, repeated surveys).

2 Estimator Definition and Properties

Let $U = (U_1, \dots, U_N)$ be a finite population. Assume that a random sample with replacement of size n is selected, $(y_i, x_{1i}, \dots, x_{ki})$ $i = 1, \dots, n$ are observed, and $S_{x_1}^2, \dots, S_{x_k}^2$, the finite population variance of the variables $x_1, \dots, x_k$ are known.

The problem is to estimate

$$S_y^2 = \frac{N}{N-1} \sigma_y^2 \quad \text{with} \quad \sigma_y^2 = \frac{1}{N} \sum_{j=1}^N (y_j - \bar{Y})^2,$$

where $\bar{Y}$ is the population mean.

Define the estimator

$$\hat{S}_{\alpha_1, \dots, \alpha_k}^2 = s_y^2 \left(\frac{S_{x_1}^2}{s_{x_1}^2} \right)^{\alpha_1} \cdots \left(\frac{S_{x_k}^2}{s_{x_k}^2} \right)^{\alpha_k}, \quad (2.1)$$

where $\alpha_1, \dots, \alpha_k \in \mathbb{R}$, and s_y^2 and $s_{x_i}^2$ $i = 1, \dots, k$ are usual estimators of S_y^2 and $S_{x_i}^2$ $i = 1, \dots, k$, respectively.

We observe that:

- if $k = 1$, the estimator $\hat{S}_{\alpha_1, \dots, \alpha_k}^2$ is the estimator proposed by *Rueda* and *Arcos* (1995),
- if $\alpha_1 = \dots = \alpha_k = 0$ the estimator $\hat{S}_{\alpha_1, \dots, \alpha_k}^2$ is the simple expansion estimator, S_y^2 ,
- if $\alpha_i = 1$ and $\alpha_j = 0, \forall i \neq j$ the estimator $\hat{S}_{\alpha_1, \dots, \alpha_k}^2$ is the ratio estimator given in *Isaki* (1983).
- if $\alpha_i = -1$ and $\alpha_j = 0, \forall i \neq j$ the estimator $\hat{S}_{\alpha_1, \dots, \alpha_k}^2$ is the univariate product estimator with x_i .

It can be shown that in a first grade of approximation

$$\text{Bias}(\hat{S}_{\alpha_1, \dots, \alpha_k}^2) = \frac{S_y^2}{n} \left(\sum_{i=1}^k \alpha_i^2 (\beta_2(x_i) - 1) - \sum_{i=1}^k \alpha_i (\theta_{yx_i} - 1) + \sum_{i \neq j} \alpha_i \alpha_j (\theta_{x_i x_j} - 1) \right). \quad (2.2)$$

In a similar way

$$\begin{aligned} \text{MSE}(\hat{S}_{\alpha_1, \dots, \alpha_k}^2) = S_y^4 \frac{1}{n} & \left[\beta_2(y) - 1 + \sum_{i=1}^k \alpha_i^2 (\beta_2(x_i) - 1) \right. \\ & \left. - 2 \sum_{i=1}^k \alpha_i (\theta_{yx_i} - 1) + \sum_{i \neq j} \alpha_i \alpha_j (\theta_{x_i x_j} - 1) \right], \end{aligned} \quad (2.3)$$

where

$$\begin{aligned} \beta_2(y) &= \frac{\mu_{40}(y, y)}{\mu_{20}^2(y, y)}; & \beta_2(x_i) &= \frac{\mu_{04}(x_i, x_i)}{\mu_{02}^2(x_i, x_i)}, \\ \theta_{yx_i} &= \frac{\mu_{22}(y, x_i)}{\mu_{20}(y, x_i) \mu_{02}(y, x_i)}; & \theta_{x_i x_j} &= \frac{\mu_{22}(x_i, x_j)}{\mu_{20}(x_i, x_j) \mu_{02}(x_i, x_j)} \end{aligned}$$

and

$$\mu_{rs}(z, t) = \frac{1}{N} \sum (z_j - \bar{Z})^r (t_j - \bar{T})^s. \quad (2.4)$$

The expression of the bias and the mean squared error are obtained considering

the variables

$$e_0 = \frac{s_y^2 - S_y^2}{S_y^2} ; \quad e_i = \frac{s_{x_i}^2 - S_{x_i}^2}{S_{x_i}^2} \quad i = 1, \dots, k ,$$

by approximating $\hat{S}_{\alpha_1, \dots, \alpha_k}^2 - S_y^2$ by its series expansion (if $|e_i| < 1$, $|\alpha_i e_i| < 1$, $i = 1, \dots, k$), and considering that in a first grade of approximation we have (see *Kendall and Stuart* (1968))

$$V(s_y^2) = \frac{1}{n} S_y^4 (\beta_2(y) - 1) ; \quad V(s_{x_i}^2) = \frac{1}{n} S_{x_i}^4 (\beta_2(x_i) - 1) \quad i = 1, \dots, k ,$$

$$\text{Cov}(s_y^2, s_{x_i}^2) = \frac{S_y^2 S_{x_i}^2}{n} (\theta_{yx_i} - 1) \quad i = 1, \dots, k ,$$

$$\text{Cov}(s_{x_i}^2, s_{x_j}^2) = \frac{S_{x_i}^2 S_{x_j}^2}{n} (\theta_{x_i x_j} - 1) \quad i, j = 1, \dots, k .$$

We must now find the optimum $(\alpha_1, \dots, \alpha_k)$ to minimize (2.3). It follows from (2.3) that

$$\text{MSE}(\hat{S}_{\alpha_1, \dots, \alpha_k}^2) = S_y^4 \frac{1}{n} [\beta_2(y) - 1 + \alpha' A \alpha - 2\alpha' A_0] ,$$

where $\alpha = \alpha_{(k \times 1)} = (\alpha_1, \dots, \alpha_k)'$, $A = A_{(k \times k)} = (a_{ij})$ with $a_{ij} = (\theta_{x_i x_j} - 1)$ and $A_0 = A_{0(k \times 1)} = (a_{0_i})$ with $a_{0_i} = (\theta_{yx_i} - 1)$.

Assuming A^{-1} exists, the value

$$\alpha_{opt} = A^{-1} A_0 ,$$

minimizes $\text{MSE}(\hat{S}_{\alpha_1, \dots, \alpha_k}^2)$. For this α , $\hat{S}_{(\alpha_{opt})}^2$ is an almost unbiased estimator, as rewriting (2.2) we get

$$\begin{aligned} \text{Bias}(\hat{S}_{\alpha_{opt}}^2) &= S_y^2 \frac{1}{n} [\alpha'_{opt} A \alpha_{opt} - 2\alpha'_{opt} A_0] \\ &= S_y^2 \frac{1}{n} [A'_0 A^{-1} A A^{-1} A_0 - 2A'_0 A^{-1} A_0] = 0 , \end{aligned}$$

and its variance is given by

$$V(\hat{S}_{\alpha_{opt}}^2) = S_y^4 \frac{1}{n} [\beta_2(y) - 1 - A_0' A^{-1} A_0] .$$

For $k = 1$ the optimum value of α is

$$\alpha_0 = \frac{\theta_{yx_1} - 1}{\beta_2(x_1) - 1} ,$$

which yields an almost unbiased ratio estimator of population variance and which has, asymptotically, the same precision as the conventional univariate regression estimator by *Isaki* (1983), as can be seen in *Rueda* and *Arcos* (1995).

3 Comparison and the Multivariate Regression Estimator

The multivariate regression estimator established by *Isaki* (1983) is given by

$$\hat{S}_{MR}^2 = s_y^2 + \sum_{i=1}^k B_i (S_{x_i}^2 - s_{x_i}^2)$$

and it can be seen that

$$V(\hat{S}_{MR}^2) = V(s_y^2) + \sum_{i=1}^k B_i^2 V(s_{x_i}^2) - 2 \sum_{i=1}^k B_i \text{Cov}(s_y^2, s_{x_i}^2) + \sum_{i \neq j} B_i B_j \text{Cov}(s_{x_i}^2, s_{x_j}^2) . \quad (3.5)$$

The partial derivative of $V(\hat{S}_{MR}^2)$ with respect to B_i is

$$2B_i V(s_{x_i}^2) - 2 \text{Cov}(s_y^2, s_{x_i}^2) + 2 \sum_{i \neq j} B_j \text{Cov}(s_{x_i}^2, s_{x_j}^2) . \quad (3.6)$$

Given a sample from a multivariate normal population $(y, x_1, \dots, x_k)$, to solve the system of equations

$$\frac{\partial V(\hat{S}_{MR}^2)}{\partial B_i} = 0 , \quad i = 1, \dots, k , \quad (3.7)$$

Isaki proposed replacing the unknown terms such as $V(s_{x_i}^2)$, $\text{Cov}(s_y^2, s_{x_i}^2)$ and $\text{Cov}(s_{x_j}^2, s_{x_i}^2)$ by $(s_{x_i}^2)^2$, $(s_{y, x_i})^2$ and $(s_{x_i x_j})^2$, respectively. In this way, if $(\hat{B}_1, \dots, \hat{B}_k)$ is the solution of the modified system of equations, under a general multivariate normal model, the estimator

$$\hat{S}_{MRE}^2 = s_y^2 + \sum_{i=1}^k \hat{B}_i (S_{x_i}^2 - s_{x_i}^2) ,$$

omitting terms of $O(n^{-3/2})$, possesses the same variance as $\hat{S}_{MR}^2$ with optimal B_i .

Now consider the proposed estimator in (2.1). From

$$\begin{aligned} \text{MSE}(\hat{S}_{\alpha_1, \dots, \alpha_k}^2) = & \left[V(s_y^2) + \sum_{i=1}^k \alpha_i^2 V(s_{x_i}^2) \frac{S_y^4}{S_{x_i}^4} - 2 \sum_{i=1}^k \alpha_i \text{Cov}(s_{x_i}^2, s_y^2) \frac{S_y^2}{S_{x_i}^2} \right. \\ & \left. + \sum_{i \neq j} \alpha_i \alpha_j \frac{\text{Cov}(s_{x_j}^2, s_{x_i}^2)}{S_{x_j}^2 S_{x_i}^2} S_y^4 \right] , \end{aligned}$$

writing $\gamma_i = \alpha_i R_i$, $i = 1, \dots, k$ where $R_i = \frac{S_y^2}{S_{x_i}^2}$, we obtain

$$\begin{aligned} \text{MSE}(\hat{S}_{\alpha_1, \dots, \alpha_k}^2) = & \left[V(s_y^2) + \sum_{i=1}^k \gamma_i^2 V(s_{x_i}^2) - 2 \sum_{i=1}^k \gamma_i \text{Cov}(s_{x_i}^2, s_y^2) \right. \\ & \left. + \sum_{i \neq j} \gamma_i \gamma_j \text{Cov}(s_{x_j}^2, s_{x_i}^2) \right] . \end{aligned}$$

The partial derivative of $\text{MSE}(\hat{S}_{\alpha_1, \dots, \alpha_k}^2)$ with respect to γ_i yields the system

$$\gamma_i V(s_{x_i}^2) - \text{Cov}(s_y^2, s_{x_i}^2) + \sum_{i \neq j} \gamma_j \text{Cov}(s_{x_i}^2, s_{x_j}^2) = 0 , \quad i = 1, \dots, k , \quad (3.8)$$

which is identical to system (3.7). Thus, the optimum γ_i , that makes $\text{MSE}(\hat{S}_{\alpha_1, \dots, \alpha_k}^2)$ minimum, γ_i^0 , and the optimum B_i that makes $V(\hat{S}_{MR}^2)$ minimum, B_i^0 , are equal:

$$B_i^0 = \gamma_i^0 = \alpha_i^0 R_i , \quad i = 1, \dots, k . \quad (3.9)$$

In this fashion, from (3.5), omitting $O(n^{-1/2})$ terms,

$$\min V(\hat{S}_{MR}^2) = \min \text{MSE}(\hat{S}_{\alpha_1, \dots, \alpha_k}^2) ,$$

and therefore asymptotically the multivariate ratio estimator of population variance $\hat{S}_{\alpha_1, \dots, \alpha_k}^2$ has the same precision as the multivariate regression estimator $\hat{S}_{MR}^2$.

4 Estimator $\hat{S}_{\alpha_1, \dots, \alpha_k}^2$ Under the Specific Multivariate Normal Model Proposed by Isaki

Assuming that $(y, x_1, \dots, x_k)$ possesses the same moments as a $1 \times (k + 1)$ multivariate normal variable up to the fourth order and assuming also that

$$\rho_{x_i x_j} = \rho_{y x_i} = \rho, \quad \forall i, j \quad (4.10)$$

and $-\frac{1}{k} < \rho < 1$, $V(\hat{S}_{MR}^2)$ is minimized for

$$B_i^0 = \frac{\rho^2 R_i}{1 + (k - 1)\rho^2} \quad i = 1, 2, \dots, k.$$

From (3.9), $\text{MSE}(\hat{S}_{\alpha_1, \dots, \alpha_k}^2)$ is minimized for

$$\alpha(k, \rho^2) = \alpha_1^0 = \dots = \alpha_k^0 = \frac{\rho^2}{1 + (k - 1)\rho^2}.$$

In many practical situations we have information about ρ . Then, we can evaluate the estimator

$$\hat{S}_{\alpha_1, \dots, \alpha_k}^2 = s_y^2 \left(\frac{S_{x_1}^2}{s_{x_1}^2} \dots \frac{S_{x_k}^2}{s_{x_k}^2} \right)^{\alpha(k, \rho^2)},$$

so that the estimator is very easy to compute and its precision increases until to be equal to that of the precision of the multivariate regression estimator, since, omitting $O(n^{-1/2})$ terms, we get

$$\text{MSE}(\hat{S}_{\alpha_1, \dots, \alpha_k}^2) = V(\hat{S}_{MR}^2) = \frac{2S_y^4}{n - 1} \left[1 - \frac{k\rho^4}{1 + (k - 1)\rho^2} \right].$$

It is common to have information about the correlation in several practical situations and in this case we can easily compute the proposed estimator and its precision.

References

- Chaudhuri A, Vos JWE (1988) Unified theory and strategies of survey sampling. North-Holland
- Cochran WG (1977) Sampling techniques, 3rd ed. New York Wiley
- Isaki CT (1983) Variance estimation using auxiliary information. *Journal of the American Statistical Association* 78 381: 117–123
- Kendall M, Stuart A (1976) The advanced theory of statistics. Vol 3, 3rd ed. London Griffin
- Krisnaiah PR, Rao CR (1988) Handbook of statistics Vol 6. North-Holland
- Olkin I (1958) Multivariate ratio estimation for finite population. *Biometrika* 45: 154–165
- Prasad B, Singh HP (1990) Some improved ratio-type estimators of finite population variance in sample surveys. *Communications in Statistics-Theory and Methods* 19 3: 1127–1139
- Prasad B, Singh HP (1992) Unbiased estimators of finite population variance using auxiliary information in sample surveys. *Communications in Statistics-Theory and Methods* 21 5: 1367–1376
- Rueda M, Arcos A (1996) Repeat substitution method: The ratio estimator for the population variance. *Metrika* 43: 101–105
- Srivastava SK (1967) An estimator using auxiliary information in sample surveys. *Calcutta Statistical Association Bulletin* 16: 121–132

Received: 26.07.1995

Book Reviews

Contents

M. A. Lifshits: Gaussian Random Functions (N. Obata)	179
D. Nualart: The Malliavin Calculus and Related Topics (L. Arnold) ...	181
R. J. Carroll, D. Ruppert, L. A. Stefanski: Measurement Error in Nonlinear Models (J. P. Buonaccorsi)	182
M. J. Schervish: Theory of Statistics (H. Witting)	184
H. Wackernagel: Multivariate Geostatistics (L. M. Bloom)	185
C. R. Rao: Was ist Zufall? Statistik und Wahrheit (J. Stark)	187
J. S. Maritz: Distribution-free Statistical Methods (E. von Collani)	188

M. A. Lifshits: Gaussian Random Functions, Kluwer Academic Publ. Dordrecht, Boston, London, 1995, XI/333 pp., Dfl245.00/US\$157.00/UK£100.00

Reviewed by N. Obata, Nagoya

This volume is intended as a coherent textbook on Gaussian random functions (the term “Gaussian random function” is used synonymously with “Gaussian family” or “Gaussian system”) with special emphasis on the convexity of measures and isoperimetric inequalities. Such geometric consideration is one of the attractive features of this book. It consists of 19 sections, each of which has reasonable length to read at a stretch and contains a few exercises, though Sect. 19 is devoted to a list of open problems. In addition, the supplementary section of Comments at the end of the book provides precise bibliographical remarks which will be extremely useful for the readers who wish to study further. The readers are presumed to have at least introductory knowledge on measure theory, probability theory, linear algebra and general topology.

Here is a brief sketch of the contents:

Background: Sections 1–5 are devoted to assembling the most basic properties of Gaussian distributions and Gaussian random functions, e.g., Fourier transform, Laplace transform, covariance, existence of a Gaussian random function and some examples including the Wiener process, multi-parametric Lévy’s Brownian motion, fractional Brownian motion and so forth.

Models of Covariant Functions: Covariant functions are of great importance as they determine a Gaussian random function completely. Section 6 discusses integral geometric realizations of covariance functions. This idea traces back to models of the Wiener process and the Wiener–Chentsov field.

Oscillations: In general, a sample function of a random function is very irregular and its oscillation becomes a random variable. For a Gaussian function, however, the oscillation function becomes nonrandom and this property is essential in various contexts. Section 7 discusses a theorem of Itô and Nisio and a few generalizations and related results.

Infinite Dimensional Gaussian Distributions: A Gaussian random function naturally induces a probability measure on an infinite dimensional vector space. In Sects. 8–10 the Gaussian distributions are defined in an (infinite dimensional) locally convex vector space, and several basic properties are mentioned, e.g., the Cameron–Martin formula, zero-one law for invariant subsets. The infinite product of Gaussian measures and the Wiener measures are also discussed.

Isoperimetric Property and Convexity: As is well known, it is a ball in the Euclidean space which has a least surface area among the sets of equal volume with respect to the Lebesgue measure. In Sect. 11 it is proved that a half plane is the solution to a parallel question when the Lebesgue measure is replaced with the Gaussian measure. Moreover, the convexity of a Gaussian measure (the Ehrhard inequality and related results) is discussed in detail. The proofs are based upon geometric consideration such as the Gaussian-symmetrization procedure.

Large Deviations and Small Deviations: The large deviation principle traces back to Varadhan and has been proved for various families of probability measures. Section 12 studies the large deviation for a Gaussian random function. For example, it is proved that the logarithmic asymptotics of the probabilities of large deviations is determined by the minimum of action functional (entropy function). In Sect. 13 the exact asymptotics is obtained by using the Laplace transform and in Sect. 14 other types of asymptotics are investigated by means of metric entropy and the comparison principle. In Sect. 18 the asymptotic behavior of small deviations is studied; however, the formalism for this question has not yet been discovered.

Regularity of Sample Functions: With the techniques and results established during the study of isoperimetric inequalities and the large deviation principle, regularity of a sample function of a Gaussian random function is studied in detail. For instance, the continuity and boundedness in terms of metric entropy in Sect. 15; the boundedness in terms of majorizing measures in Sect. 16; functional versions of the law of the iterated logarithm as a beautiful application of the isoperimetric inequality in Sect. 17.

D. Nualart: The Malliavin Calculus and Related Topics, Springer-Verlag New York, 1995, XI/266 pp., DM68.00/öS530.40/sFr65.50

Reviewed by L. Arnold, Bremen

The *Malliavin calculus* (some people prefer the name *stochastic calculus of variations*) is an infinite-dimensional calculus on the Wiener space, the canonical space of Brownian motion, the most basic object of modern probability theory. This calculus was developed by Malliavin, Krée, Stroock, Bismut, Watanabe, and others in the late 1970s and early 1980s. The original challenge was to give a probabilistic proof of *Hörmander's theorem*: Given a stochastic differential equation

$$dx = f_0(x) dt + \sum_{i=1}^m f_i(x) \circ dW_t^i, \quad x_0 = x \in \mathbb{R}^d,$$

where $f_0, \dots, f_m$ are smooth vector fields in $\mathbb{R}^d$. Let $\mathcal{L}$ be the Lie algebra of vector fields generated by $f_0, \dots, f_m$, and let $\mathcal{I} \subset \mathcal{L}$ be the ideal in $\mathcal{L}$ generated by the diffusion vector fields $f_1, \dots, f_m$. Then Hörmander's theorem states that if $\mathcal{I}(x) = \mathbb{R}^d$ for any $x \in \mathbb{R}^d$, then the probability distribution of the solution x_t has a smooth density.

In the course of time, many more applications of the calculus were discovered and developed, notably to anticipating problems which naturally arise if one considers stochastic differential equations with boundary conditions or non-adapted initial conditions. A whole catalog of anticipating problems appears in the intersection of stochastic analysis and dynamical systems, due to the *multiplicative ergodic theorem* of Oseledets.

All these developments made Malliavin calculus an established and significant part of modern probability theory.

The early systematic treatments by Bismut (1984), Watanabe (1984), Ocone (1987), and Bell (1987), though very useful and important at their time, were not yet of the kind one could easily base a course on (the reviewer has tried it). David Nualart's book is the first and badly needed book on Malliavin's calculus which can serve as a textbook. And according to the reviewer's information, it does so already throughout the world.

The book is less than 240 pages long and consists of four chapters:

- Chapter 1: Analysis on Wiener space (the core of the calculus),*
- Chapter 2: Smoothness of probability laws (the classical Hörmander's theorem, plus the author's own research on stochastic partial differential equations),*
- Chapter 3: Anticipating stochastic calculus (anticipating Itô's formula, substitution formulas, anticipating stochastic differential equations),*
- Chapter 4: Transformations of the Wiener measure (anticipating Girsanov's theorem, Markov random fields).*

This is a good and carefully written book (it had 10 years time to mature). Chapter 1, plus selected parts of the other chapters, can serve well as the basis of a graduate course (assuming basic knowledge of stochastic analysis).

R. J. Carroll, D. Ruppert, L. A. Stefanski: Measurement Error in Nonlinear Models, Chapman & Hall London, 1995, XXI/305 pp., £29.95

Reviewed by J. P. Buonaccorsi, Amherst

With the huge amount of recent and continuing work on measurement error problems, this book will be highly welcomed and appreciated by those with an interest in such problems. The authors do a nice job of providing an up to date and broad look at methods for dealing with measurement error. Just how modern this book is is reflected by the high proportion of references from the past 10 years with many from the last few years.

The main focus of the book is on problems with measurement error in the predictors in regression models. As the title indicates nonlinear regression models are of special concern, and the authors discuss a variety of methods applicable to a broad class of problems. Generalized linear models and, more specifically, logistic regression receive some special attention, but the examples are by no means limited to these models.

The book begins with an extremely well-written general overview of the problems faced with measurement error in the predictors, with seven motivating examples. This is followed by a chapter providing a brief survey of the bias resulting from naive analyses (which use the measured predictors containing error) along with a summary of some options to correct for biases in linear models. A more complete treatment of measurement error in linear models is given by Fuller (1987).

The main body of the book (Chaps. 3–9) presents a variety of methods to correct for measurement error. These utilize additional data allowing one to estimate the parameters in the measurement error model and the methods presented accounting for uncertainty arising from this estimation. Much of this work is new, having been developed over the past 10 years. The coverage includes regression calibration methods, simulation–extrapolation methods, the use of instrumental variables, functional methods, and techniques (including Bayesian methods) which invoke partial or full distributional assumptions. Of these I found the three most extensive chapters (on regression calibration, simulation extrapolation, and likelihood/quasilielihood methods) to also be the best in terms of presentation and interesting illustrations.

The remaining chapters cover miscellaneous topics including estimation with unknown link functions, issues in hypothesis testing, problems with measurement error in the response, density estimation and nonparametric regression, and a chapter entitled “*Other Topics*”, providing a brief look at various issues

ranging from case-control studies to survival analysis. This last chapter, as well as some of the preceding shorter chapters, provide a quicker look at these topics and are not quite as comprehensive as some of the earlier chapters.

An appendix to the book provides a survey of results on likelihood and quasilielihood methods, the use of estimating equations, generalized linear models, and bootstrapping. This appendix, which is very readable, is especially valuable for those approaching the book with a more limited background.

The general approach of the book is to first present a discussion of modelling, approaches to inference, and a worked example at a level which is accessible to a wide audience. The more advanced material is typically relegated to appendices or later subsections. This approach allows the reader with more applied interests to get a general appreciation for the models and methods. While this works rather well for the most important sections of the book, there are some sections which make for more challenging reading throughout. Ignoring the more theoretical developments, an appreciation of the book requires some background in probability, mathematical statistics, random vectors, and covariance matrices and prior exposure to regression and generalized linear models. To fully appreciate all of the book requires a fairly advanced background, I would say at the Ph.D. level.

I found very few typographical errors in the book. Unfortunately there is one on line 9 of the first chapter (which should read "fitting Y to (Z, W) " rather than (Z, X)) which will confuse the reader new to the subject. As with other books in this series, I like the typesetting, but I fail to understand how the editors allowed the regular occurrence of incorrectly hyphenated words; e.g., efficien-t.

Overall, I like this book and highly recommend it to anyone with an interest in measurement error problems. It would serve well as a textbook for a course, probably most suited to Ph.D. students. Many of the issues faced in treating measurement error problems are those faced in other areas of statistical analyses. A course from this book would be valuable both in addressing an important class of problems and in providing the student with some general tools.

Reference

Fuller WA (1987) Measurement error models. Wiley, New York

M. J. Schervish: Theory of Statistics, Springer Series in Statistics, Springer-Verlag, 1995, 702 pp., DM88.00/öS642.40/sFr84.50

Reviewed by H. Witting, Freiburg

Typically, textbooks of statistics either ignore the Bayesian point of view to a large extent, or they cover Bayesian topics almost exclusively. Though the author is a Bayesian, the book also discusses the elements of traditional theory, for instance the Neyman-Pearson theory of testing statistical hypotheses and the basic results in estimation like the Lehmann-Scheffé theorem or the Cramér-Rao-Bhattacharyya lower bounds. With respect to Bayesian statistics it stresses more the mathematical aspects than the philosophical ones. What makes the book attractive is the combination of Bayesian and classical results (with model assumptions adapted to the more general Bayesian framework), and the inclusion of a large number of instructive examples and counterexamples. Beyond this, the tight organization of the book, the care in choosing the terminology, and the accuracy with respect to the cross-references should be mentioned explicitly. For the reader's convenience, several results are repeated when they are used in later chapters. Almost one-third of the sections are marked to indicate that, while important, they can be skipped without interrupting the flow of ideas.

The contents of the book may be outlined by means of the headings of the nine chapters supplemented by some annotations.

Chapter 1: Probability Models (81 pages): Elements of classical and Bayesian statistics, parametric models, exchangeability, De Finetti's representation theorem, infinite dimensional parameters.

Chapter 2: Sufficient Statistics (62 pages): Concept of sufficiency from the Bayesian and non-Bayesian viewpoint; factorization theorem, minimal and complete sufficiency, ancillarity, exponential families, information concepts, Jeffreys' prior.

Chapter 3: Decision Theory (70 pages): Elements of Bayesian and Classical Decision Theory (admissibility, James-Stein estimators, minimax rules, complete classes).

Chapter 4: Hypothesis Testing (82 pages): Significance tests and P -values, Bayes solutions and Bayes factors, most powerful (unbiased) tests, nuisance parameters and Neyman structure test.

Chapter 5: Estimating Theory (48 pages): Lehmann-Scheffé theorem, Cramér-Rao- and Bhattacharyya lower bounds, elements of maximum-likelihood, Bayesian and robust estimation, confidence sets, bootstrap.

Chapter 6: Equivariance (50 pages): Location and scale problems, groups of transformations (acting on the sample space, parameter space, and action space), minimum risk equivariant (Pitman's) estimator, Haar measures; invariant tests and equivariant confidence sets.

Chapter 7: Large Sample Theory (82 pages): Convergence in probability and

convergence in distribution, Landau's O_p - and o_p -symbols, Slutsky's theorem and δ -method, large sample properties of sample quantiles, maximum-likelihood estimators, posterior distributions, likelihood ratio tests and χ^2 tests.

Chapter 8: Hierarchical Models (60 pages): Modelling of the parameters, hyper-parameter, partial exchangeability, one-way ANOVA, two-way mixed model ANOVA, nonnormal models, empirical Bayes, mixture of models, application to robustness.

Chapter 9: Sequential Analysis (34 pages): Sequential decision rules, optimality after stopping, comparison with fixed sample size procedures, sequential probability ratio test, optimality of tests.

Every chapter includes problems which range in difficulty from the easy to the challenging. The book also contains a list of about 250 references as well as a name index and a subject index. The basic tools from measure theory, integration and probability (including conditioning and martingale theory) are sketched in the appendix (91 pages). Finally, there are listings of the employed theorems from real, complex and functional analysis and listings of symbols used.

According to its introductory character the book is neither abstract in style nor does it bring more advanced results like the Hunt-Stein theorem, Le Cam's theory of asymptotical optimal tests or Hájek's convolution theorem. In addition, neither nonparametric tests nor, for instance, density estimators are discussed. Summarizing, the book can be strongly recommended as a textbook for a graduate course. It enables the instructor to make some selections or to refer to the book for proofs and supplementary readings. It is also well-suited for students who wish to learn the basics of modern mathematical statistics by private study.

H. Wackernagel: Multivariate Geostatistics, Springer-Verlag Berlin, ISBN 3-540-60127, 256 pp., DM75.00/öS540.20/sFr71.50

Reviewed by L. M. Bloom, Perth

This book is a very welcome addition to the surprisingly sparse collection of geostatistics texts. Its stated aim is to introduce geostatistics from a multivariate perspective, and in this it succeeds very well. The book itself stems from recent courses on the topic given by the author in Paris and Seattle, and in fact each of the 29 (relatively short) main chapters is intended to be regarded essentially as the text of a two-hour lecture.

The book contains five main sections together with an Introduction and an Appendix containing a range of topics. Two sections are concerned with summaries of standard concepts and techniques. *Section A: Preliminaries* (one chapter) covers standard concepts such as mean, variance, covariance and multiple linear regression, before extending the last of these into a spatial context as simple kriging. *Section C: Multivariate Analysis* (three chapters) covers the

standard multivariate statistical techniques of principal component analysis, canonical correlation analysis and correspondence analysis.

Section B: Geostatistics (fourteen chapters) provides a comprehensive overview of univariate linear geostatistics. This includes details of the random function model and the concept of stationarity, variograms, covariance functions, isotropy and anisotropy, ordinary kriging, together with its extension to a more general linear model of regionalization, and a discussion of the problem of change of support. *Section D:* Multivariate Geostatistics (eight chapters) is the main focus of the book and considers the extension of univariate linear geostatistics to the multivariate case. It proceeds from discussion of cross variograms and cross covariance functions to the linear and bilinear coregionalization models. This section includes a study of cokriging, including the cokriging of a complex variable, and details of how and when cokriging constitutes an advantage over the direct kriging of the individual variables.

Section E: Nonstationary Geostatistics (three chapters) considers the situation where the data exhibit a trend and includes a summary of universal kriging, a definition of the generalized covariance function and an overview of kriging in the presence of external drift, the special case of universal kriging where the drift variables used are not translation-invariant and need to be known at least at the sample locations of the main variable as well as at the nodes of the interpolation grid.

Exercises are included throughout the text (although not a large number and not on any regular basis) as well as in one section of the Appendix, and solutions to these are provided. The author also provides a very brief indication of some of the software packages available in this area. Three commercial packages are listed by name, but no details are given as to which topics are covered, the price range of these packages, or their overall quality and user-friendliness.

One consequence of this book arising as a set of lecture notes is that the coverage given is necessarily brief. One often feels that the exposition could be enhanced by a more extensive treatment with more of the detail being supplied, even without needing to attempt the theoretical comprehensiveness of the book by Cressie (1991). Several real applications are considered throughout. These include acoustic data, arsenic in soil data, topographic data and temperature data, and serve to illustrate the relevant concepts involved. However, the actual data are not provided in the text and the discussion itself does not dwell on the practicalities of the data analysis, in contrast say, to the book by Isaaks and Srivastava (1989). The exposition would be greatly enhanced by more detail in this area. In essence one might wish for the tutorials and the workshops as well as the lectures, and no doubt these would have accompanied the original courses.

There are a few misprints in the text but these are minor and do not detract from the material itself. Taken as a whole, this book is an insightful introduction to linear geostatistics in general and multivariate geostatistics in particular, and provides a springboard to further study and research in this very interesting area.

References

Cressie N (1991) Statistics for spatial data. Wiley, New York

Isaaks E, Srivastava R (1989) Applied geostatistics. Oxford University Press, New York

C. R. Rao: Was ist Zufall? Statistik und Wahrheit, Prentice Hall Munich, 1995, 244 pp., DM79.00/sFr74.00

Reviewed by J. Stark, Tübingen

The title of the American original version of this book is "*Statistics and Truth, Putting Chance to Work: Ramanujan Memorial Lectures*." This exact title shows that the book is based on a collection of lectures given by Rao in India, on the occasion of the centenary celebration in honour of Ramanujan. In the translation under review, which is based on the second edition of the American original version, the comments on the life and the work of the great mathematical genius Ramanujan are summarized in an introduction.

Chapter 1 ("*Unsicherheit, Zufälligkeit und Erschaffen neuen Wissens*") deals with the topic of the uncertainty met everywhere in our real world and its quantification. It is fascinating to read how Rao describes the change in our conception of the phenomenon of chance: originally seen as an unremovable obstacle, chance is today regarded more as an indispensable means of progressive development in our society. In Chap. 2, the author briefly surveys the origin of statistics, describes current developments, and speculates about its future. Passing on statistical concepts to researchers in other fields is considered one of the important tasks of the future. After a survey of the historical development of data analysis, Chap. 3 ("*Prinzipien und Strategien der Datenanalyse: Kreuzuntersuchung der Daten*") presents a new concept, which combines the ideas from K. Pearson, R. A. Fisher, and J. Tukey. Particular emphasis is laid on the analysis of data under various possible models.

Chapter 4 ("*Prinzipien und Strategien der Datenanalyse: Gewichtete und bewölkte Verteilungen*") is mathematically demanding. It describes the general theory of weighted distributions including some recent developments. Chapter 5 ("*Statistik: Ein unvermeidliches Instrument auf der Suche nach Wahrheit*") resumes the concepts concerning the extraction of information from available data. The author discusses the role of statistics in the acquisition of knowledge and in optimal decision-making under uncertainty. Chapter 6 ("*Verständnis der Statistik in der Öffentlichkeit oder Lernen von Zahlen*") is devoted to the omnipresence of statistical statements in newspapers, magazines, and other news media. As caution is often necessary in this special field, Rao pleads for the introduction of a fourth basic discipline at school, namely "statistical argumentation", to be taught at an early stage, along with reading, writing, and arithmetic.

The book under review captivates the reader. It impresses by its mostly

graphic description using many examples. Written for a large public by an author who has been substantially involved in developing fundamental concepts in statistics, the book has set an example which should be followed in the German-speaking world.

J. S. Maritz: *Distribution-free Statistical Methods*, Chapman & Hall, London, ISBN 0412552604, 1995, XII/255 pp.

Reviewed by E. von Collani, Würzburg

The second edition of Maritz's monograph that was originally published in 1981 entitled "*Distribution-free statistical methods*" has been carefully revised and, in spite of the fact that the number of pages decreased by 9, the amount of material presented has been enlarged due to the use of a slightly smaller type size. Compared with the first edition, the preface and the author index are omitted, the number of subsections enlarged, some new examples have been included, and an eighth chapter "*Miscellaneous complements*" has been added which contains Appendices A and B of the first edition. Last but not least, there are computational notes with references to statistical software packages, particularly MINITAB.

After a brief but comprehensive introduction to the basic concepts in distribution-free methods, the following cases are investigated in detail: one and two-sample problems, straight line and multiple regression, general linear models, and bivariate problems. Besides hypothesis testing, point and interval estimation is one of the major topics. The concept of estimating equations is introduced, and subsequently the method of M-estimation further elaborated. Another useful topic are efficiency considerations of the presented methods. In the last chapter, complementary remarks are made on linearization techniques, asymptotic relative efficiency, smoothing methods, kernel, and bootstrap estimation, and the concept of conditional standard error.

This is a very useful book for anyone interested in statistical data analysis; moreover it can serve as a suitable textbook for an introductory course in distribution-free methods. The mathematical formalism is kept to a justifiable extent, new concepts and methods are clearly introduced and illustrated by simple and instructive examples. Although the two editions cover more or less the same topics, the second edition would still be worthwhile purchasing even or perhaps especially if the first edition had been studied previously.

Distributions Characterized Through Conditional Expectations

K. BALASUBRAMANIAN AND ALOKE DEY

Indian Statistical Institute, Stat/Math Unit, 7, S.J.S. Sansanwal Marg, New Delhi 110016, India

Abstract: Using conditional expectations, we present results that lead to the characterization of several distributions. Both absolutely continuous random variables and discrete random variables are considered. In the case of absolutely continuous random variables, the results lead to the characterization of a family of distributions while in the case of discrete random variables, the distribution is almost uniquely determined under the stated conditions.

Key Words: Characterization, Order statistics

1 Introduction

Characterizations of distributions using conditional expectations have been studied by several authors. Let $X_1, X_2, \dots, X_n$ be a random sample of a random variable X with a continuous distribution function F which is strictly increasing over (a, b) , $-\infty \leq a < b \leq \infty$, the support of F , and $X_{1:n} \leq X_{2:n} \leq \dots \leq X_{n:n}$, the corresponding order statistics. Ferguson (1967) characterized distributions using the condition $E(X_{i:n} | X_{i+1:n} = x) = \alpha x - \beta$, while Beg and Kirmani (1974) used the condition $E(X_i | X_{n:n} = x) = \alpha x - \beta$ to characterize distributions. Here α and β are some constants. Related results can be found in Galambos and Kotz (1978), Azlarov and Volodin (1986), Beg and Balasubramanian (1990) and Balasubramanian and Beg (1992). Let g be a nonconstant continuous function over (a, b) with finite $g(a+)$ and finite expectation, $E[g(X)]$. By a suitable choice of g , Berg and Balasubramanian (1990) were able to characterize all distributions for which the explicit form of the distribution function is known, continuous and strictly increasing in its support (a, b) , through the property,

$$E \left\{ \frac{1}{s-1} \Sigma g(X_{i:n}) | X_{s:n} = x \right\} = \frac{g(x) + g(a+)}{2} \quad \forall x \in (a, b) .$$

Balasubramanian and Beg (1992) subsequently characterized distributions by conditioning on a pair of order statistics. In the last two papers the

characterizations were through the arithmetic means of the function at the end points.

In this communication, we present certain results based on conditional expectations that lead to characterizations of several distributions. We use harmonic and geometric means of the functions at the end points to arrive at our results. Specifically, we prove the following main results:

- (i) If X is a random variable with cumulative distribution function (cdf) $F(x)$ and density function $f(x)$, which is continuous in the support of the density, then

$$E[g(X)|x \leq X \leq y] = \frac{2g(x)g(y)}{g(x) + g(y)}, \quad \forall x, y, x \leq y$$

if and only if $g(x) = \sqrt{\left(\frac{b}{F(x) + a}\right)}$, where g is a positive function with continuous derivative and a and b are constants.

- (ii) If X and g are as in (i) above, then

$$E[g(X)|x \leq X \leq y] = \sqrt{\{g(x)g(y)\}}, \quad \forall x, y, x \leq y$$

if and only if

$$g(x) = \frac{b^2}{(F(x) - a)^2}.$$

- (iii) If X is a discrete random variable taking nonnegative integral values over the interval $[u, v]$ and g be a positive function with $g(i) \neq g(i + 1) \forall i \in [u, v]$, then

$$E[g(X)|a \leq X \leq b] = \frac{2g(a)g(b)}{g(a) + g(b)}, \quad \forall a, b, a \leq b$$

if and only if $\Pr(X = i) = p_i = \alpha + \beta i$ for $u \leq i \leq v$, $g(i) = c/p_i \forall i \in [u, v]$ and c is some constant.

- (iv) with X and g as in (iii) above,

$$E[g(X)|a \leq X \leq b] = \sqrt{\{g(a)g(b)\}}, \quad \forall a, b, a \leq b$$

if and only if $\Pr(X = i) = p_i = \alpha r^{i-1}$ and $g(i) = c/p_i^2$, where $r = p_{i+1}/p_i$, $i \in [u, v]$, the support of X .

The above results lead to characterizations of several continuous and discrete distributions. The proofs of the above stated results are given in Section 2. In Section 3, we give some characterizations.

2 Proofs of the Results

Theorem 1: Let X be a random variable with c.d.f. $F(x)$ and density function $f(x)$ and let $g(x)$ be continuous in the support of the density. Then

$$E[g(X)|x \leq X \leq y] = \frac{2g(x)g(y)}{g(x) + g(y)}, \quad \forall x, y \text{ } x \leq y, \quad (2.1)$$

if and only if $g(x) = \sqrt{\left(\frac{b}{F(x) + a}\right)}$, where $g(\cdot)$ is a positive function with continuous derivative and a and b are constants.

Proof: Observe that if $g(\cdot)$ is a positive function with continuous derivative then

$$E[g(X)|x \leq X \leq y] = \int_x^y \frac{g(t)f(t) dt}{F(y) - F(x)}. \quad (2.2)$$

Suppose (2.1) holds. Then equating (2.1) and (2.2), we have

$$\int_x^y g(t)f(t) dt = \{F(y) - F(x)\} \frac{2g(x)g(y)}{g(x) + g(y)}.$$

Differentiating both sides w.r.t. y , we have

$$\begin{aligned} g(y)f(y) &= \frac{\partial}{\partial y} \left[\{F(y) - F(x)\} \frac{2g(x)g(y)}{g(x) + g(y)} \right] \\ &= \frac{2\{F(y) - F(x)\}}{\{1 + g(y)/g(x)\}^2} g'(y) + \frac{2f(y)}{1/g(x) + 1/g(y)}, \end{aligned}$$

which yields

$$\frac{f(y)}{F(y) - F(x)} = \frac{2\{g(x)\}^2 g'(y)}{g(y)[\{g(y)\}^2 - \{g(x)\}^2]} \quad (2.3)$$

Note that the l.h.s. of (2.3) is the partial derivative w.r.t. y of $\ln[F(y) - F(x)]$ and the r.h.s., that of $\ln[1 - (g(x)/g(y))^2]$. Consequently,

$$F(y) - F(x) = k \left(1 - \left(\frac{g(x)}{g(y)} \right)^2 \right),$$

k being an arbitrary constant. This gives

$$g(y) = \sqrt{\left(\frac{b}{F(y) + a} \right)} \quad (2.4)$$

where a, b are constants such that $b/\{F(y) + a\} > 0$.

Conversely, suppose (2.4) holds. Then

$$\begin{aligned} \int_x^y \frac{g(t)f(t) dt}{F(y) - F(x)} &= \frac{1}{F(y) - F(x)} \int_x^y \sqrt{\left(\frac{b}{F(t) + a} \right)} f(t) dt \\ &= \frac{2g(x)g(y)}{g(x) + g(y)}. \quad \blacksquare \end{aligned}$$

Suppose $X_1, X_2, \dots, X_n$ is a sample from the random variable X , defined in Theorem 1 and let $X_{1:n} \leq X_{2:n} \leq \dots \leq X_{n:n}$ be the order statistics. Then, it is known that the conditional distribution of $X_{r+1:n}, X_{r+2:n}, \dots, X_{s-1:n}$ given $X_{r:n} = x, X_{s:n} = y$ ($r < s$) is the same as the distribution of $Y_{1:s-r-1}, Y_{2:s-r-1}, \dots, Y_{s-r-1:s-r-1}$ where $Y_{i:s-r-1}$ is the i th order statistic based on a sample of size $(s - r - 1)$ from the truncated distribution $\{X|x \leq X \leq y\}$. Using this result, we have the following corollary to Theorem 1.

Corollary 1: Let X and $g(\cdot)$ be as in Theorem 1. Then

$$E \left[\sum_{i=r+1}^{s-1} g(X_{i:n}) / (s - r - 1) | X_{r:n}, X_{s:n} \right] = \frac{2g(X_{r:n})g(X_{s:n})}{g(X_{r:n}) + g(X_{s:n})}$$

if and only if $g(x) = \sqrt{\left(\frac{b}{F(x) + a} \right)}$.

On similar lines, one can prove the following results.

Theorem 2: Let X and $g(\cdot)$ be as in Theorem 1. Then

$$E[g(X)|x \leq X \leq y] = \sqrt{\{g(x)g(y)\}} \quad (2.5)$$

if and only if

$$g(x) = \frac{b^2}{(F(x) - a)^2} \quad (2.6)$$

where a and b are constants.

Corollary 2: Let X and $g(\cdot)$ be as in Theorem 1. Then

$$E\left(\sum_{i=r+1}^{s-1} g(X_{i:n})/(s-r-1) | X_{r:n}, X_{s:n}\right) = \sqrt{g(X_{r:n})g(X_{s:n})} \quad (2.7)$$

if and only if (2.6) holds.

Remark: The result in Corollary 1 characterizes several distributions. This result can be used to test the hypothesis that the sample is from a known population. For example, if $F_0(x) = b/\{g(x)\}^2 - a$, then the hypothesis that the observations are from the distribution $F_0(x)$ can be tested by considering the difference

$$T = \sum_{i=r+1}^{s-1} g(X_{i:n})/(s-r-1) - \frac{2g(X_{r:n})g(X_{s:n})}{g(X_{r:n}) + g(X_{s:n})}.$$

Under the hypothesis, $E(T) = 0$. Hence large values of $|T|$ lead to the rejection of the hypothesis. As this test is based on a characterization property, it is likely to have large power. This inferential aspect will be dealt in a separate paper. Corollary 2 can also be used similarly. This inferential aspect makes the characterizations very useful. ■

Now suppose that X is a discrete random variable taking nonnegative integral values in the interval $[u, v]$, $v < \infty$. Suppose $g(\cdot)$ is a positive function such that $g(i) \neq g(i+1) \forall i \in [u, v]$. Then, we have

Theorem 3: With X and $g(\cdot)$ as above,

$$E[g(X)|a \leq X \leq b] = \frac{2g(a)g(b)}{g(a) + g(b)}, \quad \forall a, b \text{ } a \leq b, \quad (2.8)$$

if and only if $\Pr(X = i) = p_i = \alpha + \beta i$ and $g(i) = c/p_i \forall i \in [u, v]$, where c is some constant.

Proof: To begin with, let us fix $a = i, b = i + 1$ for some $i \in [u, v]$. Then,

$$\begin{aligned} E[g(X)|i \leq X \leq i + 1] &= \frac{g(i)p_i + g(i + 1)p_{i+1}}{p_i + p_{i+1}} \\ &= \frac{2g(i)g(i + 1)}{g(i) + g(i + 1)}, \quad \text{from (2.8)}. \end{aligned}$$

This implies that $g(i)p_i = g(i + 1)p_{i+1}$, provided $g(i) \neq g(i + 1)$.

$$\Rightarrow g(i) = c/p_i. \quad (2.9)$$

Again, let $a = i, b = i + 2$, for some $i \in [u, v]$. Then

$$\begin{aligned} E[g(X)|i \leq X \leq i + 2] &= \frac{g(i)p_i + g(i + 1)p_{i+1} + g(i + 2)p_{i+2}}{p_i + p_{i+1} + p_{i+2}} \\ &= \frac{2g(i)g(i + 2)}{g(i) + g(i + 2)}, \quad \text{from (2.8)}. \end{aligned}$$

From (2.9), we get, therefore

$$p_{i+1} = \frac{p_i + p_{i+2}}{2} \quad (2.10)$$

which implies that p_i 's are in arithmetic progression, say $p_i = \alpha + \beta i \forall i \in [u, v]$.

It is now easy to verify that for $g(i)$ given by (2.9) and p_i 's satisfying (2.10),

$$E(g(X)|a \leq X \leq b) = \frac{2g(a)g(b)}{g(a) + g(b)}. \quad \blacksquare$$

On similar lines, one can prove the following:

Theorem 4: Suppose X is a discrete random variable taking nonnegative integral values in the interval $[u, v]$, $v < \infty$ and $g(\cdot)$ is a positive function such that

$g(i) \neq g(i + 1) \forall i \in [u, v]$. Then

$$E[g(X)|a \leq X \leq b] = \sqrt{\{g(a)g(b)\}} , \quad \forall a, b \text{ } a \leq b$$

if and only if $\Pr(X = i) = p_i = \alpha r^{i-1}$ and $g(x) = c/p_i^2 \forall i \in [u, v]$, where α and c are constants and $r = p_i/p_{i+1}$.

3 Characterization of Distributions

We give two examples to illustrate how the results of the previous section help in characterization of distributions.

1. Distributions with c.d.f. of the form

$$F(x) = Ax^k + B \tag{3.1}$$

are characterized by any one of the following two conditions:-

$$\begin{aligned} \text{(a)} \quad E \left\{ \frac{1}{(s-r-1)} \sum_{i=r+1}^{s-1} \frac{1}{X_{i:n}^{k/2}} \middle| X_{r:n}, X_{s:n} \right\} &= \frac{2}{X_{r:n}^{k/2} + X_{s:n}^{k/2}}; \\ \text{(b)} \quad E \left\{ \frac{1}{(s-r-1)} \sum_{i=r+1}^{s-1} \frac{1}{X_{i:n}^{2k}} \middle| X_{r:n}, X_{s:n} \right\} &= \frac{1}{X_{r:n}^k + X_{s:n}^k}. \end{aligned}$$

Note that the family of distributions given by (3.1) includes power function distributions, Pareto distribution and Rectangular distribution.

2. Distributions with c.d.f. of the form

$$F(x) = Ae^{-\theta x^N} + B \tag{3.2}$$

are characterized by any one of the following conditions:-

$$\begin{aligned} \text{(a)} \quad E \left\{ \frac{1}{(s-r-1)} \sum_{i=r+1}^{s-1} e^{\theta X_{i:n}^N/2} \middle| X_{r:n}, X_{s:n} \right\} &= \frac{2e^{\theta(X_{r:n}^N + X_{s:n}^N)/2}}{e^{\theta X_{r:n}^N/2} + e^{\theta X_{s:n}^N/2}} \\ \text{(b)} \quad E \left\{ \frac{1}{(s-r-1)} \sum_{i=r+1}^{s-1} e^{2\theta X_{i:n}^N} \middle| X_{r:n}, X_{s:n} \right\} &= e^{\theta(X_{r:n}^N + X_{s:n}^N)}. \end{aligned}$$

The family of distribution given by (3.2) includes exponential and the Weibull distribution.

The above characterizations are for absolutely continuous distribution. But the discrete case is totally different. For discrete random variables our results do not characterize a family of distributions. Under the stated conditions, g and F are almost unique.

Acknowledgements: The authors wish to thank the referees for their constructive comments on a previous version.

References

- Azlarov TA, Volodin NA (1986) Characterization problems associated with the exponential distribution. Springer
- Balasubramanian K, Beg MI (1992) Distributions determined by conditioning on a pair of order statistics. *Metrika* 39:107–112
- Beg MI, Balasubramanian K (1990) Distributions determined by conditioning on a single order statistic. *Metrika* 37:37–43
- Beg MI, Kirmani SNUA (1974) On a characterization of exponential and related distributions. *Austral J Statist* 16:163–166
- Ferguson TS (1967) On characterizations of distributions by properties of order statistics. *Sankhya A* 29:265–278
- Galambos J, Kotz S (1978) Characterizations of probability distributions. *Lecture Notes in Mathematics* #675. Springer

On Limits of Uniquely Best Linear Estimators

LYNN ROY LAMOTTE¹

Department of Experimental Statistics, College of Agriculture, Louisiana State University,
Baton Rouge LA 70803-5606, USA

Abstract: In a general linear model, it is shown that all admissible linear estimators are limits of linear estimators that are uniquely best at some point in an extended parameter set. The principal result shows that a linear estimator that is uniquely best at a point $\underline{W}_2$ among multiple linear estimators that are best at a point $\underline{W}_1$ is the limit of uniquely best estimators at points approaching $\underline{W}_1$ along the line joining $\underline{W}_1$ and $\underline{W}_2$.

Key Words and Phrases: General linear model, admissibility, linear estimation.

1 Introduction

The admissible linear estimators in a general linear model are highly structured with respect to the parameter set. Each one can be related to a point or a finite sequence of points in the parameter set at which it, and only it, minimizes risk. When a sequence is required to reach a point at which an admissible linear estimator is uniquely best, it can happen that the parameter points in the sequence are not in the original parameter set, although they can be reached as limit points of lines extended through points in the original parameter set. Following these paths may lead, for example, to including matrices that are not nonnegative definite when the original parameter set includes only variance-covariance matrices. For this reason, several authors have established connections between admissible linear estimators and limits of estimators that are uniquely best in the original parameter set. LaMotte (1973, 1977) noted such a connection for unbiased linear estimators. Zontek and Klonecki (1990) noted this connection for invariant quadratic estimators of variance components.

¹ Research supported in part by U.S.A.F. Aerospace Research Laboratories under contract F33615-71-C-1463, summer 1973, and in part by Grant DMS-9104811 from the National Science Foundation.

Zontek (1988) established new results on limits of uniquely best linear estimators, providing sufficient conditions for them to be admissible.

The connection between admissible linear estimators and limits of uniquely best linear estimators is not always simple and direct. In some models, every limit of uniquely best linear estimators that exists is admissible, but that is not generally the case. For example, LaMotte (1985) illustrated these relations in the context of the balanced, random, single classification analysis of variance model. When estimating the error variance component by linear combinations of Between Mean Square and Within Mean Square, coefficients of the uniquely best linear estimators form a right-triangular set. On one of the legs of the triangle, though, only one point, one of the vertices, gives an admissible linear estimator. Thus, in that model, many limits of uniquely best linear estimators are not admissible.

Using a general result from statistical decision theory due to Farrell (1968), Stepniak (1987) showed that, for any admissible linear estimator, there exists a sequence of uniquely best estimators that converges to it. He did so by showing that the question in terms of linear estimation could be restated in terms of a decision game with the property that admissible decision rules are limits of Bayes rules, and hence that admissible linear estimators must have the same property. The development is not constructive: as Stepniak says, the result "is unattainable, as yet, by the algebraic way, which is actually prevalent in linear estimation."

Zontek's (1988) Theorem 2.5 appears to be the most comprehensive result to date. It describes conditions on the convergence of a sequence of parameter points such that the corresponding uniquely best linear estimators converge to an admissible linear estimator. In the same paper, Theorem 2.6 establishes that, in "regular" models, every admissible linear estimator is a limit of uniquely best linear estimators. In both theorems, the set of available estimators is a linear subspace, so the results do not extend to settings where the available estimators are restricted to affine sets, as when unbiasedness is imposed. Further, the requirements of "regular" models appear to be quite restrictive.

In this paper the setting is general. The set of linear estimators can be any affine set, and there is no restriction on the model. We shall establish that every admissible linear estimator is a limit point of linear estimators that are uniquely best at points in the minimal closed convex cone containing the original parameter set. The development is constructive; it involves constructing a sequence of points with corresponding uniquely best linear estimators that converge to the given admissible estimator.

Section 2 describes background for the general setting. The main results are presented in Section 3. Section 4 provides an example to illustrate the structure of and relations among uniquely best linear estimators.

2 Definitions and Background

The main notation and definitions are summarized in Table 1. Let Y be a random n -vector with mean vector μ and variance-covariance matrix V , with (μ, V) in an arbitrary subset $\mathcal{P}$ of the Cartesian product of Euclidean n -space $\mathbb{R}^n$ and the set of $n \times n$ symmetric nonnegative definite matrices. Let C be an $n \times t$ matrix of constants. Consider estimating $C'\mu$ by linear functions $L'Y$ with L an $n \times t$ matrix of constants. For comparing such linear estimators, we shall use Total Mean Squared Error (TMSE) as the risk function:

$$\begin{aligned} \text{TMSE}_L(V, \mu\mu') &= E\{(L'Y - C'\mu)'(L'Y - C'\mu)\} \\ &= \text{tr}\{L'VL + (L - C)'\mu\mu'(L - C)\}. \end{aligned} \quad (2.1)$$

For a matrix M , denote the transpose of M , the linear subspace spanned by the columns of M , and the null space $\{x: Mx = 0\}$ of M by M' , $\mathbf{R}(M)$, and $\mathbf{N}(M)$, respectively. Denote the trace of a square matrix M by $\text{tr}(M)$. In dealing with linear subspaces of $r \times s$ matrices, the trace inner product $\text{tr}(MH')$ will be used, along with the corresponding squared norm $\text{tr}(MM')$. A neighborhood of a point will mean an open set containing the point; the topological closure of a set A of points will be denoted $\text{cl}(A)$; the minimal linear subspace containing A by $\text{sp}(A)$; and the minimal closed convex cone containing A by $[A]$.

Affine subsets of linear estimators of the form $\mathcal{L} = \{L: AL = B\}$ play a prominent role in describing admissible linear estimators. The set of all $n \times t$

Table 1. Notation

Y is a random n -vector
$E(Y) = \mu, \text{Var}(Y) = V$
$\mathcal{P}$ is the parameter set; its elements are (μ, V)
$\mathfrak{F} = \{(\mu\mu', V): (\mu, V) \in \mathcal{P}\}$
$\Omega = \text{sp}(\mathfrak{F})$; its elements are $\underline{W} = (W_1, W_2)$, partitioned as in $\mathfrak{F}$
$\Omega_+ = \{\underline{W} = (W_1, W_2) \in \Omega: W_1 \text{ is nnd and } W_2 \text{ is nnd}\}$
C is an $n \times t$ matrix of constants
$C'\mu$ is to be estimated by $L'Y$ with L in
$\mathcal{L} = \{L \in \mathbb{R}^{n \times t}: AL = B\}$, where A and B are constant matrices
$\mathbf{B}(\underline{W})$ is an $\mathcal{L}$ -best estimator at $\underline{W}$ if $\underline{W}$ permits an $\mathcal{L}$ -best estimator
$\mathfrak{C} = \{\underline{W} \in \Omega: \mathbf{B}(\underline{W}) \text{ is unique}\}$
S , a linear subspace in Ω , is the set of trivial points for $\mathcal{L}$
$\underline{W}_0$ is a point in $[\mathfrak{F} + S] \setminus S$ that permits multiple $\mathcal{L}$ -best estimators
$\mathcal{L}_1$ is the set of $\mathcal{L}$ -best estimators at $\underline{W}_0$
$\mathbf{B}_1(\underline{W})$ is an $\mathcal{L}_1$ -best estimator at $\underline{W}$ if $\underline{W}$ permits an $\mathcal{L}_1$ -best estimator
$\mathfrak{C}_1 = \{\underline{W} \in \Omega: \mathbf{B}_1(\underline{W}) \text{ is unique}\}$
S_1 is the set of trivial points for $\mathcal{L}_1$
$\underline{W}_*$ is an element of $\mathfrak{C}_1$
$L_* = \mathbf{B}_1(\underline{W}_*)$

matrices, corresponding to unrestricted linear estimation, is such an affine subset. The set of linear estimators unbiased for $C'\mu$ is such an affine subset. And the set of linear estimators that are best at a given point in the parameter set is an affine subset. I shall refer to estimators $L'Y$ in terms of L .

Noting that TMSE_L is a real-valued linear function of V and $\mu\mu'$, let $\mathfrak{F} = \{(V, \mu\mu') : (\mu, V) \in \mathcal{P}\}$, and let $\Omega = \text{sp}(\mathfrak{F})$ be the minimal linear subspace of $\{(W_1, W_2) : W_1 \text{ and } W_2 \text{ are } n \times n \text{ symmetric matrices}\}$ containing $\mathfrak{F}$. Points in Ω will be denoted by $\underline{W} = (W_1, W_2)$: that is, the symbol denoting the point is underlined, and the same symbol without underling and with subscripts 1 and 2 denotes its components. Specific points will be designated with subscripts. For example, $\underline{W}_* = (W_{1*}, W_{2*})$ denotes a point $\underline{W}_*$ with components W_{1*} and W_{2*} .

Let $\Omega_+ = \{(W_1, W_2) \in \Omega : W_1 \text{ is nnd and } W_2 \text{ is nnd}\}$. Clearly $\mathfrak{F} \subset \Omega_+$. Extend TMSE_L to Ω as $\text{TMSE}_L(\underline{W}) = \text{tr}[L'W_1L + (L - C)'W_2(L - C)]$. Other terms are defined as detailed in LaMotte (1982). These include *better than* on subsets of Ω , *best among* $\mathcal{L}$ at points in Ω , and *admissible among* $\mathcal{L}$ on subsets of Ω . In particular, L_* is best among $\mathcal{L}$ at the point $\underline{W}_* \in \Omega$ iff $\text{TMSE}_L(\underline{W}_*) \geq \text{TMSE}_{L_*}(\underline{W}_*)$ for all $L \in \mathcal{L}$; in this case, we shall also say that L_* is $\mathcal{L}$ -best at $\underline{W}_*$.

A point $\underline{W}$ in Ω may have a unique best estimator among $\mathcal{L}$, or it may have many estimators that are best among $\mathcal{L}$ at $\underline{W}$, or it may permit no estimator that is best among $\mathcal{L}$ at $\underline{W}$. Points $\underline{W}$ in Ω such that every estimator in $\mathcal{L}$ is $\mathcal{L}$ -best at $\underline{W}$ are called *trivial points* for $\mathcal{L}$. Clearly $\underline{W} = 0$ is a trivial point, and there may be others. Denote by $\mathcal{S}$ the set of trivial points for $\mathcal{L}$. $\mathcal{S}$ is a linear subspace of Ω . Let $\mathcal{C}$ denote the set of points in Ω that have a unique $\mathcal{L}$ -best estimator.

Given matrices A and B such that the affine subset $\mathcal{L} = \{L : AL = B\}$ is not empty, let L_0 be the unique element whose columns are in $\mathbf{R}(A')$ and (if the dimension of $\mathcal{L}$ is positive) let columns of N form an orthonormal basis for $\mathbf{N}(A)$. Every $L \in \mathcal{L}$ then has a representation $L = L_0 + NZ$ for some matrix Z .

LaMotte (1982) notes that the setting may be transformed initially so that $N'(W_1 + W_2)N$ is positive definite for some points $\underline{W} = (W_1, W_2) \in \Omega$. We shall assume that this is true, then, without loss of generality.

For $\underline{W} = (W_1, W_2) \in \Omega$, LaMotte (1982, Theorem 3.1) established the following results on best-ness.

- (i) There exists an L best among $\mathcal{L}$ at $\underline{W}$ iff $N'(W_1 + W_2)N$ is nnd and $\mathbf{R}[N'W_2C - N'(W_1 + W_2)L_0] \subset \mathbf{R}[N'(W_1 + W_2)N]$.
- (ii) $L \in \mathcal{L}$ is best among $\mathcal{L}$ at $\underline{W}$ iff $N'(W_1 + W_2)N$ is nnd and $N'(W_1 + W_2)L = N'W_2C$.
- (iii) $\underline{W} \in \Omega$ is a trivial point for $\mathcal{L}$ iff $N'(W_1 + W_2)N = 0$ and $N'W_2C - N'(W_1 + W_2)L_0 = 0$.
- (iv) There is exactly one $L \in \mathcal{L}$ that is best among $\mathcal{L}$ at $\underline{W}$ iff $N'(W_1 + W_2)N$ is pd.

For each $\underline{W} \in \Omega$ that permits an $\mathcal{L}$ -best estimator, let $\mathbf{B}(\underline{W})$ be such an estimator. That “ $\mathbf{B}(\underline{W})$ is unique” will mean that $\underline{W}$ has exactly one $\mathcal{L}$ -best estimator. An important consequence of (ii) and (iii) is that the set of $\mathcal{L}$ -best estimators at $\underline{W}$ is the same as the set of $\mathcal{L}$ -best estimators at $\underline{W} + \underline{s}$ for any $\underline{s} \in \mathcal{S}$.

The set $[\mathfrak{F} + \mathcal{S}]$ can include points in Ω that are not in $\mathfrak{F}$ or $[\mathfrak{F}]$ or Ω_+ , so the connection between the points at which an estimator is best in $[\mathfrak{F} + \mathcal{S}]$ and the original parameter set $\mathfrak{F}$ can become obscure. In the next section we shall show that every admissible linear estimator is a limit of linear estimators that are uniquely best at points in $[\mathfrak{F}]$.

3 Derivations and Results

Consider an estimator L in $\mathcal{L}$ that is best among $\mathcal{L}$ at the point $\underline{W}_0 = (W_{10}, W_{20})$. Then L must satisfy

$$\begin{pmatrix} A \\ N'(W_{10} + W_{20}) \end{pmatrix} L = \begin{pmatrix} B \\ N'W_{20}C \end{pmatrix}. \quad (3.1)$$

The solution set to (3.1) is itself an affine subset $\mathcal{L}_1$; it is contained in $\mathcal{L}$, and it has lesser dimension than $\mathcal{L}$ iff $N'(W_{10} + W_{20})N$ is nonzero. See Table 1 for notation related to $\mathcal{L}_1$. The linear subspace parallel to $\mathcal{L}_1$ is $\mathbf{N}(A) \cap \mathbf{N}[N'(W_{10} + W_{20})]$. We shall be interested in cases in which $\underline{W}_0$ permits multiple best estimators, in which case the dimension of this linear subspace is positive. It is equivalent to $\mathbf{N}(A) \cap \mathbf{N}[NN'(W_{10} + W_{20})NN']$, and $\mathbf{N}(A) = \mathbf{N}(AA')$. Note that (3.1) is equivalent to

$$[A'A + NN'(W_{10} + W_{20})NN']L = A'B + NN'[W_{20}C - (W_{10} + W_{20})L_0]. \quad (3.2)$$

Note further that $\mathbf{R}(N) = \mathbf{N}(A)$ is the direct sum of

$$\mathbf{R}[NN'(W_{10} + W_{20})NN'] \quad \text{and} \quad \mathbf{N}(A) \cap \mathbf{N}[NN'(W_{10} + W_{20})NN'],$$

two linear subspaces that are orthogonal. Thus we may choose columns of N as (H, N_0) , where columns of N_0 form an orthonormal basis for $\mathbf{N}(A) \cap \mathbf{N}[N'(W_{10} + W_{20})]$ and columns of H are orthonormal eigenvectors corresponding to nonzero eigenvalues of $[NN'(W_{10} + W_{20})NN']$. Note that H and

N_0 are defined in terms of NN' , which is the unique orthogonal projection operator onto $\mathbf{N}(A)$.

As H is defined,

$$NN'(W_{10} + W_{20})NN' = HAH' , \quad H'H = I , \quad (3.3)$$

with A a diagonal matrix with nonzero diagonal elements. Note that all columns of $N = (H, N_0)$ are orthogonal.

Suppose $\underline{W}_0$ is not a trivial point for $\mathcal{L}$ and $\underline{W}_0$ permits multiple best estimators among $\mathcal{L}$. Then $N'(W_{10} + W_{20})N$ is nnd and nonzero (if it is 0 and $\underline{W}_0$ permits a best estimator, then $\underline{W}_0$ must be a trivial point), hence H has at least one column and the diagonal elements of A are positive. Further, N_0 has at least one column.

Consider now estimators in $\mathcal{L}_1$ (the solution set to (3.1)) that are best among $\mathcal{L}_1$ at points in Ω . Define $\mathbf{B}_1(\underline{W})$ in analogy to $\mathbf{B}(\underline{W})$, but with respect to $\mathcal{L}_1$. Suppose $\underline{W}_* = (W_{1*}, W_{2*})$ is a point in Ω and $\mathbf{B}_1(\underline{W}_*)$ is unique. This implies that $N'_0(W_{1*} + W_{2*})N_0$ is pd. Call this estimator L_* . Then L_* is the unique solution to

$$AL = B$$

$$N'(W_{10} + W_{20})L = N'W_{20}C \quad (3.4)$$

$$N'_0(W_{1*} + W_{2*})L = N'_0W_{2*}C .$$

Generally (except in cases that can be avoided by insisting that we consider maximal (in some sense) points at which an estimator is best), while it is uniquely best among $\mathcal{L}_1$ at $\underline{W}_*$, L_* is not uniquely best among $\mathcal{L}$ anywhere. Our objective is to show that L_* can be reached as a limit of estimators that are uniquely best among $\mathcal{L}$ along the line segment joining $\underline{W}_0$ and $\underline{W}_*$.

The line segment joining $\underline{W}_0$ and $\underline{W}_*$ is $\{\underline{W}_\lambda = (1 - \lambda)\underline{W}_0 + \lambda\underline{W}_* : 0 \leq \lambda \leq 1\}$. Given that $\underline{W}_0$ permits multiple best estimators (the set $\mathcal{L}_1$) in $\mathcal{L}$ and that $\underline{W}_*$ permits a unique best estimator (L_*) in $\mathcal{L}_1$, we shall show first that for λ sufficiently close to 0, $\underline{W}_\lambda$ permits a unique best estimator L_λ in $\mathcal{L}$. Then we shall show that as λ approaches 0, L_λ approaches L_* .

Lemma 3.1: If $\underline{W}_0 = (W_{10}, W_{20}) \in \Omega$ is not a trivial point, and $\underline{W}_0$ permits multiple best estimators in $\mathcal{L}$, and $\underline{W}_* = (W_{1*}, W_{2*}) \in \Omega$ is such that $N'_0(W_{1*} + W_{2*})N_0$ is pd, then there exists a $\lambda_* > 0$ such that for any $\lambda \in (0, \lambda_*)$, $N'[(1 - \lambda)(W_{10} + W_{20}) + \lambda(W_{1*} + W_{2*})]N$ is pd.

Proof: Let $\gamma = (1 - \lambda)/\lambda$. Then

$$\begin{aligned}
 & N'[\gamma(W_{10} + W_{20}) + (W_{1*} + W_{2*})]N \\
 &= \gamma N'NN'(W_{10} + W_{20})NN'N + N'(W_{1*} + W_{2*})N \quad (\text{since } N' = N'NN') \\
 &= \gamma(H, N_0)'H\Delta H'(H, N_0) + (H, N_0)'(W_{1*} + W_{2*})(H, N_0) \\
 &= \gamma \begin{pmatrix} \Delta & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} H'(W_{1*} + W_{2*})H & H'(W_{1*} + W_{2*})N_0 \\ N_0'(W_{1*} + W_{2*})H & N_0'(W_{1*} + W_{2*})N_0 \end{pmatrix} \\
 &= \begin{pmatrix} D_\gamma & G \\ G' & M \end{pmatrix}, \tag{3.5}
 \end{aligned}$$

where $D_\gamma = \gamma\Delta + H'(W_{1*} + W_{2*})H$. In general, it cannot be guaranteed that $H'(W_{1*} + W_{2*})H$ is nnd. Let α_* be the lesser of 0 and the least eigenvalue of $\Delta^{-1/2}H'(W_{1*} + W_{2*})H\Delta^{-1/2}$. It may then be shown that, for any $\gamma > -\alpha_*$ (corresponding to $\lambda < 1/(1 - \alpha_*)$), D_γ is pd. Note that

$$\lim_{\gamma \rightarrow \infty} D_\gamma^{-1} = 0 \quad \text{and} \quad \lim_{\gamma \rightarrow \infty} \gamma D_\gamma^{-1} = \Delta^{-1}. \tag{3.6}$$

Given that M is pd, in order to establish the lemma it is sufficient to show the existence of $\lambda_* < 1/(1 - \alpha_*)$ such that for any $\lambda \in (0, \lambda_*)$, $M^{-1} - G'D_\gamma^{-1}G$ is pd. The second term is nnd when D_γ is pd and it goes to 0 as $\gamma \rightarrow \infty$. With γ_* greater than $-\alpha_*$ and great enough that the eigenvalues of $M^{-1/2}G'D_\gamma^{-1}GM^{-1/2}$ are less than 1, the conclusion of the lemma follows with $\lambda_* \leq 1/(1 + \gamma_*)$. $\square$

Next is a derivation of an explicit representation of the unique L_* that satisfies (3.4). If L_* satisfies (3.4) then $L_* = L_0 + NZ_* = L_0 + HZ_{1*} + N_0Z_{2*}$, and

$$\begin{aligned}
 & N'(W_{10} + W_{20})L_* = N'W_{20}C \\
 & \Rightarrow NN'(W_{10} + W_{20})L_* = NN'W_{20}C \\
 & \Rightarrow NN'(W_{10} + W_{20})(L_0 + NZ_*) = NN'W_{20}C \\
 & \Rightarrow [NN'(W_{10} + W_{20})NN']NZ_* = NN'[W_{20}C - (W_{10} + W_{20})L_0] \\
 & \Rightarrow H\Delta H'NZ_* = NN'[W_{20}C - (W_{10} + W_{20})L_0]
 \end{aligned}$$

$$\begin{aligned}
&\Rightarrow H\Delta Z_{1*} = NN'[W_{20} - (W_{10} + W_{20})L_0] \quad (\text{because } H'NZ = Z_{1*}) \\
&\Rightarrow Z_{1*} = \Delta^{-1}H'[W_{20}C - (W_{10} + W_{20})L_0] \quad (\text{because } H'NN' = H') \\
&\Rightarrow HZ_{1*} = H\Delta^{-1}H'[W_{20}C - (W_{10} + W_{20})L_0] .
\end{aligned} \tag{3.7}$$

Similarly,

$$\begin{aligned}
N'_0(W_{1*} + W_{2*})L_* &= N'_0W_{2*}C \\
&\Rightarrow N'_0(W_{1*} + W_{2*})(L_0 + HZ_{1*} + N_0Z_{2*}) = N'_0W_{2*}C \\
&\Rightarrow MZ_{2*} = N'_0[W_{2*}C - (W_{1*} + W_{2*})(L_0 + HZ_{1*})] \\
&\Rightarrow N_0Z_{2*} = N_0M^{-1}N'_0[W_{2*}C - (W_{1*} + W_{2*})(L_0 + HZ_{1*})] ,
\end{aligned} \tag{3.8}$$

recalling that $M = N'_0(W_{1*} + W_{2*})N_0$ is pd. Thus

$$\begin{aligned}
L_* &= L_0 + H\Delta^{-1}H'[W_{20}C - (W_{10} + W_{20})L_0] \\
&\quad + N_0M^{-1}N'_0[W_{2*}C - (W_{1*} + W_{2*})L_0] \\
&\quad - N_0M^{-1}N'_0(W_{1*} + W_{2*})H\Delta^{-1}H'[W_{20}C - (W_{10} + W_{20})L_0]
\end{aligned} \tag{3.9}$$

is the unique solution to (3.4).

Next we show that L_* is a limit of estimators in $\mathcal{L}$ that are uniquely best among $\mathcal{L}$ along the line from $\underline{W}_*$ to $\underline{W}_0$.

Let $\lambda \in (0, \lambda_*)$ so that $N'(\underline{W}_{1\lambda} + \underline{W}_{2\lambda})N$ is pd, where $\underline{W}_\lambda = (W_{1\lambda}, W_{2\lambda}) = (1 - \lambda)\underline{W}_0 + \lambda\underline{W}_*$. Then there is a unique estimator L_λ in $\mathcal{L}$ that is best among $\mathcal{L}$ at $\underline{W}_\lambda$. Let $\gamma = (1 - \lambda)/\lambda$; L_λ is the unique solution to

$$N'[(W_{1*} + W_{2*}) + \gamma(W_{10} + W_{20})]L_\lambda = N'(\gamma W_{20} + W_{2*})C . \tag{3.10}$$

With $L_\lambda = L_0 + NZ_\lambda$, this becomes

$$\begin{aligned}
&N'[(W_{1*} + W_{2*}) + \gamma(W_{10} + W_{20})]NZ_\lambda \\
&= N'(\gamma W_{20} + W_{2*})C - N'[(W_{1*} + W_{2*}) + \gamma(W_{10} + W_{20})]L_0 .
\end{aligned} \tag{3.11}$$

Thus (see (3.5))

$$Z_\lambda = \begin{pmatrix} D_\gamma & G \\ G' & M \end{pmatrix}^{-1} \{N'(\gamma W_{20} + W_{2*})C - N'[(W_{1*} + W_{2*}) + \gamma(W_{10} + W_{20})]L_0\} . \quad (3.12)$$

Further, with $N = (H, N_0)$,

$$\begin{aligned} & N'(\gamma W_{20} + W_{2*})C - N'[(W_{1*} + W_{2*}) + \gamma(W_{10} + W_{20})]L_0 \\ &= N'[W_{2*}C - (W_{1*} + W_{2*})L_0] + \gamma N'[W_{20}C - (W_{10} + W_{20})L_0] \\ &= N'[W_{2*}C - (W_{1*} + W_{2*})L_0] + \gamma \left(H'[W_{20}C - (W_{10} + W_{20})L_0] \right. \\ & \quad \left. N'_0[W_{20}C - (W_{10} + W_{20})L_0] \right) . \end{aligned} \quad (3.13)$$

Because W_0 permits an $\mathcal{L}$ -best estimator, $N'[W_{20}C - (W_{10} + W_{20})L_0] = N'(W_{10} + W_{20})NZ$ for some Z . With $NZ = NN'NZ$ and $N'_0 = N'_0NN'$,

$$\begin{aligned} N'_0[W_{20}C - (W_{10} + W_{20})L_0] &= N'_0NN'[W_{20}C - (W_{10} + W_{20})L_0] \\ &= N'_0[NN'(W_{10} + W_{20})NN']NZ \\ &= N'_0H\Delta H'NZ = 0 , \end{aligned} \quad (3.14)$$

so that the bottom-right entry in (3.13) is 0. Thus

$$\begin{aligned} Z_\lambda &= \begin{pmatrix} K_{11} & K_{12} \\ K'_{12} & K_{22} \end{pmatrix} N'[W_{2*}C - (W_{1*} + W_{2*})L_0] \\ & \quad + \gamma \begin{pmatrix} K_{11} \\ K'_{12} \end{pmatrix} H'[W_{20}C - (W_{10} + W_{20})L_0] , \end{aligned} \quad (3.15)$$

where the K s represent the corresponding partitions of the inverse of $N'(W_{1\lambda} + W_{2\lambda})N/\lambda$ as it is represented in (3.5):

$$\begin{aligned} K_{22} &= [M - G'D_\gamma^{-1}G]^{-1} , \\ K_{11} &= D_\gamma^{-1} + D_\gamma^{-1}GK_{22}G'D_\gamma^{-1} , \quad \text{and} \\ K_{12} &= -D_\gamma^{-1}GK_{22} . \end{aligned} \quad (3.16)$$

The limits as $\gamma \rightarrow \infty$ of K_{11} , K_{12} , and K_{22} are 0, 0, and M^{-1} , respectively. The limits as $\gamma \rightarrow \infty$ of γK_{11} and γK_{12} are Δ^{-1} and $-\Delta^{-1}GM^{-1}$, respectively. Now it follows that

$$\begin{aligned} \lim_{\lambda \rightarrow 0^+} Z_\lambda &= \begin{pmatrix} \Delta^{-1}H'[W_{20}C - (W_{10} + W_{20})L_0] \\ M^{-1}N'_0[W_{2*}C - (W_{1*} + W_{2*})L_0] - M^{-1}G'\Delta^{-1}H'[W_{20}C - (W_{10} + W_{20})L_0] \end{pmatrix} \\ &= \begin{pmatrix} Z_{1*} \\ Z_{2*} \end{pmatrix}. \end{aligned} \quad (3.17)$$

At this point we have proved the following theorem.

Theorem 3.2: With the conditions stated in Lemma 3.1, let $\lambda \in (0, \lambda_*)$, $\underline{W}_\lambda = (1 - \lambda)\underline{W}_0 + \lambda\underline{W}_*$, and $L_\lambda = \mathbf{B}(\underline{W}_\lambda)$. Then

$$\lim_{\lambda \rightarrow 0^+} L_\lambda = L_* . \quad (3.18)$$

Theorem 3.2 is important in establishing the results that follow because it establishes that every estimator L_* that is uniquely best among $\mathcal{L}_1$ is a limit of estimators that are uniquely best in $\mathcal{L}$, and it describes a path of points in Ω along which uniquely best estimators in $\mathcal{L}$ approach L_* . In addition, though, we shall need to note the following results. Note that $\mathfrak{C}$ is open because it is $\{\underline{W} \in \Omega: N'(W_1 + W_2)N \text{ is pd}\}$, which is the pre-image in Ω of an open set under a linear (thus continuous) mapping. Note that $\mathbf{B}(\underline{W})$ is continuous at $\underline{W}$ if $\mathbf{B}(\underline{W})$ is unique. Finally, note that, when U is a linear subspace, $[\mathfrak{F} + U] = [[\mathfrak{F}] + U] = \text{cl}([\mathfrak{F}] + U)$.

Lemma 3.3: If $\underline{W}_* \in [\mathfrak{F} + \mathcal{S}_0]$, where $\mathcal{S}_0$ is a subspace of $\mathcal{S}$, and $\mathbf{B}(\underline{W}_*)$ is unique, then every neighborhood of $\mathbf{B}(\underline{W}_*)$ contains a point in $\{\mathbf{B}(\underline{W}): \underline{W} \in [\mathfrak{F}] \text{ and } \mathbf{B}(\underline{W}) \text{ is unique}\}$.

Proof: Let $\varepsilon > 0$. By the preceding notes, there exists a neighborhood of $\underline{W}_*$ throughout which $\mathbf{B}(\underline{W})$ is unique ($\mathfrak{C}$ is open) and $\|\mathbf{B}(\underline{W}) - \mathbf{B}(\underline{W}_*)\| < \varepsilon$ ($\mathbf{B}$ is continuous at $\underline{W}_*$). This neighborhood contains a point in $[\mathfrak{F}] + \mathcal{S}_0$, say $\underline{T} + \underline{W}_0$, with $\underline{T} \in [\mathfrak{F}]$ and $\underline{W}_0 \in \mathcal{S}_0$ ($[\mathfrak{F}] + \mathcal{S}_0 = \text{cl}([\mathfrak{F}] + \mathcal{S}_0)$). The set of $\mathcal{L}$ -best

estimators at $\underline{T} + \underline{W}_0$ is the same as the set of $\mathcal{L}$ -best estimators at $\underline{T}$; $\mathbf{B}(\underline{T} + \underline{W}_0)$ is unique, so $\mathbf{B}(\underline{T})$ is unique and $\|\mathbf{B}(\underline{T}) - \mathbf{B}(\underline{W}_*)\| < \varepsilon$. $\square$

Theorem 3.4: Suppose L_* is best among $\mathcal{L}$ at $\underline{W}_0 \in [\mathfrak{F} + \mathcal{S}]$ and $\mathbf{B}(\underline{W}_0)$ is not unique. If $\underline{W}_* \in [\mathfrak{F} + \mathcal{S}_1]$ and $L_* = \mathbf{B}_1(\underline{W}_*)$ and $\mathbf{B}_1(\underline{W}_*)$ is unique, then L_* is a limit of members of $\mathcal{L}$ that are uniquely best among $\mathcal{L}$ in $[\mathfrak{F}]$.

Proof: Let $\varepsilon > 0$. By Lemma 3.3, there exists $\underline{T}_\varepsilon \in [\mathfrak{F}]$ such that $\mathbf{B}_1(\underline{T}_\varepsilon)$ is unique and $\|\mathbf{B}_1(\underline{T}_\varepsilon) - L_*\| < \varepsilon/3$. Because $\mathbf{B}_1(\underline{T}_\varepsilon)$ is best among $\mathcal{L}$ at $\underline{W}_0$ and uniquely best among $\mathcal{L}_1$ at $\underline{T}_\varepsilon$, by Lemma 3.1 and Theorem 3.2 there exists a $\lambda > 0$ such that $L_\lambda = \mathbf{B}[(1 - \lambda)\underline{W}_0 + \lambda\underline{T}_\varepsilon]$ is unique and $\|L_\lambda - \mathbf{B}_1(\underline{T}_\varepsilon)\| < \varepsilon/3$. Note that $(1 - \lambda)\underline{W}_0 + \lambda\underline{T}_\varepsilon \in [\mathfrak{F} + \mathcal{S}]$. By Lemma 3.3 again, there exists $\underline{T}_* \in [\mathfrak{F}]$ such that $\mathbf{B}(\underline{T}_*)$ is unique and $\|L_\lambda - \mathbf{B}(\underline{T}_*)\| < \varepsilon/3$. Thus $\|L_* - \mathbf{B}(\underline{T}_*)\| < \varepsilon$. $\square$

For a finite sequence of points $\underline{W}_1, \dots, \underline{W}_{r+1}$ in Ω , define $\mathcal{S}_0$ to be $\mathcal{S}$ and $\mathcal{L}_0$ to be $\mathcal{L}$. For each $i = 1, \dots, r$, let $\mathcal{L}_i = \{L \in \mathcal{L}_{i-1} : L \text{ is best among } \mathcal{L}_{i-1} \text{ at } \underline{W}_i\}$. Denote by $\mathcal{S}_i$ the set of trivial points in Ω for $\mathcal{L}_i$, that is, $\mathcal{S}_i = \{\underline{W} \in \Omega : \text{every } L \in \mathcal{L}_i \text{ is best among } \mathcal{L}_i \text{ at } \underline{W}\}$. Define $\mathbf{B}_i(\underline{W})$ in analogy to $\mathbf{B}(\underline{W})$: for each $\underline{W} \in \Omega$ that permits a best-among- $\mathcal{L}_i$ estimator, $\mathbf{B}_i(\underline{W})$ is one such.

LaMotte (1982) showed that every member L_* of $\mathcal{L}$ that is admissible among $\mathcal{L}$ can be related to a finite sequence of points $\{\underline{W}_1, \dots, \underline{W}_{r+1}\}$ in Ω such that L_* is uniquely best among $\mathcal{L}$ with respect to the sequence in the sense that L_* is best among $\mathcal{L}_{i-1}$ at $\underline{W}_i$, $i = 1, \dots, r$, and uniquely best among $\mathcal{L}_r$ at $\underline{W}_{r+1}$. It should be understood that this sequence stops as soon as a point is reached at which L_* is uniquely best. The affine subsets $\mathcal{L}_i$ of $\mathcal{L}$ shrink by at least one dimension at each step, while the sets of trivial points $\mathcal{S}_i$ grow by at least one dimension at each step. The points $\underline{W}_i$ in the sequence are related to the original parameter set $\mathfrak{F}$ in that $\underline{W}_i \in [\mathfrak{F} + \mathcal{S}_{i-1}] \setminus \mathcal{S}_{i-1}$. With repeated applications of Theorem 3.4 it can be shown that if L_* is admissible among $\mathcal{L}$ on $\mathfrak{F}$, then L_* is a limit of estimators in $\mathcal{L}$ that are uniquely best in $[\mathfrak{F}]$.

In the proof of Theorem 3.4, note that $\mathbf{B}_1[(1 - \lambda)\underline{W}_0 + \lambda\underline{T}_\varepsilon] = \mathbf{B}_1(\underline{T}_\varepsilon)$. Clearly it is possible to identify $\underline{T}_* \in [\mathfrak{F}]$ such that $\|L_* - \mathbf{B}(\underline{T}_*)\| < \varepsilon$ and $\|L_* - \mathbf{B}_1(\underline{T}_*)\| < \varepsilon$. Thus there exists a sequence $\{\underline{T}^{(j)}, j = 1, 2, \dots\}$ of points in $[\mathfrak{F}]$ such that $\mathbf{B}(\underline{T}^{(j)})$ is unique and converges to L_* and $\mathbf{B}_1(\underline{T}^{(j)})$ is unique and converges to L_* . This construction can be repeated. Thus there exists a sequence $\{\underline{T}^{(j)}\}$ of points in $[\mathfrak{F}]$ such that $\mathbf{B}_i(\underline{T}^{(n)})$ is unique and converges to L_* , $i = 0, \dots, r$.

Zontek's (1988) proof of Theorem 2.5 is elegant and revealing. The structure there is very similar to the structure here. In fact, it appears that Theorem 3.4 and the arguments used here can be used to show that, if $\mathcal{L} = \mathbb{R}^{n \times t}$ (as in Zontek (1988)) and L_* is admissible among $\mathcal{L}$ on $\mathfrak{F}$, then there exists a sequence of points in $[\mathfrak{F}]$ satisfying the conditions in Zontek's (1988) Theorem 2.5. However, no attempt is made here to establish that result.

4 An Example

Let $Q_1 \sim (\gamma_1 + \lambda_1 \gamma_2) \chi_{r_1}^2$ and $Q_2 \sim (\gamma_1 + \lambda_2 \gamma_2) \chi_{r_2}^2$, where: $\chi_{r_1}^2$ and $\chi_{r_2}^2$ are independent, central chi-squared random variables with r_1 and r_2 degrees of freedom, respectively; $\lambda_1 = 0$ and $\lambda_2 > 0$ are known constants; and γ_1 and γ_2 are non-negative scalar parameters. In the balanced, random one-way ANOVA model, the Within Sum of Squares and the Between Sum of Squares are, under normality assumptions, distributed as Q_1 and Q_2 .

With $Q = (Q_1, Q_2)'$, we have $\mu = E(Q) = (r, (r\lambda))\gamma$, where $r = (r_1, r_2)'$, $(r\lambda) = (r_1 \lambda_1, r_2 \lambda_2)'$, and $\gamma = (\gamma_1, \gamma_2)'$. Let $v_{11} = \gamma_1^2$, $v_{22} = \gamma_2^2$, and $v_{12} = \gamma_1 \gamma_2$. Then

$$W_1 = \text{Var}(Q) = \text{diag}(2r_i(v_{11} + 2\lambda_i v_{12} + \lambda_i^2 v_{22}))$$

and

$$\begin{aligned} W_2 &= [E(Q)][E(Q)]' = (r, (r\lambda))\gamma\gamma'(r, (r\lambda))' \\ &= (r_i r_j [v_{11} + (\lambda_i + \lambda_j)v_{12} + \lambda_i \lambda_j v_{22}]) \end{aligned}$$

are linear in $v = (v_{11}, v_{22}, v_{12})$. Note that $(W_1(v), W_2(v))$ is a linear mapping that preserves closed sets (by Theorem 9.1 in Rockafellar (1970)), so (W_1, W_2) can be represented directly in terms of v . In the remainder of this description, then, v will be used in place of W . However, the notation already defined will be used. Thus $\mathfrak{F}$ will denote the parameter set in terms of v , and Ω will be $\mathbb{R}^3$. The two figures shown in this section show a cross-section of $\mathbb{R}^3$ in the plane $\{v: v_{11} + v_{22} = 1\}$, so that corresponding cross-sections of sets are represented. For example, the set labeled Ω_+ in Figure 1 is the intersection of Ω_+ and the plane of the figure.

We shall consider linear estimators $\ell'Q = \ell_1 Q_1 + \ell_2 Q_2$ of $\gamma_2 = (0, 1)\gamma = c'(r, (r\lambda))\gamma$, where $c'r = 0$ and $c'(r\lambda) = 1$. In this case, $\mathcal{L}$ is unrestricted, so $\mathcal{L} = \mathbb{R}^2$, $\ell_0 = 0$, $N = I_2$, $A = 0$, and $B = 0$. The only trivial point for $\mathcal{L}$ is 0, so $\mathcal{S} = \{0\}$.

Figure 1 shows Ω and $\mathcal{L}$ as two planes. Ω_+ is a circle with radius .5, centered at the midpoint of the line joining $(0, 1, 0)$ and $(1, 0, 0)$. That this is so may be seen by noting that W_2 is nnd iff the 2×2 matrix M with diagonal elements v_{11} and v_{22} and off-diagonal elements v_{12} is nnd, and that if M is nnd then W_1 is nnd; thus Ω_+ is the set of points in Ω where M is nnd, which is the cone generated by the circle just described. $\mathfrak{C}$ is the set of points that possess unique $\mathcal{L}$ -best estimators; it is the set in Ω for which $W_1 + W_2$ is pd.

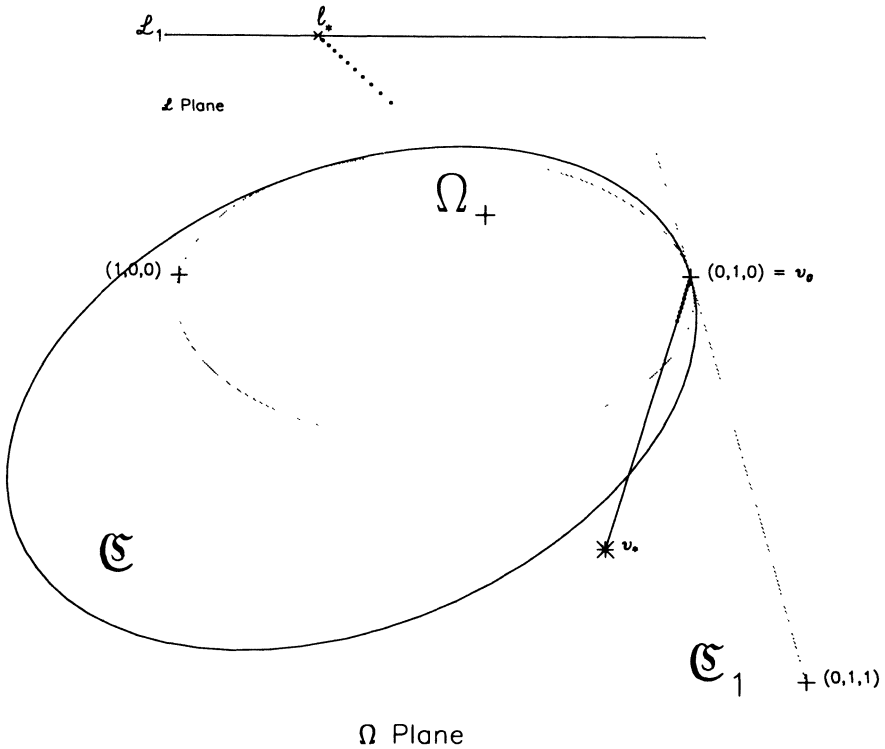


Fig. 1.

The point $v_0 = (0, 1, 0)$ is not a trivial point for $\mathcal{L}$ and it permits multiple $\mathcal{L}$ -best estimators. They are

$$\mathcal{L}_1 = \{\ell = (\ell_1, \ell_2) \in \mathbb{R}^2: \ell_2 = 1/[\lambda_2(r_2 + 2)]\}.$$

This affine set is shown in the $\mathcal{L}$ plane as a solid horizontal line.

For $\mathcal{L}_1$ estimates, the set of trivial points is $\mathcal{S}_1 = \mathbf{R}(v_0) = \mathbf{R}((0, 1, 0))$. The set of points in Ω that possess a unique $\mathcal{L}_1$ -best estimator is $\mathcal{C}_1$, which is the set of points in the half-plane to the left of the line through $(0, 1, 0)$ and $(0, 1, 1)$. The only point on the boundary of $\mathcal{C}_1$ (in the Ω plane) for which there exists an $\mathcal{L}_1$ -best estimator is the trivial point $(0, 1, 0)$.

Figure 1 illustrates Lemma 3.1 and Theorem 3.2. The point v_* has a unique $\mathcal{L}_1$ -best estimator ℓ_* ; it does not have an $\mathcal{L}$ -best estimator. Lemma 3.1 says that if v_* has a unique $\mathcal{L}_1$ -best estimator, then the line from v_* to v_0 contains an open subinterval $\{(1 - \lambda)v_0 + \lambda v_*: 0 < \lambda < \lambda_* < 1\}$ in which each point has a unique $\mathcal{L}$ -best estimator. In Figure 1, this subinterval is the line from v_0 to the bound-

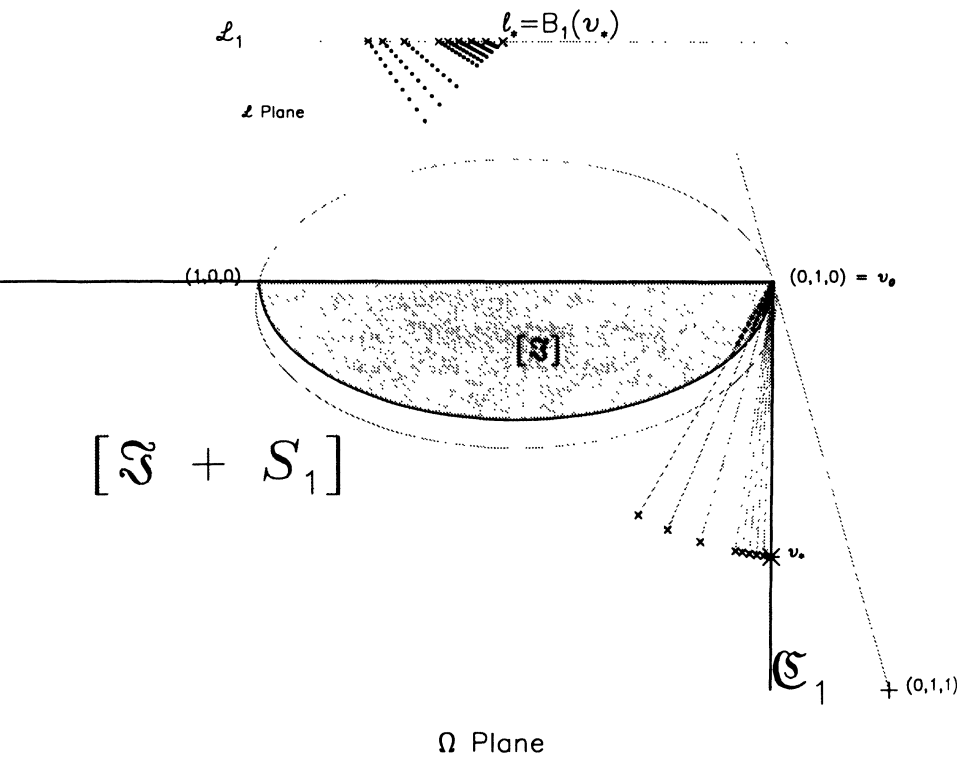


Fig. 2.

ary of $\mathfrak{C}$ in the direction of v_* . Theorem 3.2 says that ℓ_* is the limit of unique $\mathcal{L}$ -best estimators at points along this line approaching v_0 . In Figure 1, the points marked “•” approaching v_0 have the $\mathcal{L}$ -best estimators shown as points marked “•” in the $\mathcal{L}$ plane.

Figure 2 shows the same planes as Figure 1, but with the sets related to $\mathfrak{F}$ included. For this example the parameter set $\mathfrak{F}$ is the proximal arc of the dark-shaded area, so $[\mathfrak{F}]$ is the dark-shaded area. Since $S = \{0\}$, $[\mathfrak{F} + S] = [\mathfrak{F}]$. In the plane of the figure, $[\mathfrak{F} + S_1]$ is the closure of the set of points on rays from v_0 through points in $\mathfrak{F}$; it is the lightly-shaded region.

Lemma 3.3 can be illustrated in Figure 2 in terms of $\mathcal{L}_1$ and S_1 . The point v_* is in $[\mathfrak{F} + S_1]$ and $B_1(v_*) = \ell_*$ is unique. The only point in $[\mathfrak{F}]$ at which ℓ_* is $\mathcal{L}_1$ -best is the $\mathcal{L}_1$ -trivial point v_0 . The points marked “x” in the Ω plane have unique $\mathcal{L}_1$ -best estimators marked “x” in the $\mathcal{L}$ plane, and ℓ_* is a limit of them. Denote such a point in the Ω plane by v_x . $B_1(v_x)$ is unique. In fact, at every point v on the line joining v_x and v_0 , except v_0 , $B_1(v)$ is unique because the set of $\mathcal{L}_1$ -best estimator at v is the same as the set of $\mathcal{L}_1$ -best estimators at v_x . Because $v_x \in [\mathfrak{F}] + S_1$, there exist $t \in [\mathfrak{F}]$ and $s \in S_1$ such that $v_x = t + s$. Then

$\mathbf{B}_1(t) = \mathbf{B}_1(v_x)$ and $\mathbf{B}_1(t)$ is unique. All the points t in the intersection of $[\mathfrak{F}]$ and the line from v_x to v_0 , except v_0 , have this property.

Figure 2 gives a representation of Theorem 3.4, too, but a limited one because $\mathcal{S} = \{0\}$ and $[\mathfrak{F} + \mathcal{S}] = [\mathfrak{F}]$. $\mathcal{L}_1$ -best estimators are marked “ $\times$ ” and $\mathcal{L}$ -best estimators are marked “ $\bullet$ ” in the $\mathcal{L}$ plane. The unique $\mathcal{L}_1$ -best estimator ℓ_* at v_* is a limit of unique $\mathcal{L}_1$ -best estimators ($\times$) in $[\mathfrak{F}]$. Each of those is, in turn, a limit of unique $\mathcal{L}$ -best estimators ($\bullet$) in $[\mathfrak{F} + \mathcal{S}]$, which is the same as $[\mathfrak{F}]$ in Figure 2, but not in general. Thus ℓ_* is a limit of unique $\mathcal{L}$ -best estimators in $[\mathfrak{F}]$.

References

- Farrell RH (1968) Towards a theory of generalized Bayes tests. *Ann Math Statist* 39:1–22
- LaMotte LR (1973) An optimal BLUE when V is singular. Unpublished manuscript
- LaMotte LR (1977) On admissibility and completeness of linear unbiased estimators in a general linear model. *J Amer Statist Assoc* 72:438–441
- LaMotte LR (1982) Admissibility in linear estimation. *Ann Statist* 10:245–255
- LaMotte LR (1985) Admissibility, unbiasedness, and nonnegativity in the balanced, random, one-way ANOVA model. In: Caliński T, Klonecki W (Eds) *Linear Statistical Inference, Proceedings of the International Conference held at Poznań, Poland, June 4–8, 1984*, 184–199. Berlin Springer-Verlag
- Olsen A, Seely J, Birkes D (1976) Invariant quadratic estimation for two variance components. *Ann Statist* 4:878–890
- Rockafellar RT (1970) *Convex analysis*. Princeton, New Jersey, Princeton University Press
- Stepniak C (1987) A complete class for linear estimation in a general linear model. *Ann Inst Statist Math* 39 (A):563–573
- Zontek S (1988) Admissibility of limits of the unique locally best linear estimators with application to variance components models. *Probab Math Statist* 9:29–44
- Zontek S, Klonecki W (1990) Limiting admissible estimators for variance components. *Comm Statist Theory Methods* 19:2485–2508

Received 16.08.1995

A New Process Capability Index

Arjun K. GUPTA AND S. KOTZ¹

Bowling Green State University, Dept. of Mathem. and Statistics, Bowling Green,
OH43403-0221, USA

Abstract: In this paper a new process capability index (C_{pq}) has been introduced, which is easy to compute and performs well when compared with its natural competitor (C_{pm}).

Key Words and Phrases: Process capability indices; estimator; expected value; variance; worth; asymptotic distribution.

1 Introduction

The C_{pm} index of process capability was introduced in the published literature by Chan, Cheng and Spiring (1988). An elementary discussion of the index written for practitioners is given in Spiring (1991). The original definition of C_{pm} is

$$C_{pm} = \frac{USL - LSL}{6\{\sigma^2 + (\xi - T)^2\}^{1/2}} = \frac{d}{3\{E(X - T)^2\}^{1/2}}$$

where USL (LSL) is the upper (lower) specification limit established by engineering considerations, ξ is the expected value and σ is the standard deviation of the measured characteristic, X , T is the target value and $d = (USL - LSL)/2$.

The target value is commonly the midpoint of the specification interval, $m = (USL + LSL)/2$. This need not be the case but if it is not so, there can be pitfalls in the uncritical use of C_{pm} (see e.g. Pearn, Kotz and N. L. Johnson (1992) for details).

The point estimator of C_{pm} is obtained by replacing $E(X - T)^2$ by its unbiased estimator $\hat{\sigma}^2 = n^{-1} \sum (X_i - T)^2$; thus

$$\tilde{C}_{pm} = \frac{d}{3\hat{\sigma}} = \frac{d\sqrt{n}}{3} \{\sum (X_i - T)^2\}^{-1/2}.$$

¹ Work done while he was Lukacs Visiting Professor on leave from the University of Maryland.

The distribution of $n\tilde{\sigma}'^2 = \Sigma(X_i - T)^2$ is that of

$\sigma^2 \times$ (noncentral χ^2 with n degrees of freedom and non-centrality parameter $\lambda = n(\xi - T)^2\sigma^{-2}$) .

The mean value is given by

$$E(\tilde{C}_{pm}) = \frac{d\sqrt{n}}{3\sigma} e^{-\lambda/2} \sum_{j=0}^{\infty} \frac{(\lambda/2)^j}{j!} \frac{\Gamma((n-1)/2 + j)}{2^{1/2} \Gamma\left(\frac{1}{2}n + j\right)} \quad (1.1)$$

and the variance is

$$\text{Var}(\tilde{C}_{pm}) = \left(\frac{d\sqrt{n}}{3\sigma} \right)^2 e^{-\lambda/2} \sum_{j=0}^{\infty} \frac{(\lambda/2)^j}{j!} \frac{1}{n-2+2j} - \{E(\tilde{C}_{pm})\}^2 . \quad (1.2)$$

T. Johnson (1992) introduces the concept of maximal worth W_T of an item corresponding to characteristic X having the target value T . As deviation of X from T increases, the worth becomes less eventually becoming 0 and then negative. The worth function is of the form:

$$W_T - K(X - T)^2 \quad \text{for } W_T > K(X - T)^2$$

and it becomes zero when $|X - T| = \sqrt{W_T/K}$. The ratio $\frac{W_T - K(X - T)^2}{W_T} = 1 - \frac{(X - T)^2}{\Delta^2}$ with $\Delta = \sqrt{W_T/K}$ is called the *relative worth*, and $(X - T)^2/\Delta^2$ is the *relative loss* (see T. Johnson, 1992). It is then easy to see that the expected relative loss, $L_e = E\left(\frac{(X - T)^2}{\Delta^2}\right)$, is given by:

$$L_e(C_{pm}) = \left(\frac{d}{3\Delta}\right)^2 C_{pm}^{-2} \quad (1.3)$$

yielding an inverse functional relationship between the loss function concepts and the process capability notions.

Calculation of $E(\tilde{C}_{pm})$ and $\text{Var}(\tilde{C}_{pm})$ present substantial difficulties and the index itself is not easy to calculate under stressful conditions of daily activities at a plant. It is therefore suggested in this paper to utilize an approximation to

C_{pm} denoted by C_{pq} which retains all the positive properties of C_{pm} for small and moderate deviation of ξ from the target value T plus the attractive computational simplicity.

A detailed numerical comparison is presented in Section 3.

2 The C_{pq} Index and Its Estimator

Definition: In terms of the basic index $C_p = \frac{d}{3\sigma}$, C_{pm} can be expressed as

$C_{pm} = \frac{C_p}{\sqrt{1 + \frac{(\xi - T)^2}{\sigma^2}}}$. Expanding the function $(1 + x^2)^{-1/2}$ we obtain the first approximation

$$C_{pq} = C_p \left[1 - \frac{1}{2} \left(\frac{\xi - T}{\sigma} \right)^2 \right] \quad (2.1)$$

and its estimator

$$\begin{aligned} \hat{C}_{pq} &= \hat{C}_p \left(1 - \frac{1}{2} \frac{(\bar{X} - T)^2}{S^2} \right) \\ &= \frac{d}{3S} \left(1 - \frac{1}{2} \frac{(\bar{X} - T)^2}{S^2} \right). \end{aligned} \quad (2.2)$$

Properties of C_{pq} and $\hat{C}_{pq}$.

Besides being easily computable, the index C_{pq} has the following properties.

- (i) $C_{pq} < C_p$.
- (ii) $\frac{C_{pq}}{C_{pm}} = \left[1 + \left(\frac{\xi - T}{\sigma} \right)^2 \right]^{1/2} \cdot \left[1 - \frac{1}{2} \left(\frac{\xi - T}{\sigma} \right)^2 \right]$.
- (iii) The distribution of $\hat{C}_{pq}$ is that of

$$\begin{aligned} &\frac{d}{3} \left[\frac{\sqrt{n-1}}{\sigma \chi_{n-1}} - \frac{(n-1)^{3/2}}{2\sigma^3 \chi_{n-1}^3} \left\{ N \left(\zeta - T, \frac{\sigma^2}{n} \right) \right\}^2 \right] \\ &= \frac{\sqrt{n-1} d}{3\sigma} \left[\frac{1}{\chi_{n-1}} - \frac{(n-1)}{2\chi_{n-1}^3} \left\{ N \left(\frac{\zeta - T}{\sigma}, \frac{1}{n} \right) \right\}^2 \right]. \end{aligned} \quad (2.3)$$

This follows from (2.2) by observing that independently, $\bar{X} \sim N(\zeta, \sigma^2/n)$ and $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$. Consequently, substituting $\frac{1}{S} \sim \frac{\sqrt{n-1}}{\sigma\chi_{n-1}}$ and $\frac{(\bar{X} - T)^2}{S^3} \sim \frac{(n-1)^{3/2}}{2\sigma^3\chi_{n-1}^3} \left\{ N\left(\zeta - T, \frac{\sigma^2}{n}\right) \right\}^2$, in (2.2) we obtain (2.3)

(iv) The expected relative loss of C_{pq} is given by:

$$L_e(C_{pq}) = 3\sigma^2 \left(\frac{\zeta - T}{\sigma} \right)^{-2} \left[1 - \frac{2\sigma}{d} C_{pq} \right] \quad (2.4)$$

and hence can be easily computed. From (1.3) and (2.4) it is noted that $L_e(C_{pm}) \propto C_{pm}^{-2}$ where as $L_e(C_{pq}) \propto (-C_{pq})$. Thus the expected relative loss of C_{pm} increases as C_{pm} decreases, it being inversely proportional to the square of C_{pm} . The expected relative loss of C_{pq} also increases as C_{pq} decreases, but the relationship is linear.

(v) The expected value and the variance of $\hat{C}_{pq}$ are given by

$$E(\hat{C}_{pq}) = \frac{3}{3\sigma} \left[a_1 - \frac{1}{2} a_3 \left(\frac{1}{n} + \left(\frac{\zeta - T}{\sigma} \right)^2 \right) \right] \quad (2.5)$$

and

$$\begin{aligned} \text{Var}(\hat{C}_{pq}) = & \frac{d^2}{9\sigma^2} \left[a_2 - a_4 \left(\frac{1}{n} + \left(\frac{\zeta - T}{\sigma} \right)^2 \right) \right. \\ & + \frac{1}{4} a_6 \left(\frac{3}{n^2} + \frac{6}{n} \left(\frac{\zeta - T}{\sigma} \right)^2 + \left(\frac{\zeta - T}{\sigma} \right)^4 \right) \\ & \left. - \left(a_1 - \frac{1}{2} a_3 \left(\frac{1}{n} + \left(\frac{\zeta - T}{\sigma} \right)^2 \right) \right) \right], \quad (2.6) \end{aligned}$$

where $a_j = \frac{(n-1)^{j/2} \Gamma[(n-1-j)/2]}{2^{j/2} \Gamma[(n-1)/2]}$, $j = 1, 2, \dots$

The expressions (2.5) and (2.6) are derived from the representation (2.3) by computing the required moments of the associated chi- and normal-random variables which are independent. Note that

$$a_j = E\left(\frac{\sigma}{S}\right)^j = E\left[\frac{\sqrt{n-1}}{\chi_{n-1}}\right]^j = \frac{(n-1)^{j/2} \Gamma\left(\frac{n-1-j}{2}\right)}{2^{j/2} \Gamma\left(\frac{n-1}{2}\right)}, \quad (2.7)$$

and

$$E\left(\frac{\bar{X} - T}{\sigma}\right)^k = \sum_{r=0}^k \binom{k}{r} \left(\frac{\zeta - T}{\sigma}\right)^{k-r} C_r, \quad (2.8)$$

where $C_r = 0$, for r odd, $C_r = \frac{(r-1)(r-3)\dots 3 \cdot 1}{n^{r/2}}$ for r even and $C_0 = 1$.

Substituting from (2.7) and (2.8) in the expressions for $E(\hat{C}_{pq})$ and $\text{Var}(\hat{C}_{pq}) = E(\hat{C}_{pq}^2) - \{E(\hat{C}_{pq})\}^2$ gives the results (2.5) and (2.6), respectively.

(vi) The asymptotic distribution of $\hat{C}_{pq}$ is given by

$$\sqrt{n}(\hat{C}_{pq} - C_{pq}) \xrightarrow{\mathcal{D}} X \quad \text{as } n \rightarrow \infty \quad (2.8)$$

where $X \sim N(0, \sigma_{pq}^2)$, and

$$\begin{aligned} \sigma_{pq}^2 = c_p^2 & \left[\left(\frac{\zeta - T}{\sigma} \right)^2 + \frac{1}{4}(\beta_2 - 1) \left(1 - \frac{3}{2} \left(\frac{\zeta - T}{\sigma} \right)^2 \right)^2 \right. \\ & \left. + \sqrt{\beta_1} \left(\frac{\zeta - T}{\sigma} \right) \left(1 - \frac{3}{2} \left(\frac{\zeta - T}{\sigma} \right)^2 \right) \right], \end{aligned} \quad (2.9)$$

and β_1 and β_2 are the measures of skewness and kurtosis respectively.

The result (2.8) is an application of the δ -method (see Serfling, 1980, p. 122). Since $\sqrt{n}(\bar{X}_n - \mu, S_n^2 - \sigma^2) \xrightarrow{\mathcal{D}} N(0, \Sigma)$ where

$$\Sigma = \begin{pmatrix} \sigma^2 & \mu_3 \\ \mu_3 & \mu_4 - \sigma^4 \end{pmatrix} = (\sigma_{ij}), \quad \text{say,}$$

and

$$g(\bar{X}_n, S_n^2) = \frac{d}{3S} - \frac{d}{6S^3}(\bar{X} - T)^2$$

is totally differentiable, therefore

$$\sqrt{n}[g(\bar{X}_n, S_n^2) - g(\mu, \sigma^2)] \xrightarrow{\mathcal{D}} N(0, D\Sigma D')$$

where $D = \left(\frac{\partial g(u, v)}{\partial u} \Big|_{\mu, \sigma^2}, \frac{\partial g(u, v)}{\partial v} \Big|_{\mu, \sigma^2} \right) \neq (0, 0)$. Now it is easy to see that $\frac{\partial g(u, v)}{\partial u} \Big|_{\mu, \sigma^2} = -\frac{d}{3\sigma^3}(\mu - T)$, and $\frac{\partial g(u, v)}{\partial v} \Big|_{\mu, \sigma^2} = -\frac{d}{6\sigma^3} + \frac{d(\mu - T)^2}{4\sigma^5}$. From these values we can verify that $D\Sigma D' = \sigma_{pq}^2$ as given in (2.9)

Remark: If sampling is from the normal population then $\beta_1 = 0$, $\beta_2 = 3$ and

$$\sigma_{pq}^2 = c_p^2 \left[\left(\frac{\zeta - T}{\sigma} \right)^2 + \frac{1}{2} \left(1 - \frac{3}{2} \left(\frac{\zeta - T}{\sigma} \right)^2 \right)^2 \right].$$

Furthermore if the target value is the same as the population mean i.e. $\zeta = T$, then $\sigma_{pq}^2 = \frac{1}{2} c_p^2 = \sigma_{pm}^2$, where σ_{pm}^2 is the asymptotic variance of $\hat{C}_{pm}$. Consequently approximate (asymptotic) $100(1 - \alpha)\%$ confidence interval for C_{pq} is

$$\hat{C}_{pq} \pm z_{\alpha/2} \hat{\sigma}_{pq}$$

where $\hat{\sigma}_{pq}^2$ is obtained from σ_{pq}^2 by substituting $\bar{X}$ for ζ and S for σ .

(vii) $\hat{C}_{pq}$ is a consistent estimator of C_{pq} . Since $S_n^2 \xrightarrow{P} \sigma^2$, $\bar{X}_n \xrightarrow{P} \mu$, and $\hat{C}_{pq}$ is a continuous function of S_n^2 and $\bar{X}_n$, the result follows.

3 Numerical Comparison

Tables 1 and 2 present values of $\hat{C}_{pq}$ and $\tilde{C}_{pm}$ respectively for the sample sizes $n = 10(10)50$, $d/\sigma = 2(1)6$ and $\frac{|\zeta - T|}{\sigma} = 0(1).5$ together with the corresponding

theoretical values of C_{pq} and C_{pm} . Clearly for $\frac{\zeta - T}{\sigma} = 0$ the *theoretical* values

coincide both being equal to the $C_p = \frac{d}{3\sigma}$. However for this case of the target

value being at the expected value of the process, the bias of $\hat{C}_{pq}$ is by far smaller than the bias of $\tilde{C}_{pm}$ and the variance of $\hat{C}_{pq}$ is slightly less than that of $\tilde{C}_{pm}$ except for the small sample size of $n = 10$. The effect is more pronounced as d/σ (i.e. C_p) becomes large.

The situation for value of parameters $\frac{\xi - T}{\sigma} = .1$, i.e. the deviation of the target value from the mean of the process is 10% of the standard deviation is as follows: The bias of $\hat{C}_{pq}$ remains to be substantially smaller than that of $\tilde{C}_{pm}$ (roughly to same extent as the $\xi = T$ case) even for the case of $n = 50$. Similarly the variance of $\hat{C}_{pq}$ remains slightly smaller than that of $\tilde{C}_{pm}$ for $n \geq 20$; however the variance of $\hat{C}_{pm}$ is smaller than that of $\tilde{C}_{pq}$ for $n = 10$ to a greater extent than in the $\xi = T$ case. In fact the variance of $\tilde{C}_{pm}$ unlike that of $\hat{C}_{pq}$ decreases as $\frac{|\xi - T|}{\sigma}$ increases from 0 to .5, while that of $\hat{C}_{pq}$ remains remarkably constant for $n \geq 30$ even for large d/σ .

The behaviour of $\hat{C}_{pq}$ and that of $\tilde{C}_{pm}$ for $\frac{\xi - T}{\sigma} = .2$ now follows the same pattern described above.

The value of $\hat{C}_{pq}$ remains consistent with a very small bias. Remarkably the bias is slightly larger for larger values of n , while that of $\tilde{C}_{pm}$ carries a substantial bias especially for smaller n . As to variance, $\tilde{C}_{pm}$ becomes noticeably superior over $\hat{C}_{pq}$ for $n = 10$ and catches up with $\hat{C}_{pq}$ for the large values of n , where the variances are almost identical.

We note that here we observe the *transition* from the positive bias to the negative bias of $\hat{C}_{pq}$. For $|\xi - T| = .36$ we observe the interesting phenomenon that $\hat{C}_{pq}$ becomes actually unbiased for all $n \geq 20$, the bias of $\tilde{C}_{pm}$ diminishes but it is still quite noticeable especially for $n \leq 20$. The variances of the estimators remain practically identical for $n \geq 20$, but variances of $\hat{C}_{pq}$ far exceed that of $\tilde{C}_{pm}$ for $n = 10$ (almost double).

The situation for $|\xi - T| = .4\sigma$ indicates a stabilizing pattern. The bias of $\hat{C}_{pq}$ monotonically decreases with the sample size, the estimator becoming practically unbiased for $n = 50$ for all $d/\sigma = 2(1)6$ and the bias remains to be negligible for practical purposes even for $n = 20$, while noticeable for $n = 10$. As in the case of $|\xi - T| = .3\sigma$ the variances are almost the same for $n \geq 30$, however even though for $n = 20$ and $d/\sigma \geq 3$, the variance of $\tilde{C}_{pm}$ is noticeably smaller than that of $\hat{C}_{pq}$, the variance of $\hat{C}_{pq}$ exceeds that of $\tilde{C}_{pm}$ (almost thrice!) for $n = 10$.

Finally in the case $|\xi - T| = .5\sigma$, $E(\hat{C}_{pq}) < C_{pq}$ for all $n \geq 10$, (unlike the situation for $|\xi - T| \leq .2\sigma$, say) with $C_{pq} - E(\hat{C}_{pq})$ becoming negligible for $d/\sigma \leq 3$ and $n \geq 30$, and rather substantial for n small and large d/σ . On the other hand, the *positive bias* of $\tilde{C}_{pm}$ is relatively less pronounced and becoming comparable to that of $\hat{C}_{pq}$. As far as variances are concerned, as in the previous case, $\text{Var}(\tilde{C}_{pm}) < \text{Var}(\hat{C}_{pq})$ for all the values of the parameters but the differences are of practical significance only for $n = 10$ where the variance of $\hat{C}_{pq}$ attains unacceptably high values. (It must be emphasized that except for this situation both $\text{Var}(\tilde{C}_{pm})$ and $\text{Var}(\hat{C}_{pq})$ are *remarkably* small for all values of d/σ , n and $|\xi - T|/\sigma$ investigated in this paper.)

This detailed analysis suggests that both estimators are very stable while the smaller bias of $\hat{C}_{pq}$ makes this “sample” index and its “population” indicator C_{pq}

Table 1. Expected values (EV) and variances (VAR) of $\hat{C}_{pq}$ for $n = 10(10)50; d/\sigma = 2(1)6$ and $\frac{|\zeta - T|}{\sigma} = 0(1).5$

n	d/σ	0.0			0.1			$ \zeta - T /\sigma$ 0.2			0.3			0.4			0.5		
		EV		VAR	EV		VAR	EV		VAR	EV		VAR	EV		VAR	EV		VAR
		EV	VAR	EV	VAR	EV	VAR	EV	VAR	EV	VAR	EV	VAR	EV	VAR	EV	VAR	EV	VAR
10	2	0.675	0.036	0.669	0.038	0.653	0.044	0.626	0.056	0.587	0.076	0.538	0.109						
	3	1.012	0.082	1.004	0.086	0.979	0.099	0.938	0.125	0.881	0.171	0.807	0.245						
	4	1.350	0.146	1.339	0.153	1.306	0.176	1.251	0.223	1.174	0.304	1.076	0.436						
	5	1.687	0.227	1.673	0.238	1.632	0.275	1.564	0.348	1.468	0.475	1.345	0.682						
	6	2.024	0.327	2.008	0.343	1.959	0.396	1.877	0.502	1.762	0.685	1.614	0.981						
	20	2	0.674	0.013	0.670	0.013	0.657	0.013	0.637	0.013	0.608	0.014	0.571	0.015					
3	1.011	0.029	1.005	0.029	0.986	0.029	0.955	0.029	0.912	0.031	0.856	0.035							
4	1.348	0.051	1.340	0.051	1.315	0.051	1.274	0.052	1.216	0.055	1.142	0.062							
5	1.685	0.080	1.674	0.080	1.643	0.080	1.592	0.081	1.520	0.086	1.427	0.097							
6	2.022	0.116	2.009	0.115	1.972	0.115	1.910	0.117	1.824	0.123	1.712	0.139							
30	2	0.672	0.008	0.668	0.008	0.657	0.008	0.637	0.008	0.611	0.008	0.576	0.008						
	3	1.008	0.018	1.002	0.018	0.985	0.018	0.956	0.018	0.916	0.018	0.865	0.019						
	4	1.344	0.033	1.336	0.032	1.313	0.032	1.275	0.031	1.221	0.032	1.153	0.034						
	5	1.680	0.051	1.670	0.051	1.417	0.050	1.594	0.049	1.527	0.050	1.441	0.053						
	6	2.015	0.073	2.004	0.073	1.970	0.072	1.912	0.071	1.832	0.071	1.729	0.076						
	40	2	0.671	0.006	0.667	0.006	0.656	0.006	0.637	0.006	0.612	0.006	0.579	0.006					
3	1.006	0.013	1.000	0.013	0.984	0.013	0.956	0.013	0.918	0.013	0.868	0.013							
4	1.341	0.024	1.334	0.024	1.312	0.023	1.275	0.023	1.223	0.023	1.157	0.024							
5	1.677	0.037	1.667	0.037	1.640	0.036	1.594	0.035	1.529	0.035	1.446	0.037							
6	2.012	0.054	2.001	0.053	1.968	0.052	1.912	0.051	1.853	0.051	1.736	0.053							

2	0.670	0.005	0.666	0.005	0.655	0.005	0.637	0.004	0.612	0.004	0.580	0.005
3	1.005	0.011	0.999	0.010	0.983	0.010	0.956	0.010	0.918	0.010	0.870	0.010
4	1.340	0.019	1.333	0.019	1.311	0.018	1.275	0.018	1.224	0.018	1.159	0.018
5	1.675	0.029	1.666	0.029	1.639	0.029	1.594	0.028	1.530	0.027	1.449	0.028
6	2.010	0.042	1.999	0.042	1.966	0.041	1.912	0.040	1.837	0.040	1.739	0.041
	C_{pq}		C_{pq}		C_{pq}		C_{pq}		C_{pq}		C_{pq}	
2	0.667		0.663		0.653		0.637		0.613		0.583	
3	1.000		0.995		0.980		0.955		0.920		0.875	
4	1.333		1.327		1.307		1.273		1.227		1.167	
5	1.667		1.658		1.633		1.592		1.533		1.458	
6	2.000		1.990		1.960		1.910		1.840		1.750	

2	0.677	0.005	0.674	0.005	0.664	0.005	0.648	0.004	0.628	0.004	0.605	0.004
3	1.105	0.011	1.010	0.011	0.996	0.010	0.972	0.010	0.942	0.009	0.908	0.008
4	1.354	0.019	1.347	0.019	1.327	0.018	1.297	0.017	1.257	0.016	1.210	0.015
5	1.692	0.030	1.684	0.030	1.659	0.029	1.621	0.027	1.571	0.025	1.513	0.023
6	2.031	0.043	2.021	0.043	1.991	0.041	1.945	0.039	1.885	0.037	1.815	0.033
	C_{pm}		C_{pm}		C_{pm}		C_{pm}		C_{pm}		C_{pm}	
2	0.667		0.663		0.654		0.639		0.619		0.596	
3	1.000		0.995		0.981		0.958		0.928		0.894	
4	1.333		1.327		1.307		1.277		1.238		1.193	
5	1.667		1.658		1.634		1.596		1.547		1.491	
6	2.000		1.990		1.961		1.916		1.857		1.789	

extremely attractive when the deviation of the target value from the mean of the process does not exceed roughly $1/3$ of the standard deviation and highly recommended for $n \geq 20$.

Acknowledgement: The authors would like to acknowledge the help of Mr. Yue Liu in the computations for $\hat{C}_{pq}$ and Professor Norman L. Johnson for providing the corresponding computations for $\hat{C}_{pm}$. They are also thankful to Professor Uhlmann and an anonymous referee for useful comments on an earlier version of this paper.

References

- Chan LK, Cheng SW, Spiring FA (1988) A new measure of process capability: C_{pm} Journ Qual Techn 20:162–175
- Johnson T (1992) The relationship of C_{pm} to squared error loss. Journ Qual Techn 24 4:211–215
- Pearn WL, Kotz S, Johnson NL (1992) Distributional and inferential properties of process capability indices. Journ Qual Techn 24 4:216–231
- Serfling RJ (1980) Approximation theorems of mathematical statistics. Wiley, New York
- Spiring FA (1991) The C_{pm} index. Quality Progress 24(2):57–61

Received 15.02.1994

Revised version 24.05.1995

Remarks on the Strong Law of Large Numbers for a Triangular Array of Associated Random Variables

ISHA DEWAN AND B. L. S. PRAKASA RAO

Indian Statistical Institute, 7, S.J.S. Sansanwal Marg, New Delhi-110016, India

Abstract: A strong law of large numbers for a triangular array of strictly stationary associated random variables is proved. It is used to derive the pointwise strong consistency of kernel type density estimator of the one-dimensional marginal density function of a strictly stationary sequence of associated random variables, and to obtain an improved version of a result by Van Ryzin (1969) on the strong consistency of density estimator for a sequence of independent and identically distributed random variables.

Key Words: Strong law of large numbers, associated random variables, kernel type density estimation.

1 Introduction

The concept of association was introduced by Esary, Proschan and Walkup (1967): a finite family $\{X_1, \dots, X_m\}$ of random variables is said to be associated if

$$\text{Cov}(h(X_1, \dots, X_m), g(X_1, \dots, X_m)) \geq 0$$

for any componentwise nondecreasing functions h, g on R^m such that the covariance exists. An infinite family of random variables is said to be associated if every finite subfamily is associated.

Independent random variables represent one example of associated random variables (Esary, Proschan and Walkup (1967)). There are several examples in Reliability and Survival Analysis where the random variables of interest are not independent but are associated. Some common examples of associated random variables are: positively correlated normal random variables (Pitt (1982)), components of the multivariate exponential distribution due to Marshall and Olkin (1967) etc.

Some probabilistic and statistical aspects of associated random variables have been extensively discussed in recent years, see for example, Birkel (1988 a, b, 1989), Cox and Grimmett (1984), Newman (1984), Bagai and Prakasa Rao

(1991, 1995), Prakasa Rao (1993) among others. Strong law of large numbers for a stationary sequence and an arbitrary sequence of associated random variables have been obtained by Newman (1984) and Birkel (1989), respectively.

In what follows we prove a strong law of large numbers for a triangular array of associated random variables and give an application to kernel-type density estimation for a strictly stationary associated sequence of random variables. We also prove an improved version of a result by Van Ryzin (1969) on the strong consistency of density estimator for a sequence of independent and identically distributed (i.i.d.) random variables.

2 SLLN for Triangular Array

Consider a triangular array of random variables $\{X_{nj}, 1 \leq j \leq k_n, n \geq 1\}$ where the random variables in each row are strictly stationary and associated.

Let c denote a generic positive constant which may vary from one step to another. We denote $\text{Var}(X)$ as $\text{Cov}(X, X)$ for convenience.

Lemma 2.1: Let $\{X_{nj}, 1 \leq j \leq k_n, n \geq 1\}$ be a strictly stationary associated triangular array of random variables with $\text{Var}(X_{n1}) < \infty$ for all n . Let $S_{n,t} = \sum_{j=1}^t X_{nj}$. Suppose that there exists $c > 0$ such that for every $n \geq 1$,

$$\sum_{j=1}^{k_n} \text{Cov}(X_{n1}, X_{nj}) \leq c. \quad (2.1)$$

Then, for $n \geq 1$,

$$\text{Var}(S_{n,k_n}) \leq ck_n. \quad (2.2)$$

Proof: Note that, for $n \geq 2$,

$$\begin{aligned} \text{Var}(S_{n,k_n}) &= \text{Var} \left[\sum_{j=1}^{k_n} X_{nj} \right] \\ &= \sum_{j=1}^{k_n} \text{Var}(X_{nj}) + 2 \sum_{1 \leq i < j \leq k_n} \text{Cov}(X_{ni}, X_{nj}) \end{aligned}$$

$$\begin{aligned}
&\leq 2 \left[\sum_{j=1}^{k_n} \text{Var}(X_{nj}) + \sum_{1 \leq i < j \leq k_n} \text{Cov}(X_{ni}, X_{nj}) \right] \\
&= 2 \left[\sum_{j=1}^{k_n} \text{Var}(X_{nj}) + \sum_{j=2}^{k_n} \text{Cov}(X_{n1}, X_{nj}) \right. \\
&\quad \left. + \sum_{j=3}^{k_n} \text{Cov}(X_{n2}, X_{nj}) + \cdots + \text{Cov}(X_{n, k_n-1}, X_{nk_n}) \right] \\
&= 2 \left[\sum_{j=1}^{k_n} \text{Cov}(X_{n1}, X_{nj}) + \sum_{j=2}^{k_n} \text{Cov}(X_{n2}, X_{nj}) + \cdots \right. \\
&\quad \left. + \sum_{j=k_n-1}^{k_n} \text{Cov}(X_{n, k_n-1}, X_{nj}) + \text{Cov}(X_{n, k_n}, X_{n, k_n}) \right] \\
&= 2 \left[\sum_{j=1}^{k_n} \text{Cov}(X_{n1}, X_{nj}) + \sum_{j=1}^{k_n-1} \text{Cov}(X_{n1}, X_{nj}) + \cdots \right. \\
&\quad \left. + \sum_{j=1}^2 \text{Cov}(X_{n1}, X_{nj}) + \text{Cov}(X_{n1}, X_{n1}) \right] \quad (\text{by stationarity of } X_{nj}) \\
&\leq ck_n \quad (\text{by (2.1)}) .
\end{aligned}$$

Remark 2.2: The condition that $\{X_{nj}\}$ is associated is not necessary for the above lemma. It is sufficient to assume that $\text{Cov}(X_{ni}, X_{nj}) \geq 0$ for all $1 \leq i < j \leq k_n$, $n \geq 1$.

Theorem 2.3: Let $\{X_{nj}, 1 \leq j \leq k_n, n \geq 1\}$ be a triangular array of strictly stationary associated random variables with $E[X_{n1}] = 0$ and $\text{Var}(X_{n1}) < \infty$ for all n . Suppose that $k_n = O(n^\gamma)$ for some $0 \leq \gamma < \frac{3}{2}$ and the condition (2.1) holds. Further suppose that

$$E \left[\max_{n^2 < j \leq (n+1)^2} |S_{j, k_j} - S_{n^2, k_{n^2}}| \right]^2 = O(n^{4-\delta}) \quad (2.3)$$

for some $\delta > 1$. Then,

$$\frac{S_{n, k_n}}{n} \rightarrow 0 \quad \text{a.s. as } n \rightarrow \infty . \quad (2.4)$$

Proof: Using Chebychev's inequality and Lemma 2.1, we note that

$$\Pr[|S_{n^2, k_{n^2}}| > n^2 \varepsilon] \leq \frac{ck_{n^2}}{n^4 \varepsilon^2} .$$

Note that $\sum_n \Pr[|S_{n^2, k_{n^2}}| > n^2 \varepsilon] < \infty$ for all $\varepsilon > 0$, and hence, by the Borel-Cantelli lemma, it follows that

$$\frac{S_{n^2, k_{n^2}}}{n^2} \rightarrow 0 \quad \text{a.s. as } n \rightarrow \infty . \quad (2.5)$$

Let

$$D_n = \max_{n^2 < j \leq (n+1)^2} |S_{j, k_j} - S_{n^2, k_{n^2}}| .$$

Then

$$\frac{|S_{j, k_j}|}{j} \leq \frac{|S_{n^2, k_{n^2}}| + D_n}{n^2} \quad \text{for } n^2 < j \leq (n+1)^2 .$$

If

$$\frac{D_n}{n^2} \rightarrow 0 \quad \text{a.s. as } n \rightarrow \infty \quad (2.6)$$

then, from (2.5) and (2.6), it would follow that,

$$\frac{S_{n, k_n}}{n} \rightarrow 0 \quad \text{a.s. as } n \rightarrow \infty . \quad (2.7)$$

Thus it is sufficient to prove that

$$\sum_n \Pr \left\{ \max_{n^2 < j \leq (n+1)^2} |S_{j, k_j} - S_{n^2, k_{n^2}}| \geq n^2 \varepsilon \right\} < \infty$$

for all $\varepsilon > 0$. By Chebychev's inequality

$$\Pr \left\{ \max_{n^2 < j \leq (n+1)^2} |S_{j,k_j} - S_{n^2,k_{n^2}}| \geq n^2 \varepsilon \right\} \leq \frac{1}{n^4 \varepsilon^2} E \left[\max_{n^2 < j \leq (n+1)^2} |S_{j,k_j} - S_{n^2,k_{n^2}}| \right]^2 .$$

The problem is to estimate

$$E \left[\max_{n^2 < j \leq (n+1)^2} |S_{j,k_j} - S_{n^2,k_{n^2}}| \right]^2 .$$

Since this term is $O(n^{4-\delta})$ for some $\delta > 1$ by assumption, the result holds by the Borel-Cantelli lemma.

Remark 2.4: Cox and Grimmett (1984) and Birkel (1988a,b), among others, observed that in any asymptotic study of a sequence of associated random variables, the covariance structure plays an important role. Cox and Grimmett (1984), while considering asymptotic normality of a triangular array of associated random variables, assume that there exists a function,

$$u = \{0, 1, \dots\} \rightarrow R, \quad u(r) \rightarrow 0 \quad \text{as } r \rightarrow \infty$$

such that

$$\sum_{j: |\ell-j| \geq r} \text{Cov}(X_{nj}, X_{ni}) \leq u(r) \quad \text{for all } \ell, n, r \geq 0. \quad (2.8)$$

It can be easily seen that (2.8) implies (2.1) if $\{X_{ni}\}$ is a strictly stationary associated sequence for each n .

3 Applications

Let $\{X_n, n \geq 1\}$ be a strictly stationary sequence of associated random variables with one-dimensional unknown marginal density function f for X_1 . Let

$$f_n(x) = \frac{1}{nh_n} \sum_{j=1}^n K \left(\frac{x - X_j}{h_n} \right), \quad x \in R \quad (3.1)$$

be a kernel type estimator for $f(x)$ where h_n is a bandwidth and $K(\cdot)$ is a kernel satisfying the following conditions:

- (A₁) $K(\cdot)$ is a bounded density function of bounded variation on R . Hence $K(x) = K_1(x) - K_2(x)$, where $K_1(x)$ and $K_2(x)$ are two monotonic functions. Suppose that (i) $\lim_{|x| \rightarrow \infty} |x| K(x) = 0$, (ii) $\int_{-\infty}^{\infty} u^j K_i^2(u) du < \infty$ $i = 1, 2$, $j = 0, 1, 2$;
- (A₂) $K_i(x)$ are differentiable with $\sup_x |K'_i(x)| \leq c < \infty$, $i = 1, 2$.

Further, assume that the covariance structure of $\{X_n\}$ satisfies the condition:

- (A₃) for all ℓ and r

$$\sum_{j: |\ell-j| \geq r} \text{Cov}(X_j, X_\ell) \leq u(r)$$

where $u(r) = e^{-\alpha r}$ for some $\alpha > 0$.

- (A₄) In addition, assume that f is thrice differentiable and $\sup_x |f'''(x)| < \infty$.

Remark 3.1: The assumption that $K(\cdot)$ is of bounded variation is used to ensure that the transformed variables $K_j\left(\frac{x - X_i}{h_n}\right)$, $1 \leq i \leq n$ are associated for $j = 1, 2$. This is also used in the classical density estimation for i.i.d. random variables (cf. Prakasa Rao (1983), pp 37). In view of (A₂), Newman's (1980) inequality (see Lemma 3.2) can be used to estimate covariance of functions of associated random variables. Importance of (A₃) is discussed in Remark 2.4 and assumption (A₄) is used to obtain a bound on the variance of the estimator as in the classical i.i.d. case (cf. Praskasa Rao (1983)).

The following inequality due to Newman (1980) is used later in the paper.

Lemma 3.2: (Newman (1980)). Let (Z_1, Z_2) be associated random variables with finite variance. Then, for any two differentiable functions h and g ,

$$|\text{Cov}(h(Z_1), g(Z_2))| \leq \sup_x |h'(x)| \sup_y |g'(y)| \text{Cov}(Z_1, Z_2) \quad (3.2)$$

where h' and g' denote the derivatives of h and g respectively.

Let

$$\begin{aligned}
X_{ni} &= \frac{1}{h_n} \left\{ K \left(\frac{x - X_i}{h_n} \right) - EK \left(\frac{x - X_i}{h_n} \right) \right\} \\
&= \frac{1}{h_n} \left\{ \left[K_1 \left(\frac{x - X_i}{h_n} \right) - EK_1 \left(\frac{x - X_i}{h_n} \right) \right] \right. \\
&\quad \left. - \left[K_2 \left(\frac{x - X_i}{h_n} \right) - EK_2 \left(\frac{x - X_i}{h_n} \right) \right] \right\} \\
&= \psi_n^{(1)}(x, X_i) - \psi_n^{(2)}(x, X_i) \quad (\text{say}) .
\end{aligned} \tag{3.3}$$

Then, for fixed x , $\{\psi_n^{(j)}(x, X_i)\}$, $j = 1, 2$, being monotonic functions of associated random variables are associated. Using Lemma 3.2 it follows, that for $j = 1, 2$,

$$\begin{aligned}
&\sum_{j: |\ell-j| \geq r} \text{Cov}(\psi_n^{(j)}(x, X_j), \psi_n^{(j)}(x, X_\ell)) \\
&\leq \sup_y \left\{ \frac{\partial}{\partial y} \psi_n^{(j)}(x, y) \right\}^2 \sum_{j: |\ell-j| \geq r} \text{Cov}(X_j, X_\ell) .
\end{aligned} \tag{3.4}$$

In view of (A₁) to (A₃) and Remark 2.4, the conditions stated in Lemma 2.1 hold for $\psi_n^{(j)}(x, X_i)$, $j = 1, 2$. Using (A₄) it is easy to see that, for $j = 1, 2$,

$$\text{Var}(\psi_n^{(j)}(x, X_1)) \leq \frac{1}{h_n} [f(x)\beta_0^{(j)} - f'(x)h_n\beta_1^{(j)} + f''(x)h_n^2\beta_2^{(j)} + O(h_n^3)] \tag{3.5}$$

where

$$\beta_i^{(j)} = \int_{-\infty}^{\infty} x^i K_f^2(x) dx , \quad i = 0, 1, 2 . \tag{3.6}$$

Furthermore, for $j = 1, 2$,

$$\sum_{1 \leq i} \sum_{< j \leq n} \text{Cov}\{\psi_n^{(j)}(x, X_i), \psi_n^{(j)}(x, X_j)\} \leq \frac{cn}{h_n^4} \tag{3.7}$$

by (A₃) and stationarity. Let

$$T_j^{(n)} = \sum_{i=1}^j X_{ni} . \tag{3.8}$$

Then

$$\begin{aligned}
 & E \left[\max_{n^2 < j \leq (n+1)^2} |T_j^{(n)} - T_{n^2}^{(n)}|^2 \right] \\
 & \leq \sum_{j=n^2+1}^{(n+1)^2} E \left\{ \left(\sum_{i=n^2+1}^j X_{ni} \right)^2 \right\} \\
 & = \sum_{j=n^2+1}^{(n+1)^2} E \left\{ \left(\sum_{i=n^2+1}^j \psi_n^{(1)}(x, X_i) - \sum_{i=n^2+1}^j \psi_n^{(2)}(x, X_i) \right)^2 \right\} \\
 & \leq 2 \sum_{j=n^2+1}^{(n+1)^2} \left\{ E \left(\sum_{i=n^2+1}^j \psi_n^{(1)}(x, X_i) \right)^2 + E \left(\sum_{i=n^2+1}^j \psi_n^{(2)}(x, X_i) \right)^2 \right\} \\
 & = 2 \sum_{j=n^2+1}^{(n+1)^2} \left\{ \text{Var} \left(\sum_{i=n^2+1}^j \psi_n^{(1)}(x, X_i) \right) + \text{Var} \left(\sum_{i=n^2+1}^j \psi_n^{(2)}(x, X_i) \right) \right\} \\
 & \leq 4n \left[\text{Var} \left(\sum_{i=n^2+1}^{(n+1)^2} \psi_n^{(1)}(x, X_i) \right) + \text{Var} \left(\sum_{i=n^2+1}^{(n+1)^2} \psi_n^{(2)}(x, X_i) \right) \right] \\
 & \quad (\text{by the associativity of } \{\psi_n^{(1)}\} \text{ and } \{\psi_n^{(2)}\}) \\
 & = 4n \left\{ O \left(\frac{n}{h_n} \right) + O \left(\frac{n}{h_n^4} \right) \right\} \quad (\text{using (3.4) and (3.6)}) \\
 & = O(n^2 h_n^{-4}) . \tag{3.9}
 \end{aligned}$$

Note that, if $h_n = O(n^{-\alpha})$, where $\alpha > 0$, then

$$E \left[\max_{n^2 < j \leq (n+1)^2} |T_j^{(n)} - T_{n^2}^{(n)}|^2 \right] = O(n^{2+4\alpha}) \tag{3.10}$$

and the condition (2.3) holds provided $\alpha < \frac{1}{4}$. In particular, for the optimal choice $h_n = O(n^{-1/5})$ (cf. Bagai and Prakasa Rao (1995)) the condition (2.3) holds.

Theorem 3.1: Suppose that the conditions (A1) to (A4) hold and $h_n = O(n^{-\alpha})$ where $0 < \alpha < \frac{1}{4}$. Define $f_n(x)$ by (3.1). Then,

$$f_n(x) \rightarrow f(x) \quad \text{a.s. as } n \rightarrow \infty \tag{3.11}$$

at all continuity points x of f .

Proof: It is clear from the calculations given above that the conditions stated in Theorem 2.3 hold. Hence $f_n(x) - Ef_n(x) \rightarrow 0$ a.s. as $n \rightarrow \infty$. Since $E[f_n(x)] \rightarrow f(x)$ as $n \rightarrow \infty$ at the continuity points x of $f(\cdot)$ under $(A_1)(i)$, it follows that (3.10) holds, proving Theorem 3.1.

Remark 3.2: Strong consistency for a kernel-type density estimator was proved for a sequence of i.i.d. random variables by Van Ryzin (1969) and for mixing sequences by Roussas (1988).

In the case of a sequence of i.i.d. random variables $\{X_n\}$ with finite second moment, condition that $K(\cdot)$ is of bounded variation in (A_1) and assumption (A_2) can be removed. Assumption (A_3) holds in view of independent $\{X_n\}$. Suppose assumption (A_4) is satisfied. The following result on strong consistency for density estimators for i.i.d. random variables holds improving upon the earlier result of Van Ryzin (1969) in the i.i.d. case.

Theorem 3.3: Let $\{X_i, i \geq 1\}$ be i.i.d. with density f . Suppose $K(\cdot)$ is a bounded probability density function satisfying (i) $\lim_{|x| \rightarrow \infty} |x| K(x) = 0$, (ii) $\int_{-\infty}^{\infty} x^j K^2(x) dx < \infty$ $j = 0, 1, 2$ and (A_4) holds. Further assume that $h_n = O(n^{-\alpha})$ where $0 < \alpha < 1$. Then

$$f_n(x) \rightarrow f(x) \quad \text{a.s. as } n \rightarrow \infty \quad (3.12)$$

for all continuity points x of f .

Proof: This follows immediately from the fact that $X_{ni} = \frac{1}{h_n} \left\{ K\left(\frac{x - X_i}{h_n}\right) - EK\left(\frac{x - X_i}{h_n}\right) \right\}$, $1 \leq i \leq n$, $n \geq 1$ form a triangular array of strictly stationary associated random variables. Let $T_j^{(n)}$ be as defined in (3.8). It is easy to check that

$$E \left[\max_{n^2 < j \leq (n+1)^2} |T_j^{(n)} - T_{n^2}^{(n)}|^2 \right] = O\left(\frac{n^2}{h_n}\right) = O(n^{2+\alpha})$$

if $h_n = O(n^{-\alpha})$ and the condition (2.3) holds for $\alpha < 1$. Hence

$$f_n(x) - E[f_n(x)] \rightarrow 0 \quad \text{a.s. as } n \rightarrow \infty.$$

Since $E[f_n(x)] \rightarrow f(x)$ as $n \rightarrow \infty$ at the continuity points x of $f(\cdot)$ under $(A_1)(i)$, it follows that (3.12) holds, proving Theorem 3.3.

Acknowledgement: The authors thank the referee for comments and suggestions.

References

- Bagai I, Prakasa Rao BLS (1991) Estimation of the survival function for stationary associated processes. *Statist Probab Letters* 12:385–391
- Bagai I, Prakasa Rao BLS (1995) Kernel-type density and failure rate estimation for associated sequences. *Ann Inst Statist Math* 47:253–266
- Birkel T (1988a) Moment bounds for associated sequences. *Ann Probab* 16:1184–1193
- Birkel T (1988b) On the convergence rate in the central limit theorem for associated processes. *Ann Probab* 16:1685–1698
- Birkel T (1989) A note on the strong law of large numbers for positively dependent random variables. *Statist Probab Letters* 7:17–20
- Cox JT, Grimmett G (1984) Central limit theorems for associated random variables and the percolation model. *Ann Probab* 12:514–528
- Esary J, Proschan F, Walkup D (1967) Association of random variables with applications. *Ann Math Statist* 38:1466–1474
- Marshall AW, Olkin I (1967) A multivariate exponential distribution. *J Amer Statist Assoc* 62:30–44
- Newman CM (1980) Normal fluctuations and the FKG inequalities. *Comm Math Phys* 74:119–128
- Newman CM (1984) Asymptotic independence and limit theorems for positively and negatively dependent random variables. In: Tong YL (Ed) *Inequalities in Statistics and Probability*. IMS, Hayward, CA 127–140
- Pitt L (1982) Positively correlated normal variables are associated. *Ann Probab* 10:496–499
- Prakasa Rao BLS (1983) *Nonparametric functional estimation*. Academic Press, New York
- Prakasa Rao BLS (1993) Bernstein-type Inequality for Associated Sequences. In: Ghosh JK, Mitra SK, Parthasarathy KR, Prakasa Rao BLS (Eds) *Statistics and Probability: A Raghu Raj Bahadur Festschrift*. Wiley Eastern, New Delhi 499–509
- Roussas GG (1988) Nonparametric estimation in mixing sequences of random variables. *J Statist Plann Inference* 18:135–149
- Van Ryzin J (1969) On strong consistency of density estimates. *Ann Math Statist* 40:1765–1772

Received 06.11.1995

On Estimating the Reliability After Sequentially Estimating the Mean: The Exponential Case

Nitis MUKHOPADHYAY

Department of Statistics, University of Connecticut, Storrs, CT 02629, USA

A. R. PADMANABHAN

Department of Mathematics, Monash University, Clayton, Victoria 3168, Australia

T. K. S. SOLANKY

Department of Mathematics, University of New Orleans, New Orleans, LA 70148, USA

Abstract: Sequential estimation problems for the mean parameter of an exponential distribution has received much attention over the years. Purely sequential and accelerated sequential estimators and their asymptotic second-order characteristics have been laid out in the existing literature, both for minimum risk point as well as bounded length confidence interval estimation of the mean parameter. Having obtained a data set from such sequentially designed experiments, the paper investigates estimation problems for the associated *reliability function*. Second-order approximations are provided for the bias and mean squared error of the proposed estimator of the reliability function, first under a general setup. An ad hoc bias-corrected version is also introduced. Then, the proposed estimator is investigated further under some specific sequential sampling strategies, already available in the literature. In the end, simulation results are presented for comparing the proposed estimators of the reliability function for moderate sample sizes and various sequential sampling strategies.

Key Words and Phrases: Exponential distribution; reliability function; estimation; bias correction; mean square error; sequential sampling.

1 Introduction

Suppose that $X_1, X_2, \dots$ are independent and identically distributed (i.i.d.) random variables representing failure times of a certain component, having the probability density function (p.d.f.)

$$f(x; \lambda) = \lambda^{-1} \exp(-x/\lambda) I(x > 0) . \quad (1.1)$$

Here, $I(\cdot)$ stands for the indicator function of $(\cdot)$ and $0 < \lambda < \infty$ represents the mean of the distribution. Sequential estimation problems for the parameter λ

have drawn much attention in the literature over the years. Sequential minimum risk point estimation of λ was addressed in Starr and Woodroffe (1972), and later in Woodroffe (1977). A review appeared in Mukhopadhyay (1988). Subsequently, multi-sample extensions and second-order characteristics associated with such methodologies were provided in Mukhopadhyay and Chattopadhyay (1991). More recently, sequential fixed-width confidence interval problems for λ and associated second-order properties have been provided in Mukhopadhyay and Datta (1996). Also, analogous accelerated sequential estimation techniques and other formulations have received attention in Mukhopadhyay (1993) and Mukhopadhyay and Datta (1995, 1996). Refer to Basu (1991) for many other related details.

In general, in the area of sequential estimation, one arrives with an appropriate stopping variable N and the data set gathered thus far, that is $X_1, \dots, X_N$. At this stage, one customarily estimates λ by means of the sample mean, $\bar{X}_N$. The properties of this estimator largely depend on intricate relationship between N and $(X_1, \dots, X_N)$.

In this note, our goal is to estimate the *reliability function*, that is

$$\begin{aligned} R(t) &= P(X > t) \\ &= \exp(-t/\lambda) \ , \end{aligned} \tag{1.2}$$

for *some fixed* $t(> 0)$, having obtained some relevant sequential data such as $(N, X_1, \dots, X_N)$. In Section 2, we first provide an estimator of $R(t)$ in general and enumerate the *second-order expansions* of the *bias* and *mean squared error* of such an estimator. We also attempt to adjust for the bias in this estimator. Section 3 provides some specific stopping rules and the second-order properties of the associated estimators of $R(t)$, the reliability function. Stopping rules are included from both the point as well as interval estimation formulations while pursuing estimation of $R(t)$ after the original sampling process stops gathering any more data. Section 4 summarizes moderate sample-size characteristics of these estimators, obtained via computer simulations.

2 Estimator of $R(t)$ in a General Sequential Setup

First suppose that, under some preassigned, fixed-accuracy consideration, one opts for an appropriate sequential design and ends up with the data $(N, X_1, \dots, X_N)$. Here, N is a stopping variable that often mimics an expression such as $n^*(= a\lambda^s$ where $a(> 0)$ and $s(> 0)$ are assumed known). The expression of n^* is customarily obtained via fixed-sample-size optimization for estimating the

mean parameter λ under given loss and/or cost considerations. Since n^* turns out to be unknown, the stopping variable N adaptively estimates n^* and at the stopped stage λ is estimated by $\bar{X}_N (= N^{-1} \sum_{i=1}^N X_i)$. Hence, $R(t)$ is proposed to be estimated by means of

$$\hat{R}_N(t) = \exp(-t/\bar{X}_N) , \quad (2.1)$$

while the rest of this section is primarily devoted to evaluations of both the bias and mean squared error of this estimator.

One could perhaps start with a loss function taking into consideration $R(t)$ and its “appropriate” estimator, and thereby propose some adaptive sequential sampling design under the given optimization criteria on hand. But, then, such sequential sampling strategies would naturally depend on the choice of t , though held fixed. At some later point, if the experimenter’s interests shift to estimation of the reliability function $R(t')$, say, for some $t'(> 0)$, $t' \neq t$, then the earlier stopping variable and the gathered data, that depends on t but not on t' , may not seem very appropriate to use. The approach we have is somewhat different though. With respect to some loss/cost considerations, we look at some of the available sequential designs that give us estimators such as $\bar{X}_N$ of λ having appropriate first-/second-order properties, and then use this information to estimate $R(t)$ by means of $\hat{R}_N(t)$ given by (2.1) for arbitrary, but fixed $t(> 0)$. Practically speaking, this approach is very useful and flexible because the sampling strategy does not depend on one’s particular choice of t .

Now, we state a few regularity conditions on $(N, \bar{X}_N)$ as follows. Recall that $n^* = a\lambda^s$ for some known $a(> 0)$ and $s(> 0)$. Then, as $n^* \rightarrow \infty$, suppose that

$$N/n^* \xrightarrow{P} 1 , \quad n^{*(1/2)}(\bar{X}_N - \lambda) \xrightarrow{\mathcal{L}} N(0, \lambda^2) ; \quad (2.2)$$

$$|n^{*(1/2)}(\bar{X}_N - \lambda)|^\omega \quad \text{is uniformly integrable for some } \omega(> 0) ; \quad (2.3)$$

$$E[\bar{X}_N - \lambda] = An^{*-1} + o(n^{*-1}) , \quad \text{with some constant } A . \quad (2.4)$$

Note that the second part in (2.2) actually follows from Anscombe’s (1952) Theorem in view of the first part. We, however, include the second part in (2.2) for completeness. The following theorem provides explicit properties of the estimator $\hat{R}_N(t)$.

Theorem: For arbitrary, but fixed $t(> 0)$, we have as $n^* \rightarrow \infty$:

- i) $n^{*(1/2)}\{\hat{R}_N(t) - R(t)\} \xrightarrow{\mathcal{L}} N(0, t^2 R^2(t) \lambda^{-2})$ under condition (2.2);
- ii) $E\{\hat{R}_N(t) - R(t)\} = (A - \lambda + \frac{1}{2}t\lambda^{-2}R(t)n^{*-1} + o(n^{*-1}))$, under conditions (2.2), (2.3) with $\omega = 2$, and (2.4);

iii) $E[\{\hat{R}_N(t) - R(t)\}^2] = t^2 \lambda^{-2} R^2(t) n^{*-1} + o(n^{*-1})$, under conditions (2.2), and (2.3) with $\omega = 4$;

where $\hat{R}_N(t)$ comes from (2.1).

Before we set out to prove these results, let us make an observation. The real number “ A ” that appears in (2.4) and part (ii) of the Theorem may sometimes be known. In such a situation when A is known, an intuitive bias-corrected (up to $o(n^{*-1})$) estimator of $R(t)$ can be proposed as

$$\hat{R}_N^*(t) = \{1 + (A - \bar{X}_N + \frac{1}{2}t)t\bar{X}_N^{-2}N^{-1}\}^{-1} \hat{R}_N(t). \quad (2.5)$$

The estimator seems reasonable in view of part (ii) of the Theorem. We shall plug-in $\hat{A}$ in (2.5) if A depends on λ .

Proof: (i) Let $g(x) = e^{-t/x}$, $x > 0$, then $\hat{R}_N(t) = g(\bar{X}_N)$. In view of (2.2), the Mann-Wald Theorem would let us conclude that $n^{*(1/2)}\{g(\bar{X}_N) - g(\lambda)\} \xrightarrow{\mathcal{L}} N(0, \lambda^2(g'(\lambda))^2)$, and hence the result follows since $g'(\lambda) = t\lambda^{-2}R(t)$.

(ii) Let $g(x)$ be as before and note that $g'(x) = tx^{-2}e^{-t/x}$ and $g''(x) = t(t - 2x)x^{-4}e^{-t/x}$. Thus, with some random variable ξ between λ and $\bar{X}_N$, we write via Taylor expansion,

$$\hat{R}_N(t) = R(t) + t\lambda^{-2}e^{-t/\lambda}(\bar{X}_N - \lambda) + \frac{1}{2}(\bar{X}_N - \lambda)^2 t(t - 2\xi)\xi^{-4}e^{-t/\xi}. \quad (2.6)$$

The expectation of the term in the middle is immediate in view of (2.4). In order to evaluate the expectation of the last term in (2.6), note that

$$\begin{aligned} & |n^*(\bar{X}_N - \lambda)^2 t(t - 2\xi)\xi^{-4}e^{-t/\xi}| \\ & \leq |n^{*(1/2)}(\bar{X}_N - \lambda)|^2 t(t + 2\xi)\xi^{-4}e^{-t/\xi} \\ & \leq K |n^{*(1/2)}(\bar{X}_N - \lambda)|^2 \{th(\xi^{-1}|t, 4) + 2h(\xi^{-1}|t, 3)\}, \end{aligned} \quad (2.7)$$

where $h(x|p, q) = e^{-px}x^q$ for $x > 0$, $p > 0$ and $q > 0$, and also K is a positive generic constant that is independent of ξ . Since the h -function is bounded for $p = t$, $q = 3$ and 4 , in view of (2.3) with $\omega = 2$, we claim the uniform integrability of the left hand side in (2.7). Obviously, $\xi \xrightarrow{P} \lambda$ as $n^* \rightarrow \infty$, and hence under the condition (2.2), we obtain

$$n^*(\bar{X}_N - \lambda)^2 t(t - 2\xi)\xi^{-4}e^{-t/\xi} \xrightarrow{\mathcal{L}} t(t - 2\lambda)\lambda^{-2}R(t)\lambda_1^2 \quad \text{as } n^* \rightarrow \infty.$$

Under uniform integrability of the left hand side, we can write

$$E\{(\bar{X}_N - \lambda)^2 t(t - 2\xi)\xi^{-4} e^{-t/\xi}\} = t(t - 2\lambda)\lambda^{-2} R(t)n^{*-1} + o(n^{*-1}) , \quad (2.8)$$

as $n^* \rightarrow \infty$. Now, we combine (2.4), (2.6) and (2.8) to obtain the desired result as follows:

$$\begin{aligned} E\{\hat{R}_N(t) - R(t)\} &= t\lambda^{-2} R(t)E\{\bar{X}_N - \lambda\} + \frac{1}{2}E\{(\bar{X}_N - \lambda)^2 t(t - 2\xi)\xi^{-4} e^{-t/\xi}\} \\ &= t\lambda^{-2} R(t)(An^{*-1} + o(n^{*-1})) + \frac{1}{2}t(t - 2\lambda)\lambda^{-2} R(t)n^{*-1} + o(n^{*-1}) \\ &= (A - \lambda + \frac{1}{2}t)t\lambda^{-2} R(t)n^{*-1} + o(n^{*-1}) . \end{aligned}$$

(iii) Observe that from (2.6), one has

$$\begin{aligned} \{\hat{R}_N(t) - R(t)\}^2 &= t^2 \lambda^{-4} R^2(t)(\bar{X}_N - \lambda)^2 + t^2 \lambda^{-2} R(t)(\bar{X}_N - \lambda)^3 (t - 2\xi)\xi^{-4} e^{-t/\xi} \\ &\quad + \frac{1}{4}t^2 (\bar{X}_N - \lambda)^4 (t - 2\xi)^2 \xi^{-8} e^{-2t/\xi} . \end{aligned} \quad (2.9)$$

Under (2.2) and (2.3) with $\omega = 2$, one gets

$$E\{(\bar{X}_N - \lambda)^2\} = \lambda^2 n^{*-1} + o(n^{*-1}) , \quad (2.10)$$

whereas the next two terms in (2.9) can be evaluated as follows:

$$\begin{aligned} &|n^{*(3/2)}(\bar{X}_N - \lambda)^3 (t - 2\xi)\xi^{-4} e^{-t/\xi}| \\ &\leq |n^{*(1/2)}(\bar{X}_N - \lambda)|^3 (t + 2\xi)\xi^{-4} e^{-t/\xi} \\ &\leq K |n^{*(1/2)}(\bar{X}_N - \lambda)|^3 \{th(\xi^{-1}|t, 4) + 2h(\xi^{-1}|t, 3)\} \\ &\leq K |n^{*(1/2)}(\bar{X}_N - \lambda)|^3 , \end{aligned} \quad (2.11)$$

where, as before, $K(> 0)$ is a generic constant, independent of ξ . That is, under (2.3) with $\omega = 3$, the left hand side of (2.11) is uniformly integrable as before, and hence with distributional convergence, we claim that

$$E\{(\bar{X}_N - \lambda)^3(t - 2\xi)\xi^{-4}e^{-t/\xi}\} = O(n^{*-(3/2)}) = o(n^{*-1}) . \quad (2.12)$$

Similarly, the last term's expectation in (2.9) can be shown to be $O(n^{*-2}) = o(n^{*-1})$, provided that (2.3) holds with $\omega = 4$. Now, combining this with (2.9)–(2.12), we complete the desired simplification. ■

3 Few Specific Stopping Rules and Estimation of $R(t)$

In this section, we consider stopping rules for both minimum risk point estimation as well as fixed-width confidence interval estimation of λ . In these contexts, we include both purely sequential as well as accelerated sequential stopping times for estimating λ . The data thus obtained, namely, $(N, \bar{X}_N)$ is then utilized in order to estimate $R(t)$ along the lines of Section 2.

3.1 Stopping Rules From Point Estimation of the Mean

Having recorded $X_1, \dots, X_n$ suppose that the loss function in estimating λ by $\bar{X}_n$ is taken as

$$L_n = (\bar{X}_n - \lambda)^2 + cn , \quad (3.1)$$

where $c(> 0)$ is the known “cost” per unit sample. The associated risk, $E(L_n)$, is minimized when $n = n^* = \lambda c^{-1/2}$ and the minimum risk is given by $E(L_{n^*}) = 2cn^*$. Throughout, we disregard the fact that n^* may not be an integer. Since n^* turns out to be unknown, Starr and Woodroffe (1972) proposed a purely sequential stopping rule that was later studied further by Woodroffe (1977). Mukhopadhyay (1993) has recently developed an accelerated version in order to estimate λ . The goal in these investigations has centered around achieving the fixed-sample minimum risk, namely $2cn^*$, asymptotically (as $c \rightarrow 0$), while estimating λ . Our target is, of course, estimation of $R(t)$.

3.1.1 Purely Sequential Sampling

One starts with $X_1, \dots, X_m$, $m \geq 1$, and proceeds by taking one observation at a time according to the stopping rule

$$N = N(c) = \inf\{n \geq m: n \geq c^{-1/2} \bar{X}_n\} , \quad (3.2)$$

and since $P(N < \infty) = 1$, one then estimates λ by means of $\bar{X}_N$, obtained from the data $(N, X_1, \dots, X_N)$. Woodroffe (1977) had shown that

$$E[L_N] = 2cn^* + 3c + o(c) , \quad (3.3)$$

as $c \rightarrow 0$, if $m \geq 3$. That is, the discrepancy between the sequential risk and the optimal fixed-sample risk amounts to the cost of three additional observations up to $o(c)$, as $c \rightarrow 0$.

Now, Chow-Robbins' (1965) and Anscombe's (1952) results would verify (2.2) for $m \geq 1$. Theorem 2 of Chow et al. (1979) ensures the validity of (2.3) for $m \geq 1$, for all $\omega \geq 1$, since $(N/n^*)^\omega$ is uniformly integrable for all $\omega \geq 1$. Also, from the Theorem [equation (2.10)] of Mukhopadhyay and Chattopadhyay (1991), one has the validity of (2.4) for $m \geq 2$, with $A = -\lambda$.

In other words, parts (i), (ii) and (iii) of the Theorem hold respectively when $m \geq 1$, $m \geq 2$ and $m \geq 1$. The bias-corrected form given in (2.5) now looks like

$$\hat{R}_N^*(t) = \{1 + (\frac{1}{2}t - 2\bar{X}_N)t\bar{X}_N^{-2}N^{-1}\}^{-1} \hat{R}_N(t) \quad (3.4)$$

instead.

3.1.2 Accelerated Sequential Sampling

Choose and fix a number $\rho \in (0, 1)$ such that ρ^{-1} is an integer. Now, start with $X_1, \dots, X_m$, $m \geq 1$, and proceed as earlier, according to the stopping rule

$$\tau = \inf\{n \geq m: n \geq \rho c^{-1/2} \bar{X}_n\} , \quad (3.5)$$

and then define

$$N = \rho^{-1} \tau . \quad (3.6)$$

The stopping variable τ estimates ρn^* , a fraction of n^* , purely sequentially, while N estimates n^* . One now samples the difference $(N - \tau)$ in *one single batch*. The final data set is $(N, X_1, \dots, X_N)$ and observe that the accelerated estimation procedure (3.5)–(3.6) requires approximately 100 $\rho\%$ of the sampling operations

set out in terms of the full sequential scheme (3.2), while one again estimates λ by means of $\bar{X}_N$. Mukhopadhyay (1993) had shown that

$$E[L_N] = 2cn^* + (2 + \rho^{-1})c + o(c) , \quad (3.7)$$

as $c \rightarrow 0$, if $m \geq 3$.

Again, (2.2) and (2.3) are both verifiable for $m \geq 1$. Also, (2.3) holds for all $\omega \geq 1$. See Mukhopadhyay (1993) for details. In order to verify (2.4), let us proceed as follows. We write $\bar{X}_\tau = \tau^{-1} \sum_{i=1}^\tau X_i$ and $\bar{X}_\tau^* = (N - \tau)^{-1} \sum_{i=\tau+1}^N X_i$, and thus $\bar{X}_N = \rho \bar{X}_\tau + (1 - \rho) \bar{X}_\tau^*$. Hence,

$$\begin{aligned} E[(\bar{X}_N - \lambda)] &= \rho E[(\bar{X}_\tau - \lambda)] + (1 - \rho) E[(\bar{X}_\tau^* - \lambda)] \\ &= \rho E[(\bar{X}_\tau - \lambda)] . \end{aligned} \quad (3.8)$$

Now, we use the following trick. Pretend that we are working under the pseudo loss function $L_n^* = \rho^2 (\bar{X}_n - \lambda)^2 + cn$. Then, the risk is minimized when $n = n^{**} = \rho n^*$ and the corresponding full sequential stopping variable would then coincide with “ τ ”. Thus, from (2.10) of Mukhopadhyay and Chattopadhyay (1991), we can claim that $E[(\bar{X}_\tau - \lambda)] = -\lambda n^{**^{-1}} + o(n^{**^{-1}})$, so that (3.8) simplifies to $E[(\bar{X}_N - \lambda)] = -\rho \lambda n^{**^{-1}} + o(n^{**^{-1}}) = -\lambda n^{*-1} + o(n^{*-1})$, if $m \geq 2$. That is, (2.4) holds with $A = -\lambda$, if $m \geq 2$.

Hence, parts (i), (ii) and (iii) of the Theorem hold as before. One needs $m \geq 1$ in parts (i) and (iii), and needs $m \geq 2$ in part (ii). The bias-corrected estimator of $R(t)$ has the same expression (3.4) where the stopping variable N now corresponds to (3.6).

3.2 Stopping Rules From Interval Estimation of the Mean

Given $0 < \alpha < 1$ and $d(> 0)$, we consider estimating λ by means of confidence intervals having the length at most $2d$ and the associated confidence coefficient (approximately) at least $(1 - \alpha)$. We consider the confidence interval $I_n^* = [(\bar{X}_n - d) \vee 0, \bar{X}_n + d]$ for λ , having recorded the observations $X_1, \dots, X_n$. Now, $P\{\lambda \in I_n^*\} \geq 1 - \alpha$ if $E\{(\bar{X}_n - \lambda)^2/d^2\} \leq \alpha$, that is if $n \geq n^* = \lambda^2(\alpha d^2)^{-1}$. But, since n^* is unknown, we opt for the purely sequential and accelerated sequential procedures of Mukhopadhyay and Datta (1996). We stay away from Chow and Robbins' (1965) nonparametric techniques via approximations through the central limit theorem because we can achieve sharper rate $[o(d^2)]$ of the remainder term in the approximation for the converge probability in the new approach. Towards the end of Section 4, we provide brief remarks on some of the related issues.

3.2.1 Purely Sequential Sampling

Define $D = \sum_{n=1}^{\infty} \frac{1}{2} n^{-1} E\{(\chi_{2n}^2 - 3n \vee 0)\}$ and let $\varepsilon = -16.5 - 2D$. Start with the observations $X_1, \dots, X_m$, $m \geq 1$, and proceed sequentially by taking one observation at a time according to the stopping rule

$$N = N(d) = \inf\{n \geq m: (n + \varepsilon)\alpha d^2 \geq \bar{X}_n^2\} , \quad (3.9)$$

and since $P(N < \infty) = 1$, we then estimate λ by means of $I_N^* = [(\bar{X}_N - d) \vee 0, \bar{X}_N + d]$, obtained from the data $(N, X_1, \dots, X_N)$. Mukhopadhyay and Datta (1996) have shown that

$$P\{\lambda \in I_N^*\} \geq (1 - \alpha) + o(d^2) , \quad (3.10)$$

as $d \rightarrow 0$, if $m \geq 7$.

Again, (2.2) and (2.3) are verifiable for $m \geq 1$. Also, (2.3) holds for all $\omega \geq 1$. In the Appendix, we show that (2.4) holds where

$$A = -1.625\lambda , \quad \text{if } m \geq 6 . \quad (3.11)$$

Thus, one needs $m \geq 1$ for parts (i) and (iii) in our Theorem, whereas m needs to be at least six for part (ii) to hold. The bias-corrected estimator of $R(t)$ has the following expression

$$\hat{R}_N^*(t) = \{1 + (\frac{1}{2}t - 2.625\bar{X}_N)t\bar{X}_N^{-2}N^{-1}\}^{-1}\hat{R}_N(t) \quad (3.12)$$

where the stopping variable now corresponds to (3.9).

3.2.2 Accelerated Sequential Sampling

Choose and fix a number $\rho \in (0, 1)$ such that ρ^{-1} is an integer. Now, start with $X_1, \dots, X_m$, $m \geq 1$, and proceed as earlier [in (3.9)] according to the stopping rule

$$\tau = \inf\{n \geq m: (n + \varepsilon)\alpha d^2 \geq \rho \bar{X}_n^2\} , \quad (3.13)$$

where $\varepsilon = -16.5 - 2D$, as before. Then, define

$$N = \rho^{-1}\tau, \quad (3.14)$$

and we sample the difference $(N - \tau)$ in *one single batch*. Finally, we estimate λ by means of the confidence interval $I_N^* = [(\bar{X}_N - d) \vee 0, \bar{X}_N + d]$, based on the data $(N, X_1, \dots, X_N)$. It was shown in Mukhopadhyay and Datta (1996) that

$$P\{\lambda \in I_N^*\} \geq (1 - \alpha) + B\lambda^{-2}d^2 + o(d^2)$$

as $d \rightarrow 0$, if $m \geq 7$, where B is a computable real number.

Again, (2.2) and (2.3) are both verifiable for $m \geq 1$. Also, (2.3) holds for all $\omega \geq 1$. See Mukhopadhyay and Datta (1996) for details. In order to verify (2.4), first write $\bar{X}_N = \rho\bar{X}_\tau + (1 - \rho)\bar{X}_\tau^*$ as in Section 3.1.2 and recall that $E[(\bar{X}_N - \lambda)] = \rho E[(\bar{X}_\tau - \lambda)]$. Now, pretend that we replace d by $d\rho^{-1/2}$ in (3.9) and then our τ from (3.13) would match with N from (3.9) whereas we would be mimicing the fixed-sample optimal size $n^{**} = \rho n^*$. Hence, $E[(\bar{X}_\tau - \lambda)] = -1.625\lambda n^{**^{-1}} + o(n^{**^{-1}})$, that is $E[\rho(\bar{X}_\tau - \lambda)] = -1.625\lambda n^{*-1} + o(n^{*-1})$, as $c \rightarrow 0$, if $m \geq 6$, in view of (3.11). That is (2.4) holds with $A = -1.625\lambda$, if $m \geq 6$. Hence, one needs $m \geq 1$ for parts (i) and (iii) in our Theorem, whereas m needs to be at least six for part (ii) to hold. The bias-corrected estimator of $R(t)$ has the same expression (3.12) where the stopping variable now corresponds to (3.14).

4 Moderate Sample Performances Via Simulations

This section presents summary findings regarding moderate sample performances of the estimator of $R(t)$, following sequential estimation of λ , as discussed in Section 3. The IBMRS/6000 computer and programs written in Fortran were used for simulation purposes at the University of New Orleans. The subroutine RNEXP from IMSL (1989, Version 10) is used to generate the exponential distribution (1.1) with $\lambda = 5$. Throughout this section, **T stands for the number of independent replicates** used to obtain a particular row in any of the tables presented here, and **we fix T = 10,000**.

First, we consider the point estimation problem of Section 3.1 and the associated purely sequential procedure (3.2). We consider $m = 10, n^* = 50, 100, 200, 400, 800$ and obtain $c = (\lambda/n^*)^2$ with $\lambda = 5$. For a particular choice of n^* , let n_i be the value of N , $\bar{X}_{n_i}$ be the value of $\bar{X}_N$ and $L_{n_i} = (\bar{X}_{n_i} - 5)^2 + cn_i$ for the i^{th} replication, $i = 1, \dots, T$. We also fix $t = 10, 15, 25$ and 40 . When the i^{th} replication stops sampling, $R(t)$ is estimated by $\hat{R}_{n_i}(t) = \exp\{-t/\bar{X}_{n_i}\}$ while

its bias-corrected form given in (3.4) turns out to be $\hat{R}_n^*(t) = \{1 + (\frac{1}{2}t - 2\bar{X}_{n_i})t\bar{X}_{n_i}^{-2}n_i^{-1}\}^{-1}\hat{R}_n(t)$, $i = 1, \dots, T$. Now, we obtain $\bar{n} = T^{-1}\sum_{i=1}^T n_i$, $\bar{L} = T^{-1}\sum_{i=1}^T L_{n_i}$, $\bar{R} = T^{-1}\sum_{i=1}^T \hat{R}_n(t)$ and $\bar{R}^* = T^{-1}\sum_{i=1}^T \hat{R}_n^*(t)$. We also obtain $s^2(\bar{n}) = (T^2 - T)^{-1}\sum_{i=1}^T (n_i - \bar{n})^2$, $s^2(\bar{L}) = (T^2 - T)^{-1}\sum_{i=1}^T (L_{n_i} - \bar{L})^2$, $s^2(\bar{R}) = (T^2 - T)^{-1}\sum_{i=1}^T (\hat{R}_n(t) - \bar{R})^2$, and $s^2(\bar{R}^*) = (T^2 - T)^{-1}\sum_{i=1}^T (\hat{R}_n^*(t) - \bar{R}^*)^2$. The quantities $\bar{n}$ and $\bar{L}$ respectively estimate $E(N)$ and $E(L_N)$. Observe that $E(L_N)$ is supposed to be very close to $2cn^*$ in view of (3.3). Also, $\bar{R}$ estimates $R(t)$, whereas $\bar{R}^*$ is the corresponding bias-corrected form. The quantities $s(\bar{n})$, $s(\bar{L})$, $s(\bar{R})$ and $s(\bar{R}^*)$ respectively provide the values of the estimated standard errors of the estimators $\bar{n}$, $\bar{L}$, $\bar{R}$ and $\bar{R}^*$. Table 1 summarizes the findings in the case of the purely sequential point estimation rule (3.2).

From Table 1, we observe that $\bar{n}$ and $\bar{L}$ values are respectively very close to n^* and $2cn^*$, with reasonable magnitudes of the standard errors. It is also clear that $\bar{n}$ underestimates n^* very slightly. The $\bar{R}$ values are usually not very close to $R(t)$, for small values of n^* , however, $\bar{R}$ seems to estimate $R(t)$ fairly well for $n^* > 100$. The bias-corrected estimator $\bar{R}^*$ seems to perform very well throughout.

Table 2 summarizes our findings for the accelerated sequential point estimation rule (3.5)–(3.6) with $\rho = \frac{1}{2}$. We considered $t = 10, 15, 25, 30, 40$ and 50 . For brevity, we present our results only for the cases when $t = 10, 15, 25$ and 40 . The results here are fairly similar to those in Table 1. There is a slight difference though. Table 1 pertains to purely sequential sampling all the way, but Table 2 pertains to purely sequential sampling part of the way followed by batch sampling. That is, we would expect little extra variabilities in the estimators derived from the corresponding accelerated methodology. When one compares the values of the estimated standard errors in Tables 1 and 2, this feature becomes quite apparent.

Next, we provide summaries of our findings in the case of estimation of $R(t)$, preceded by interval estimation of λ . We fix $\alpha = .05$, $\lambda = 5$ and $n^* = 50, 100, 200, 400, 800$ and obtain $d = \lambda/(\alpha n^*)^{1/2}$. Also, $D = .7117564$ and $\varepsilon = -16.5 - 2D$. In order to estimate the coverage probability, for the i^{th} replication, we define $p_i = 1$ (or 0) according as $5 \in (\notin) I_{n_i}^* = [(\bar{X}_{n_i} - d) \vee 0, \bar{X}_{n_i} + d]$, $i = 1, \dots, T$. Let $\bar{p} = T^{-1}\sum_{i=1}^T p_i$ and $s^2(\bar{p}) = \bar{p}(1 - \bar{p})/T$. We expect $\bar{p}$ to be larger than .95 with small values of $s(\bar{p})$. In the purely sequential case (Table 3), we considered $m = 10$ and $t = 10, 15, 20, 40, 50$, but we present the results only for $t = 10, 15, 40, 50$ for brevity. The corresponding accelerated sequential version (3.13)–(3.14) was implemented with $\rho = \frac{1}{2}$ and $t = 10, 15, 25, 40, 50$, but in Table 4 we include only $t = 10, 15, 40, 50$. In Tables 3 and 4, the entries are self-explanatory. Observe that the bias-corrected version $\bar{R}^*$ is now obtained using (3.12). But one feature will catch reader's attention. The $\bar{p}$ values are almost always 1.0 out of 10,000 replications. This is due to a combination of the following facts. First, note that the $\bar{n}$ values are also considerably larger than n^* values throughout Tables 3 and 4. Note that the original sampling techniques were designed so that one could claim the result (3.10) and this needed one to fine-tune the sample size upwards compared with n^* .

Table 1. Estimation of $R(t)$ following the purely sequential point estimation procedure (3.2)
 $m = 10 \quad \lambda = 5$

n^*	$\bar{n}$ ($s(\bar{n})$)	$2cn^*$	$\bar{L} \times 10^4$ ($s(\bar{L}) \times 10^4$)	$\bar{R}$ ($s(\bar{R}) \times 10^9$)	$\bar{R}^*$ ($s(\bar{R}^*) \times 10^9$)
$t = 10 \quad R(t) = .135335281$					
50	49.73 (.0728)	1.0000	10351 (83)	.130201325 (331022)	.135398075 (390509)
100	99.87 (.1027)	0.5000	5147 (37)	.133043781 (274366)	.135701835 (277432)
200	199.59 (.1434)	0.2500	2540 (18)	.133784115 (192903)	.135124281 (193929)
400	399.44 (.2002)	0.1250	1252 (9)	.134446397 (135055)	.135119110 (135402)
800	799.52 (.2826)	0.0625	625 (4)	.134922713 (95471)	.135259971 (95592)
$t = 15 \quad R(t) = .049787067$					
50	49.74 (.0733)	1.0000	10412 (85)	.048587348 (203861)	.049966019 (212011)
100	99.57 (.0998)	0.5000	5008 (35)	.048830349 (144585)	.049539223 (147515)
200	199.57 (.1416)	0.2500	2508 (17)	.049294215 (104191)	.049657807 (105240)
400	399.63 (.2011)	0.1250	1258 (9)	.049562663 (74469)	.049746931 (74843)
800	799.72 (.2838)	0.0625	627 (4)	.049692038 (52755)	.049784798 (52887)
$t = 25 \quad R(t) = .006737947$					
50	49.59 (.0745)	1.0000	10617 (84)	.007050047 (47948)	.006792133 (48643)
100	99.76 (.1011)	0.5000	5062 (37)	.006914603 (33492)	.006765207 (33774)
200	199.77 (.1421)	0.2500	2516 (18)	.006826754 (23674)	.006747217 (23781)
400	399.92 (.2002)	0.1250	1252 (9)	.006794671 (16779)	.006753758 (16819)
800	799.96 (.2833)	0.0625	626 (4)	.006767979 (11910)	.006747236 (11925)
$t = 40 \quad R(t) = .000335463$					
50	49.67 (.0739)	1.0000	10505 (85)	.000438685 (4648)	.000359717 (4209)
100	99.61 (.1016)	0.5000	5093 (37)	.000385079 (2996)	.000340230 (2813)
200	199.72 (.1410)	0.2500	2497 (18)	.000360928 (1987)	.000336491 (1917)
400	399.79 (.2013)	0.1250	1259 (9)	.000349192 (1389)	.000336396 (1363)
800	799.80 (.2849)	0.0625	630 (5)	.000342482 (971)	.000335932 (962)

Table 2. Estimation of $R(t)$ following the accelerated sequential point estimation procedure (3.5)–(3.6) $m = 10$ $\lambda = 5$ $\rho = .5$

n^*	$\bar{n}$ ($s(\bar{n})$)	$2cn^*$	$\bar{L} \times 10^4$ ($s(\bar{L}) \times 10^4$)	$\bar{R}$ ($s(\bar{R}) \times 10^9$)	$\bar{R}^*$ ($s(\bar{R}^*) \times 10^9$)
$t = 10$ $R(t) = .135335281$					
50	49.23 (.1061)	1.0000	10562 (82)	.129692912 (390436)	.135055766 (400299)
100	99.28 (.1472)	0.5000	5090 (38)	.132403731 (271426)	.135098413 (274405)
200	199.48 (.2028)	0.2500	2534 (18)	.133980736 (192694)	.135330170 (193703)
400	399.05 (.2841)	0.1250	1255 (9)	.134410799 (135310)	.135086432 (135658)
800	799.24 (.4062)	0.0625	631 (4)	.135015264 (96492)	.135353148 (96614)
$t = 15$ $R(t) = .049787067$					
50	49.09 (.1052)	1.0000	10410 (80)	.048074011 (204860)	.049473815 (213550)
100	99.31 (.1474)	0.5000	5133 (38)	.049067851 (148550)	.049790736 (151605)
200	199.52 (.2041)	0.2500	2544 (19)	.049283903 (105546)	.049649186 (106619)
400	399.35 (.2838)	0.1250	1249 (9)	.049588323 (74048)	.049773280 (74422)
800	799.21 (.3991)	0.0625	627 (4)	.049641650 (52800)	.049734332 (52933)
$t = 25$ $R(t) = .006737947$					
50	49.22 (.1062)	1.0000	10547 (85)	.006984927 (47554)	.006719213 (48281)
100	99.41 (.1451)	0.5000	5062 (37)	.006879702 (33328)	.006728138 (33615)
200	199.31 (.2044)	0.2500	2539 (19)	.006812863 (23939)	.006732770 (24050)
400	398.69 (.2844)	0.1250	1264 (9)	.006738944 (16833)	.006697730 (16873)
800	799.72 (.4024)	0.0625	627 (4)	.006761963 (11921)	.006741163 (11936)
$t = 40$ $R(t) = .000335463$					
50	49.15 (.1066)	1.0000	10532 (82)	.000426738 (4643)	.000348136 (4184)
100	99.52 (.1472)	0.5000	5112 (38)	.000391653 (3001)	.000345868 (2815)
200	199.21 (.2030)	0.2500	2523 (18)	.000359567 (1981)	.000335049 (1911)
400	399.69 (.2877)	0.1250	1260 (9)	.000348688 (1381)	.000335872 (1355)
800	799.58 (.3980)	0.0625	622 (4)	.000342501 (956)	.000335938 (946)

Table 3. Estimation of $R(t)$ following the purely sequential interval estimation procedure (3.9)
 $m = 10 \quad \lambda = 5 \quad \alpha = .05$

n^*	$\bar{n}$ ($s(\bar{n})$)	d	$\bar{p} \times 10^4$ ($s(\bar{p}) \times 10^4$)	$\bar{R}$ ($s(\bar{R}) \times 10^9$)	$\bar{R}^*$ ($s(\bar{R}^*) \times 10^9$)
$t = 10 \quad R(t) = .135335282$					
50	73.77 (.1150)	3.1623	10000 (0)	.130858973 (312860)	.136797950 (315874)
100	123.03 (.1813)	2.2361	9999 (1)	.131770059 (247671)	.135326862 (248089)
200	221.92 (.2708)	1.5811	9999 (1)	.132685810 (185207)	.134657085 (185080)
400	422.33 (.3969)	1.1180	9998 (1)	.134113222 (135009)	.135153592 (134885)
800	822.32 (.5593)	0.7906	10000 (0)	.134721860 (94869)	.135256574 (94810)
$t = 15 \quad R(t) = .049787067$					
50	73.64 (.1148)	3.1623	10000 (0)	.048199248 (167006)	.050392557 (172472)
100	122.88 (.1803)	2.2361	9999 (1)	.048358891 (132473)	.049676336 (134776)
200	222.94 (.2675)	1.5811	9999 (1)	.049062856 (99124)	.049802739 (99954)
400	422.42 (.3900)	1.1180	10000 (0)	.049317457 (72505)	.049710728 (72796)
800	822.54 (.5612)	0.7906	10000 (0)	.049561407 (52293)	.049764659 (52395)
$t = 40 \quad R(t) = .000335463$					
50	73.46 (.1164)	3.1623	10000 (0)	.000387864 (3431)	.000349438 (3310)
100	123.18 (.1814)	2.2361	10000 (0)	.000366589 (2514)	.000341363 (2454)
200	222.42 (.2705)	1.5811	9999 (1)	.000348903 (1828)	.000333968 (1803)
400	422.70 (.3928)	1.1180	9999 (1)	.000344010 (1325)	.000335736 (1316)
800	821.71 (.5575)	0.7906	10000 (0)	.000338116 (935)	.000333757 (932)
$t = 50 \quad R(t) = .000045400$					
50	73.38 (.1162)	3.1623	10000 (0)	.000059820 (663)	.000048798 (593)
100	122.96 (.1826)	2.2361	10000 (0)	.000053856 (464)	.000046712 (430)
200	222.77 (.2711)	1.5811	9999 (1)	.000050089 (325)	.000045819 (310)
400	422.43 (.3931)	1.1180	10000 (0)	.000047754 (230)	.000045379 (224)
800	822.93 (.5580)	0.7906	10000 (0)	.000046742 (161)	.000045477 (159)

Table 4. Estimation of $R(t)$ following the accelerated sequential interval estimation procedure (3.13)–(3.14) $m = 10$ $\lambda = 5$ $\alpha = .05$ $\rho = .5$

n^*	$\bar{n}$ ($s(\bar{n})$)	d	$\bar{p} \times 10^4$ ($s(\bar{p}) \times 10^4$)	$\bar{R}$ ($s(\bar{R}) \times 10^9$)	$\bar{R}^*$ ($s(\bar{R}^*) \times 10^9$)
$t = 10$ $R(t) = .135335282$					
50	98.38 (.1396)	3.1623	10000 (0)	.132850617 (269717)	.142233312 (276168)
100	147.40 (.2300)	2.2361	10000 (0)	.132941723 (224082)	.139118835 (225921)
200	245.99 (.3638)	1.5811	10000 (0)	.133699432 (176229)	.137364432 (176275)
400	445.68 (.5375)	1.1180	10000 (0)	.134296611 (129672)	.136298880 (129527)
800	843.48 (.7869)	0.7906	10000 (0)	.134621322 (93318)	.135672137 (93209)
$t = 15$ $R(t) = .049787067$					
50	98.45 (.1398)	3.1623	10000 (0)	.049044590 (147669)	.052568793 (156247)
100	147.57 (.2282)	2.2361	10000 (0)	.049084589 (119613)	.051408973 (123585)
200	245.90 (.3627)	1.5811	10000 (0)	.049114272 (95373)	.050492402 (96963)
400	444.17 (.5418)	1.1180	9999 (1)	.049253609 (70721)	.050012249 (71282)
800	845.38 (.7814)	0.7906	10000 (0)	.049603064 (51215)	.050002098 (51414)
$t = 40$ $R(t) = .000335463$					
50	98.30 (.1391)	3.1623	10000 (0)	.000387705 (3035)	.000330018 (2843)
100	147.13 (.2329)	2.2361	10000 (0)	.000363356 (2306)	.000322098 (2201)
200	245.89 (.3658)	1.5811	10000 (0)	.000350465 (1749)	.000323934 (1702)
400	444.29 (.5498)	1.1180	10000 (0)	.000340789 (1305)	.000325367 (1287)
800	844.59 (.7728)	0.7906	10000 (0)	.000339468 (931)	.000331038 (924)
$t = 50$ $R(t) = .000045400$					
50	98.61 (.1400)	3.1623	10000 (0)	.000058718 (570)	.000042971 (472)
100	147.49 (.2291)	2.2361	10000 (0)	.000054313 (431)	.000042943 (375)
200	245.68 (.3618)	1.5811	10000 (0)	.000048956 (307)	.000041734 (280)
400	445.59 (.5461)	1.1180	10000 (0)	.000047769 (224)	.000043425 (213)
800	845.17 (.7794)	0.7906	10000 (0)	.000046415 (158)	.000044017 (154)

Instead of (3.9), one may be tempted to implement the following Chow-Robbins (1965) type purely sequential estimation procedure. Let

$$N = N(d) = \inf\{n \geq m: n \geq Z_{\alpha/2}^2 \bar{X}_n^2/d^2\} \text{ ,} \tag{4.1}$$

where $Z_{\alpha/2}$ is the upper 50 $\alpha\%$ point of the standard normal distribution. Once sampling is terminated, we estimate λ by means of $I_N = [\bar{X}_N - d, \bar{X}_N + d]$. Here, the corresponding $n^* = Z_{\alpha/2}^2 \lambda^2/d^2$, and we fixed $m = 10, t = 10, 20, 40, \alpha = .05, n^* = 50, 100, 200, 400, 800, \lambda = 5$ with $d = Z_{\alpha/2} \lambda/n^{*(1/2)}$. Observe that d -values are different here compared with those in Tables 3 and 4. In Table 5, we summarize our findings when $t = 10$ and 40 with $T = 10,000$. Theoretically, we can claim here that $P\{\lambda \in I_N\} = (1 - \alpha) + O(d^{1/2-\gamma})$ for $0 < \gamma < \frac{1}{2}$, whereas the conclusion given in (3.10) is much too sharp in comparison. From Table 5, one can see easily that in the case of the Chow-Robbins (1965) methodology, $\bar{p}$ values are fairly jumpy and these are not exceeding .95 for the most part. Along the lines of our Appendix, we can again verify (3.11) in this case, and hence we

Table 5. Estimation of $R(t)$ following the Chow-Robbins' purely sequential interval estimation procedure (4.1) $m = 10 \quad \lambda = 5 \quad \alpha = .05$

n^*	$\bar{n}$ ($s(\bar{n})$)	d	$\bar{p} \times 10^4$ ($s(\bar{p}) \times 10^4$)	$\bar{R}$ ($s(\bar{R}) \times 10^9$)	$\bar{R}^*$ ($s(\bar{R}^*) \times 10^9$)
$t = 10 \quad R(t) = .135335282$					
50	46.95 (.1569)	1.3859	8752 (33)	.120940298 (448169)	.129664138 (453847)
100	97.06 (.2200)	0.9800	9184 (27)	.128324270 (308398)	.132749334 (308756)
200	198.07 (.2933)	0.6930	9348 (25)	.132578611 (201217)	.134793773 (200748)
400	398.13 (.3994)	0.4900	9477 (22)	.134002224 (136216)	.135106131 (136022)
800	798.29 (.5690)	0.3465	9483 (22)	.134696260 (96568)	.135246813 (96495)
$t = 40 \quad R(t) = .000335463$					
50	46.92 (.1586)	1.3859	8754 (33)	.000379856 (3992)	.000332660 (3801)
100	96.97 (.2184)	0.9800	9154 (27)	.000355753 (2723)	.000326575 (2644)
200	197.88 (.2942)	0.6930	9327 (25)	.000349733 (1954)	.000333262 (1926)
400	397.47 (.4078)	0.4900	9413 (23)	.000340739 (1360)	.000332028 (1350)
800	797.57 (.5659)	0.3465	9466 (23)	.000337978 (947)	.000333495 (943)

studied the behaviors of $\hat{R}_N(t)$ and $\hat{R}_N^*(t)$, respectively given by (2.1) and (3.12) where N is taken from (4.1). These seem to be reasonable estimators of $R(t)$ in this situation as well.

Appendix Verification of (3.11)

First observe that the stopping variable N from (3.9) has the same representation as in (2.4) of Mukhopadhyay (1988) with $h = \lambda n^{*-1/2}$, $\delta = \frac{3}{2}$, $L_0 = \frac{\varepsilon}{2}$, $\theta = \lambda$, $\tau^2 = \lambda^2$, $\beta^* = 2$, $n_0^* = n^*$, $p = 1$ and $b = 1$.

Thus $(n^*/N)^\omega$ is uniformly integrable for $\omega > 0$ if $m > 2\omega$. Now, let $\mathcal{R}$ be the excess over the boundary, that is $\mathcal{R} = N(N + \varepsilon)^{1/2}(\alpha d^2)^{1/2} - \sum_{i=1}^N X_i$, and hence

$$\{\bar{X}_N - \lambda\} = \lambda \left\{ \left(\frac{N + \varepsilon}{n^*} \right)^{1/2} - 1 \right\} - \frac{1}{n^*} \mathcal{R} \left(\frac{n^*}{N} \right). \quad (\text{A.1})$$

From Woodroffe (1977), it follows that all positive powers of $\mathcal{R}$ are uniformly integrable and $\mathcal{R}$ has an asymptotic distribution whose mean turns out to be $\lambda(1.25 - D) = v$, say. Thus, the expectation of the second term in (A.1) simplifies to

$$-vn^{*-1} + o(n^{*-1}) \quad \text{if } m \geq 3. \quad (\text{A.2})$$

Now, we utilize Taylor expansions to write

$$E \left[\left(\frac{N + \varepsilon}{n^*} \right)^{1/2} \right] = E[(N/n^*)^{1/2}] + \frac{1}{2} \varepsilon n^{*-1} E[(n^*/N)^{1/2}] + o(n^{*-1}). \quad (\text{A.3})$$

From Theorem 3 of Mukhopadhyay (1988), we get

$$E[(N/n^*)^{1/2}] = 1 + (\eta - .25)(2n^*)^{-1} + o(n^{*-1}) \quad \text{if } m \geq 6 \quad (\text{A.4})$$

where $\eta = 2v\lambda^{-1} - 2(\varepsilon/2) - 3 = 16$. Also,

$$E[(n^*/N)^{1/2}] = 1 + o(1) \quad \text{if } m \geq 2. \quad (\text{A.5})$$

Now, we combine (A.3)–(A.5) to obtain

$$E\left[\left(\frac{N + \varepsilon}{n^*}\right)^{1/2}\right] = 1 - (.75 + 2D)(2n^*)^{-1} + o(n^{*-1}), \quad (\text{A.6})$$

if $m \geq 6$. Thus, from (A.1)–(A.2) and (A.6), we get

$$\begin{aligned} E[(\bar{X}_N - \lambda)] &= -\lambda(.75 + 2D)(2n^*)^{-1} - \lambda(1.25 - D)n^{*-1} + o(n^{*-1}) \\ &= -1.625\lambda n^{*-1} + o(n^{*-1}), \quad \text{if } m \geq 6. \end{aligned}$$

This verifies (3.11).

Acknowledgement: The research of Dr. Solanky was supported by the LEQSF grant RD-A-31.

References

- Anscombe FJ (1952) Large sample theory of sequential estimation. *Proc Camb Phil Soc* 48:600–607
- Basu AP (1991) Sequential methods in reliability and life testing. In: Ghosh BK, Sen PK (Eds) *Handbook of Sequential Analysis*. Marcel Dekker Inc, New York 581–592
- Chow YS, Hsiung C, Lai T (1979) Extended renewal theory and moment convergence in Anscombe's Theorem. *Ann Prob* 7:304–318
- Chow YS, Robbins H (1965) On the asymptotic theory of fixed width sequential confidence intervals for the mean. *Ann Math Statist* 36:457–462
- Mukhopadhyay N (1988) Sequential estimation problems for negative exponential populations. *Commun Statist (Reviews Section), Theory & Methods* 17:2471–2506
- Mukhopadhyay N (1993) An alternative formulation of accelerated sequential procedures with applications. *Dept Statist Tech Rep #93-27*, Univ of Connecticut Storrs
- Mukhopadhyay N, Chattopadhyay S (1991) Sequential methodologies for comparing exponential mean survival times. *Sequential Analysis* 10:139–148
- Mukhopadhyay N, Datta S (1995) On fine-tuned bounded risk sequential point estimation of the mean of an exponential distribution. *S Afr Statist J* 29:9–27
- Mukhopadhyay N, Datta S (1996) On sequential fixed-width confidence intervals for the mean and second-order expansions of the associated coverage probabilities. *Ann Inst Statist Math* 48:497–507
- Starr N, Woodroffe M (1972) Further remarks on sequential estimation: The exponential case. *Ann Math Statist* 43:1147–1154
- Woodroffe M (1977) Second order approximations for sequential point and interval estimation. *Ann Statist* 5:984–995

A Relationship between Efficiencies of Marginal Designs and the Product Design

RAINER SCHWABE¹

Freie Universität Berlin 1. Mathematisches Institut, Arnimallee 2–6, 14 195 Berlin, Germany

WENG KEE WONG

Department of Biostatistics, UCLA, Los Angeles, CA 90095, USA

Abstract: We derive the D - and G -efficiencies of product designs in a multifactor experiment in terms of the D - and G -efficiencies of the designs in the marginal models.

Key Words and Phrases: Approximate design, D -, G -optimal design, marginal models, interaction, multi-factor experiments.

1 Introduction

Multi-factor experiments are becoming increasingly common in practice. In this paper, we study the relationship between the efficiency of a design in a multi-factor experiment and the marginal efficiencies of each of its factors for a class of linear models. We focus on the D - and G -efficiencies of product designs since these designs are intuitively appealing and easy to construct, see for example Book and Thibodeau (1980), Lau (1988), Lim and Studden (1988), and Wong (1994). Moreover in the settings under consideration every design is dominated by the product of its marginals (see Schwabe, 1996b).

All designs considered here are approximate designs or probability measures defined on a given compact design region T . The models are linear with a known regression function $F(t)$ and errors are assumed to be independent and homoscedastic. Following convention, we measure the worth of the design ξ by its information matrix:

$$M(\xi) = \int_T F(t)F(t)^\top \xi(dt) ,$$

¹ Work supported by grants Ku 719/2 and 477/645/96 of the Deutsche Forschungsgemeinschaft

which will be assumed to be non-singular throughout. The variance of the predicted response at the point t using the design ξ is given by $d(t, \xi) = F(t)^\top M(\xi)^{-1} F(t)$, apart from a multiplicative constant. If the dimension of $F(t)$ is p and ξ_D denotes the D -optimal design, which simultaneously maximizes the determinant $|M(\xi)|$ and minimizes the maximal variance $\sup_{t \in T} d(t, \xi)$ the D - and G -efficiencies of ξ are respectively given by

$$\text{eff}_D(\xi) = |M(\xi_D)^{-1} M(\xi)|^{1/p} \quad \text{and} \quad \text{eff}_G(\xi) = p \left/ \sup_{t \in T} d(t, \xi) \right. .$$

These efficiencies are numbers between zero and one by definition, and they measure reciprocally, the number of times the design ξ has to be replicated in order to do as well as the optimal design. For further background reading, see Fedorov (1972).

2 Main Result

Consider a multi-factor experiment involving k factors defined on a set $T = \times_{i=1}^k T_i$, where each T_i is compact. An element in T is denoted by $t = (t_1, t_2, \dots, t_k)^\top$, $t_i \in T_i$, $i = 1, 2, \dots, k$. The class of models of interest here is that of complete m -factor interactions

$$E y^{(m)}(t) = \beta^\top f^{(m)}(t), \quad 1 \leq m \leq k, \quad t \in T \quad (1)$$

where $f^{(m)}(t) = \{ \bigotimes_{j \in J} f_j(t_j); J \subset \{1, 2, \dots, k\}, |J| \leq m \}$ with $\bigotimes_{j \in \emptyset} = 1$. Here $|J|$ denotes the cardinality of the set J while the integer parameter $m (\leq k)$ corresponds to the complexity of interactions in the models. Each of the regression functions f_i is assumed to be known and has dimension p_i , $i = 1, 2, \dots, k$. Model (1) covers several commonly used models. For example, $m = 1, 2$ and k give the additive model, the first-order interaction model, and the full Kronecker product model, respectively. These models are popular in practice; Kronecker product models are mathematically convenient to study and permit multi-faceted interactions among the factors (Lau 1988, Lim and Studden 1988, Rafajłowicz and Myszka, 1988). See also Draper, Gaffke and Pukelsheim (1991) and Draper and Pukelsheim (1994) for related model embeddings. Additive models describe situations in which no interactions occur among factors and they have been used by Clark (1965), Cook and Thibodeau (1980), and more recently in Schwabe and Wierich (1995) and Schwabe (1996a). First-order interaction models provide an intermediate modelling choice between these two extremes.

They are more flexible than additive models and are appropriate where some interactions among the factors are anticipated and model parsimony is important. Clearly, the dimensions p of the regression functions for the additive, first order interaction and Kronecker product models are $p^A = \sum_{i=1}^k p_i + 1$, $p^{1st} = \sum_{i=1}^k \sum_{j=i+1}^k p_i p_j + \sum_{i=1}^k p_i + 1$ and $p^K = \prod_{i=1}^k (p_i + 1)$, respectively.

The associated marginal or single-factor models

$$E y_i(t_i) = \beta_0 + \beta_i^\top f_i(t_i), \quad t_i \in T_i, \quad i = 1, 2, \dots, k,$$

are obtained by keeping all but one factor fixed. The corresponding efficiencies of a design ξ_i on T_i are denoted by $\text{eff}_{D,i}(\xi_i)$ and $\text{eff}_{G,i}(\xi_i)$, respectively, in the marginal models.

Model (1) can be reparametrized as $E y^{(m)}(t) = \tilde{\beta}^\top \tilde{f}^{(m)}(t)$ where the components $\tilde{f}_j(t_j) = f_j(t_j) - \int_{T_j} f_j d\xi_j$ of $\tilde{f}^{(m)}$ are centered with respect to the components ξ_j of the product design $\xi = \xi_1 \times \dots \times \xi_k$. The resulting transformed information matrix

$$\tilde{M}(\xi_1 \times \dots \times \xi_k) = \text{diag} \left(\bigotimes_{j \in J} C_j(\xi_j); J \subset \{1, 2, \dots, k\}, |J| \leq m \right)$$

is block diagonal where $C_j(\xi_j) = \int_{T_j} f_j f_j^\top d\xi_j - \int_{T_j} f_j d\xi_j \int_{T_j} f_j^\top d\xi_j$ (see Schwabe 1995, 1996b). As this reparametrization leaves the determinant and the variances of the predicted response unchanged, a straightforward calculation yields

$$|M_m(\xi_1 \times \xi_2 \times \dots \times \xi_k)| = \prod_{|J| \leq m} \prod_{j \in J} |C_j(\xi_j)|^{p_J/p_j} = \prod_{i=1}^k |C_i(\xi_i)|^{(p - q_{m,i})/p_i}$$

where $p_J = \prod_{j \in J} p_j$, $q_{m,i} = \sum_{|J|=m; J \not\ni i} p_J$, and

$$d((t_1, t_2, \dots, t_k), \xi_1 \times \xi_2 \times \dots \times \xi_k) = \sum_{|J| \leq m} \prod_{j \in J} \{d_j(t_j, \xi_j) - 1\}$$

with $\prod_{j \in \emptyset} = 1$, where $d_i(t_i, \xi_i) = 1 + \tilde{f}_i(t_i)^\top C_i(\xi_i)^{-1} \tilde{f}_i(t_i)$ is the variance of the predicted response at t_i in the i th marginal model. Now, let $\xi_{D,i}$ be the D -optimal design in the i th marginal model. It has been shown in Schwabe (1995, 1996b) that the resulting product design $\xi_{D,1} \times \xi_{D,2} \times \dots \times \xi_{D,k}$ is simultaneously D -optimal and, hence, G -optimal for all models with complete m -factor interactions.

This leads to the following key result relating the efficiencies of marginal designs to that of the product design:

Theorem 2.1: Let the dimension of β in model (1) of complete m -factor interactions be p . Then

$$(i) \quad \text{eff}_D(\xi) = \prod_{i=1}^k \text{eff}_{D,i}(\xi_i)^{\alpha_{m,i}}$$

where $\alpha_{m,i} = 1 - \frac{1}{p} \sum_{1 \leq i_1 < i_2 < \dots < i_m \leq k; i_v \neq i} p_{i_1} \cdot \dots \cdot p_{i_m}$, and

$$(ii) \quad \text{eff}_G(\xi) = p \left/ \left\{ \sum_{n=0}^m (-1)^{m-n} \binom{k-n-1}{m-n} \sum_{|J|=n} \prod_{j \in J} (p_j + 1) \text{eff}_{G,j}(\xi_j)^{-1} \right\} \right.$$

for every product design $\xi = \xi_1 \times \xi_2 \times \dots \times \xi_k$.

For instance, for $\xi = \xi_1 \times \xi_2 \times \dots \times \xi_k$, we have $\text{eff}_D(\xi) = \prod_{i=1}^k \text{eff}_{D,i}(\xi_i)^{(p_i+1)/p^k}$ and $\text{eff}_G(\xi) = p^k / \{ \sum_{i=1}^k (p_i + 1) \text{eff}_{G,i}(\xi_i)^{-1} - (k-1) \}$ for the additive model ($m = 1$), $\text{eff}_D(\xi) = \prod_{i=1}^k \text{eff}_{D,i}(\xi_i)^{(p_i+1)(p^k-p_i)/p^{1st}}$ and $\text{eff}_G(\xi) = p^{1st} / \{ \sum_{i=1}^k \sum_{j=i+1}^k (p_i + 1)(p_j + 1) \text{eff}_{G,i}(\xi_i)^{-1} \text{eff}_{G,j}(\xi_j)^{-1} - (k-2) \sum_{i=1}^k (p_i + 1) \text{eff}_{G,i}(\xi_i)^{-1} + (k-1)(k-2)/2 \}$ for the first-order interaction model ($m = 2$), and $\text{eff}_D(\xi) = \prod_{i=1}^k \text{eff}_{D,i}(\xi_i)$ and $\text{eff}_G(\xi) = \prod_{i=1}^k \text{eff}_{G,i}(\xi_i)$ for the full Kronecker product model ($m = k$).

In case all the marginals are indential the D -efficiency relationships simplify to $\ln \text{eff}_D(\xi) = \left\{ 1 + \frac{k-1}{kp_1+1} \right\} \ln \text{eff}_{D,1}(\xi_1)$ for the additive model, $\ln \text{eff}_D(\xi) = \left\{ 2 + \frac{2(k-2)(kp_1+1)}{k(k-1)p_1^2+2kp_1+2} \right\} \ln \text{eff}_{D,1}(\xi_1)$ for the first-order interaction model and $\ln \text{eff}_D(\xi) = k \ln \text{eff}_{D,1}(\xi_1)$ for the full Kronecker product model.

The results obtained here can be applied to study robustness properties of product designs in multifactor polynomial models. If we denote the D -optimal design for the polynomial of degree n_i by ξ_{n_i} , Dette and Wong (1995) showed that if $n_i \geq p_i$, the D - and G -efficiencies of ξ_{n_i} relative to the polynomial of degree p_i are given by

$$\begin{aligned} & \text{eff}_{D,i}(\xi_{n_i})^{p_i+1} \\ &= \left(\frac{n_i(2p_i-1)}{p_i(2n_i-1)} \right)^{p_i} \prod_{j=2}^{p_i} \left\{ \frac{(n_i-j+1)^2(2p_i-2j+1)(2p_i-2j+3)}{(p_i-j+1)^2(2n_i-2j+1)(2n_i-2j+3)} \right\}^{p_i-j+1} \end{aligned}$$

and

$$\text{eff}_{G,i}(\xi_{n_i}) = \frac{n_i p_i + n_i}{2n_i p_i - p_i^2 + n_i},$$

respectively. The D - and G -efficiencies of $\xi = \xi_{n_1} \times \xi_{n_2} \times \cdots \times \xi_{n_k}$ can now be found directly from the marginal efficiencies using Theorem 2.1.

Acknowledgement: The authors wish to express their thanks to the referee for his comments which helped to shorten and strengthen the presentation by including the general treatment of complete m -factor interaction models.

References

- [1] Clark V (1965) Choice of levels in polynomial regression with one or two variables. *Technometrics* 7:325–333
- [2] Cook RD, Thibodeau LA (1980) Marginally restricted D -optimal designs. *J Amer Statist Assoc* 75:366–371
- [3] Dette H, Wong WK (1995) On G -efficiency calculation for polynomial models. *Ann Statist* 23:2081–2101
- [4] Draper NR, Gaffke N, Pukelsheim F (1991) First and second order rotatability of experimental designs, moment matrices, and information surfaces. *Metrika* 38:129–161
- [5] Draper NR, Pukelsheim F (1994) On third order rotatability. *Metrika* 41:137–161
- [6] Fedorov VV (1972) *Theory of optimal experiments*. Academic Press, New York
- [7] Lau TS (1988) D -optimal designs on the unit q -Ball. *J Statist Plann Inference* 19:299–315
- [8] Lim YB, Studden WJ (1988) Efficient D_s -optimal designs for multivariate polynomial regression on the q -cube. *Ann Statist* 16:1225–1240
- [9] Rafajłowicz E, Myszka W (1988) Optimum experimental design for a regression on a hypercube – generalization of Hoel's result. *Ann Inst Statist Math* 40:821–827
- [10] Schwabe R, Wierich W (1995) D -optimal designs of experiments with non-interacting factors. *J Statist Plann Inference* 44:371–384
- [11] Schwabe R (1995) Experimental design for linear models with higher interaction terms. In: Mammitzsch V, Schneeweiß H (eds) *Symposia Gaussiana. Proc. 2nd Gauss Symposium, München 1993. Conf. B: Statistical Sciences*. DeGruyter, Berlin, 281–288
- [12] Schwabe R (1996a) Optimal designs for additive linear models. *Statistics* 27:267–278
- [13] Schwabe R (1996b) Optimum designs for multi-factor models. *Lectures Notes in Statistics* 113. Springer, New York
- [14] Wong WK (1994) Multi-factor G -optimal designs with heteroscedastic errors. *J Statist Plann Inference* 40:127–133

Received 21.08.1995

Change Point Tests Based on U-Statistics with Applications in Reliability

EMAD-ELDIN A. A. ALY¹

Department of Statistics and Operations Research, Kuwait University, P.O. Box 5969, Safat 13060, Kuwait and Department of Mathematical Sciences, The University of Alberta, Edmonton, Alberta, Canada T6G 2G1

SUBHASH C. KOCHAR²

Indian Statistical Institute, 7, SJS Sansanwall Marg, New Delhi, India

Abstract: We consider the problem of testing the null hypothesis of no change against the alternative of exactly one change point. The proposed tests are based on generalized two-sample U-statistic processes. We derive the limiting null distributions of the proposed tests. Some applications in Statistical Reliability are given.

1 Introduction

Change point problems have received considerable attention in the last two decades. They are often encountered in the fields of quality control, reliability and survival analysis. A typical situation occurs in quality control when one observes the output of a production line and is interested in detecting any change in the quality of the product.

Let $X_1, X_2, \dots, X_n$ be independent random variables with respective distribution functions $F_1, F_2, \dots, F_n$. We may be interested in the problem of testing the null hypothesis of no change

$$H_0: F_1 = F_2 = \dots = F_n = F, \quad \text{where } F \text{ is unknown}$$

against the alternative of exactly one change point

$$H_1': F_1 = \dots = F_{[n\tau]} = F \quad \text{and} \quad F_{[n\tau]+1} = \dots = F_n = G,$$

$$\text{where } F \neq G \quad \text{and} \quad \tau \in (0, 1) \quad \text{are unknown},$$

¹ This research was supported by an NSERC Canada grant at the University of Alberta.

² Part of this research was done while visiting the University of Alberta supported by the NSERC Canada grant of the first author.

where $[x]$ is the integer part of x . In H'_1 we may consider various types of restrictions on the nature of the difference between F and G , parametric as well as nonparametric. In this article we will only consider the nonparametric approach. For additional results and references on change point analysis we refer to Zacks (1983), Bhattacharyya (1984), Csörgő and Horváth (1988a), Sen (1988), Lombard (1989) and Hušková and Sen (1989).

Most of the previous work in this area is focused on possible location changes, that is, when G is of the form $G(x) = F(x - \theta)$ for all x and for some unknown θ . We note here that the problem of testing H_0 against H'_1 is like a two-sample problem based on two independent samples, $X_1, X_2, \dots, X_{[n\tau]}$ and $X_{[n\tau]+1}, \dots, X_n$, with unknown $\tau \in (0, 1)$. Motivated by this observation Sen and Srivastava (1975) proposed the test statistics $S_n^{(1)}$ and $S_n^{(2)}$ for the one-sided and the two-sided location alternatives, respectively, where

$$S_n^{(1)} = \max_{1 \leq k \leq n-1} \left(\frac{V_{k,n-k} - E_0(V_{k,n-k})}{\sqrt{\text{var}_0(V_{k,n-k})}} \right),$$

$$S_n^{(2)} = \max_{1 \leq k \leq n-1} \left| \frac{V_{k,n-k} - E_0(V_{k,n-k})}{\sqrt{\text{var}_0(V_{k,n-k})}} \right|,$$

$V_{k,n-k}$ is the usual Mann-Whitney form of the Wilcoxon statistics based on the two samples $X_1, X_2, \dots, X_k$ and $X_{k+1}, \dots, X_n$ and $E_0(V_{k,n-k})$ (resp. $\text{var}_0(V_{k,n-k})$) is the expectation (resp. variance) of $V_{k,n-k}$ under H_0 . We note here that $S_n^{(1)}$ and $S_n^{(2)}$ are extensions of the Wilcoxon-Mann-Whitney test statistic to the change point setup. Csörgő and Horváth (1988b) obtained the limiting distributions of statistics of the above type when the associated U-statistics are based on kernels of degree (1, 1), which include the above case. Pettitt (1979), Sen (1982), Csörgő and Horváth (1988b), and Hawkins (1989) proposed tests for H_0 against H'_1 which are based on one-sample U-statistics.

It is well known that the Wilcoxon-Mann-Whitney test has good power properties for detecting location differences for moderate tailed distributions and for testing against stochastic ordering alternatives in some other cases. However, several two-sample tests based on generalized U-statistics of degrees other than (1, 1) are available in the literature which are good competitors of the Wilcoxon-Mann-Whitney test and it would be of interest to examine their extensions to the change point setup. The investigation of change point test procedures which are based on generalized U-statistics of degrees other than (1, 1) is also motivated by the following example.

Example 1: Let F be a life distribution function, $\bar{F} = 1 - F$ and f be the corresponding survival and density functions, respectively. The hazard rate function of F is $r_F(\cdot) = f(\cdot)/\bar{F}(\cdot)$. Suppose that the quality of items produced in an assembly line is measured by their hazard rates. An item with a smaller hazard rate would

tend to survive longer. Due to wear and tear of the machinery it might be of interest to test H_0 against H'_1 where the difference between F and G is expressed as $F \prec_{hr} G$, i.e., F is less than G in the hazard rate ordering which holds if and only if, $r_F(\cdot) \geq r_G(\cdot)$. In the corresponding two-sample problem, Kochar (1979) proposed a testing procedure based on a generalized U-statistic of degree (2, 2). A change point extension of Kochar (1979)'s test will be discussed in the sequel.

In this article we consider test procedures based on generalized U-statistics of degrees other than (1, 1). These procedures can be used to test H_0 against

$$H_1: H'_1 \text{ holds with } F \prec_{\rho} G \text{ and } F \neq G ,$$

where $\prec_{\rho}$ is a specified partial ordering of the considered family of distribution functions like, for example, the stochastic and hazard rate orderings.

Next we give some basic definitions of generalized two-sample U-statistics of degree (m, m) . Let $Y_1, Y_2, \dots, Y_{n_1}$ and $Z_1, Z_2, \dots, Z_{n_2}$ be two independent samples of sizes n_1 and n_2 drawn from the Y and the Z populations, respectively. Let $\phi(y_1, y_2, \dots, y_m; z_1, z_2, \dots, z_m)$ be a kernel of degree (m, m) , $1 \leq m \leq \min(n_1, n_2)$. Without any loss of generality we may assume that ϕ is symmetric in its first (resp. last) m arguments. The roles of these two sets of arguments need not be (and frequently are not) symmetric. A kernel ϕ of degree (m, m) is said to be skew-symmetric if

$$\phi(y_1, y_2, \dots, y_m; z_1, z_2, \dots, z_m) = -\phi(z_1, z_2, \dots, z_m; y_1, y_2, \dots, y_m) .$$

In most of the applications we employ skew-symmetric kernels. For this reason we will confine our attention to this type of kernels.

The generalized two-sample U-statistic based on the Y and Z samples and the kernel ϕ of degree (m, m) is defined as

$$U_{n_1, n_2} = \left\{ \binom{n_1}{m} \binom{n_2}{m} \right\}^{-1} \sum \sum \phi(Y_{i_1}, \dots, Y_{i_m}; Z_{j_1}, \dots, Z_{j_m}) , \quad (1)$$

where the above summations extend over $1 \leq i_1 < i_2 < \dots < i_m \leq n_1$ and $1 \leq j_1 < j_2 < \dots < j_m \leq n_2$. Since the kernels are assumed to be skew-symmetric, it follows that under H_0 , $E(U_{n_1, n_2}) = 0$. For the kernel ϕ of (1) we define

$$\phi_1(y) = E\phi(y, Y_2, \dots, Y_m; Z_1, Z_2, \dots, Z_m) . \quad (2)$$

We will need the following condition

$$E|\phi(Y_1, Y_2, \dots, Y_m; Z_1, Z_2, \dots, Z_m)|^v < \infty , \quad \text{for some } v \geq 2 . \quad (3)$$

Let

$$\sigma_1^2 = E\phi_1^2(Y_1) \quad \text{and} \quad \sigma_2^2 = E\phi_1^2(Z_1) .$$

Condition (3) insures that σ_1, σ_2 and $E|\phi_1(Y_1) \cdot \phi_1(Z_1)|$ are finite.

2 Tests Against the One-Change Point Alternative

Assume first that the change point τ is known and $[n\tau] = k, m \leq k \leq n - m$. Let $U_{k,n-k}$ be a generalized two-sample U-statistic based on a skew-symmetric kernel ϕ of degree (m, m) and the two samples $X_1, \dots, X_k$ and $X_{k+1}, \dots, X_n$. The kernel ϕ is selected in such a way that $U_{k,n-k}$ is an appropriate test statistic for testing H_0 against H'_1 when τ is known and $[n\tau] = k$ and large values of $U_{k,n-k}$ are significant. The two-sample U-statistic tests proposed here are based on the U-statistic process $\{U_n(t), 0 \leq t \leq 1\}$, where

$$U_n\left(\frac{k}{n}\right) = m^{-1}n^{-2}k(n-k) \cdot U_{k,n-k} , \quad m \leq k \leq n - m , \quad (4)$$

$U_n(0) = U_n(1) = 0$ and $U_n(\cdot)$ is defined elsewhere by interpolation. One of the following test statistics may be used for testing H_0 against H_1 ,

$$T_{1n} = \sum_{k=m}^{n-m} U_n^2\left(\frac{k}{n}\right) ,$$

$$T_{2n} = \sqrt{n} \max_{m \leq k \leq n-m} U_n\left(\frac{k}{n}\right) ,$$

$$T_{3n} = \sqrt{n} \max_{m \leq k \leq n-m} \left| U_n\left(\frac{k}{n}\right) \right| ,$$

$$T_{4n} = n^2 \sum_{k=m}^{n-m} \frac{U_n^2\left(\frac{k}{n}\right)}{k(n-k)} ,$$

$$T_{5n} = n^{3/2} \max_{m \leq k \leq n-m} \frac{U_n\left(\frac{k}{n}\right)}{\sqrt{k(n-k)}} ,$$

$$T_{6n} = n^{3/2} \max_{m \leq k \leq n-m} \frac{\left| U_n \left(\frac{k}{n} \right) \right|}{\sqrt{k(n-k)}} .$$

The limiting null distributions of T_{in} , $i = 1, 2, \dots, 6$ are given in the following Theorem which is proved in Section 4.

Theorem 1: Assume that ϕ is skew-symmetric of degree (m, m) , $B(\cdot)$ is a Brownian bridge, H_0 and condition (3) hold and let $\sigma^2 = E\phi_1^2(X_1)$.

1. If $v = 2$ in condition (3), then as $n \rightarrow \infty$,

$$T_{1n}/\sigma^2 \xrightarrow{D} \int_0^1 B^2(s)ds := t_1 ,$$

$$T_{2n}/\sigma \xrightarrow{D} \sup_{0 < s < 1} B(s) := t_2 ,$$

and

$$T_{3n}/\sigma \xrightarrow{D} \sup_{0 < s < 1} |B(s)| := t_3 .$$

2. If $v > 2$ in condition (3), then as $n \rightarrow \infty$,

$$T_{4n}/\sigma^2 \xrightarrow{D} \int_0^1 \frac{B^2(s)}{s(1-s)} ds := t_4 ,$$

$$P\{T_{5n}/\sigma \leq a(y, \log n)\} \rightarrow \exp\{-\exp - y\} , \quad -\infty < y < \infty$$

and

$$P\{T_{6n}/\sigma \leq a(y, \log n)\} \rightarrow \exp\{-2 \exp - y\} , \quad 0 < y < \infty ,$$

$$\text{where } a(y, s) = (2 \log s)^{-1/2} \{y + 2 \log s + \tfrac{1}{2} \log \log s - \tfrac{1}{2} \log \pi\}.$$

The critical values of t_1 , t_3 and t_4 are respectively given in Tables 4, 1 and 5 of Section 8 of Chapter 3 of Shorack and Wellner (1986). The critical values of t_2 are obtained from the well known result $\Pr\{t_2 \geq x\} = \exp\{-2x^2\}$, $x > 0$.

3 Applications

Example 2: This is a continuation of Example 1 in which we wish to test H_0 against

$$H_{1,hr}: H'_1 \text{ holds with } F \underset{hr}{\prec} G \text{ and } F \neq G .$$

In the corresponding two-sample problem, Kochar (1979) proposed the kernel

$$\phi(x_1, x_2; y_1, y_2) = \begin{cases} -1 & \text{if } xxxy \text{ or } yxyx \\ 0 & \text{if } xyxy \text{ or } yxyx \\ 1 & \text{if } yyxx \text{ or } xyyx , \end{cases} \quad (5)$$

where, for example, $yyxx$ represents $y_1 \leq y_2 \leq x_1 \leq x_2$, $y_2 \leq y_1 \leq x_2 \leq x_1$, $y_2 \leq y_1 \leq x_1 \leq x_2$ or $y_1 \leq y_2 \leq x_2 \leq x_1$. It is shown in Kochar (1979) that the two-sample test based on the kernel ϕ of (5) performs better than the Wilcoxon test for detecting hazard rate ordering.

Tests for H_0 against $H_{1,hr}$ can be based on the process $U_n(\cdot)$ of (4) with the kernel ϕ of (5) which is skew symmetric of degree (2, 2). Under H_0 and condition (3) we obtain the results of Theorem 1 with $\sigma^2 = \frac{8}{105}$ (see Theorem 2.1 of Kochar (1979) for details).

Example 3: Testing Against a Change in the Location Parameter

We are interested in testing H_0 against H'_1 with $G(x) = F(x + \theta)$, $\theta \geq 0$ and F is unknown. In the corresponding two-sample problem Deshpandé and Kochar (1982) proposed the kernel

$$\phi(x_1, \dots, x_m; y_1, \dots, y_m) = \begin{cases} 1 & \text{if } \min(y_1, \dots, y_m) \leq \min(x_1, \dots, x_m) \\ & \text{and } \max(y_1, \dots, y_m) \leq \max(x_1, \dots, x_m) \\ -1 & \text{if } \min(x_1, \dots, x_m) \leq \min(y_1, \dots, y_m) \\ & \text{and } \max(x_1, \dots, x_m) \leq \max(y_1, \dots, y_m) \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

and showed that large values of m are more suitable for thin tailed distributions. Note that when $m = 1$, the kernel ϕ of (6) reduces to the Wilcoxon-Mann-Whitney kernel.

Tests for H_0 against a change in the location parameter can be based on the process $U_n(\cdot)$ of (4) with the kernel ϕ of (6) which is skew-symmetric of degree (m, m) . Under H_0 and condition (3) we obtain the results of Theorem 1 with

$$\sigma^2 = 2m^2(2m-1)^{-2}(4m-1)^{-1} \left\{ 1 - \frac{1}{\binom{4m-1}{2m-1}} \right\},$$

(see Theorem 2.1 of Deshpandé and Kochar (1982) for details).

Example 4: Testing Against a Change in the Scale Parameter

We consider here the problem of testing H_0 against H'_1 with $G(x) = F(\theta x)$, $\theta \geq 1$ and F is symmetric around zero and unknown. In the corresponding two-sample problem Kochar and Gupta (1986) proposed tests based on the kernel

$$\phi(x_1, \dots, x_m; y_1, \dots, y_m) = \begin{cases} 1 & \text{if min as well as max of } (x_1, \dots, x_m, y_1, \\ & \dots, y_m) \text{ are some } x\text{'s} \\ -1 & \text{if min as well as max of } (x_1, \dots, x_m, y_1, \\ & \dots, y_m) \text{ are some } y\text{'s} \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

Kochar and Gupta (1986) showed that the corresponding U-statistic when $m = 2$ is equivalent to the two-sample Mood's statistic for the scale problem. For other choices of m these tests perform better for thin tailed distributions. For additional properties of these tests we refer to Kochar and Woodworth (1992).

Tests for H_0 against a change in the scale parameter can be based on the process $U_n(\cdot)$ of (4) with the kernel ϕ of (7) which is skew-symmetric of degree (m, m) . Under H_0 and condition (3) we obtain the results of Theorem 1 with

$$\sigma^2 = \frac{2m^2}{(2m-1)^2} \left\{ \frac{1}{4m-1} - \frac{1}{2m^2} + \frac{\{(2m-1)!\}^2}{(4m-1)!} \right\},$$

(see Corollary 2.1 of Kochar and Gupta (1986) for details).

4 Proofs

Theorem 1 follows from the results of Section 4 of Csörgő and Horváth (1988b) and the following result.

Lemma 1: Assume that ϕ is skew symmetric of degree (m, m) and H_0 and condition (3) hold. Let $\sigma^2 = E\phi_1^2(X_1)$ and $\xi_i = \sigma^{-1}\phi_1(X_i)$, $i = 1, 2, \dots, n$, where $\phi_1(\cdot)$ is as in (2). Then,

$$\max_{m \leq k \leq n-m} \left| \sqrt{n} \sigma^{-1} U_n \left(\frac{k}{n} \right) - n^{-1/2} \left\{ \sum_{i=1}^k \xi_i - \frac{k}{n} \sum_{i=1}^n \xi_i \right\} \right| \stackrel{P}{=} O(n^{-1/2}) . \quad (8)$$

Proof: The proof for $m > 2$ is a routine extension of that of $m = 2$ which is given below.

Recall that

$$U_n \left(\frac{k}{n} \right) = m^{-1} n^{-2} k(n-k) U_{k, n-k}^* \left\{ \binom{k}{2} \binom{n-k}{2} \right\}^{-1} , \quad (9)$$

where

$$U_{k, n-k}^* = \sum_{1 \leq i < j \leq k} \sum_{k+1 \leq r < s \leq n} \phi(X_i, X_j; X_r, X_s) . \quad (10)$$

Let $\Phi(\cdot; \cdot): \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ be such that $\Phi(\mathbf{x}; \mathbf{y}) = \phi(x_1, x_2; y_1, y_2)$, $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$. Define $\mathbf{Z} = (X_1, X_2)$, $K = \binom{k}{2}$, $N - K = \binom{n-k}{2}$ and

$$\Phi_1(\mathbf{x}) = E\Phi(\mathbf{x}; \mathbf{Z}) = -E\Phi(\mathbf{Z}; \mathbf{x}) .$$

Let $\mathbf{Z}_1, \dots, \mathbf{Z}_K$ be the pairs $\{(X_i, X_j), 1 \leq i < j \leq k\}$ in lexicographic (dictionary) ordering and $\mathbf{Z}_{K+1}, \dots, \mathbf{Z}_N$ be the pairs $\{(X_r, X_s), k+1 \leq r < s \leq n\}$ in lexicographic ordering. Note that

$$U_{k, n-k}^* = \sum_{1 \leq i \leq K} \sum_{K+1 \leq j \leq N} \Phi(\mathbf{Z}_i; \mathbf{Z}_j) .$$

Following Csörgő and Horváth (1988b), we write

$$U_{k,n-k}^* = U_N^{(1)} - U_K^{(2)} - U_K^{(3)} , \quad (11)$$

where

$$U_N^{(1)} = \sum_{1 \leq i < j \leq N} \Phi(\mathbf{Z}_i; \mathbf{Z}_j) ,$$

$$U_K^{(2)} = \sum_{1 \leq i < j \leq K} \Phi(\mathbf{Z}_i; \mathbf{Z}_j) ,$$

and

$$U_K^{(3)} = \sum_{K+1 \leq i < j \leq N} \Phi(\mathbf{Z}_i; \mathbf{Z}_j) .$$

By (2.9) of Janson and Wichura (1983) and the equation immediately preceding it (see also (4.4)–(4.7) of Csörgő and Horváth (1988b)), we obtain

$$\max_{2 \leq k \leq n-2} \left| U_N^{(1)} - \sum_{i=1}^N (N - 2i + 1) \Phi_1(\mathbf{Z}_i) \right| \stackrel{P}{=} O(n^2) \quad (12)$$

$$\max_{2 \leq k \leq n-2} \left| U_K^{(2)} - \sum_{i=1}^K (K - 2i + 1) \Phi_1(\mathbf{Z}_i) \right| \stackrel{P}{=} O(n^2) \quad (13)$$

and

$$\max_{2 \leq k \leq n-2} \left| U_K^{(3)} - \sum_{i=K+1}^N (N + K - 2i + 1) \Phi_1(\mathbf{Z}_i) \right| \stackrel{P}{=} O(n^2) . \quad (14)$$

By (11)–(14) we obtain

$$\begin{aligned} \max_{2 \leq k \leq n-2} \left| U_{k,n-k}^* - \left\{ \binom{n-k}{2} \sum_{i=1}^K \Phi_1(\mathbf{Z}_i) - \binom{k}{2} \sum_{i=K+1}^N \Phi_1(\mathbf{Z}_i) \right\} \right| \\ \stackrel{P}{=} O(n^2) . \end{aligned} \quad (15)$$

Next, we note that

$$\sum_{i=1}^K \Phi_1(\mathbf{Z}_i) = \sum_{1 \leq i < j \leq k} \Phi_1((X_i, X_j)) , \quad (16)$$

$$\sum_{i=K+1}^N \Phi_1(\mathbf{Z}_i) = \sum_{k+1 \leq i < j \leq n} \Phi_1((X_i, X_j)) , \quad (17)$$

and

$$\phi_1(x) = E\Phi_1((x, X_1)) = E\Phi_1((X_1, x)) . \quad (18)$$

By (16)–(18) and Theorem 1 of Hall (1979) we obtain

$$\max_{2 \leq k \leq n-2} \left| \sum_{i=1}^K \Phi_1(\mathbf{Z}_i) - \binom{k-1}{1} \sum_{i=1}^k \phi_1(X_i) \right| \stackrel{P}{=} O(n) , \quad (19)$$

and

$$\max_{2 \leq k \leq n-2} \left| \sum_{i=K+1}^N \Phi_1(\mathbf{Z}_i) - \binom{n-k-1}{1} \sum_{i=k+1}^n \phi_1(X_i) \right| \stackrel{P}{=} O(n) . \quad (20)$$

By (15), (19) and (20) we obtain

$$\begin{aligned} \max_{2 \leq k \leq n-2} \left| U_{k,n-k}^* - \binom{n-k}{2} \binom{k-1}{1} \sum_{i=1}^k \phi_1(X_i) + \binom{k}{2} \binom{n-k-1}{1} \sum_{i=k+1}^n \phi_1(X_i) \right| \\ \stackrel{P}{=} O(n^2) . \end{aligned} \quad (21)$$

By (9) and (21) we obtain (8).

References

- [1] Bhattacharyya GK (1984) Tests for randomness against trend or serial correlations. In: Krishnaiah PR, Sen PK (Eds) Handbook of Statistics V4, Nonparametric Methods. North-Holland, Amsterdam
- [2] Csörgő M, Horváth L (1988a) Nonparametric methods for changepoint problems. In: Krishnaiah PR, Rao CR (Eds) Handbook of Statistics V7, Quality Control and Reliability. North-Holland, Amsterdam

- [3] Csörgő M, Horváth L (1988b) Invariance principles for changepoint problems. *J Mult Anal* 27:151–168
- [4] Deshpandé JV, Kochar SC (1982) Some competitors of Wilcoxon-Mann-Whitney test for the location alternatives. *J Indian Statist Assoc* 19:9–18
- [5] Hall P (1979) On the invariance principle for U-statistics. *Stoch Proces Appl* 9:163–174
- [6] Hawkins DL (1989) Retrospective and sequential tests for change in an estimable parameter based on U-statistics. *Sankhya B*, 51:271–286
- [7] Hušková M, Sen PK (1989) Nonparametric tests for shift and change in regression at an unknown time point. In: Hackl P (Ed) *Statistical Analysis and Forecasting of Economic Structural Change*. Springer Verlag, New York
- [8] Janson S, Wichura MJ (1983) Invariance principles for stochastic area and related stochastic integrals. *Stoch Proces Appl* 16:71–84
- [9] Kochar SC (1979) Distribution-free comparison of two probability distributions with reference to their hazard rates. *Biometrika* 66:437–441
- [10] Kochar SC, Gupta RP (1986) Some competitors of the Mood test for the two-sample problem. *Commun Statist – Theory and Meth* 15:231–239
- [11] Kochar SC, Woodworth G (1992) Rank order probabilities for the two sample dispersion problem. *Statist Prob Letters* 13:203–208
- [12] Lombard F (1989) Some recent developments in the analysis of changepoint data. *South African Statist J* 23:1–21
- [13] Pettitt AN (1979) A non-parametric approach to the change-point problem. *Appl Statist* 28:126–135
- [14] Sen A, Srivastava MS (1975) On tests for detecting changes in mean. *Ann Statist* 3:98–108
- [15] Sen PK (1982) Tests for change points based on recursive U-statistics. *Sequential Anal* 1:263–284
- [16] Sen PK (1988) Robust tests for change-point models. In: Kotz S, Johnson NL (Eds) *Encyclopedia of Statistical Sciences* 8. John Wiley and Sons, New York
- [17] Shorack GR, Wellner JA (1986) *Empirical processes with applications to staistics*. John Wiley and Sons, New York
- [18] Zacks S (1983) Survey of classical and Bayesian approaches to the change-point problem: Fixed sample and sequential procedures of testing and estimation. In: Rizvi MH, Rustagi JS, Siegmund D (Eds) *Recent Advances in Statistics*, Academic Press, New York

Received 16.08.1995

Revised version 23.01.1996

Book Reviews

Contents

D. Stauffer, A. Aharony: Perkolationstheorie. Eine Einführung (H.-O. Georgii)	271
L. Alonso, R. Schott: Random Generation of Trees (H. Prodinger)	272
A. M. Mathai, S. B. Provost, T. Hayakawa: Bilinear Forms and Zonal Polynomials (Y. Chikuse)	274
V. S. Borkar: Probability Theory. An Advanced Course (H. Rost)	275
E. B. Andersen: The Statistical Analysis of Categorical Data (St. E. Fienberg)	278
F. Weisz: Martingale Hardy Spaces and Their Application in Fourier Analysis (P. Imkeller)	279

D. Stauffer, A. Aharony: Perkolationstheorie. Eine Einführung, deutsche Übersetzung der Originalausgabe: Introduction to Percolation Theory, Revised Second Edition, VCH Publishing Group Weinheim, 1995, 202 pp., DM58.00

Reviewed by H.-O. Georgii, Munich

Percolation theory developed from the following simple model introduced independently by P. J. Flory (1941) for polymer gelation and J. M. Hammersley (1957) for the passage of a fluid through a porous medium. Consider an infinite connected regular graph such as the integer lattice $\mathbb{Z}^d$ of any dimension $d \geq 2$ with edges between nearest neighbors. Suppose the vertices are independently removed with probability $1 - p$ and retained with probability p . The retained vertices form an infinite subgraph which splits into connected components called clusters. Let $\theta(p)$ be the probability that the origin belongs to an infinite cluster. The basic observation is that there exists a threshold $0 < p_c < 1$ (depending on the dimensions d) such that $\theta(p) = 0$ when $p < p_c$ and $\theta(p) > 0$ when $p > p_c$. One is interested in the size distribution and structure of the clusters for p below, above and, in particular, near p_c . Since percolation is a relatively simple spatial model showing a critical phenomenon at the threshold p_c , it can serve as a testing ground for theories and techniques dealing with the thermal phase transitions in the spatial models of statistical mechanics.

The book under review approaches its subject from a purely phenomenological point of view. Written by two theoretical physicists for students of theoretical physics (as opposed to mathematical physics), its main emphasis is on the

conclusions which can be drawn from Monte-Carlo simulations on the basis of natural, simplifying, but mostly unproven assumptions. A central theme are the critical exponents, i.e., the exponents determining a power law behavior of various quantities such as $\theta(p)$, the mean cluster size, etc. as functions of $p < p_c$ or $p > p_c$ near p_c . One reason for the interest in these critical exponents is that these are believed (and experimentally shown) to be universal, in that they only depend on the global structure of the underlying lattice (such as the dimension) rather than its specific local properties (which determine the value of p_c). Further themes are the fractal structure of the incipient infinite cluster at p_c , the finite-size effect of the observed window, electric conductivity properties, dynamical models such as random walks on percolation clusters, and the analogies and applications to thermal phase transitions in diluted and nondiluted Ising models.

The book is convenient to read. It provides a variety of interesting concepts and a clear picture of what can be observed in computer simulations. A mathematical treatment lies beyond its scope. So, in the end, a mathematically inclined reader will ask what is really known. He then may take, for example, the monograph by G. Grimmett (Percolation, Springer, 1989) who presents a completely different aspect of the subject: a wealth of fascinating probabilistic and geometric ideas and results which (among many other things) prove or, at least, set a firm basis for the issues discussed in the book under review.

L. Alonso, R. Schott: Random Generation of Trees, Kluwer Academic Publishers Boston Dordrecht London, 1995, XI/208 pp., Dfl180.00/US\$95.00/UK£64.75

Reviewed by H. Prodinger, Vienna

This book touches without any doubt on a very interesting subject. Historic cornerstones in the area of the (random) generation of combinatorial objects are the books by Nijenhuis and Wilf, "*Combinatorial algorithms*" (second edition, 1969), Wilf's update from 1989, "*Non-uniform random variate generation*" by Devroye (1986), and the recent fundamental study "*A calculus for random generation of combinatorial structures*" by Flajolet, Zimmermann, and van Cutsem [Theoretical Computer Science 132 (1994): 1–35] which develops a general machinery – based on some fundamental observations – that gives very good (or even optimal) algorithms for trees as *special cases*! (As a bonus, it provides an automatic analysis.) A first realization of it ran under the name Gaia, and it is now incorporated in the system MAPLE V.4 as COMBSTRUCT.

The present text is in no way encyclopedic and covers a random selection of topics. The first two chapters are introduction and notation; then comes the generating of binary trees, increasing trees, and Cayley trees. For that, the encoding by (Prüfer) sequences is used (the name "Prüfer" is not mentioned). Chapter 4 includes something about encoding trees by languages, much in the French tradition started by M. Schützenberger. Chapter 5 deals with all kinds of forests, and Chap. 6 with colored trees. This family of trees is a bit esoteric,

and it is not too clear why they are important. The next chapter is nice; it covers material of Barucci, Pisani and Sprugnoli. They have invented a rejection method in order to generate Motzkin trees, and they in turn are useful for the generation of so-called animals. Some credits here go to X. Viennot and his school in Bordeaux. Chapter 8 deals with arborescences (rooted trees). It describes RANRUT, an algorithm from the Nijenhuis/Wilf book and contains a new result, from the thesis of Alonso, that the complexity of RANRUT is $O(n \log n)$. Chapter 9 works out an idea attributed to P. Zimmermann about binary trees with height restrictions by means of using small precision arithmetic. The last chapter briefly discusses parallel algorithms. Then comes appendix 1 about the Horton-Strahler numbers. This could have been a regular chapter if it had been elaborated a bit more. Then there are some algorithms coded in C, followed by some pictures of random trees.

Apparently the typesetting was done by the authors themselves, with not too much expertise. True, the text should not be judged by the typesetting, but several things simply look unpleasant. What is much worse is the waste of space! My guess is that “normal” typesetting would reduce the number of pages by at least 30%! I got my copy for free, but I can imagine that the book (Kluwer!) is not cheap, so this generous layout is directly proportional to the price. The material is partially based on Alonso’s doctoral thesis, and for instance on p. 135, several French words “survived” from the original text. This adds to the general appearance of sloppiness, and one would wish that the authors had spent one more year writing a more appealing book. Alonso’s thesis, the paper of Barucci et al., and the above-mentioned general calculus for random generation all appeared at the beginning of the 1990s. This shows that this field is currently seeing some new and challenging ideas. Due to these parallel developments, some material did not make it into the book, and undoubtedly the near future will see even more changes and new ideas (better ranking algorithms for example).

When it comes to asymptotics – very important for the analysis of the described algorithms – the authors usually say that it is too complicated for the present book and can be found in the original papers. We wholeheartedly disagree with this: Yes, we can find everything in original papers, but why then write this book?

We do not want to give a list of typos (many!), but there are several things too strange *not* to be mentioned: $\ln(x)$ is the authors’ way of writing the dyadic (base 2) logarithm! Page 13 mentions “these functions, useful for asymptotics”, but it is not said what they are. I assume it means that generating functions are useful for the asymptotics of the coefficients, via Darboux’s method or, more generally, singularity analysis. Page 113 has the “Marty algorithm”, but this is not a researcher named “Marty”, but rather it is dedicated to “Marion Marty”! On page 156, the asymptotic behavior is derived from the study of some recurrence relations, but this study is neither performed nor are there references to the literature. (I think a pointer to some papers on singularity analysis of generating functions and the height of binary trees would be in order here.)

The index is a bit short, and the list of references biased and unbalanced; Polya's pioneering paper from 1937 and the book of Devroye do not appear, but Er and Pallo are there.

The book does not make bad reading – since the subject is lovely – and if you see it in a library, have a look. You might then feel the urge to read and learn more about the area, and that means the book has fulfilled its function. As a rigorous introduction, I cannot recommend it; for seminars it might be O.K. Too bad, since the community would definitely welcome a thorough textbook.

A. M. Mathai, S. B. Provost, T. Hayakawa: Bilinear Forms and Zonal Polynomials, Springer, 1995, XII/376 pp., DM 59.00/öS 460.20

Reviewed by Y. Chikuse, Takamatsu

This book aims to present the theoretical developments from basic principles to the current research level in the following two subjects “bilinear and quadratic forms in real random vectors and matrices” and “zonal polynomials”. The latter subject is well applied in the derivations of distributional results on the former one, particularly for the generalized quadratic forms in matrix variates.

Let X and Y be a $p \times 1$ and a $q \times 1$ random vector, and A , b and c , and d be a $p \times q$ matrix, a $p \times 1$ and a $q \times 1$ vector, and a scalar of constant elements, respectively. Then $Q_1(X, Y) = X'AY$ and $Q_2(X) = X'AX$, with A symmetric, are defined as a bilinear form and a quadratic form, respectively, whose generalizations are $Q_1^*(X, Y) = X'AY + b'X + c'Y + d$ and $Q_2^*(X) = X'AX + b'X + d$, with A symmetric. Thus bilinear forms can be considered to be particular cases of quadratic forms in the $(p + q) \times 1$ random vector $Z = (X'; Y')$. Hence the theory of quadratic forms dealt with in the previous book by Mathai and Provost [1] may be applicable to the bilinear forms as well. However, this procedure becomes too complicated when dealing with bilinear forms, and alternative procedures are developed in this book. The material in this book thus complements that in the previous book.

The book starts in Chap. 1 with the preliminaries needed in later chapters. Chapter 2 deals with bilinear forms in random variables and particularly in singular and nonsingular normal random vectors, discussing distributional aspects and Laplacianess, a concept analogous to chi-squaredness for quadratic forms. In Chap. 3, various distributional results on quadratic forms in elliptically contoured vectors and matrices are presented, some of which also apply to bilinear forms. Chapter 4 is devoted to a systematic development of the theory of zonal polynomials, including worked examples and related topics. Many of these results are used in Chap. 5. Invariant polynomials, which are an extension of zonal polynomials, are discussed in the Appendix. The final chapter, Chap. 5, is concerned with some distributional problems associated with

generalized quadratic forms in normal matrix variates and some of their applications in multivariate analysis.

Most of the theorems are accompanied by detailed proofs and illustrative examples, and exercises are provided at the end of each chapter. The book is thus made easily readable in spite of the difficult topics treated, and self-contained, being well-written for self-study. The book deserves to be received as a textbook for a graduate course in mathematical statistics as well as a valuable reference source in the areas concerned.

Reference

1. Mathai AM, Provost SB (1992) Quadratic forms in random variables: theory and applications. Marcel Dekker, New York

V. S. Borkar: Probability Theory. An Advanced Course, Springer, New York Berlin Heidelberg 1995, XIV/138 pp., DM 48.00/öS 350.40/sFR 48.00

Reviewed by H. Rost, Heidelberg

First of all, we should put straight expectations that might arise from the title. The decisive word is “course”, not “probability theory”; one should also keep in mind the name of the series: “Universitext”. What we have here is not a comprehensive textbook of *the* probability theory, comparable in scope to the works of Feller (1950/66) or Breiman (1968), but notes of a course of one year (or perhaps less) duration, written for students who just have decided to specialize in probability.

Whenever a new book appears on the market, two questions naturally come up: the first concerns its concept and the special needs the author hopes to satisfy; the second concerns its quality. To be clear from the outset, both tests yield a negative result. First, I cannot approve the concept (if there is any) and, second, I know of books which address the same readership, but are written much better. As for the quality statement, it may suffice to mention a single title which is better than the present book in every respect, including layout and bookbinding; namely R. Durrett *Probability: theory and examples*. Wadsworth, Belmont, CA, (1991). Durrett’s book was also originally based on lectures for graduate students; but in contrast to what apparently happened here, the author *formed* the material and created a comprehensible book that can exist on its own and which is more than just an aide-mémoire for people who have taken a certain course.

The concept of Borkar’s book is outlined – to say “expounded” would cer-

tainly be an exaggeration – in the “Preface”. The author says that he is more interested in presenting the measure-theoretical side of probability theory than in guiding his readers to what is usually called “probabilistic thinking”. Of course, the author is free to decide which of the two sides seems to him more interesting or more important with regard to training in applied mathematics; but even if one shares his preferences (I do not), one has to admit that the title is misleading. The book by Neveu (1964), for example, which was read by a whole generation of probabilists, named things correctly, saying in its title: “*Bases mathématiques du calcul des probabilités*”. (By the way, it contained a great deal of probabilistic substance beyond pure measure theory.)

Leaving aside the title and preface, let us try to find out the author’s concept empirically, by studying the text. What one sees is, to a large extent, standard material from measure theory and basic probability. By the way, things are just told, not really *explained*. One sees neither remarkable emphasis on special features nor an opening up of new perspectives. The central paradigmata of Brownian motion and the Poisson process (just the simple counting process on the real axis, let alone the concept of a random geometric object in space) occur only by name-dropping – incredible in view of the title. On the other hand, the notion of separability is exhumed, more than 40 years after Doob, *nota bene* in a book for beginners in higher stochastics, and the martingale inequality by Burkholder, Davis, and Gundy is formulated in a general and proven in a special version. I wonder where the author wants to lead his students: in which field of probability are they supposed to specialize afterwards, or is the book designed as a basis for professional training? There is not the slightest indication as to which of these is the author’s objective. Martingale inequalities do not seem to be the final goal, they remain too isolated and without consequences. A pessimistic interpretation of the author’s intention is that students have just to *learn* whatever the teacher proposes to them, definitions (e.g., of conditional expectation) and theorems (e.g., the law of the iterated logarithm). (How central is the law of the iterated logarithm from today’s viewpoint, and what is its role in an introductory course?) To summarize, what we see is essentially the n -th rehash of an old brew which always has the same ingredients. As an illustration of what I mean by saying that a scientific text is recognized by its perspectives, let me mention another book in advanced stochastics, namely, D. W. Stroock *Probability theory, an analytic view*. Cambridge University Press, Cambridge (1993).

On the back cover we read that one advantage of Borkar’s book is that it presents new material, like Etemadi’s strong law of large numbers or the tree representation of the invariant measure of a Markov chain. To this I can say the following: Etemadi’s theorem also appears in Durrett’s book. Besides, I do not believe that this theorem (it says that for the strong law identical distributions and pairwise independence suffice) is as important as Khintchine’s classical law of large numbers, which requires independence in the usual sense, because pairwise independence is a somehow artificial notion. What is more serious is the neglect of Birkhoff’s ergodic theorem by Borkar (in a book which pretends to stress the measure-theoretic aspect!). Here is the natural generalization and

the framework for understanding the strong law. The tree representation, however, is a nice discovery for which we shall be grateful to the author, regardless of whether it is of practical use for numerical calculations or not. A slight criticism: Borkar does not ascribe that theorem to a concretely named author. But he gives a reference, namely, V. Anantharam, P. Tsoucas *A proof of the Markov chain tree theorem*. Stat Prob Letters 8: 189–192 (1989); those who want to learn more about the subject may look there.

The didactical value of the book is poor. The lack of explanation and motivation has already been pointed out, but there is also a lack of historical references. More serious is, however, the almost complete absence of interesting examples; Brownian motion and Poisson process are not the only items one tries to find in vain. It is strange that the simple common sense wisdom “A theory is as good as the examples it helps to explain” has not yet found its way to the author. He seems to believe that a young person has to learn things like Fourier transform, entropy function, Markov chain, etc., from the bare definition. Chapter 6, dealing with the Skorohod topology of the space $D[0, 1]$, is completely without life because it contains not a single motivating example. That last chapter, and so the whole book, is crowned by the hardly exciting statement that the right continuous version of a certain process is a $D[0, 1]$ -valued random variable, but one looks in vain for a single example of a weakly convergent (in the sense of Skorohod) sequence of processes or measures. Convergence in $C[0, 1]$ is not treated much better: Donsker’s classical invariance principle is not even mentioned, neither here nor in the context of the central limit theorem, let alone the asymptotics of Kolmogorov–Smirnov. (A glance into the book by Stroock is enough to show how general analytic theorems arise naturally from concrete problems.)

A last point of criticism: the appearance of the typed pages is bad. The breaks in the text are not clearly indicated by spacings between paragraphs, only “Definition”, “Theorem”, “Lemma” are typed in separate blocks. In addition to this technicality, the author has a habit of not explicitly stating the assumptions of a theorem, but hiding them in the preceding current text. In order to find them, detective work is sometimes necessary, impeded by the fact there are no formal definitions. (I must correct myself: there are two of them, but only in Chap. 6. The reader wonders why the author, and later the technical editor, were not able to make the style uniform over the whole book.) All the other definitions are part of the “*unendliche Melodie*”. On the back cover page we read not only about Etemadi and tree representation, but also that the book addresses (among others) scientists from outside mathematics who want to enlarge their knowledge in probability. On which specific features of the text this particular claim is based is Springer’s secret. Happily, there are practically no misprints.

In brief, the book was unnecessary. On this occasion a general remark: Usually we, the reviewers, praise or blame the authors; they are primarily creating the books, and they are our peers. But when doing it, we should not forget the publishers and their responsibility for the quality standards in scientific publication.

E. B. Andersen: *The Statistical Analysis of Categorical Data*, Third Edition, Springer, Berlin Heidelberg New York, 1994, XI/532 pp., DM 185.00/öS 1.443.00/sFr 185.00

Reviewed by St. E. Fienberg, Pittsburgh

Erling Andersen has made important contributions to the theory and application of methods for the analysis of categorical data, especially those associated with the Rasch and other latent class models. His earlier book (Andersen 1980) was one of the few sources where one could go to see the linkages between the literature on loglinear/logit models and their close latent class relatives. In the present book, designed as text for a graduate course in categorical data analysis, he presents an authoritative account of loglinear/logit models that encompasses most of the important developments in the field during the 1980s. This includes topics such as RC-association models, correspondence analysis, a bit on graphical models, and the critical work of Tjur and others linking the Rasch model and conditional likelihood estimation to the more traditional loglinear models. For the models he examines, Andersen emphasizes almost exclusively maximum likelihood estimation and standard measures of goodness of fit, but he does include some information and reference on diagnostics. The book has been thoughtfully organized, and it builds up to the final 89-page chapter on latent structure analysis.

Despite the proliferation of texts on categorical data analysis over the past 15 years, there are few that rival Andersen in their coverage, technical accuracy, and reliance on motivation with real examples. Most of the examples are, in fact, from Danish sources and appear to be associated with the author and his collaborators. This means that the reader gets his own valuable insights from the illustrative analyses.

This book is described as a third edition. It was first published in 1990, and the "second edition" corrected a number of errors and incorporated a brief final chapter of nine pages on computer programs. This "third edition" provides a few additional corrections and new versions of the figures. I found only a few residual errors, e.g., Chen and Fienberg (1986) should really be (1976). The references stop with the completion by the author of the first edition in 1989. I have used and gladly referred others to the original edition. In reading the third edition for this review, I discovered again why I liked the book and originally found it to be such a useful reference.

My biggest quibble is with the way in which Andersen introduces association graphs in Chap. 3. Only later, in Chap. 4, does he explain that several loglinear models have the same association graph. Then he follows this by a careful explanation of graphical models: if one wishes to define a model solely in terms of the conditional independence associated with the absence of edges, then there is a unique graphical model associated with each association graph. Nothing here is incorrect, but I found the use of association graphs to be pedagogically misleading. This is a minor complaint in an otherwise admirable book.

What I do take issue with is the way manner that the book has been produced and priced by the publisher. The book appears to have be reproduced directly from typescript supplied, with enormous care, by the author. Further, the price of the book is prohibitively high and thus it cannot compete with many other inferior but affordable texts on the topic. In this day and age of computerized typesetting, surely Springer can afford to prepare a properly typeset edition at a price that will yield the readership that this book justly deserves.

Reference

Andersen EB (1980) Discrete statistical models with social science applications. North Holland, Amsterdam

F. Weisz: Martingale Hardy Spaces and Their Application in Fourier Analysis, Springer, Berlin Heidelberg New York, 1994, VIII/217 pp., DM 52.00/öS 405.50/sFr 52.00

Reviewed by P. Imkeller, Berlin

It is by now classical that solutions of the Dirichlet problem, for example on the unit disc and Brownian martingales, are closely related. Take Brownian motion started inside the disc, stop it when it reaches the boundary; then the expectation of the boundary value function, taken at the place where the Brownian motion leaves the disc, yields the harmonic function solving the classical problem. The resulting correspondence between harmonic functions and martingales gives classical Hardy spaces a probabilistic companion, the "martingale Hardy spaces" investigated in the book under review.

In the past 3 decades, several books have dealt with the subject. The classic "*Martingale inequalities*" by Garsia, and Durrett's "*Brownian motion and martingales in analysis*" were followed more recently by Long's book on martingale Hardy spaces, and Bass's "*Probabilistic techniques in analysis*". All these books contain, as the book under review, a more or less detailed account of central topics of the field, such as martingale inequalities, martingale transforms, and the duality theory of martingale spaces along with spaces of martingales of bounded oscillation. Weisz' book is the only one in which besides harmonic functions and martingales also pluriharmonic functions and multiparameter martingales are treated, and which contains fine results about applications of tree martingales to Walsh and Vilenkin series.

And the book has another special feature. Although the theory of multi-parameter and tree martingales presents many difficulties not encountered for one-parameter martingales, Weisz provides a method of investigation which gives a basis for a common treatment of the three different kinds: he further develops the technique of “atomic decompositions” due to Herz, Bernard, and Maisonneuve.

Chapter 1 is preliminary and introduces Vilenkin systems of orthogonal functions. Based upon the technique of atomic decompositions which is presented in the first sections, in Chaps. 2 and 3 a parallel presentation of the theory for one- and two-parameter martingales becomes possible. After proving the central inequalities, Weisz introduces the various martingale spaces, and investigates “bounded oscillation” and with the help of this notion the dual martingale spaces. He finally studies previsible transforms and locates them in the hierarchy of his spaces. In Chap. 4, tree martingales are investigated. As far as applications to Fourier analysis are concerned, this is the heart of the book. For example, the martingale results obtained allow convergence results to be deduced for Walsh–Fourier and Vilenkin–Fourier systems. Chapter 5 is devoted to interpolation of martingale Hardy spaces, and applied in Chap. 6 to multi-dimensional systems of the Walsh or Vilenkin type.

In spite of the highly technical and specialized level, the book should be accessible to both probabilists and analysts and is readable with only a few prerequisites. It is carefully written, although in places a little concentrated and dry. Especially analysts reading this book will enjoy making the acquaintance of an elegant and powerful tool from another area of mathematics.